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DEPARTMENT OF DEFENSE

HANDBOOK

AIRCRAFT FUEL SYSTEM SERVICE LIFE EXTENSION

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This handbook provides a systematic process to assess the Aircraft Fuel System (AFS) for design, overall condition, extend service life, and sustain airworthiness. It is aligned with the Mechanical Equipment and Sub-Systems Integrity Program (MECSIP), MIL-STD-1798, and makes extensive use of lessons learned from military and industry sources. This handbook provides basic AFS descriptions, common component failure modes, failure mechanisms, causes, and mitigation strategies that can be applied to AFS design. This handbook also defines approaches to address sustainment issues, which will promote the continued Operational Safety, Suitability, and Effectiveness (OSS&E) of the fuel systems throughout all phases of the aircraft's life when applied correctly.

3. Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB OH, 45433-7017, or emailed to Engineering.Standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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TABLE OF CONTENTS

FOREWORD ..... **II**

1. SCOPE **1**

1.1 SCOPE 1
1.2 PURPOSE. 1

2. APPLICABLE DOCUMENTS **2**

2.1 GENERAL. 2
2.2 GOVERNMENT DOCUMENTS. 2
2.2.1 Specifications, standards, and handbooks. 2
2.2.2 Other Government documents, drawings, and publications. 3
2.3 NON-GOVERNMENT PUBLICATIONS. 4

3. DEFINITIONS **5**

3.1 ACRONYMS. 5
3.2 DEFINITION OF TERMS. 7

4. CORE PROCESS TASKS **8**

4.1 INTRODUCTION. 8
4.2 BACKGROUND. 8
4.2.1 Generic fuel system overview. 9
4.3 APPROACH. 9
4.4 PROGRAM OVERVIEW. 9
4.5 APPLICABILITY. 10
4.6 TASK 1: AIRCRAFT FUEL SYSTEM & COMPONENT IDENTIFICATION. 10
4.7 TASK 2: COLLECT AND ANALYZE AFS FAILURE AND MAINTENANCE DATA. 11
4.8 TASK 3: REVIEW COMMON FAILURE MODES OF KEY AFS COMPONENTS. 12
4.9 TASK 4: APPLY GUIDELINES FOR DESTRUCTIVE AND NON-DESTRUCTIVE AFS COMPONENT ASSESSMENT. 12
4.10 TASK 5: RISK ASSESSMENT. 13
4.11 TASK 6: PLAN OF ACTION. 13

5. DETAILED REQUIREMENTS DEVELOPMENT **13**

5.1 APPENDIX A. 13
5.2 APPENDIX B. 13
5.3 APPENDIX C. 14
5.4 APPENDIX D. 14
5.5 APPENDIX E. 14
5.6 APPENDIX F. 14
5.7 APPENDIX G. 14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

5.8 APPENDIX H. 14
5.9 APPENDIX I. 14

6. NOTES 14

6.1 INTENDED USE. 14
6.2 SUBJECT TERM (KEY WORD) LISTING. 14
6.3 ADDITIONAL RESOURCES. 15

APPENDIX A 16

A.1 SCOPE 16

A.1.1 SCOPE 16
A.1.2 SOURCE INFORMATION. 16

A.2 AFS COMPONENTS AND FUNCTIONS 17

A.2.1 FUEL PUMPS FUNCTIONAL. 17
A.2.2 TRANSFER PUMPS. 17
A.2.2.1 Air pressure powered fuel transfer. 17
A.2.3 PUMP CATEGORIES. 17
A.2.4 FUEL LINES. 18
A.2.5 CONNECTIONS/SEALING. 18
A.2.6 TRANSFER VALVES. 18
A.2.7 FIRE SHUTOFF VALVES. 18
A.2.8 CHECK VALVES. 18
A.2.9 FUEL VENTING VALVES. 18
A.2.10. FUEL QUANTITY INDICATION SYSTEM. 18
A.2.10.1 Float level. 18
A.2.10.2 Capacitance probe fuel quantity instrumentation system. 19
A.2.10.3 Densitometer probe fuel quantity instrumentation system. 19
A.2.11 ULTRASONIC FUEL QUANTITY INSTRUMENTATION SYSTEM. 19
A.2.12 CENTER OF GRAVITY (CG) MANAGEMENT. 19
A.2.13 FUEL TANK. 19
A.2.14 FUEL TANK INERTING AND FIRE SUPPRESSION SYSTEMS. 19
A.2.15 RETICULATED FOAM. 20
A.2.16 REFUELING. 20
A.2.17 BOOM AND HOSE REEL AERIAL REFUELING. 21
A.2.18 DOUBLE BARRIERS. 21
A.2.19 FUEL LINE AND COMPONENT SEPARATION FROM POTENTIAL IGNITION SOURCES. 21
A.2.20 THERMAL MANAGEMENT. 21
A.2.21 DRAINAGE. 21
A.2.22 COMPATIBILITY WITH APPROVED FUELS. 21
A.2.23 FUEL TANK COATINGS. 21

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

A.3 SYSTEM SAFETY ASSESSMENT 22

A.3.1 FAILURE MODES, EFFECTS, AND CRITICALITY ANALYSIS (FMECA). 22
A.3.2 FAILURE FAULT TREES. 22
A.3.3 FISH BONES. 22
A.3.4 SINGLE POINT FAILURE. 22
A.3.5 CATASTROPHIC PHYSICAL FAILURE ASSESSMENT. 22
A.3.6 AFS SECONDARY EFFECTS OF FAILURE. 23
A.3.6.1 Flexible fuel hoses. 24
A.3.6.2 Fuel pump cooling. 24
A.3.6.3 F-16 Fuel shutoff valve 25
A.3.6.4 TWA 800. 25

A.4 COMPONENT CLASSIFICATION LIST 26

A.5 TASK RESULTS 26

APPENDIX B 27

B.1 SCOPE 27

B.1.1 SCOPE 27
B.1.2 MINING OF MAINTENANCE DATA. 27
B.1.3 COMPARING SUBSYSTEM RELIABILITY ON MULTIPLE PLATFORMS. 27
B.1.4 MAINTENANCE DATA TREND ANALYSIS APPROACH. 28

B.2 DATA SOURCES 28

B.3 DETERMINING SCALE OF EXISTING MAINTENANCE 29

B.4 VISUAL OBSERVATION 29

B.5 ANALYSIS 29

B.5.1 RELIABILITY ANALYSIS. 29
B.5.2 COMPARATIVE ANALYSIS. 30
B.5.3 STATISTICAL ANALYSIS. 30

B.6 ANALYSIS PROCESS 30

B.6.1 DATA MINING PROCESS. 30
B.6.1.1 Data gathering example. 30

B.7 FIELD, DEPOT, AND ENGINEERING SURVEYS 32

B.8 GOALS FOR FILTERING OF DATA 32

B.9 MAINTENANCE DATABASES 32

B.10 IDENTIFY AVAILABLE DATA 33

B.11 KEYWORD SEARCH 34

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

B.11.1 DATA SELECTION. 34
B.11.2 DATA REVIEW. 34
B.11.2.1 Quality of gathered data. 35
B.11.2.2 Tier 1 data review. 35
B.11.3 TIER 2 DATA REVIEW. 36
B.12 DATA ANALYSIS 36
B.12.1 ANALYSIS TECHNIQUES. 36
B.12.2 FISH BONE DIAGRAMS. 36
B.12.3 PARTITIONING OF DATA. 38
B.12.4 U-CHART. 38
B.12.5 RELIABILITY PREDICTIVE ANALYSIS. 38
B.12.6 CURRENT RELIABILITY PERFORMANCE AND INITIAL RELIABILITY. 39
B.12.7 MAINTAINABILITY ANALYSIS. 39
B.12.8 AVAILABILITY ANALYSIS. 39
B.12.9 SUPPORTABILITY ANALYSIS. 39
B.12.10 PRODUCT QUALITY DEFICIENCY REPORTING ANALYSIS. 40
B.12.11 DATA FITTING. 40
B.12.12 CURRENT MAINTENANCE PLAN. 40
B.13 ADDITIONAL REFERENCES 40
APPENDIX C 41
C.1 SCOPE 41
C.1.1 SCOPE 41
C.1.2 OVERVIEW. 41
C.2 SAFETY CONCERNS WITH AGING SYSTEMS 41
C.2.1 COMMON SAFETY CHALLENGES. 41
C.2.2 SOURCE OF ACCIDENTS. 41
C.2.3 AIRCRAFT MISHAPS. 42
C.2.3.1 KC-135 fuel pump. 43
C.2.3.2 F-16 fuel leak/flame out. 43
C.2.3.3 B-1B mishap caused by failure of a fuel system control panel. 43
C.2.3.4 Nimrod. 44
C.2.3.5 TWA 800. 44
C.3 FOAM IN FUEL TANKS SUMMARY 45
C.4 FUEL TANK IGNITION SOURCES AND MITIGATION STRATEGIES 45
C.5 OTHER DESIGN CONCEPTS 45
C.5.1 DESIGN CRITERIA FOR IN-DIRECT FAILURE MODE EFFECTS. 46
C.5.2 COMPONENTS OPERATING IN A FLAMMABLE ZONE. 46

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

C.5.3 COMPONENTS OPERATING IN A FLAMMABLE FLUID LEAKAGE ZONE. 47
C.5.4 FLAMMABLE FLUID CARRYING LINE AND WIRE HARNESSES. 47
C.5.5 SERVICE LIFE ASSESSMENT OF HYDROMECHANICAL SYSTEMS. 47
C.5.6 FUEL CHARACTERISTICS AND COMPATIBILITY. 47
C.5.7 AIRCRAFT FUEL COMPATIBILITY WITH FUEL TANK MATERIALS. 48
C.5.7.1 Cadmium. 49
C.5.8 LIMITATIONS ON USE OF PROTECTIVE METALLIC COATINGS - MIL-STD-1568. 50
C.5.8.1 Cadmium. 50
C.5.8.2 Silver. 51
C.5.8.3 Solders. 51

C.6 REFERENCES 51

C.7 FAILURE MECHANISM DESCRIPTIONS 51

C.7.1 FUEL BOOST PUMP. 53
C.7.2 SEPARATION OF LINES CARRYING FLAMMABLE FLUIDS. 54
C.7.3 FUEL TANK FQIS. 56
C.7.4 NOZZLES. 58
C.7.5 FUEL BLADDER/FUEL CELL. 60
C.7.6 FUEL SHUTOFF VALVE ACTUATOR. 62
C.7.7 ACTUATOR, ELECTROMECHANICAL LINEAR (LVT). 64
C.7.8 A/R MASTER SWITCH. 65
C.7.9 HOSE ASSEMBLIES. 66
C.7.10 TOPCOAT DEGRADATION. 68
C.7.11 FUEL SHUTOFF VALVE. 70
C.7.12 VALVE, FUEL PRESSURE REGULATOR. 72
C.7.13 VALVE ASSEMBLY, FUEL SWING CHECK (TRANSFER VALVE). 73
C.7.14 VALVE, SWING CHECK (CWT OVERRIDE PUMP). 74
C.7.15 VALVE, SWING CHECK (A/R PUMP). 75
C.7.16 ACTUATOR, ELECTROMECHANICAL LINEAR (VALVES 1&2). 76
C.7.17 INERTING SYSTEM AND FLAMMABILITY. 78
C.7.18 REFUELING BOOM SIGNAL COIL SYSTEM. 80
C.7.19 FAILURE MODE: CONNECTOR CORROSION. 82
C.7.20 APU PUMP. 83
C.7.21 FAILURE MODE: ELECTRICAL ARCING AT HOUSING INTERFACES. 85
C.7.22 FAILURE MODE: NOZZLE LEAKS. 87
C.7.23 COMPONENT: O-RINGS. 89
C.7.24 FAILURE MODE: FOD. 91
C.7.25 REFERENCE JSSG-2009. 93
C.7.26 BYPASS VALVE. 95
C.7.27 FUEL TUBE FAILURE. 97

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

C.8 TEST ASSESSMENTS 99

C.8.1 TEST ASSESSMENT: DIELECTRIC ASSESSMENT. 100
C.8.2 TEST ASSESSMENT: BONDING. 102

C.9 CASE STUDY ON KC-135 FUEL TANK EXPLOSION 103

C.9.1 INTRODUCTION. 103
C.9.2 INVESTIGATION. 104
C.9.3 RECOMMENDATIONS. 107
C.9.4 OBSERVATIONS. 108

C.10 F-16 MAIN FUEL SHUTOFF VALVE MISHAPS 110

C.10.1 INTRODUCTION. 110
C.10.2 BACKGROUND. 110
C.10.3 INVESTIGATION. 110
C.10.3.1 Connector assessment. 111
C.10.4 MITIGATION. 112
C.10.5 SUMMARY. 113

C.11 B-1B MISHAP CAUSED BY FAILURE OF A FUEL SYSTEM CONTROL PANEL 114

C.11.1 INTRODUCTION. 114
C.11.2 ACCIDENT INVESTIGATION. 114
C.11.3 CIRCUIT CARD EVALUATION. 115
C.11.4 ACTIONS TAKEN. 117
C.11.5 SUMMARY. 118
C.11.6 REFERENCES. 118

C.12 NIMROD CASE STUDY 119

C.12.1 INTRODUCTION. 119
C.12.2 MISHAP INVESTIGATION. 119
C.12.3 DESIGN FLAWS. 120
C.12.4 CONCLUSIONS FROM INVESTIGATING BOARD. 121

C.13 TWA 800 INFLIGHT FUEL TANK EXPLOSION MISHAP 123

C.13.1 INTRODUCTION. 123
C.13.2 INVESTIGATION. 123
C.13.3 IGNITION. 124
C.13.4 RECOMMENDATIONS. 126

C.14 C-21 TRANSPORT AIRCRAFT FUEL IMBALANCE MISHAP 127

C.14.1 CASE INTRODUCTION. 127
C.14.2 AIRCRAFT INFORMATION. 127

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

C.14.3 USAF FINDINGS. 127
C.14.3.1 Crew. 127
C.14.3.2 Maintenance personnel. 128
C.14.4 RECOMMENDATIONS. 128
C.14.5 COMPONENT ASSESSMENT. 129
C.14.5.1 On-site evaluation. 129
C.14.5.2 Pump. 130
C.14.5.3 Fuel pump control relay. 130
C.14.5.4 Component assessment conclusion. 132
C.14.6 DESIGN EVALUATION. 133
C.14.7 MAINTENANCE EVALUATION. 133
C.14.8 CONCLUSIONS. 133

C.15 FOAM IN FUEL TANKS SUMMARY 135

C.15.1 INTRODUCTION. 135
C.15.2 REFERENCES. 137

C.16 B-1B FUEL LEAK 138

C.16.1 INTRODUCTION. 138
C.16.2 BACKGROUND. 138
C.16.3 INVESTIGATION. 140
C.16.4 MITIGATION. 141

APPENDIX D 143

D.1 SCOPE 143

D.1.1 SCOPE 143

D.2 COMPONENT HANDLING OVERVIEW 143

D.2.1 RULES OF THUMB. 143
D.2.2 TYPICAL PRACTICES. 143

D.3 AFS COMPONENT AGING ANALYSIS 144

D.4 PROJECT FLOW FOR FAILURE ANALYSIS 144

D.5 FAILURE ANALYSIS PROCESS FOR A FUEL PUMP 146

D.5.1 CONDUCT NONDESTRUCTIVE EVALUATION. 146
D.5.2 PUMP FAILURE ANALYSIS TEARDOWN: 146
D.5.3 CONDUCT ELECTRICAL TESTS: 146
D.5.4 REVIEW RESULTS AND PROCEED WITH TEAR DOWN. 147

APPENDIX E 148

E.1 SCOPE 148

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

E.1.1 SCOPE 148
E.2 RCM INPUTS 148
E.3 EVALUATION EXAMPLE 148
E.3.1 SCOPE 149
E.3.2 FAILURE COUNT AND IMPACT. 149
E.3.3 MISSION IMPACT. 150
E.4 MAINTENANCE ACTION CODE 151
E.5 TOTAL FAILURE EVALUATION 152
E.6 REFERENCES 156
APPENDIX F 158
F.1 SCOPE 158
F.1.1 SCOPE 158
F.2 RISK MITIGATION STRATEGIES 158
F.2.1 MATERIALS CHANGE. 158
F.2.2 DESIGN CHANGES. 158
F.2.2.1 Development and validation of physical failure mitigation strategies. 158
F.2.3 REPLACEMENT. 159
F.2.4 MAINTENANCE CHANGES. 159
F.3 COMPONENT REPAIR PROCEDURE CHANGES 159
F.4 FORECASTING REPLACEMENT 159
F.5 PERIODIC UPDATE PROCESS 159
F.5.1 AFS COMPONENT REASSESSMENT. 159
F.5.2 DEVELOP ASSESSMENT METRICS. 160
F.5.3 ACTION PLAN IMPLEMENTATION ASSESSMENT. 160
F.5.4 PERIODIC REASSESSMENT. 160
F.5.5 CHANGE IN RISK TOLERANCES. 160
F.5.6 DESIGN LIFE REVIEW. 160
F.5.7 SYSTEM UPGRADES. 160
F.5.8 MISSION CHANGE. 160
APPENDIX G 161
G.1 SCOPE 161
G.1.1 SCOPE 161
G.2 INSTALLATION – FEDERAL AVIATION REGULATION (FAR) 25.901 161

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Check the source to verify that this is the current version before use.

G.3 FUEL TANK IGNITION PREVENTION – FEDERAL AVIATION REGULATION (FAR) PART 25.981 162

G.3.1 ADVISORY CIRCULARS (AC). 162

G.4 ELECTRICAL CONNECTORS 163

G.5 COMPONENTS 164

G.5.1 CONDUITS, WIRING, AND SYSTEMS. 164
G.5.2 CIRCUIT BOARDS. 166
G.5.3 CIRCUIT BREAKERS. 166
G.5.4 PUMPS. 167
G.5.5 MOTOR-OPERATED VALVES 169
G.5.6 FUEL QUANTITY PROBES 169
G.5.7 FLOAT SWITCHES 169
G.5.8 FUEL TUBES AND COUPLINGS. 170
G.5.9 BONDING & STRAPS. 170
G.5.10 PNEUMATIC SYSTEM. 171

G.6 FUEL AND ADDITIVES 171

G.6.1 FLAMMABILITY. 172
G.6.2 FUEL HEATING. 173

G.7 LIGHTNING & EMI 173

G.7.1 FLOW-INDUCED STATIC ELECTRICITY 174

G.8 ADDITIONAL MITIGATIONS AND TESTING 175

APPENDIX H 177

H.1 SCOPE 177

H.1.1 SCOPE 177

H.2 BONDING STANDARDS 177

H.2.1 MIL-STD-464. 177
H.2.1.1 Mechanical interfaces. 177
H.2.1.2 Shock, fault, and ignitable vapor protection. 178

H.3 MIL-STD-1798. 178

H.4 COMMERCIAL APPROACH TO BONDING 178

H.4.1 BONDING MEASUREMENTS. 179
H.4.2 UNDERSTANDING BONDING. 179
H.4.3 NEED FOR HIGH CURRENT 180
H.4.4 HIGH CURRENT APPLICATIONS. 182

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Check the source to verify that this is the current version before use.

APPENDIX I 184

I.1 SCOPE 184

I.1.1 SCOPE 184
I.1.2 INTRODUCTION 184

CONCLUDING MATERIAL 212

LIST OF TABLES

TABLE I- I. Gravity transfer valve. 185
TABLE I- II. Pump assembly. 186
TABLE I- III. Hose assembly. 187
TABLE I- IV. Line valves. 188
TABLE I- V. Fuel pressure regulator. 189
TABLE I- VI. Valve assembly, fuel swing check. 190
TABLE I- VII. Actuator. 191
TABLE I- VIII. APU fuel pump. 192
TABLE I- IX. Bladders. 194
TABLE I- X. Boom equipment. 195
TABLE I- XI. FQIS. 205

LIST OF FIGURES

FIGURE C- 1. USAF Class A aircraft mishap causes 2008-2013 compiled from publically available accident reports. 42
FIGURE C- 2. Chart from http://www.PlaneCrashInfo.com provides info on worldwide commercial accidents. 42
FIGURE C- 3. Indirect failure modes. 46
FIGURE C- 4. Typical fuel compatibility summary table from AFRL/RXS 07-085 (limited distribution). P44 is a low temperature fuel additive. 49
FIGURE C- 5. Example of cadmium plated fasters in a fuel tank that have been degraded by fuel (see AFRL/RXS 10-097). 50
FIGURE C- 6. Remains of the KC-135 after the aircraft exploded. 103

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Check the source to verify that this is the current version before use.

FIGURE C- 7. Example of the boost fuel pump suspected to have initiated the aircraft explosion. 104
FIGURE C- 8. Exemplar boost pump. 105
FIGURE C- 9. Radiograph of a fuel pump removed from inventory. 105
FIGURE C- 10. Mishap pump from exploded aircraft. 106
FIGURE C- 11. Arced copper wire and enlarged flame arrestor hole from the mishap pump which contained a nugget of aluminum and copper. 106
FIGURE C- 12. Lab simulation of an arc inside an aluminum housing (same alloy and thickness as the mishap pump housing) with a 1/32 inch flame arrestor hole 107
FIGURE C- 13. Fuel off valve connector showing corrosion on the tin-plated steel electrical pins. Note the internal O-ring is intact. 111
FIGURE C- 14. Three of the six tin plated steel pins have been corroded and broken away from the connector. Note the internal O-ring is missing (see photo above). 112
FIGURE C- 15. Fire warning extinguisher panel (FWEP). 114
FIGURE C- 16. Mishap PWB with melting and arcing of copper traces and carbonized epoxy which resulted in 28VDC being applied to the fuel cut-off valve control relays. 115
FIGURE C- 17. Close-up of the carbonized PWB and location of traces that control the fuel cut off relays. 116
FIGURE C- 18. Diagram of assessed connector. 116
FIGURE C- 19. Cross-Feed duct installed in aircraft between 1969-1975. Also note location of No. 7 dry bay. 120
FIGURE C- 20. [Left] No. 7 tank dry bay (lower panel). [Right] No. 7 tank dry bay (internal drain holes) 121
FIGURE C- 21. Comparison of center wing tank ullage temperatures. 123
FIGURE C- 22. Diagram of fuel probe system. 124
FIGURE C- 23. Photos of damaged wire and analysis. 124
FIGURE C- 24. Fuel probe terminal block removed from retired Boeing747 aircraft. 125
FIGURE C- 25. Reprinted from Flight Safety Foundation, Accident Prevention, Fuel Imbalance Cited in Learjet 35A Control Loss, Vol. 57 No. 2, prepared by FSF Editorial staff. 129
FIGURE C- 26. [Left] Left hand standby pump. [Right] Intact mishap right hand standby pump. 130

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Check the source to verify that this is the current version before use.

FIGURE C- 27. Right side standby fuel pump control relay. 131
FIGURE C- 28. [Left] The mishap right pump control relay. 131
FIGURE C- 29. Scanning electron micrographs of the mishap right pump control relay contacts showing melting of the contacts but no evidence of welding. 132
FIGURE C- 30. Typical foam installation in a fuel tank. 135
FIGURE C- 31. Fuel spill B-1B TN 85-0081. 138
FIGURE C- 32. B-1B fuel tube configuration representation. 139
FIGURE C- 33. W700-series flexible fuel couplings. 139
FIGURE C- 34. Tube with sweged fitting ferrule 140
FIGURE C- 35. Tube with machined integral ferrule. 140
FIGURE C- 36. Fuel tube. 140
FIGURE C- 37. Axial misalignment. 141
FIGURE C- 38. Scalloping. 142

FIGURE E- 1. Average monthly total non-mission capable maintenance (TNMCM) time. 150
FIGURE E- 2. Fuel system total non-mission capable maintenance (TNMCM). 150
FIGURE E- 3. In-flight and ground aborts caused by failure of system (2009 – 2010). 151
FIGURE E- 4. Top ten drivers of aborts by failure mode (2009 – 2010). 151
FIGURE E- 5. Type of maintenance. 152
FIGURE E- 6. Prominent component unscheduled maintenance. 152
FIGURE E- 7. Distribution and count of failures. 153
FIGURE E- 8. Prominent failure modes and number of failures. 153
FIGURE E- 9. Top ten failure modes by cost per flight hour. 154
FIGURE E- 10. System work unit code accuracy. 154
FIGURE E- 11. Component work unit code accuracy. 155
FIGURE E- 12. FMECA compilation spreadsheet. 155
FIGURE E- 13. Failure risk index (FRI). 156
FIGURE E- 14. Level of analysis matrix 156

FIGURE H- 1. Typical fuel pump with multiple bonding interfaces. Note fastener bolts and safety wire (potential FOD). 180

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Check the source to verify that this is the current version before use.

FIGURE H- 2. Fault current vs maximum allowed resistance for bonding between equipment and structure..... 181
FIGURE H- 3. Four-point probe setup for accurately measuring the bonding resistance..... 182

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1. SCOPE

1.1 Scope.

This handbook was developed using information primarily from an aircraft tanker. The basic concepts should also apply to most manned transports, rotary wing aircraft, fighters, bombers, and Unmanned Aircraft Systems (UAS).

1.2 Purpose.

This handbook provides a systematic process to assess the AFS for design, overall condition, extend service life, and sustain airworthiness. For purposes of this handbook, the AFS includes electrical and hydraulically driven fuel pumps, jet pumps, valves, control relays, fuel lines, fuel filters, fuel properties and impact on AFS components, fuel tank inerting systems, fuel quantity probes and measurement system, fuel tank electrical bonding, and refueling system components. It is aligned with the Mechanical Equipment and Sub-Systems Integrity Program (MECSIP), MIL-STD-1798, is based on Reliability Centered Maintenance (RCM) principles, and makes extensive use of lessons learned from military and industry sources. Along with the extensive lessons learned provided herein, this handbook contains a framework to achieve and maintain the physical and functional integrity of the AFS. Processes will normally be tailored to meet specific platform, program office, system and/or subsystem, requirements, and constraints. The process and core tasks should also be tailored relative to platform status; whether in design, newly fielded, or based on years in sustainment. By using this process, program offices and maintenance organizations should have the necessary information to perform a Service Life Extension Program (SLEP) and initiate appropriate trades relative to the cost of modification or integrity initiatives versus required performance, maintenance and mission impact, total operating cost, and aircraft availability.

Airworthiness is addressed in different ways by each of service. In the USAF, per AFI 63-101/20-101, the Program Manager is responsible for OSS&E of the system managed. The USAF Technical Airworthiness Authority reviews the design based airworthiness of USAF aircraft (including modifications to those aircraft) and provides the airworthiness release documents (e.g. military type certificate) under which aircraft operate.

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2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5578	Tanks, Fuel, Aircraft, Self-Sealing
MIL-DTL-6396	Tanks, Fuel, Oil, Cooling Fluids, Internal, Removable Non-Self-Sealing
MIL-F-17874	Fuel Systems, Aircraft, Installation and Test of
MIL-DTL-25988	Rubber, Fluorosilicone Elastomer, Oil-and-Fuel Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes
MIL-DTL-27422	Tank, Fuel, Crash-Resistant, Ballistic-Tolerant, Aircraft
MIL-DTL-62011	Pump, Fuel, Electrically-Operated: 24-Volt DC, In-Tank Type, General Specification for
MIL-PRF-81975	Couplings, Regulated, Aerial Pressure Refueling Type MA-2, Type MA-3 and Type MA-4
MIL-DTL-83054	Baffle and Inerting Material, Aircraft Fuel Tank (Inactive for new design)
MS24354	Drogue Cone, Nozzle and Reception Coupling - Type MA-2 Flight-Pressure-Refueling System, Assembly of (Inactive for new design)
JSSG-2009	Air Vehicle Subsystems

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202-301	Method 301, Dielectric Withstanding Voltage
MIL-STD-202-302	Method 302, Insulation Resistance
MIL-STD-202-303	Method 303, Dc Resistance
MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1798	Mechanical Equipment and Subsystems Integrity Program

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-516	Airworthiness Certification Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications.
The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

AIR FORCE TECHNICAL ORDER

T.O. 1-1-3	Technical Manual, Inspection and Repair of Aircraft Integral Tanks
T.O. 6R9-10-25-3	Overhaul Instructions, Motor Operated Gate Valve AV-16B Series, P/N AV-16B1139A and Similar Valves
T.O. 8D1-14-52-3	Overhaul, Actuator Assembly, Part Numbers,102842A, 102842AT, 102842AW,102842B, 102842BM, 102842CN, 102843AH, 102843AQ, 102843T, 102843Y

(Copies of these documents are available online at https://etims.cce.af.mil/ETIMS/index.jsp.)

DATA ITEM DESCRIPTION

DI-SESS-82329	Mechanical Equipment and Subsystems Integrity Program (MECSIP) Master Plan (MP)

(Copies of this document are available online at https://quicksearch.dla.mil.)

CODE OF FEDERAL REGULATIONS

14CFR Appendix M to Part 25	Fuel Tank System Flammability Reduction Means
14CFR Appendix N to Part 25	Fuel Tank System Flammability Exposure and Reliability Analysis
14CFR FAR Part 25.863, Subpart E, Fire Protection	Flammable Fluid Fire Protection
14CFR FAR Part 25.901, Subpart E, Powerplant	Installation
14CFR FAR Part 25.954, Subpart E, Fuel System	Fuel System Lightning Protection
14CFR FAR Part 25.981, Subpart E, Fuel System	Fuel Tank Explosion Prevention
14CFR FAR Part 25.1707, Subpart H, Electrical Wiring Interconnection Systems (EWIS)	System Separation: EWIS
14CFR FAR Part 26, Subpart D, Fuel Tank Flammability	Fuel Tank Flammability
14CFR FAR Part 26.5, Subpart A, General	Applicability Table
14CFR FAR Part 26.33, Subpart D, Fuel Tank Flammability	Holders of Type Certificates: Fuel Tank Flammability
14CFR FAR Part 21, Special Federal Aviation Regulation (SFAR)-88	Fuel Tank System Fault Tolerance Evaluation Requirements

(Copies of these documents are available online at http://www.ecfr.gov.)

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FEDERAL AVIATION ADMINISTRATION (FAA)

AC 25-8	Auxiliary Fuel Systems Installations
AC 25.981-1	Fuel Tank Ignition Source Prevention Guidelines
AC 25.981-2	Fuel Tank Flammability
AC 25.1309-1	System Design and Analysis
AC 25.1701-1	Certification of Electrical Wiring Interconnection Systems on Transport Category Airplanes

(Copies of these documents are available online at https://www.faa.gov.)

2.3 Non-Government publications.
The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM D149	Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
ASTM D3755	Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials under Direct-Voltage Stress

(Copies of these documents are available online at http://www.astm.org.)

RTCA

RTCA DO-160	Environmental Conditions and Test Procedures for Airborne Equipment

(Copies of these documents are available online at http://www.rtca.org/.)

SAE INTERNATIONAL

SAE AIR1662	Minimization of Electrostatic Hazards in Aircraft Fuel Systems
SAE AIR4170	Reticulated Polyurethane Safety Foam Explosion Suppressant Material for Fuel Systems and Dry Bays
SAE AIR5128	Electrical Bonding of Aircraft Fuel Systems
SAE-AMS-P-5315	Acrylonitrile-butadiene (NBR) Rubber For Fuel-Resistant Seals 60 to 70
SAE ARP594	Fuel Pump Thermal Safety Design
SAE ARP6179	Evaluation of Gas Turbine Engine Lubricant Compatibility with Elastomer O-Rings
SAE ARP8615	Fuel System Components: General Specification For
SAE AS568	Aerospace Size for O-Rings

(Copies of these documents are available online at http://www.sae.org/.)

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3. DEFINITIONS

3.1 Acronyms.

AC	Advisory Circular
ACSG	Aircraft Sustainment Group
AD	Airworthiness Directives
AFB	Air Force Base
AFI	Air Force Instruction
AFRL	Air Force Research Laboratory
AFS	Aircraft Fuel System
AFSAS	Air Force Safety Automated System
AIR	Aerospace Information Report
AMARG	Aerospace Maintenance and Regeneration Group
AR	Aerial refueling
ARP	Aerospace Recommended Practice
ARR	Aerial refueling Receiver
AT	Action Taken
AWB	Airworthiness Bulletin
CCA	Common Cause Assessment
CCL	Component Classification List
CDCCL	Critical Design Configuration Control Limitations
CG	Center of Gravity
CSI	Critical Safety Items
DCC	Durability Critical Components
DLA	Defense Logistics Agency
DOT	Department of Transportation
EA	Engineering Assignments
ECS	Environmental Control Systems
EMI	Electromagnetic Interference
EWIS	Electrical Wire Interconnection System
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulation
FEET	Flammability Exposure Evaluation Time
FHA	Failure Hazard Assessment
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes, Effects, and Criticality Analysis
FOD	Foreign Object Damage
FSIP	Functional Systems Integrity Plan
FQIS	Fuel Quantity Indication System
FWEP	Fire Warning Extinguisher Panel
GCSS	Global Combat Support System
GFI	Ground Fault Interrupter
HA	Hazard Analysis
HIRF	High Intensity Radiated Fields
HMAL	How Malfunction
HRI	Hazard Risk Index

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IHS	Information Handling Standards
IMDS	Integrated Maintenance Data System
JDRS	Joint Deficiency Reporting System
KPI	Key Performance Issue
LIMS-EV	Logistics Installation Mission Support Enterprise View
LFL	Lower Flammability Limit
LRU	Line Replaceable Unit
MC	Mission Capable
MCC	Mission Critical Components
MECSIP	Mechanical Equipment and Subsystems Integrity Program
MFHBF	Mean Flying Hours Between Failure
MICAP	Mission Impaired Capability Awaiting Parts
MRA	Mission Risk Assessment
MSG	Maintenance Steering Group
NALDA	Naval Aviation Logistics Data Analysis
NMCS	Non-Mission Capable for Supply
NMCM	Non-Mission Capable for Maintenance
NMCB	Non-Mission Capable for Both Supply and Maintenance
NSN	National Stock Number
PWIG	Product Improvement Working Group
PDM	Programmed Depot Maintenance
P/N	Part Number
PSSA	Preliminary System Safety Assessment
QPA	Quantity Per Aircraft
RCM	Reliability Centered Maintenance
REMIS	Reliability Maintenance Information System
RMS	Root Mean Squared
RTCA	Radio Technical Commission for Aeronautics
SMART	Supply Management Analysis Reporting Tool
SCC	Safety Critical Component
SFAR	Special Federal Aviation Regulation
SLAP	Service Life Assessment Program
SLEP	Service Life Extension Program
STD	Standard
SIB	Safety Investigation Board
SSA	System Safety Assessment
TAA	Technical Airworthiness Authority
TCTO	Time Compliant Technical Order
TNMCM	Total Non-Mission Capable Maintenance
T.O.	Technical Order
UFL	Upper Flammability Limit
USAF	United States Air Force
VAC	Voltage Alternating Current
VDC	Voltage Direct Current
WD	When Discovered
WUC	Work Unit Code

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3.2 Definition of terms.

Aerial Refueling Receiver (ARR) System:	A fuel system subsystem of tubing, valves, and controls used to fill and distribute fuel among tanks from an aerial source.
Aerial Refueling Tanker (ART) System:	A fuel system subsystem of pumps, tubing, valves, and controls used to offload fuel to a receiver aircraft.
Defuel System:	A fuel system subsystem of pumps, tubing, valves, and controls used to offload fuel to a ground system truck.
Drain System:	A fuel system subsystem of tubing and valves used to drain tanks and tubing by gravity.
Feed System:	A fuel system subsystem of tubing, couplings, pumps, valves, electrical components, and controls used to provide fuel to one or more engines, up to and including the engine fire shutoff valve.
Fish Bone:	A diagram with a horizontal line connecting radiating tree structure of lines, where each is labeled with a fault cause in a progressively higher level of detail.
Fuel Dump System:	A fuel system subsystem of pumps, tubing, valves, and controls used to offload fuel in-flight, to the atmosphere.
Fuel Quantity Indication System (FQIS):	A fuel system subsystem of sensors, wiring, processors and gages used to indicate fuel quantity in tanks.
Fuel System:	Tubing, couplings, pumps, valves, gages, sensors, electrical and hydraulic components, and controls used to provide fuel to one or more engines, up to and including the engine fire shutoff valve.
Fuel Transfer System:	A fuel system subsystem of tubing, valves, and controls used to redistribute fuel among tanks.
Fuel Vent System:	A system that allows air and fuel to flow into and out of fuel tanks to maintain pressure within safe limits. During refueling operation this system may act as a means to maintain fuel tank pressures within limits if the refueling system fails to stop the refueling operation when the maximum tank capacity is reached.
Gravity Transfer System:	A fuel system subsystem of tubing and valves used to redistribute fuel among tanks by gravity.
Refueling System:	A fuel system subsystem of tubing, valves, and controls used to fill and distribute fuel among tanks. The source may be ground or aerial.
Storage System:	A fuel system subsystem of tanks, bladders (or cells), external tanks, fuel seal plane, “top coat” or paint, and sealant. The tank metal is not included, since it is part of Structure. Sealant is usually controlled on Structure drawings, but is in the purview of Fuel Systems due to sensitivity to fuel type.

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4. CORE PROCESS TASKS

4.1 Introduction.

The processes/tasks in this section support an approach to assess fuel systems design, maintenance, and service life extension, and to establish/sustain continued airworthiness. The tasks should be tailored to specific program office needs and may include all or a combination of these tasks as necessary. Much of the information contained in the appendices have been derived from experience and assessments of the 2013 REMIS database.

There have been several well documented aircraft incidents that have been caused by the failure of AFS components, poor maintenance of these components, and/or poor system design. From each of these failures the hard lessons learned can be applied to improve the airworthiness of new and existing aircraft. Previous aircraft incidents are identified/provided as part of Task 3 for consideration to prevent similar problems within the design process or sustainment processes.

The approach discussed in this handbook has been reviewed by members of the MECSIP organization.

4.2 Background.

The AFS is the largest liquid carrying system on the aircraft. It powers the engine, and is a potential ignition source that has the potential to destroy the aircraft when not properly designed or maintained. Regular maintenance and sustainment activities of the aircraft fuel system are necessary for aircraft airworthiness. There have been major aircraft mishaps related to fuel systems, yet there is no formal military or commercial document dedicated to maintaining the air worthiness of the AFS. The AFS is included in the Mechanical Subsystems Integrity Program (MECSIP) standard, MIL-STD-1798.

This document includes a process for equipment assessment, which can be tailored to specific aircraft types. Most of the information has been derived from a KC-135 tanker, yet can be applied to other aircraft types. This handbook primarily focuses on fuel component failure modes, failure mechanisms, and the failure impact on performance and overall safety of an aircraft.

Single page guides (see APPENDIX C) provide a fuel system component review for most aircraft types. In each of these single page guides is information that can be used to direct design, sustainment activities, and references that can be used for determining equipment operation and best sustainment practices.

Single page guides typically include the following sections:

a. Common Failure Modes: The common failure modes section provides a short description of the common failures found in the particular equipment based on in-service maintenance reports.

b. Potential Causes: Based on the items identified in the common failure modes, the potential causes section lists items identified to have produced the failure event. In addition, information is included as to when these types of failures are likely to occur in the equipment lifecycle.

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c. Failure Mitigation Strategies: This section contains information on lessons learned for AFS sustainment. This will often include references and quotes for sustainment technical orders (T.O.s).
d. Other Information: This contains a set of pertinent information associated with the component.
e. Equipment Sustainment: This contains a brief summary of tasks that could be performed for equipment overhaul and sustainment.
f. References: List of references that can be used for equipment sustainment, assessment, or existing maintenance practices for this equipment.

4.2.1 Generic fuel system overview.

The fuel feed subsystem delivers fuel delivers fuel to the engines:

1. Engine feed
2. Fuel transfer
3. Refuel/Defuel
4. Fuel vent
5. Fuel quantity gauging

The system sometimes includes:

1. Aerial refuel subsystem
2. Active center-of-gravity management subsystem

The fuel feed subsystem includes the fuel feed subsystem which delivers fuel to the engines. Jet transport aircraft typically supply fuel to the engines via in-tank fuel boost pumps, usually two per tank. They pump fuel under pressure through a fire shutoff valve for each engine. The use of crossfeed valves provides the means for any tank to supply any engine. Boost pump bypasses allow fuel flow via engine pump suction, should a boost pump pair fail when the transfer system is not being utilized. A potential single-point failure in the engine feed system is the fire shutoff valve in the fuel line. This valve needs to be open to supply fuel to the engine.

Each system component is discussed more in APPENDIX A.

4.3 Approach.

The overall approach includes six tasks described more fully in section 4.

4.4 Program overview.

This handbook provides a systematic process to assess the Aircraft Fuel System (AFS) for overall condition, extend service life, and sustain airworthiness. It is aligned with the Mechanical Equipment and Sub-Systems Integrity Program (MECSIP), MIL-STD-1798 and makes extensive use of lessons learned from military and industry sources. This handbook contains a framework, to achieve and maintain the physical and functional integrity of the AFS. Processes in this handbook should be tailored to meet specific platform, program office, system and/or subsystem, requirements, and constraints. Process and core tasks should also be tailored relative to platform status; whether in design, newly fielded or based on years in sustainment. By using this process, program offices and maintenance organizations should have the necessary information to perform an AFS Service Life Extension Program (SLEP) and initiate appropriate trades relative to the cost

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of modification or integrity initiatives versus required performance, maintenance and mission impact, total operating cost, and aircraft availability.

4.5 Applicability.

This handbook applies to military aircraft tankers, transports, fighters, bombers, and Remotely Piloted Vehicles (RPAs). For purposes of this handbook, the aircraft fuel system includes electrical and hydraulically driven fuel pumps, jet pumps, valves, control relays, fuel lines, fuel filters, fuel properties and impact on AFS components, fuel tank inerting systems, fuel quantity probes and measurement system, and refueling system components. Stakeholders include MECSIP managers, commodity engineers, program engineers, depots, failure analysts, acquisition engineers, OEM design engineers, and field engineers.

Implementation of a program supporting service life extension efforts should begin early in the service life assessment process as the outputs from these tasks will provide the necessary information for data-driven decisions. Through the implementation of this process, the changes to maintenance procedures, component replacement, and/or design changes necessary for service life extension will become clearer.

Tasks 1 and 3 are of direct benefit to those involve involved in the design process for a new aircraft or aircraft modification/sustainment. Each task provides insights into system design, lessons learned, and factors that should be considered during part selection.

Each task has a corresponding appendix providing details on each of the identified subtasks. The appendices include examples of analysis and evaluations that have been performed supporting various AFS related efforts. It is highly recommended the appendices be reviewed prior to implementation of task.

4.6 Task 1: Aircraft fuel system & component identification.

This task focuses on the AFS engineering assessment to gather fuel system component information and ascertain the potential causes and severity of failure. This analysis may require multiple assessments if significant variations exist in the fleet. The potential failure causes are then used in subsequent subtasks as a guide for maintenance data evaluation (Task 2), component assessment (Tasks 3 and 4), and subsequent risk mitigation (Task 5), if applicable. The following six subtasks are further discussed in APPENDIX A.

1. Identify AFS components and materials and all safety critical fault paths. This requires a thorough review of the aircraft AFS and its components and will help to provide a better understanding of scale the AFS assessment.
2. Identify all safety critical functions and components that impact AFS function and physical integrity of surrounding systems, structure, aircraft and availability. The failure consequence of AFS components may impact other systems and potentially eliminate the safety margins gained by redundant systems.
3. Identify AFS failure fault trees. Failure fault trees help to both identify the potential causes of functional failure and what combination of events may be necessary to create a hazardous or catastrophic event (see A.3.2).
4. Identify AFS fish bones (see A.3.3).

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5. Identify AFS FMECAs for fuel systems and primary components. The best publicly available guidance for AFS FMECA comes from the FAA and is summarized in APPENDIX A.3.1. An example of an AFS FMECA is included in APPENDIX I.
6. Identify AFS single point failures. The combination of results from the primary and secondary failure impact analysis will help to identify AFS single-point failures. This will help to identify those components/functions that are to be more closely examined in subsequent tasks.

4.7 Task 2: Collect and analyze AFS failure and maintenance data.

The gathering and review of AFS maintenance and failure data is the core of Task 2. The goal is to gather data on component reliability to focus efforts in the subsequent tasks. Several potential data sources are discussed, as are methods for the data assessment. The ten subtasks are thoroughly discussed in APPENDIX B.

1. Document how the aircraft AFS failure and maintenance data is collected and analyzed. Though there is no one best way to mine data, there are some "best practices" available. However, for the sake of congruence between projects, a standard basic process is recommended. Guidance on gathering maintenance data is included in APPENDIX B.10.
2. Review and assess mishap and maintenance databases and applicable AFS Directives.
3. Interview maintenance and engineering support staff.
4. Use data mining approaches to examine maintenance and failure data. Maintenance actions are recorded in fleet specific databases. Database systems are continually updated and enhanced since they are integral to managing aircraft operationally. The USAF and USN have developed new data management systems that allow organizations to query aircraft platform maintenance data. Guidance for gathering this data is provided in APPENDIX B.11.
5. Review findings, maintenance actions, discrepancies, and repairs carried out as part of mandatory or voluntary inspections. The individual review of findings and maintenance actions can help to identify underlying issues with AFS components. It is likely that these details will be missed if only arrogate data analysis is performed.
6. Organize data by zone/station, probability, and seriousness of failure. To gather the most information from the maintenance data, it is recommended that the data is examined from several viewpoints. By grouping and examining the data by location, function, and/or failure severity, it may be possible to gain additional insights into AFS component reliability and maintenance trends.
7. Review failure analysis reports of previously document failure modes.
8. Track part usage through DLA. The history of component usage and maintenance can provide insights into the current condition of AFS components and may help to identify sources of problems.
   Review data in various systems such as the Air Force Safety Automated System (AFSAS) and Global Combat Support System (GCSS) USAF maintenance data.
9. Review data in FAA Service Difficulty Reports (SDRs). FAA SDRs provide maintenance information gathered from reports made by private maintainers; this can provide additional maintenance data for those seeking to extend the service life of commercial derivative aircraft.

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10. Review data in FAA Airworthiness Directives (ADs). All ADs should be reviewed when the maintenance program is being developed.

4.8 Task 3: Review common failure modes of key AFS components.

Task 3 seeks to use the data gathered from Tasks 1 and 2 to direct engineering review of AFS components. Lessons learned are used to direct teardown analysis of components (Task 4) and AFS risk assessment (Task 5). See APPENDIX C for more task information.

1. Review Encyclopedia of Mechanisms Section. A compilation of dozens of AFS component descriptions, failure modes, and sustainment recommendations are summarized in APPENDIX C.7. These provide high-level insight on what can be done to determine the condition of AFS components. Also included on each of these single-page guides is recommended further reading.
2. Identify areas of applicability to system descriptions of each AFS failure component and diagnostic mechanism.
3. Compare lessons learned to current system design. The lesson gathered on AFS components come from all types of aircraft and should be reviewed for applicability to current aircraft design. Options for teardown analysis and/or improved maintenance should be identified by the outcome of this task.

Also, other items that should be considered when reviewing the aircraft components and identifying potential safety recommendations should come from aircraft investigations. Some of the following incidents have had a significant impact on current engineering requirements and maintenance activities.

1. C.10. F-16 Main Fuel Shutoff Valve Mishaps 110
2. C.11. B-1B Mishap Caused by Failure of a Fuel System Control Panel 114
3. C.12. Nimrod Case Study 119
4. C.13. TWA 800 Inflight Fuel Tank Explosion Mishap 123
5. C.14. C-21 Transport aircraft Fuel Imbalance Mishap 127
6. C.15. Foam in Fuel Tanks Summary 135

4.9 Task 4: Apply guidelines for destructive and non-destructive AFS component assessment.

Select AFS components are removed from aircraft for in-lab evaluations with the objective of identifying life limiting conditions that may require proactive maintenance actions. Component assessment methods are discussed.

1. Identify a process for analyzing and developing an overall risk and life assessment of the AFS using the findings from the first three tasks. The objective is to examine parts of AFS components that have been identified as reducing component reliability (e.g. pump seals) and the reasons for the degradation. The degradation may be from normal use, but the analysis may identify other potential causes.
2. Perform inspection/analysis of AFS components. Several methods for component assessment are included in APPENDIX D.5. Not all of these are applicable for each

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component, but they provide a baseline for assessment methods. The selection of tests should be based on the component and its possible failure modes.
3. Lay out an approach to proactively identify emerging failure modes before they adversely impact safety or readiness.
4. Provide input to component incoming inspection and repair processes. The output from this work will directly support the decisions made in the risk assessment (see Task 5) and the action plan (see Task 6).

4.10 Task 5: Risk assessment.

Using the data collected from the previous four tasks, Task 5 focuses on developing the necessary information for a Reliability Centered Maintenance (RCM) approach. Examples are provided to guide the analysis. This task is fully covered in APPENDIX E.

1. Combine maintenance data review with teardown assessment for component condition and life prediction estimates.
2. Use failure severity information paired with component condition to generate risk assessment. The outcome from this task will be a risk ranking of AFS components and recommendation as to what should be considered for risk mitigation (see Task 6). Guidance on risk assessment performance is included in APPENDIX E.2.

4.11 Task 6: Plan of action.

Tailor and apply an action plan to reassess AFS and to ensure desired outcomes have been achieved and maintained. The tailoring and iterative application should also consider changes in aircraft configuration, specific program direction, and changes to operational requirements such as service life, mission, etc. Details are included in APPENDIX F.

1. Identify effective risk mitigation strategies. Those high risk AFS components identified in Task 5 are reviewed for potential risk reduction options. These changes may be to maintenance procedures, component service life, and/or design changes.
2. Establish a process for periodic updates. To ensure the effectiveness of this AFS service life extension effort, this process should be periodically reviewed to ensure the recommendations have been implemented. This will also provide feedback on the effectiveness of the recommendations generated by this work.
3. Identify which depot repair / overhaul shops should be scrutinized to enable improvements in safety and aircraft availability.

5. DETAILED REQUIREMENTS DEVELOPMENT

5.1 Appendix A.

Appendix A describes the elements for documenting the overall AFS and identify the mission and safety critical components.

5.2 Appendix B.

Appendix B describes methods for maintenance data analysis and data mining to determine poor performing AFS functions/components.

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5.3 Appendix C.
Appendix C describes methods for AFS component evaluation. This appendix also contains summaries of aircraft incident investigations.

5.4 Appendix D.
Appendix D provides an overview and guidance of component teardown analysis.

5.5 Appendix E.
Appendix E describes the task for development of a RCM risk assessment of the AFS.

5.6 Appendix F.
Appendix F describes the tasks for the development of a mitigation and action plan.

5.7 Appendix G.
Appendix G contains information on fuel tank ignition suppression based on FAA guidance from SFAR-88.

5.8 Appendix H.
Appendix H describes considerations for electrical bonding inside of fuel tanks.

5.9 Appendix I.
Appendix I provides an example of a FMECA performed on a tanker aircraft.

6. NOTES

6.1 Intended use.
This handbook is intended to serve as a practical reference guide to provide a systematic process to assess the Aircraft Fuel System (AFS) for design, overall condition, extend service life, and sustain airworthiness. This handbook also defines approaches to address sustainment issues which will promote the continued Operational Safety, Suitability, and Effectiveness (OSS&E) of the fuel systems throughout all phases of the aircraft's life when applied correctly.

6.2 Subject term (key word) listing.

Aging
Bleed
Bonding
Breaker
Circuit
Degraded
Degradation
Electrical
Error
Explosion
Fuel
Fuel system best practices
Fuel system lessons learned
Ignition

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Leak
Mishap
Nozzle
Switch
Temperature
Wear

6.3 Additional resources.

ECIA RS-186-4	Standard Test Methods for Passive Electronic Component Parts, Method 4: Dielectric Test (Withstanding Voltage)

(Copies of this document are available online at https://www.ecianow.org/eia-technical-standards.)

FAA Report DOT/FAA/AR-98/26	A Review of the Flammability Hazard of Jet A Fuel Vapor in Civil Transport Aircraft Fuel Tanks (June 1998)
FAA Report DOT/FAA/AR-00/19	The Cost of Implementing Ground-Based Fuel Tank Inerting in the Commercial Fleet. (May 2000)
FAA Report DOT/FAA/AR-01/6	Inerting of a Vented Aircraft Fuel Tank Test Article with Nitrogen Enriched Air (December 2000)
FAA Report DOT/FAA/AR-05/8	Fuel Tank Flammability Assessment Method User's Manual (May 2008)
FAA Report DOT/FAA/AR-07/30	Jet A Volatility Survey, (July 2007)
FAA Report DOT/FAA/AR-TN02/79	Limiting Oxygen Concentrations Required to Inert Jet Fuel Vapors Existing at Reduced Fuel Tank Pressures (April 2003)

(Copies of these documents are available online at https://fire.tc.faa.gov/Reports/reports.asp or https://fire.tc.faa.gov/systems.asp.)

US Air Force Technical Report AFAPL-TR-70-65	Effects of Fuel Slosh and Vibration on the Flammability Hazards of Hydrocarbon Turbine Fuels Within Aircraft Fuel Tanks (November 1970), Edwin E. Ott

(Copies of this document are available online at https://www.dtic.mil/ or https://discover.dtic.mil/.)

Report No. AFRL/RXS 09-026 Evaluation Report (SA103002)	KC-135 Refueling Pressure Regulator Butterfly Valve Fracture (Failure Analysis) (5 May 2009)

(Copies of this document are available via email request to brett.jordan@us.af.mil.)

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APPENDIX A

TASK 1 – AIRCRAFT FUEL SYSTEM & COMPONENT IDENTIFICATION

A.1 SCOPE

A.1.1 Scope.

The task objective is to identify and provide a general description of the common fuel systems components.

A.1.2 Source information.

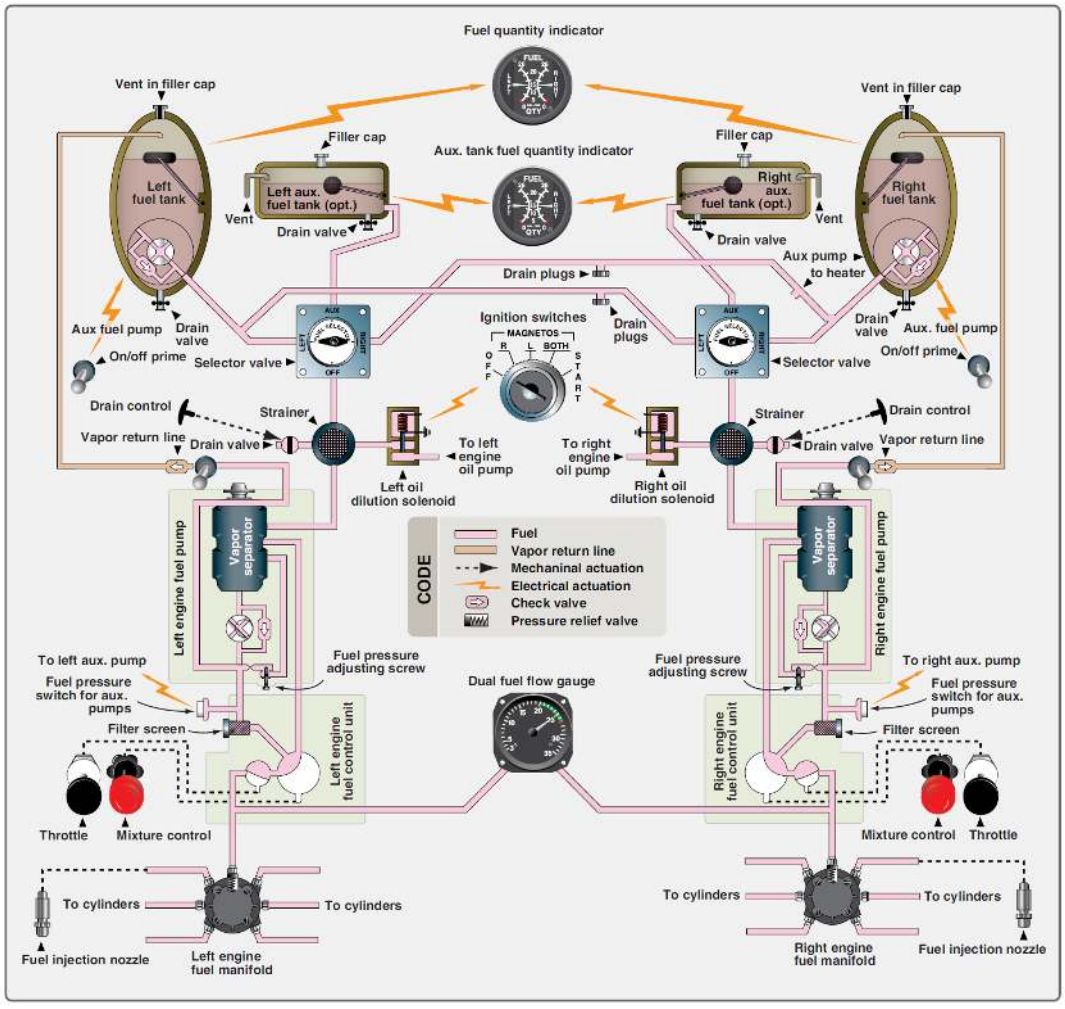


FIGURE A- 1. Low wing, twin engine, light aircraft fuel system.

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A.2 AFS COMPONENTS AND FUNCTIONS

A.2.1 Fuel pumps functional.

Fuel pumps enable the movement of fuel from a tank. There are several types of pumps that aid in the transmission of fuel. A few types of pumps are explained in the followed section.

A.2.2 Transfer pumps.

Fuel is transferred between tanks via the transfer pump. Electric motor driven and motive flow powered ejector pumps are used to transfer fuel, as well as hydraulic-powered pumps. Transfer pump selection is based upon its specific application on aircraft. The benefits of using the motor pump include higher performance rating under high altitudes and temperature, while ejector pumps are more reliable.

A.2.2.1 Air pressure powered fuel transfer.

In some aircraft fuel system designs (such as the P-8 fuel system), an air pressure fuel transfer system is provided to transfer wing and external tank fuel to the fuselage cells.

A.2.2.2 Boost pumps.

Boost pumps are used to deliver fuel to the engine on aircraft. There are two common types of boost pumps used today: motor-driven and motive flow pumps. The benefits of using a motor-driven pump include high-pressure output, and a motive flow source is not needed. Motive flow boost pumps rely on a motive flow source and have high reliability, with no ignition hazards.

A.2.2.3 Scavenge pumps.

Scavenge pumps are typically motive flow pumps. These pumps remove fuel, water and contamination from limited access areas. Water contamination is commonly addressed with scavenge pumps because multiple inlets can be used in the AFS where contamination is prevalent, and pump weight is low. Scavenge pumps breakdown removed water into small droplets that can be safely be fed with fuel to the engines.

A.2.3 Pump categories.

The aircraft fuel system is comprised of many subsystems. Each subsystem requires several types of components to meet the requirements of each individual subsystem. Hydraulic components and electric components are two categories discussed in the following section.

A.2.3.1 Venturis.

A venturi is utilized for each ejector pump in order to create a low pressure for suction. A scavenge pump is an ejector pump used for scavenging water and contaminants.

A.2.3.2 Hydraulically / electrically driven.

Pumps and valves require hydraulically or electrically driven components to deliver fuel. There are benefits to using both hydraulically and electrically driven components. Electrically driven pumps and valves are more common. Hydraulically driven pumps are often employed for very high flow fuel pumps. Hydraulically driven components provide more power and have been present in aircraft longer than electrically driven components. However, hydraulic systems require additional hardware, which can lead to increased maintenance needs.

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A.2.4 Fuel lines.

Fuel lines are connected to the aircraft engine to enable transfer of fuel between the fuel tanks and to the engine.

A.2.5 Connections/sealing.

Connections consist of either welded flanges or flexible couplings. Electrical continuity is maintained by use of jumpers (at flanges) or by integral clips and springs. At least two independent electrical paths to ground should be available at each location. Sealing should be accomplished by double (redundant) O-rings or gaskets. Connections in passenger areas should be shrouded and these areas should be drained, so that the exposure time for calculating hazard probability is minimized to one flight. Legacy aircraft may not follow this current standard.

A.2.6 Transfer valves.

Transfer valves are used to control transfer fuel between fuel tanks (see 3.2).

A.2.7 Fire shutoff valves.

These valves control the inflow of fuel to the engine via fuel lines. These relay controlled valves can be manually switched to the open or closed position. These valves typically close to terminate fuel flow when the engine fire handle is pulled and, on some aircraft types, when the engine control is set to cutoff.

A.2.8 Check valves.

Check valves are also frequently in application where one-way fuel flow is desired, such as on the output of fuel tank boost/transfer pumps. Another common application for check valve is to ensure that fuel does not overflow when the refuel adapter connection fails.

A.2.9 Fuel venting valves.

Fuel venting valves can be used for a variety of applications on the AFS. A common application for fuel venting is the use of valves to allow the inflow of air through the vent lines and close selected valves before it begins to spill fuel. Fuel systems require fuel quantity indication systems, as described in 3.2. The most common types of indication systems on AFS are described in the following section.

A.2.10 Fuel quantity indication system.

Fuel systems require fuel quantity indication systems, as described in 3.2. The most common types of indication systems on AFS are described below/

A.2.10.1 Float level.

The float level indicator sensor is a common method used to detect fuel level. The magnetic float moves up and down on a circular stem to indicate the fuel level. A few disadvantages to using the flow level sensor include large size, low reliability in turbulent fuel, and low durability. Floats are not commonly used as a primary fuel quantity gauging system component. Floats are often used in the following applications:

a. Low fuel warning, independent of the fuel quantity gauging system.
b. Automatic termination of fuel dump function.
c. Fuel tank high level shutoff trigger.

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A.2.10.2 Capacitance probe fuel quantity instrumentation system.

Capacitance probes are commonly used to measure fuel level in the various tanks. The fuel level is determined by the dielectric properties of air and fuel, as well as fuel density. The length of the probe immersed in the fuel changes capacitance in relation to the level of fuel.

A.2.10.3 Densitometer probe fuel quantity instrumentation system.

Densitometer probes use a spring-mass resonance to determine density in real time, providing an indication of fuel density over aircraft fuel tank temperature ranges. One end of a probe is driven torsionally to resonance using an oscillating electrical signal, the other end of the spring vibrates freely at its natural frequency.

A.2.11 Ultrasonic fuel quantity instrumentation system.

Ultrasonic systems are used to gauge fuel level by measuring the travel time of sound to the surface at the top end, and back to the receiving transducer at the bottom. The speed of sound is inversely proportional to fuel temperature, and is also related to fuel type. This type of fuel quantity gauging system is not currently in wide use across the USAF.

A.2.12 Center of gravity (CG) management.

Fuel tanks are generally located in the wings of an aircraft, and often in the fuselage, and affect the CG as fuel is depleted during flight. To maintain stability of the aircraft during flight, the fuel may be moved between tanks. This is done also in order to maintain the center of gravity and reduce the strain on the engines and flight surfaces. When fuel CG is not optimal, fuel efficiency is degraded. The CG management function can be flight critical.

A.2.13 Fuel tank.

Fuel tank safety is a vital issue to be addressed on AFS. Fuel tanks have a high severity risk factor, which can lead to the destruction of aircraft in the worst-case scenario. As a result, proper designs and mitigation strategies were established to prevent flame propagation on aircraft.

Sealant between adjacent metal walls (spars and ribs) and wing skins prevents fuel leakage. Sealant is also required around fastener penetrations. Leaks can become a fire hazard. Fuel vents can become ignition sources if vapors are exposed to lightning or ground fires. Any electrical components in the tank can become ignition sources. Dual electrical bonding is required to prevent static discharge or lightning sparks. Refueling inlets should be located at the bottom of the tank to avoid splashing or spraying, which creates static charge and ideal vapor ratios for ignition. Fuel slosh can couple with aircraft motion to accentuate motion, or expose inlets, so baffles are sometimes used. Where bladders are used, the containment space (cavity) should be drained, so that the exposure time for calculating hazard probability is minimized to one flight. Legacy aircraft may not follow this current standard. Bladder size at installation should be carefully controlled, so that newly manufactured bladders do not become smaller over time, resulting in torn interconnects.

A.2.14 Fuel tank inerting and fire suppression systems.

Several types of fire suppression systems have been used to reduce the risk of fire in fuel tanks since the 1950's. The most common system in use in modern aircraft fuel systems are On Board Inert Gas Generating Systems (OBIGGS). These systems separate the nitrogen from oxygen in

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engine bleed air and send the nitrogen enriched air to the fuel tanks, reducing the fuel tank oxygen below the level required to support combustion. The USAF usually specifies the fuel tank oxygen be kept below nine percent or a percentage as a function of altitude to prevent any combustion reaction, regardless of the energy of the ignition source. The FAA considers a fuel tank to be inert at oxygen levels below 12 percent. Other fire suppression systems in use include:

a. Liquid nitrogen – In use on the C-5 aircraft. A drawback of liquid nitrogen is the logistics burden of providing a source of liquid nitrogen at each aircraft operating location, especially for deployments.
b. Halon 1301 – In use on the F-16 aircraft. This system is only activated for combat operations. A disadvantage of Halon is that Halon 1301 is no longer produced, making availability limited to remaining stock.

A.2.15 Reticulated foam.
Reticulated foam is a passive explosion protection system. It works by absorbing energy from the combustion reaction faster than energy is generated, resulting in cessation of the combustion reaction. Reticulated foam installations work through one of two design approaches:

a. Packing the fuel tank volume 80 percent or more full with foam (typically coarse pore). The flame front within the tank will burn a large volume, but the total pressure energy developed will be very low, easily within the structural capability of the fuel tank.
b. Highly voided concepts. This approach uses fine pore foam to isolate small compartments within the fuel tank. The flame front is not able to penetrate the fine pore foam, resulting in cessation of the combustion reaction and keeping the overpressure low.

While reticulated foam decreases risk of fuel tank explosions, it has many disadvantages. A five-year replacement interval has been shown to reduce fuel foam related failures on F-15 aircraft. The foam has a limited operational lifetime and adds weight to aircraft. The service life of the foam begins on the date of foam manufacture. Service life limitation is a cumulative effect of its storage life exposure and its exposures in aircraft fuel tanks. Service life for foam baffles are also dependent upon how often the foam is removed and replaced in an aircraft. Although it is anticipated that foams under this specification will have a long, useful service life, determination of the practical limit of the service life by vigilant periodic inspection and testing, is the responsibility of each aircraft's maintenance and engineering organizations. It also requires high maintenance to remove foam in order to access hardware on aircraft. If not maintained/cycled, foam may generate FOD and get caught in check valves and pumps.

A.2.16 Refueling.
Refueling is the process by which the aircraft receives fuel through a refueling station in an airport, or tanker aircraft. Pressure refueling is the most common type of refueling method used by commercial and military aircraft because it allows for quick refueling, a large aircraft can receive fuel at 600 gallons per minute per single point refueling receptacle. Some large aircraft (e.g. C-5 and C-17) are equipped with two single point refueling receptacles. Some aircraft have fuel caps located on the top of the wing, which may be removed to refuel on the ground by inserting a hose nozzle; this is referred to as "over-the-wing" or "gravity" refueling.

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A.2.17 Boom and hose reel aerial refueling.
Aerial refueling, a commonly used technique for military aircraft, is the process of transferring fuel from one aircraft to another in flight. There are two types of aerial refuel systems commonly used today: 'Probe and Drogue' and 'Flying Boom' systems. Advantages of using the 'Probe and Drogue' system include a tanker requiring less maintenance, no boom operator needed, and a tanker can refuel more than one receiver simultaneously, improving efficiency. The advantages of using the 'Flying Boom' include higher fuel transfer (1000 gpm vs 200 gpm), and lower susceptibility to human error and harsh weather conditions. Some KC-135, KC-10 and KC-46 tankers (plus other foreign tankers) offer both systems, where the 'Probe and Drogue' pods are mounted near each with tip and/or on the centerline.

A.2.18 Double barriers.
Fuel system components, tanks, lines when practical should have a drained double barrier to prevent fuel or vapors from reaching inhabited areas or potential ignition zones. Coatings have been used on fuel tank surfaces as a secondary barrier so fuel and vapors will not enter the pressurized areas of the aircraft. Inspection and maintenance of these coatings is necessary to prevent leakage into the pressurized areas.

A.2.19 Fuel line and component separation from potential ignition sources.
Fuel lines should be routed below electrical harness by at least 0.5 inch to prevent leakage from impinging on the harnesses and potential for chafing between the two.

A.2.20 Thermal management.
Fuel is frequently used as a heat sink for the thermal management system (see APPENDIX C.7.5). Heat exchanges and fuel recirculation systems are commonly used for this purpose. Ensure the fuel system and interface is designed to prevent leakage between the two systems, and has sufficient capacity to prevent overheating of the system or the fuel.

A.2.21 Drainage.
Drainage, vent, and fuel dump (jettison) outlets are located such that normal or accidental leakage is routed to a safe location outside the air vehicle and not re-ingested into the engine, hazardous ignition areas, or the Environmental Management System.

A.2.22 Compatibility with approved fuels.
Verify fuel system functions safely under all specified conditions with all approved fuels and additives.

A.2.23 Fuel tank coatings.
Ensure the fuel tank coatings used to prevent corrosion are compatible with the approved fuels and additives for the life of the airframe. Fuel tank coating incompatibility issues resulting from additives in JP-8+100 such as Di-ethylene glycol monomethyl ether (DiEGME) and switch to Jet-A for CONUS aircraft have led to fuel tank top-coat degradation/peeling resulting in engine flame-out on KC-135, B-52, and other aircraft.

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A.3 SYSTEM SAFETY ASSESSMENT

FAR 25.1309 requires that a safety assessment be made for commercial transport aircraft, and AC 25.1309 provides guidance, which may serve as military platform guidance. Under the Acceptable Techniques paragraph, seven techniques are described: 1) Functional Hazard Assessment; 2) Analysis of Minor Failure Conditions; 3) Analysis of Major Failure Conditions; 4) Analysis of Catastrophic Failure Conditions; (5) Operational or Environmental Conditions; 6) Latent Failures; 7) Acceptable Means of Compliance with 25.1309(c).

A.3.1 Failure Modes, Effects, and Criticality Analysis (FMECA).

The Failure Modes, Effects, and Criticality Analysis (FMECA) is discussed in MIL-STD-1798, which references FAA AC 25.1701-1. Within the FMECA, it is the "end effect" that should be considered for component criticality determination. Component classification shall be performed at the LRU and Weapons Replaceable Assembly (WRA) level, if applicable. In addition, AC 25.1309 allows that a FMEA may be used to support the Analysis of Major Failure Conditions. It may also support the Qualitative and Quantitative Assessments. The FMECA should address the loss of function as well as the potential secondary effects such as an inadvertent ignition source. Included in 0 is an example of an AFS FMECA.

A.3.2 Failure fault trees.

Air Circular (AC) AC 25.1309 allows that a Fault Tree may be used to support the Analysis of Major Failure Conditions. It may also support the Qualitative and Quantitative Assessments.

A.3.3 Fish bones.

See 3.2 for a definition. While these may be requested by the government, the other methods listed here are preferred, such as the Fault Tree.

A.3.4 Single point failure.

A single point of failure is a failure that does not require any other failures to manifest, and that has one or more secondary affects. These failures are generally addressed by aircraft design, and are rarely addressed in safety analysis. An example would be a piece of primary structure, such as a wing spar. If that one item fails, it would likely be catastrophic. However, experience has shown that the safety factors used are sufficient to preclude failure. Another example would be an engine rotor burst in a single engine aircraft. Again, safety factors used are sufficient to preclude failure for those aircraft. Because these failures are part of a simple failure mode, a structured approach is not warranted.

A.3.5 Catastrophic physical failure assessment.

No ignition source may be present at each point in the fuel tank or fuel tank system where catastrophic failure could occur due to ignition of fuel or vapors. This needs to be shown by (see FAR 25.981):

1. Determining the locations subject to highest temperatures and allowing a safe margin below the lowest expected auto-ignition temperature of the fuel in the fuel tanks.
2. Demonstrating that no temperature at each place inside each fuel tank, where fuel ignition is possible, will exceed a safe operating temperature. This needs to be verified under all

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probable operating, failure, and malfunction conditions of each component whose operation, failure, or malfunction could increase the temperature inside the tank.

3. Demonstrating that an ignition source could not result from each single failure, from each single failure in combination with each latent failure condition not shown to be extremely remote, and from all combinations of failures not shown to be extremely improbable. The effects of manufacturing variability, aging, wear, corrosion, and likely damage need to be considered.

A.3.6 AFS Secondary effects of failure.

The following sections taken from MIL-HDBK-516 contain guidelines for assessing the airworthiness of the AFS:

a. Verify that the fuel system is safely compatible with other system interfaces.
b. Verify adequate information is available to notify operator(s) of the system operating conditions.
c. Verify that all fuel system critical failure modes and hazards have acceptable risk levels.
d. Verify the safe installation of the fuel system and components.
e. Verify that fuel transfer flow rates meet the operational ground and flight envelope requirements.
f. Verify that the air vehicle center-of-gravity limits are not exceeded during all fuel system and air vehicle functions, including release of stores, aerial refueling (if applicable), fuel transfer, fuel dumping operations, wing sweep operations, catapult launches, arrested landings, and engine feed.
g. Verify that the system is designed to withstand the hazards associated with lightning, static electricity, fuel leaks, and the introduction of electrical power into fuel tanks.
h. Verify that secondary fuel and vapor tight barriers are provided between fuel tanks, fire hazard areas, and inhabited areas.

The physical failures, both direct and single point failure causes, should be documented and, if necessary, possible mitigation strategies for unsatisfactory results. The selection of mitigation technique used is determined in Task 6.

Use the aircraft level Failure Hazard Assessment (FHA) results to guide the system level FHA. Incorporate AFS failures into the system level and aircraft level FHA, and as necessary, the Preliminary System Safety Assessment (PSSA), the Common Cause Assessment (CCA), and the System Safety Assessment (SSA). These analyses are performed to satisfy requirements such that the occurrence of any failure condition, which would prevent the continued safe flight and landing of the aircraft, is extremely improbable. Use results of these analyses to update the AFS definition.

Using the severity categories taken from the MIL-STD-882 (see TABLE A- I), examine the AFS components identified in the previous sections of this document. These components needs to be associated with a failure consequence.

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TABLE A- I. Severity categories (Source categories (source MIL-STD-882)).

Description	Severity Category (CAT)	Mishap Result Criteria
Catastrophic	1	Could result in one or more of the following: death, permanent total disability, irreversible significant environmental impact, or monetary loss equal to or exceeding $10M.
Critical	2	Could result in one or more of the following: permanent partial disability, injuries or occupational illness that may result in hospitalization of at least three personnel, reversible significant environmental impact, or monetary loss equal to or exceeding $1M but less than $10M.
Marginal	3	Could result in one or more of the following: injury or occupational illness resulting in one or more lost work day(s), reversible moderate environmental impact, or monetary loss equal to or exceeding $100K but less than $1M.
Negligible	4	Could result in one or more of the following: injury or occupational illness not resulting in a lost work day, minimal environmental impact, or monetary loss less than $100K.

At this stage in the assessment, the failure consequence should focus on direct AFS component loss. Examples of failures to consider include but are certainly not limited to the following:

A.3.6.1Flexible fuel hoses.
Corrosion of braided stainless steel flexible fuel hoses led to two lost single engine fighters in FMS service. Four of 40 lines are single point failure locations (i.e. without redundant backup systems). Loss of the primary function, fuel delivery to the engine, in turn led to loss of aircraft (see AFRL/MLSA 98-047; 98-0159).

A.3.6.2Fuel pump cooling.
Inadequate cooling of fuel pumps led to overheating in service (for example: relying on fuel to cool the pumps). At least two aircraft losses were attributed to this cause. The pumps had been designed for periodic usage but in practice, were run much longer and even allowed to run dry. Investigation revealed overhaul technicians had observed evidence of thermal distress (i.e. bluing of the shafts) but had not recognized the significance of the observation.

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A.3.6.3F-16 Fuel shutoff valve.
Per findings of a 1993 Tiger Team, several F-16 aircraft have been lost potentially, due to suspected un-commanded closure of the fuel system Main Fuel Shut Off Valve (MFSOV). Un-commanded closure of a fuel valve has occurred on other aircraft fuel systems due to design flaws, manufacturing defects, use of improper materials, poor maintenance practices, or aging of the component (see APPENDIX C.10 for additional information).

A.3.6.4TWA 800.
On July 17, 1996, Trans World Airlines, Inc. (TWA) Flight 800, a Boeing 747, crashed in the Atlantic Ocean near East Moriches, New York. The National Transportation Safety Board (NTSB) determined that the probable cause of the TWA Flight 800 accident was an explosion of the center wing fuel tank (CWT), resulting from ignition of the flammable fuel/air mixture in the tank. The source of ignition energy for the explosion could not be determined with certainty, but, of the sources evaluated by the investigation, the most likely was a short circuit outside of the CWT that allowed excessive voltage to enter it through electrical wiring associated with the fuel quantity indication system.

Contributing factors to the accident were the design and certification concept that fuel tank explosions could be prevented solely by precluding all ignition sources and the design and certification of the Boeing 747 with heat sources located beneath the CWT with no means to reduce the heat transferred into the CWT or to render the fuel vapor in the tank nonflammable.

The recommendations from the NTSB included the following:

1. “Examine manufacturers design practices with regard to bonding of components inside fuel tanks and require changes in those practices, as necessary, to eliminate potential ignition hazards.”
2. “Review the design specifications for aircraft wiring systems of all U.S.-certified aircraft and (1) identify which systems are critical to safety and (2) require revisions, as necessary, to ensure that adequate separation is provided for the wiring related to those critical systems.”
3. “Require the development and implementation of corrective actions to eliminate the ignition risk posed by silver-sulfide deposits on fuel quantity indication system components inside fuel tanks.”
4. “[T]he need for improved training of maintenance personnel to ensure adequate recognition and repair of potentially unsafe wiring conditions.”
5. “[T]he need for improved documentation and reporting of potentially unsafe electrical wiring conditions.”
6. “[T]he need to incorporate the use of new technology, such as arc-fault circuit breakers and automated wire test equipment.”

Information taken from NTSB report PB2000-910403.

Inside fuel tanks, circuits should either be:

a. Intrinsically safe – energy below the minimum fuel ignition energy of 0.2 millijoules. Discuss how to analyze circuits to verify the energy is below this threshold value and

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emphasize it is critical to not run intrinsically safe circuits in the same wire harness as higher energy circuits.

b. For some applications (e.g. a fuel boost pump located inside the fuel tank) it is not possible to avoid running high-energy circuits inside fuel tanks. For these installations, the wires need to be contained inside conduits that are capable of containing the overpressure of a fuel conflagration. It is important in cases such as these to include a periodic inspection of the conduit to insure integrity.

A.4 COMPONENT CLASSIFICATION LIST

As an alternative, or in addition to the above, improvement projects may be identified through the prioritized Component Classification List (CCL), Functional Systems Integrity Plan (FSIP) Finding Reviews, MECSIP, AVIP, and via Customer request. Key steps in the execution plan for each system are: (1) Identify Safety Critical Components (SCCs), Mission Critical Components (MCCs), and Durability Critical Components (DCCs) IAW MIL-STD-1798. SCCs will be derived from the Critical Safety Items (CSI) list. (2) Populate the CCL and enter the values required to rank the components.

Include a listing of fuel system driven loss of aircraft events such as:

a. B-1B – over wing fairing fires (see APPENDIX C.11).
b. Fuel line icing resulting in loss of thrust (commercial flight from China to England).
c. B-52 icing at fuel strainer.
d. B-52 fuel slosh resulting in loss of controlled flight following CG shift (early 60s).
e. British Nimrod in Afghanistan (see APPENDIX C.12).
f. TWA-800 (see APPENDIX C.13).
g. USAF E-3 (fuel vent blocked by FOD left following PDM) during aerial refueling.

A.5 TASK RESULTS

Determine whether the potential for AFS failure needs to be mitigated. Any AFS failures requiring mitigation should be identified and included in the Task 1 report. Strategies for mitigation will be developed and verified in Task 6. If no mitigation is needed, complete the appropriate safety assessment.

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APPENDIX B

TASK 2 – DATA ANALYSIS

B.1 SCOPE

B.1.1 Scope.

The first step is to determine AFS condition and extent of problems (or Root Cause). If problems are identified, formulating the solutions is normally a more straightforward process. The following analytical methodologies have been used to identify conditions and/or performance gaps.

Different analytical methods are available for use; each one is valid for a given situation or environment. A thorough understanding of the methods listed below is essential for the analyst to accurately use statistical and cost analysis to diagnose AFS related issues.

B.1.2 Mining of maintenance data.

Maintenance data sources such as Reliability Maintenance Information System (REMIS) may be valuable to the MECSIP engineer. Monitoring the reliability trends of MECSIP components may allow a MECSIP engineer to identify emerging issues before availability is severely affected. Also, comparing the relative reliability of one specific subsystem of a given platform against other platforms may enable a MECSIP engineer to identify which platforms feature more robust design approaches. By contacting their counterparts on the other platforms, MECSIP engineers can help to ensure that lessons learned are conveyed across all DoD and commercial fleets.

B.1.3 Comparing subsystem reliability on multiple platforms.

It may be possible for MECSIP engineers to gain insight into the relative reliability of their platforms subsystems with respect to that of other platforms, by reviewing maintenance data contained in databases such as REMIS or similar. This approach may allow a MECSIP engineer to identify those platforms not plagued with the same issues. By contacting their counterparts on the other platforms, they may gain insight into how to reduce their reliability burdens.

To use this approach, care needs to be taken to collect data not biased by the inherent database structure. Entries in REMIS may be redundant. For instance, once a pilot reports a discrepancy, one line in REMIS will be generated, with a specific When Discovered (WD) code. The same discrepancy may have other lines in REMIS associated with corrective actions implemented by maintenance personnel. By selecting limited WD codes, this bias can be eliminated. Also, the inherent difference in fleet sizes will affect the total number of write-ups per subsystem. In order to correct for this issue, the number of write-ups should be normalized by fleet size and/or utilization.

To eliminate redundant entries for one issue, it may be helpful to ensure only one entry per Job Control Number (JCN) is counted. Alternatively, it may be possible to filter by codes such as WD codes or Action Taken Code. For example, if WD codes of only A-E are selected, only the pilot's initial write-ups will be listed. Listings of the codes (How Malfunctioned, When Discovered,

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Action Taken and Work Unit Codes) are typically listed in each aircraft's -06 Technical Order, for example TO 1A-10A-06.

In summary, this approach should include the steps. An example is provided in APPENDIX B.10.

1. Download the system entries from REMIS.
2. Eliminate the duplicates.
3. Focus on keywords that are relevant to the component evaluated.
4. Review the remaining entries for relevant listing.
5. Perform data analysis on results.

B.1.4 Maintenance data trend analysis approach.

Other actionable information may be extracted from careful review of maintenance data sources, in addition to determining how the reliability of a specific subsystem on a given platform compares to that on a different platform. In particular, it may be possible to identify from trends in availability emerging issues before they become so pronounced as to affect overall system availability.

For example, a MECSIP manager may be able to drill down to the 5-digit Work Unit Code (WUC) data from REMIS to show how the number of write-ups is trending over time. Tracking trends in the top 50 5-digit WUCs for a given 3-digit WUC over varying time scales could alert the MECSIP manager to projected non-availability due to that specific system. Identifying these trends would enable the MECSIP manager to pursue activities to better understand the cause of the increase in write-ups, and, in-turn, devise and implement corrective action strategies early enough to prevent readiness limitations.

Strategy to forestall non-availability include proactive condition assessments. The Maintenance Steering Group (MSG-3) Reliability Centered Maintenance (RCM) program includes a task entitled 'Age Exploration'. In this task, components of interest are torn down to better understand emerging unanticipated failure modes. Such an approach is similar to the Analytical Condition Inspections (ACIs) which are a part of the Aircraft Structural Integrity Program (ASIP). ACIs involve tearing down high time assets to characterize the aircraft structure condition, specifically assessing fatigue and corrosion damage. In addition to these two modes, MECSIP components may also show other unanticipated failure modes, such as permanent set of springs, wear of moving parts, and seal and bearing deterioration. Additional examples of how these proactive condition assessments can positively affect availability rates are contained in APPENDIX C.

B.2 DATA SOURCES

There are several sources from which the data can be gathered. Aircraft/fleet data may be gathered from systems such as REMIS, G081, Integrated Maintenance Data System (IMDS), DECKPLATE, Joint Deficiency Reporting System (JDRS), and/or the Air Force Safety Center data. Other data sources are discussed throughout the following sections.

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B.3 DETERMINING SCALE OF EXISTING MAINTENANCE

The relative maintenance effort and impact on aircraft availability is a good starting point for the analysis. FIGURE B- 1 is an example of the Total Non-Mission Capable Maintenance (TNMCM); this indicates that the fuel system has the third greatest impact on mission availability.

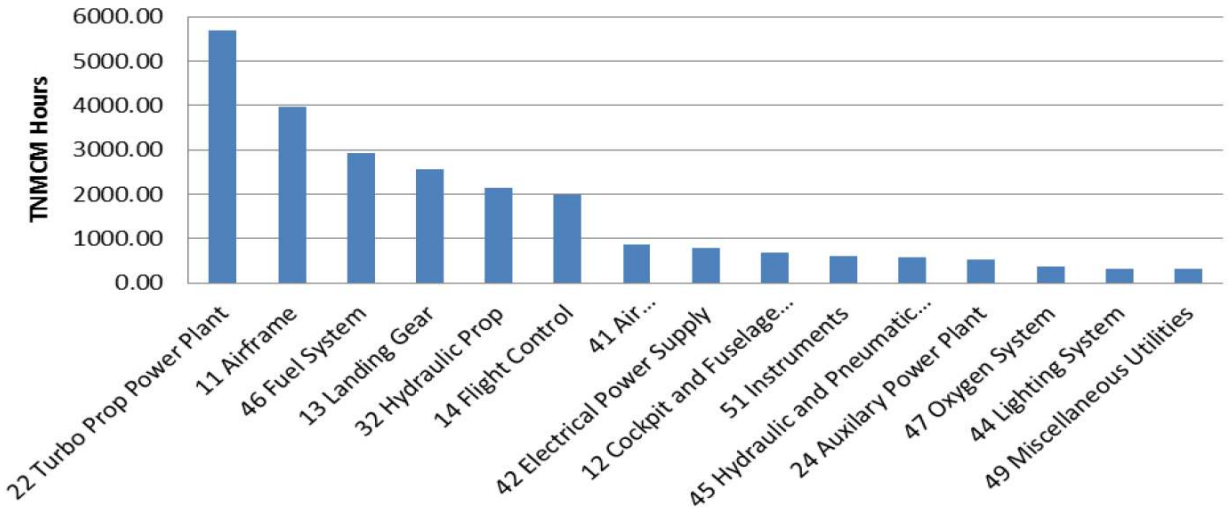


FIGURE B- 1. Example analysis of all maintenance actions broken down by system for C-130.

Starting with this information will help with setting bounds and expectations on total maintenance actions for the given aircraft.

B.4 VISUAL OBSERVATION

Visual observation is valid under certain conditions; however, it depends largely on the observer's experience and knowledge and degree of access. This type of observation provides limited information and may not be useful or meaningful for complex situations. With this limitation as a stand-alone activity, it should be considered useful when combined with the other methods.

B.5 ANALYSIS

B.5.1 Reliability analysis.

Reliability is assessed by examining inherent failures (Type 1) in two ways. First, by measuring "hard" assembly failure rates as reflected by REMIS (or equivalent) LRU remove & replace actions. These are used to determine relative component endurance and future requirements. Second, by examining all inherent failures, including those requiring only adjustment or minor parts replacement and sorting these by How Malfunction (HMAL) and Action Taken (AT) codes, the results may be used to confirm the efficacy of existing maintenance schedules or to develop modified maintenance plans. How Malfunction Codes and/or Discrepancy Narratives are helpful in sorting out categories leading to all maintenance not requiring assembly replacement. Of special interest is Type 1 (inherent failures), Action Taken "F" (repaired), "G" (replaced minor hardware), "L" (adjusted), or "V" (cleaned).

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B.5.2 Comparative analysis.

Comparative analysis may be performed statistically or visually and involves comparing two or more like processes or items to identify variations or differences. Though a viable stand-alone method, it is normally used in conjunction with other methods.

B.5.3 Statistical analysis.

Statistical analysis is the methodical study of data, used to reveal facts, correlations, and trends about data. It is useful in conjunction with comparative and visual analysis.

B.6 ANALYSIS PROCESS

The analysis process is the methodical conversion of data into information for managerial decision-making and control. It amalgamates the three previous methods into a single coherent analytical process. This combination makes it the most robust of these problem-solving processes.

B.6.1 Data mining process.

Though there is no one best way to mine data, there are some "best practices" available. However, for the sake of congruence between projects, a standard basic process is recommended. It needs to be remembered however, that the data mining process will vary based on the way the project is being conducted.

Data is usually downloaded from REMIS, G081, IMDS, DECKPLATE, JDRS, and/or the Air Force Safety Center data into a program that allows for filtering and analysis of the data. To maintain an accurate and reliable product, some measures are advised; such as system data being prepared (or "scrubbed"), then separated into units based on WUC.

B.6.1.1 Data gathering example.

This approach should include the steps shown in the following example, in which the fuel quantity indicator system (FQIS) reliability is assessed for multiple platforms.

1. Download all system 46 REMIS entries (on some A/C, FQIS is contained in system 51 - instrumentation). In this example, approximately 11,000 write-ups were found on all MDSs over a 5-year period.
2. Eliminate duplicate entries for example by selecting only WD codes of A-E.
3. Review entries for discrepancy keywords, which seem relevant to the issue at hand. In this case, fluctuating, erratic, off scale, are good indicators of FQIS issues.
4. Sort the listings so that only the cases of interest are listed. In Excel, this can be accomplished using conditional formatting, highlight cells containing a given text string (see FIGURES B- 2, B- 3, and B- 4). Custom sort on cell color will bring the relevant cases to the top of the list.

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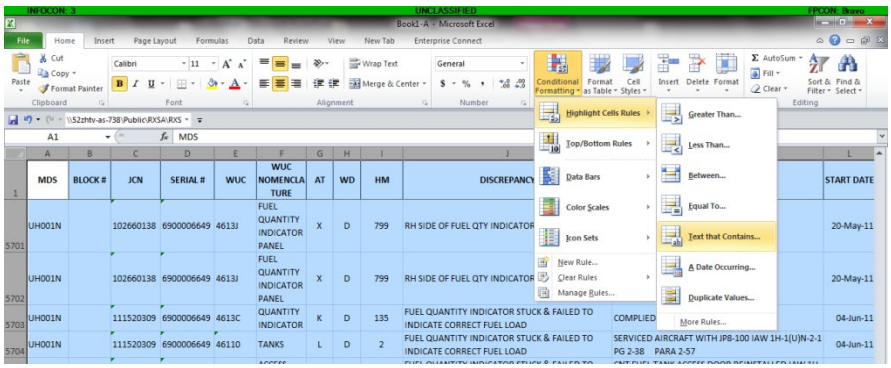


FIGURE B- 2. Part 1 of conditional data formatting to highlight critical REMIS items.

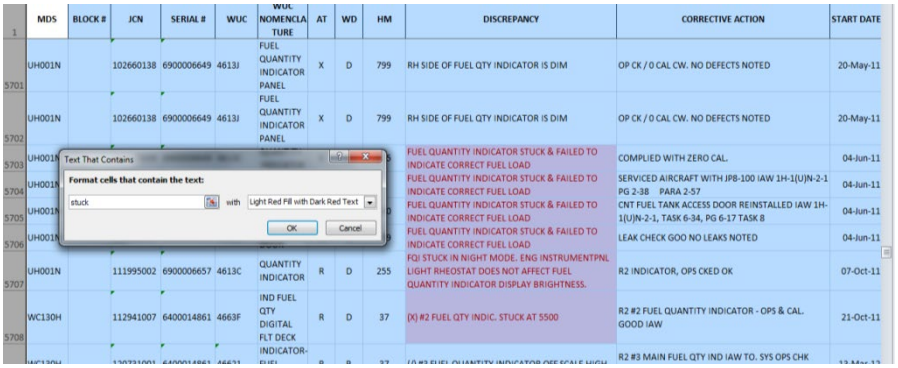


FIGURE B- 3. Part 2 of conditional data formatting.

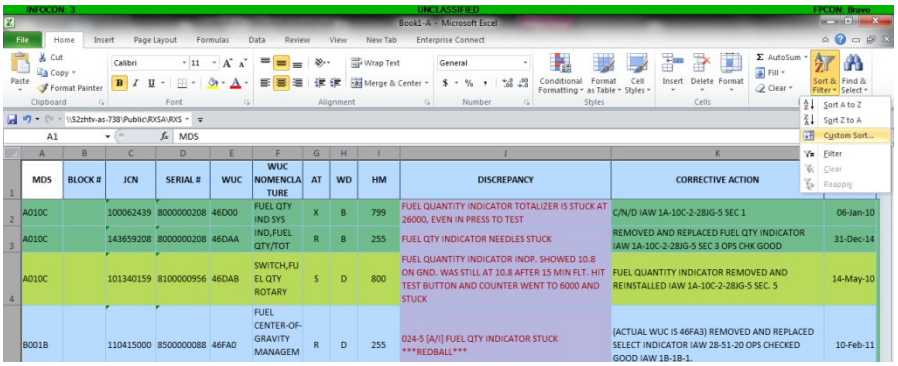


FIGURE B- 4. Example of filtered REMIS data.

5. Filter the relevant listings by Mission Design Series (MDS) and count occurrences. Normalize these results by fleet size to assess how significant the issue is on each of the platforms surveyed. To determine the number of aircraft remaining in service for a particular MDS, run a REMIS query for the specific Mission Design Series with TERMINATION DATE filtered by IS NULL. An example data evaluation is shown on FIGURE B- 5.

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	# of Occurrances														
MDS	A010	AC130	B001B	B052	C005	C017	C130	E003	EC130	F015	F016	HC130	HH060	MC130	WC130
NEEDLE PROBLEMS	17				4			1		25	12				
FQI INOP	21	4	6	8	4	13	69	9	3	11	2	11	2	11	2
Fleet Size	422	37	82	181	110	222	501	31	21	634	1537	48	100	83	20
NEEDLE PROBLEMS	4.0%				3.6%			3.2%		3.9%	0.8%				
FQI INOP	5.0%	10.8%	7.3%	4.4%	3.6%	5.9%	13.8%	29.0%	14.3%	1.7%	0.1%	22.9%	2.0%	13.3%	10.0%

FIGURE B- 5. Example normalization of results across multiple platforms.

Review of such data may enable a MECSIP manager on one platform to seek out the experiences of a MECSIP manager on a platform with fewer such issues.

B.7 FIELD, DEPOT, AND ENGINEERING SURVEYS

These surveys address issues concerning AFS components (e.g. Fuel Boost Pump) and subsystems as well as, related issues such as management, technical orders, and training. These questionnaires are directed at all personnel involved with the fuel system sustainment process and should be completed at all levels of the logistics organization. For example, questions concerning power plant maintenance (WUC 23000) would be directed to everyone involved in flight line maintenance. All bases with the related airframes and AFS systems would be requested to participate in the survey.

Different questions would be sent to the depot units as they have a different type of maintenance environment, geared primarily towards preventative and heavy maintenance.

System program engineers, logistics personnel, and systems safety would receive questions concerning requests for assistance and various issues they were required to resolve.

B.8 GOALS FOR FILTERING OF DATA

The goals of the data analysis are:

a. Identity where AFS issues exist in the fleet,
b. Determine the size of AFS related issues,
c. Provide insight into how AFS failures impact aircraft systems electrically and physically and they relate to the risk assessment process.

B.9 MAINTENANCE DATABASES

Maintenance and service databases are filled with entries from maintenance personnel. Historically, in the USAF, these databases were primarily designed to track part usage and manage inventory. As a side benefit, the databases can provide useful engineering data. While there is guidance provided to fleet maintainers to use common terminology in these databases, there are often misspellings, incorrect term usage, improper system identification, incorrect or nonexistent WUC, and multiple entries for the same issue. An example of a maintenance data report is shown on FIGURE B- 6 and how the data can be used to focus on one system such as the fuel system is given on FIGURE B- 7. Note the fuel system is a major contributor to maintenance man hours for this aircraft.

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FUEL SYSTEM 46XXX MAINTENANCE MANHOURS BY CALENDAR YEAR

WUC	WUC Description (Historical)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12	Total
46000	FUEL SYSTEM	18949	22817	25860	18753	13905	21326	15078	17998	36664	20885	15807	13731	241773.1
46130	NO 1 MAIN	6497	5870	6719	5012	6212	5251	4081	3748	16145	36426	33150	16039	145150.2
46110	NO 1 RESERVE	5841	5214	7335	5370	5345	6752	6643	5773	10518	12236	9157	9788	89971.3
46280	NO 4 RESERVE	3649	4218	3785	3962	4004	5265	6388	6989	8968	12224	9324	8892	77666.2
46240	NO 4 MAIN	6391	5080	7319	3480	4756	3785	3676	3573	7618	10506	7690	8182	72056.3
46210	NO 3 MAIN	4713	4403	4044	4458	3310	2990	2332	2915	6987	7203	6766	12205	62325.8
46170	NO 2 MAIN	4310	3799	4681	2777	3240	2617	4005	2604	5091	7500	7334	8302	56258.2
46410	FWD BODY	5563	4133	3850	3345	2920	3331	3256	3512	5590	4871	3474	2919	46765.4
46471	CELL NR 1	4902	6111	3789	2916	3633	3707	3118	2694	3942	3778	3485	3174	45247.8
46421	CELL NR 0	3773	4003	4318	3007	2657	2880	1963	3215	3262	3570	2929	3055	38634.6
46460	AFT BODY	4186	5852	4063	1911	2115	2899	2612	2448	2666	3111	3534	2187	37582.6
46427	CELL NR 3	2506	2031	3107	2793	2645	3093	1492	1728	2867	3912	2691	2393	31256.3
46310	CENTER WG TANK LH	2117	2309	2276	1292	1260	930	2289	819	5190	4206	4359	3754	30800.1
46423	CELL NR 1	3258	3253	2361	1403	3151	1683	1306	2108	2544	3094	2563	2565	29288.6
46340	CENTER WING RH	2955	2060	2741	1529	1282	1395	991	1388	2577	3307	3613	4641	28480.3
467A0	DROGUE KIT COMP.	963	1391	1755	2213	2645	1675	1527	1785	2209	2639	4050	2008	24860.0
46440	CENTER WING	1630	1537	1727	649	967	799	2165	1536	3035	3481	2452	3141	23118.4
46771	NOZZLE ASSY	1550	2087	1875	1509	2289	2088	1640	1658	2087	2141	1929	2182	23034.6
46473	CELL NR 2	1547	1794	2170	1477	1499	1656	2102	1947	2347	2463	1749	2020	22769.8
46100	INTEGRAL TANKS	1391	776	1463	1608	1686	1164	449	814	1485	4227	2929	4400	22390.2

FIGURE B- 6. Example fuel system maintenance man-hours per calendar year for representative fleet.

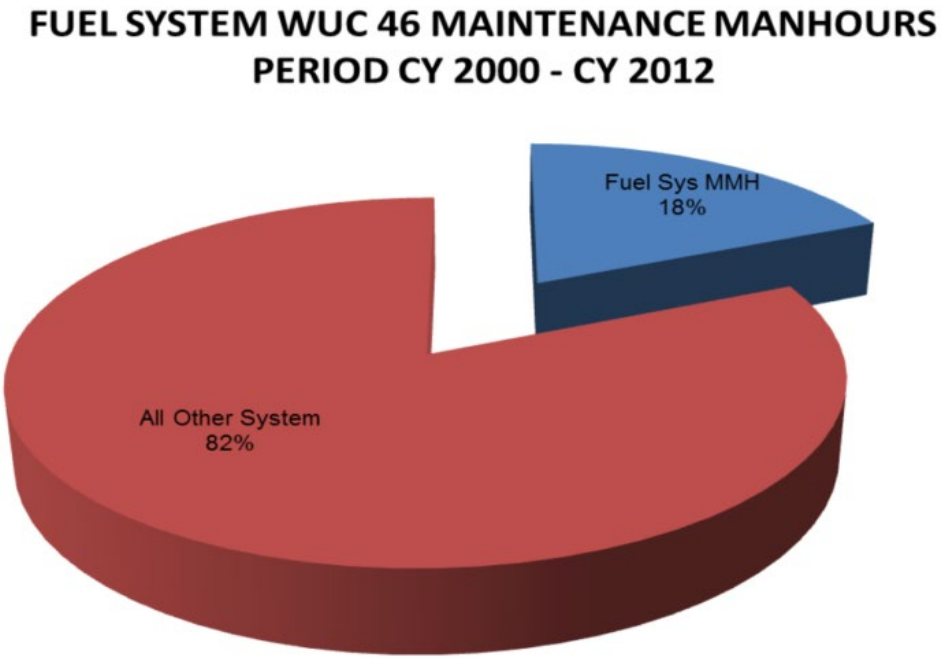


FIGURE B- 7: Example breakdown of AFS man-hours as a total of representative fleet.

The data entry errors, mentioned above, can cause a simple keyword search to produce incomplete and/or invalid results and lead to the incorrect conclusions from the data.

B.10 IDENTIFY AVAILABLE DATA

Maintenance actions are recorded in fleet specific databases. Database systems are continually updated and enhanced since they are integral to managing aircraft operationally. The USAF and USN have developed new data management systems that allow organizations to query aircraft platform maintenance data. The Air Force uses the IMDS and the Navy uses DECKPLATE (accesses the Naval Aviation Logistics Data Analysis (NALDA) system). Examples of specific databases include REMIS, G081 (Maintenance Data collection system for Air Force Mobility Aircraft), JDRS, and the Air Force Safety Automated System. These databases store information

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on aircraft maintenance actions using categories such as the WUC, a narrative with a brief problem description, or a corrective action description.

Unfortunately, many records have the wrong work unit codes assigned and the problem and correction action descriptions are poorly written due to misspellings, generalizations, and shorthand notation. Because of these reasons, mining information from the databases requires an exhaustive combination of keyword searches and ultimately a subject matter expert to manually read and interpret each maintenance record. This last action introduces subjectivity and can lead to more erroneous conclusions.

The following data sources should be considered. The information may be found in multiple locations:

a. Failure Analysis Reports of Failure Modes
b. FAA Service Difficulty Reports
c. Joint Deficiency Reporting Systems (JDRS)
d. Global Combat Support System (GCSS) USAF Maintenance Data
e. USAF Deficiency Reporting System (DR)

B.11 KEYWORD SEARCH

B.11.1 Data selection.

When querying the data system, a large number of data fields are available and this can be overwhelming from an analysis perspective. The data fields returned from the query can be reduced to a subset. It is recommended that, at a minimum, the following fields be selected when performing the analysis:

a. Dates fields (Date created, updated, closed, etc.),
b. Unique ID (this is the unique identifier for all maintenance records),
c. Work Unit code,
d. Narrative,
e. System and/or subsystem,
f. Problem description,
g. Corrective action,
h. Location.

Other fields that may be of use include: flight hours, scheduled/unscheduled maintenance, component model, and stage of operation.

NOTE: It is easier to hide or remove data fields from a query than to add them later in the process.

B.11.2 Data review.

Once the keywords have been selected and the desired data identified, run the query and store the results. The results are returned in a tabular form with each record on an individual line.

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Given that maintenance systems are regularly updated, it is recommended the original data be stored for further reference and the filter/review process be documented. This allows for direct traceability through the process and, if necessary, identifying possible analysis errors.

When going through the entries, a two-tier approach is recommended. The first tier, performed by a junior engineer or data analyst with some familiarity of the aircraft or maintenance reports, examines each item in the data, reads the details, and tags the entries. The second tier, performed by a senior engineer or senior technician, reviews the results and makes determinations on items that are unclear.

B.11.2.1 Quality of gathered data.

Review of maintenance database data will find that much of the data is miscategorized. As such, the information in each WUC should not be accepted and reported upon without further evaluation. FIGURE B- 8 shows the results of manual assessment for a fleet and was used to identify the specific assembly or part within an end item on which an action was performed. The figure points out the low percentage of WUC accuracy extracted in the analysis data. Only 23 percent of the components were correctly identified.

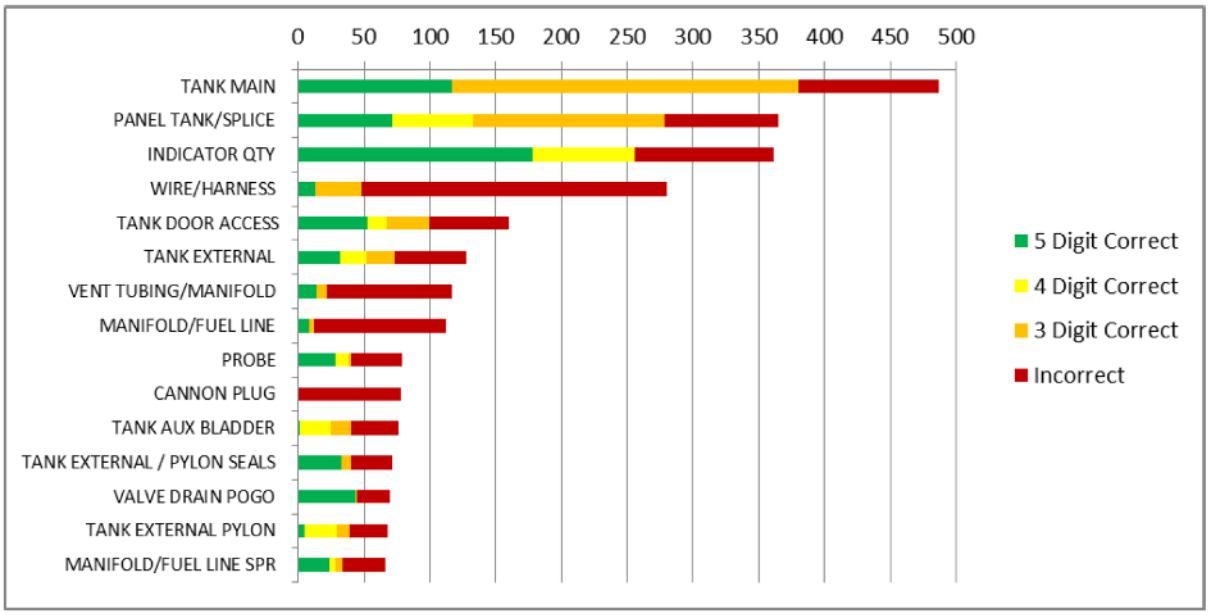


FIGURE B- 8. Correctness of WUC in maintenance database.

B.11.2.2 Tier 1 data review.

The evaluation of the maintenance data requires manual data review. The manual record examination process should be to separate the records into one of three categories:

1. The record does not involve an AFS component, or it was indirectly related (e.g. hydraulic failure caused damage to wires);
2. The record clearly is associated with AFS items;
3. It cannot be deciphered if the record is an AFS related event or insufficient information was provided.

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The results should be reviewed regularly to ensure the records are being accurately partitioned.

In addition to sorting, when going through the maintenance actions, AFS related items should be examined for correct work unit code assignment, component type, and any other fields that may be useful in the data analysis.

Notes for Maintenance Data review:

a. Do not delete any data. If data is not applicable, then it should be filtered out, or a column should be added identifying the information as 'Not Applicable'.
b. Do not overwrite original data. Again, this is for traceability of the data. If the report is associated with the wrong work unit code, then add a new column 'Correct work unit code' and apply the correct code.
c. During the data examination, generate a list of keywords to be included in another query on the maintenance data. Once this has been complied, rerun the report and merge the data. Repeat as necessary.

B.11.3 Tier 2 data review.
The Tier 2 assessment is performed by a senior engineer or technician to review the items marked as 'Unknown'. These 'Unknown' entries from Tier 1 are to be broken into either an AFS related issue or non-AFS issue. At the completion of this work, the data is ready for analysis.

B.12 DATA ANALYSIS

Data analysis begins after the data has been gathered, sorted, and cleaned up. There are multiple methods for analyzing the data, and some are presented here. These methods are intended to provide insight into the fuel system related maintenance actions that are performed on the fleet.

B.12.1 Analysis techniques.
Pareto Analysis: By concentrating on 20 percent of the factors (the top drivers or the biggest problems), 80 percent of the issues will be resolved.
Use Scatter Diagrams to correlate data. Scatter diagrams are plots of two variables used to show ; the relationships between the two variables; whether they are positive or negative. For example, when trying to determine the relationship between environmental exposure and connector corrosion; a Scatter Diagram will show a positive correlation between environmental exposure and connector corrosion.

Use Cause and Effect (Fishbone) diagrams to narrow causes. When relating a Key Performance Issue (KPI) to its potential causes, a powerful method is to develop a Cause and Effect Diagram, also known as a "Fishbone" Diagram. The main performance issue (or gap) is labeled as the fish's head, the major categories as structural bones, and the likely specific causes as ribs. The analysis identifies the major categories of potential problem causes.

B.12.2 Fish bone diagrams.
Fishbone diagrams (see FIGURE B- 9) are often cited as one of the seven basic tools of quality management and failure analysis. The fishbone diagram is often used early in a project to identify the potential factors that can cause an operational defect. It is recommended that the process

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should not be considered until after some data has been gathered on the topic as this will help to limit the amount of time spent investigating all of the brainstormed options.

The main performance issue (or gap) is labeled as the fish's head, the major categories as structural bones, and the likely specific causes as ribs. The analysis identifies the major categories of potential problem causes.

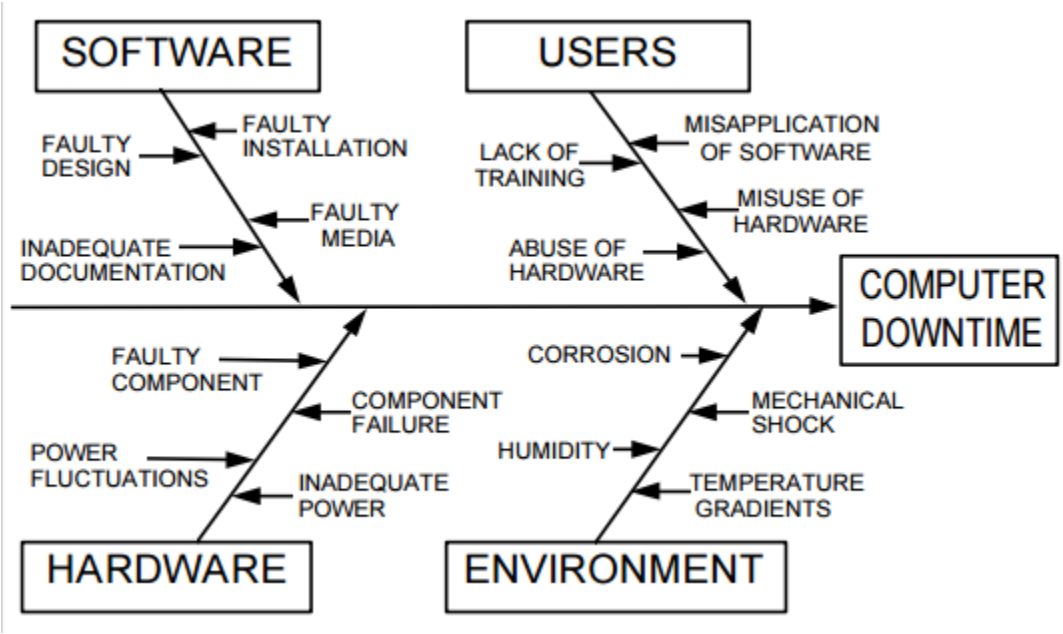


FIGURE B- 9. Example of a fishbone diagram. (Source: Center for Strategic Leadership Studies).

The use of the fishbone diagram is an ideal method for developing ideas, particularly when the problems (or potential problems) are elusive. After the ideas have been gathered, pruning of the diagram then becomes useful to help identify which areas should be ignored and those that may require more data.

When developing fishbone diagrams, there are some pitfalls of which to be aware. When gathering the ideas for filling in the diagram in a group setting, differences of opinion can cause issues and develop too fully a certain viewpoint. Also, it is common to have disagreements as to what may be part of the active failure or potential defect. Furthermore, if not properly managed, a great deal of time may be spent deliberating over whether or not to rule out unlikely factors rather than channeling resources toward evaluating likely factors.

Perform Trend Analysis. This is merely determining the direction and/or magnitude of a trend (increasing or decreasing; strong or weak). A time-series chart can also be considered a trend measurement tool. When measuring trends on fuel systems, various metrics such as sortie rate needs to be calculated and factored in when determining failure and/or cost trends.

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Identify "Hot Spots" and/or provide solutions. The last stage of AFS analysis is the identification of "hot spots" or failure clusters. These are areas of consistent weakness characterizes by high levels of maintenance and/or operational traffic, or excessive environmental exposure.

B.12.3 Partitioning of data.
In addition to the direct analysis of the data, other mechanisms for reviewing the data include:

a. Hot spot analysis partitioned by system, subsystem, location, zone, device type,
b. Number of maintenance actions that are allocated to the wrong work unit code,
c. If the fleet consists of multiple versions, it may be advantageous to partition the data by model. This can help identify potential issues that are experienced by a subgroup within the fleet.
d. If the fleet is operated by different MAJCOMs (e.g. AMC and ANG), partition by MAJCOM.

An example data analysis is shown on FIGURE B- 10. This analysis shows the in-flight and ground aborts caused by fuel system failure over a two-year period. In this case, the analysis partitioned the data by fuel system function.

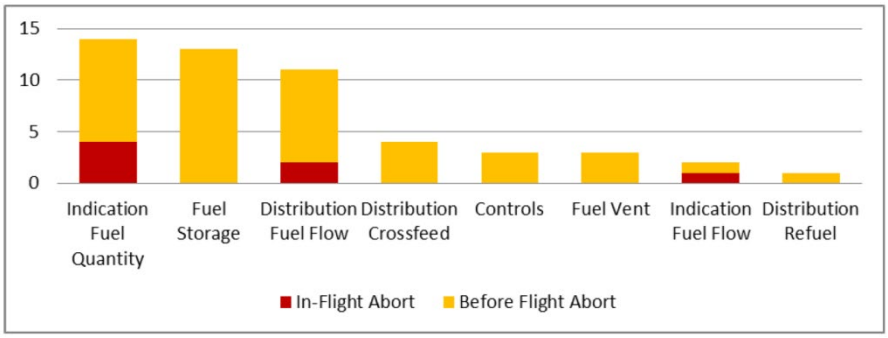


FIGURE B- 10. In-flight and ground aborts caused by fuel system failure.

B.12.4 U-Chart.
The U-Chart plots failures per unit of time vs. date for a given Work Unit Code (WUC). The U-Chart is used to detect emerging component reliability problems. The data used in the chart spans a two-year interval and are all derived from REMIS: 1) Failure counts are validated REMIS "hard" assembly failures, (i.e. Type 1, inherent failure, Action Taken "R", remove and replace action. 2) fleet flying hours. 3) Quantity Per Aircraft (QPA)).

B.12.5 Reliability predictive analysis.
The methods employed for predictive reliability analysis are Poisson or Weibull depending on the nature of the component data. Generally, when component failure times are known, a Weibull analysis is applied. Component failure time can only be calculated when a specific component can be tracked from installation to failure. This can normally only be determined when a scheduled replacement occurs at Programmed Depot Maintenance (PDM) and the field failure replacement is recorded in REMIS. The Poisson process yields the following: 1) λ, failures per flight hour, and

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its reciprocal Mean Flying Hours Between Failure (MFHBF); 2) Reliability performance trends; 3) Predicted failures for the next 12 months.

B.12.6 Current reliability performance and initial reliability.
In this section, the current reliability performance is compared to the initial assembly specification. Sources may include endurance testing requirements or reliability metrics as defined in source control documents or associated military specifications. Another source for reliability information will be the initial qualification test reports. These reports can be obtained from the manufacturer or sometimes the program office. Military specifications and their change history can be viewed online using Acquisition Streamlining and Standardization Information System (ASSIST), database through https://quicksearch.dla.mil , or Information Handling Standards (IHS), Inc., with a paid subscription, "Standards Expert," an online service providing mission critical industry standards and military/federal specifications needed for research and design, purchasing, bidding, logistics, and related applications.

B.12.7 Maintainability analysis.
Maintainability is assessed by examining unscheduled man-hours attributed to a WUC for trends and possible issues contributing to these hours, e.g. lengthy troubleshooting, significant time expended on adjustment, off aircraft intermediate level overhaul, or no scheduled item replacement. A bar chart may be produced which shows total scheduled and unscheduled man-hours by year.

B.12.8 Availability analysis.
The Air Force calculates aircraft availability as the percentage of Mission Capable (MC) aircraft in the Total Aircraft Inventory (TAI). The algorithms used to determine MC and TAI involve possessed, depot possessed, unit possessed not reported, NMCM, NMCS, and NMCB times. Because the impact a given assembly has on aircraft availability using these calculations becomes unclear, a simpler measure of an assembly's impact on aircraft availability is Non-Mission Capable (NMC) time charged to the subject assembly during a given time period. Non-mission capable time can be Non-Mission Capable for Supply (NMCS), Non-Mission Capable for Maintenance (NMCM), or Non-Mission Capable for Both Supply and Maintenance (NMCB). NMCS is an indicator of supportability performance issues such as spare supply, overhaul facility output, or limited sources of supply. NMCM is an indicator of maintainability performance issues such as trouble shooting duration, maintenance manning levels, or extended times for removal, adjusting, overhaul, or operational checks. Combining NMCS, NMCM, and NMCB into total NMC hours provides an overall impact the subject assembly has on aircraft availability. A bar chart may be produced which shows the total of each of these versus year.

B.12.9 Supportability analysis.
Assembly supportability can be ascertained by reviewing its MICAP history, any Aerospace Maintenance and Regeneration Group (AMARG) requisitions, the Problem Item Data Sheet, and Haystack Gold (Haystack Gold is available through the IHS website). When a part fails, a spare is not readily available, and it impairs the capability of an aircraft to perform its mission, it is backordered and designated as Mission Impaired Capability Awaiting Parts (MICAP). MICAP parts can be an indicator of supply issues such as required spares, overhaul facility capacity, or source of supply. A monthly MICAP summary can be obtained from the Supply Management

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Analysis Reporting Tool (SMART) and input to a database that can summarize MICAPs by WUC, P/N, or NSN from January 2006 to July 2011. From July 2011 forward, MICAP information is obtained from the Logistics Installation Mission Support-Enterprise View (LIMS-EV).

B.12.10 Product quality deficiency reporting analysis.

Product quality deficiency reports are available on repairable parts. These reports document deficiencies discovered by users on newly overhauled parts. They also contain action summaries by the repair facility as well as 24-month deficiency trend analysis. These reports are accessed using Joint Deficiency Reporting System (JDRS), a cross-service web-enabled automated tracking system, available at https://www.jdrs.mil. Reports can be queried by P/N, NSN, WUC, or report number. This section will summarize applicable quality deficiency reports on the subject assembly. Significant quality deficiency reporting negatively impacts reliability performance and may help identify needed modifications to overhaul procedures. The quality deficiency reporting system is outlined in T.O. 00-35D-54.

B.12.11 Data fitting.

The data may also be normalized. This process helps to account for increased operations tempo or an overall increase in maintenance actions. Normalization can be based on the total number of sorties flown, hours of flight, number of total maintenance actions, or total maintenance man-hours.

B.12.12 Current maintenance plan.

In this section, the current maintenance plan will be annotated. The maintenance plan will describe what maintenance actions, both scheduled and unscheduled, are currently being performed on the component. The maintenance plan will also identify at what level the current actions are being performed. For purposes of this report, the maintenance levels are defined by AFI 21-101, Aircraft and Equipment Maintenance Management, and are as follows: Organizational, Intermediate, and Depot.

B.13 ADDITIONAL REFERENCES

1. Integrity Program: Reliability Centered Maintenance Analysis, 151700-12020-F1.

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APPENDIX C

TASK 3 – COMPONENT FAILURE MODES

C.1 SCOPE

C.1.1 Scope.

The information gathered here is based on field experience, teardown reports, incident investigations and research conducted over the last 50 years. This information has been summarized and presented in this section with additional references included in each subtopic.

C.1.2 Overview.

Task 3 seeks to use the data gathered from Tasks 1 and 2 to direct engineering review of AFS components. Lessons learned are used to direct teardown analysis of components (Task 4) and AFS risk assessment (Task 5). The following section covers several general lessons learned on AFS system components, considerations for system level assessment, and dozens of single-page summaries of AFS components including common failure modes and recommended sustainment practices.

C.2 SAFETY CONCERNS WITH AGING SYSTEMS

C.2.1 Common safety challenges.

The information presented in this handbook does address some platform or mission specific components. In general, there is far more commonalities between platforms than there are differences. The mishaps have occurred on aircraft from across the fleet, the lessons learned are often applicable to most other platforms. The components used in the fuel systems are typically designed, installed, and maintained based on the same/similar type of specification.

C.2.2 Source of accidents.

It is well known in the aviation industry, pilot and/or crew error causes most aircraft accidents. The two charts (see FIGURES C- 1 and C- 2) below were compiled from USAF Class A mishap accident reports from 2008-2013, and from http://www.PlaneCrashInfo.com commercial aircraft accidents. Both show most aircraft accidents are attributed to the crew. Fuel imbalance has caused a number of military and commercial accidents and while crew error is typically a major contributor, there can be fuel system component failures that can significantly contribute to a mishap.

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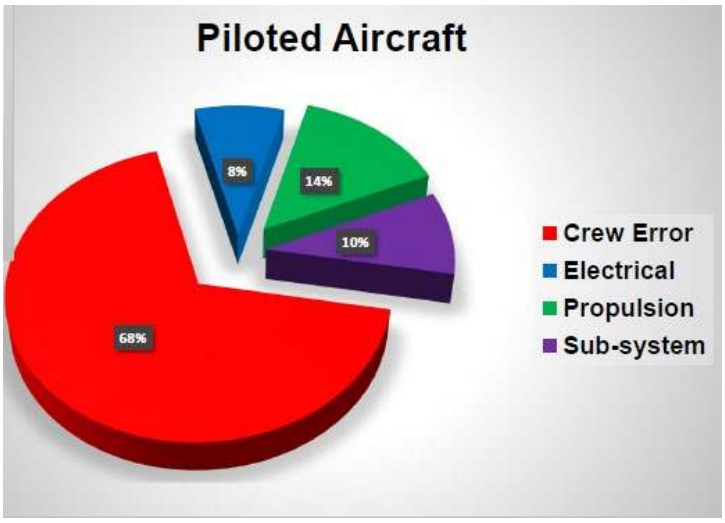


FIGURE C- 1. USAF Class A aircraft mishap causes 2008-2013 compiled from publically available accident reports.

Causes of Fatal Accidents by Decade (percentage)

Cause	1950s	1960s	1970s	1980s	1990s	2000s	All
Total Pilot Error	58	63	44	57	55	57	53
Pilot Error	43	33	25	29	29	34	32
Pilot Error (weather related)	9	18	14	16	21	18	16
Pilot Error (mechanical related)	7	4	5	2	5	5	5
Other Human Error	2	8	9	5	8	6	6
Weather	15	12	14	14	8	6	12
Mechanical Failure	19	19	20	21	18	22	20
Sabotage	5	4	11	12	10	9	8
Other Cause	0	2	2	1	1	0	1

The table above was compiled from the PlaneCrashInfo.com accident database and represents 1,015 fatal accidents involving commercial aircraft, world-wide, from 1950 thru 2010 for which a specific cause was known. Aircraft with 18 or less passengers aboard, military aircraft , private aircraft and helicopters were excluded.

"Pilot error (weather related)" represents accidents in which pilot error was the cause but brought about by weather related phenomena. "Pilot error (mechanical related)" represents accidents in which pilot error was the cause but brought about by some type of mechanical failure. "Other human error" includes air traffic controller errors, improper loading of aircraft, fuel contamination and improper maintenance procedures. Sabotage includes explosive devices, shoot downs and hijackings. "Total pilot error" is the total of all three types of pilot error (in yellow). Where there were multiple causes, the most prominent cause was used.
Source: PlaneCrashInfo.com database

FIGURE C- 2. Chart from http://www.PlaneCrashInfo.com provides info on worldwide commercial accidents.

C.2.3 Aircraft mishaps.

To highlight the importance of proper fuel system maintenance, design, and risk mitigation, the following mishaps shown here are brief summaries of events where the fuel system led to hazardous or catastrophic events. Each incident shown here had a different root cause that could be traced back to equipment failure, poor design, poor maintenance, and/or poor system management.

a. KC-135 - Fuel Pump Failure (see C.2.3.1).
b. F-16 - Fuel Leak/Flame Out (see C.2.3.2).

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c. B-1B - Failure of Fuel System Control (see C.2.3.3).
d. Nimrod - Midair Fire (see C.2.3.4).
e. TWA 800 - Fuel Tank Ignition (see C.2.3.5).
f. C-21 - Transport Aircraft Fuel Imbalance Mishap (see C.14).

Each mishap occurred on a different platform type and happened to a different part of the fuel system. Further discussion for each of these can be found in the appendices.

C.2.3.1KC-135 fuel pump.
This discussion concerns a KC-135 accident that occurred in 1993 in which six USAF airmen were killed and the aircraft destroyed by a fuel tank explosion and ensuing fire. The airmen were performing routine maintenance to troubleshoot an increase in the amount of fuel in the forward fuel tank which was reported after the previous night's mission. Mishap investigators determined ignition occurred as a result of uncontained arcing in an electrically driven center tank fuel boost pump which was activated during the maintenance operation. The cause of the failure was further attributed to an improper overhaul process that occurred at an Air Logistics Center (ALC).

Further discussion of this incident is in APPENDIX C.9. Directed inspection/sustainment recommendations are discussed in Component Failure Mode #01.

C.2.3.2F-16 fuel leak/flame out.
A number of F-16 aircraft have been lost due to suspected un-commanded closure of the fuel system Main Fuel Shut Off Valve (MFSOV) as a result of improper materials selection and safety critical single point failure condition. In most incidents (at least six documented mishaps), the pilot was forced to eject and the aircraft impacted the ground due to partial or complete fuel starvation of this single engine aircraft. Un-commanded closure of a fuel valve has occurred on other aircraft fuel systems due to design flaws, manufacturing defects, use of improper materials, poor maintenance practices or component aging. Further discussion of these incidents can be found in APPENDIX C.10.

C.2.3.3B-1B mishap caused by failure of a fuel system control panel.
This case study involves the failure of a printed circuit board within the Fire Warning Extinguisher Panel (FWEP) which caused the crash of a B-1B aircraft. The system failed in a way that prevented the crew from recovering the aircraft. Furthermore, the mishap panel was a redesign managed by an Air Force depot in order to improve the aircraft reliability. The incident findings resulted in a complete shift in the way the USAF conducts engineering design reviews and airworthiness assessments. This resulted in development and implementation of the Operational Safety, Suitability, and Effectiveness (OSS&E) process.

On the morning of February 18, 1998, a B-1B bomber took off from its base in Texas to conduct a routine training mission over the central U.S. About three and one-half hours later, as the bomber's four-man crew was commencing the final portion of the mission, the pilot was alerted to a possible Auxiliary Drive System failure on engine No. 3. Following standard procedure, he initiated a shut-down of that engine using the FWEP. As soon as the switch was enabled to cut-off fuel to the engine, the fire warning lights for the remaining three engines immediately illuminated, and thick smoke filled the cockpit. All four engines shut down without action by the

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crew, and in turn, disabled the B-1B's electrical and hydraulic systems. The loss of the aircraft was estimated at $310 million.

Further discussion of this incident can in APPENDIX C.11.

C.2.3.4Nimrod.
On September 2, 2006, an RAF Nimrod XV230 was on a routine mission over the Helmand Province in Southern Afghanistan when, only minutes after completing air-to-air refueling, the aircraft suffered a catastrophic mid-air fire which led to the total loss of the aircraft and the death of all 14 on board. Following an initial mishap investigation, a broader independent review was instigated.

The investigating board focused on why the Nimrod program failed to recognize design flaws over many decades. The presence of hot ducting and fuel pipes in two Tank Dry Bays, and the threat from fuel from the starboard tank blow-off valve entering the fuselage, represented a serious, failure condition to the Nimrod fleet for several decades. Many competent and dedicated engineers had worked on the Nimrod program and had observed fuel pipes and couplings within inches of hot ducting, yet took no action. Apparently, the engineers and technicians assumed the aircraft was designed properly and delivered in an airworthy condition.

The following are examples where the program made incorrect assumptions:

1. The Nimrod program had assessed the potential for fuel system leak as 'improbable' yet a review of Nimrod maintenance data showed that there was an average of 40 fuel leaks each year.
2. The program had determined the Cross-Feed Duct was only pressurized during engine start. Testing determined the Cross-Feed Duct is actually pressurized for long periods (at up to 420 °C (788 °F)).
3. The program noted fire detection and suppression systems as mitigation for the No. 7 Tank Dry Bay, yet neither a detection nor suppression existed within the bay.

In summary, this case illustrates how dangerous such fundamental "embedded" design defects can be, and why a robust MECSIP program is critical to maintaining air worthiness and aircraft safety (further reading in APPENDIX C.12).

C.2.3.5TWA 800.
On July 17, 1996, Trans World Airlines (TWA), Inc. Flight 800, a Boeing 747, crashed in the Atlantic Ocean near East Moriches, New York. The NTSB determined the probable cause of the TWA Flight 800 accident was a center wing fuel tank (CWT) explosion resulting from ignition of the flammable fuel/air mixture in the tank. The source of explosion ignition energy could not be determined with certainty, but, of the sources evaluated, the most likely was a short circuit outside of the CWT that allowed excessive voltage to enter it through electrical wiring associated with the fuel quantity indication system (further reading in APPENDIX C.13).

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C.3 FOAM IN FUEL TANKS SUMMARY

The function of fuel tank foam baffle systems is to suppress explosion reactions and prevent overpressure damage to the fuel tank. This is accomplished by controlling the rate of burning after ignition of fuel vapors through removal of energy from the combustion process by absorption of heat and by mechanical interference of the foam. Tanks containing foam are less likely to explode when pierced or ruptured by ground fire, arcing as the result of wiring/component malfunction, or as a result of lightning strike. The foam will also dampen fuel sloshing which can have an adverse impact on safety. Foam is specified in MIL-PRF-87260 (conducting foam). Additional information is included in APPENDIX C.15.

C.4 FUEL TANK IGNITION SOURCES AND MITIGATION STRATEGIES

Aircraft fuel tank system requirements are derived from MIL-STD-1798. This standard describes the general process to achieve and maintain the physical and functional integrity of the mechanical elements of aircraft systems.

The following is a brief summary of considerations:

a. Proper bonding of electrical equipment is critical to mitigate fuel tank ignition (see APPENDIX H).
b. There are several guidance documents from the FAA addressing fuel tank ignition (see APPENDIX G).
c. There are tools available to help with the risk assessment of fuel tanks.
d. All Electrical Wire Interconnection System (EWIS) components (including but not limited to: wiring, connectors, and electrical conduits attached to or near fuel system components.
e. Pumps can generate friction sparks that should be considered in failure safety assessments.
f. Fuel Quantity Indication System (FQIS) wiring needs to be given special attention as contamination can create arc path between probe and fuel tank walls.
g. Intrinsically safe circuits.
h. Hermetically sealed electric circuits.
i. Flame arrestors (used on electric motors in fuel tanks and on fuel vent overboard outlets).

Additional information on fuel tank ignition mitigation strategies is included in APPENDIX G.

C.5 OTHER DESIGN CONCEPTS

For those safety-critical components (and mission-, and durability-critical components determined to be feasible economically to prevent in-service failure) where damage tolerance design concepts do not apply or are not practical, other design approaches are available and needs to be utilized. Use guidance provided in AWB-013 and NAVAIRINST 5000.21B to assess the component's rate of failure, the consequence of failure, and determine the hazard risk index (HRI) value. Risks that cannot be acceptably mitigated may be reviewed and accepted by the appropriate risk acceptance authority within the program risk management process. The following other design concepts are available to prevent in-service critical failures:

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Inherently improbable occurrence of failure per AWB-013 and NAVAIRINST 5000.21B. This concept is most applicable to generic classes of components which are very similar across a wide range of aircraft (including installation loading and environment).

Examples of components for which "Improbable Occurrence" fits best are: fuel lines, fuel line couplings, and wire harnesses. In cases where "Improbable Occurrence" is used, the program should still look at potential for induced damage to the safety-critical parts from damage mechanisms such as chaffing. This look can be accomplished through actions such as recurring inspections or other appropriate means. Use of an "inherently improbable" approach for a MECSIP component needs to be identified in the MECSIP Master Plan (see DI-SESS-82329).

C.5.1 Design criteria for in-direct failure mode effects.

The subsystem design and mitigations implemented needs to reduce the predicted rate of Class A mishaps resulting for indirect failure mode effects to less than $1x10^{-7}$ per flight hour. Indirect failure mode effects are those in which the failed part is not directly involved in delivering a lost safety-critical function, but a collateral effect of failure interferes with a safety-critical function.

An example is: An Auxiliary Power Unit experiences an uncontained failure sending hot turbine material into a fuel tank, resulting in a fire, which causes loss of aircraft. The System Hazard Analysis (SHA) is a critical tool available to the MECSIP engineer to identify the hazards associated with each subsystem and define mitigations for those hazards. A list of indirect failure modes and the appropriate design criteria to control those hazards are shown on FIGURE C- 3.

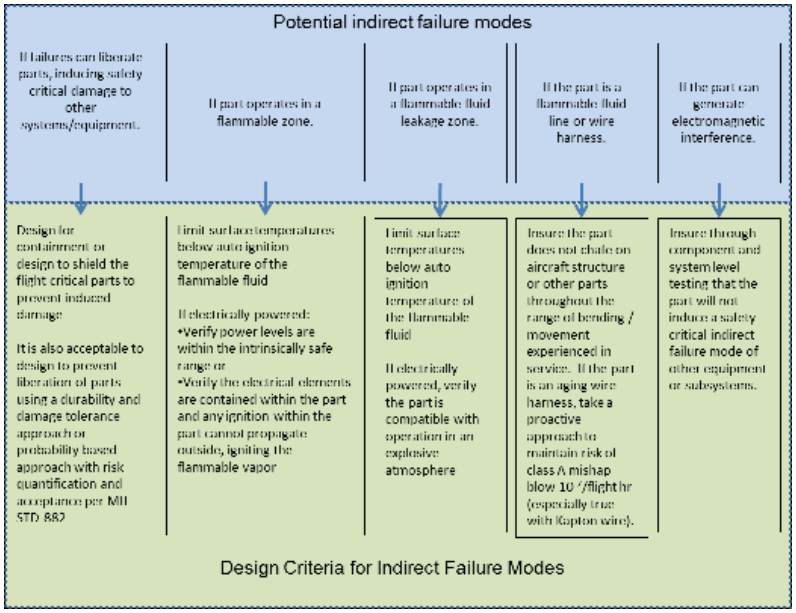


FIGURE C- 3. Indirect failure modes.

C.5.2 Components operating in a flammable zone.

For components operating in a flammable zone (e.g., inside a fuel tank), surface temperatures needs to be maintained under all conditions (including failure conditions) to less than the auto-ignition temperature of the flammable fluid. If the part is electrically powered, verify power levels

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are within the intrinsically safe range and/or verify the electrical elements are contained within the part and any ignition within the part cannot propagate outside, igniting the flammable vapor.

C.5.3 Components operating in a flammable fluid leakage zone.
For components operating in a flammable fluid leakage zone (e.g., dry bay next to a fuel tank) surface temperatures needs to be maintained under all conditions (including failure conditions) to less than the auto-ignition temperature of the flammable fluid. If the part is electrically powered, verify the part is compatible with operation in an explosive atmosphere.

C.5.4 Flammable fluid carrying line and wire harnesses.
Insure the line/wire harness does not chafe on aircraft structure or other components throughout the range of bending/movement experienced in service.

C.5.5 Service life assessment of hydromechanical systems.
Many Mechanical Equipment and Subsystems Integrity Program (MECSIP) components, particularly those with existing life limits, do not require assessment beyond investigation of continued availability of sources of repair. For MECSIP components, which are overhauled, Service Life Assessment Program (SLAP) needs to look at the overhaul procedures to see if any changes are warranted as the aircraft life is extended. Other MECSIP equipment (e.g., fuel/hydraulic/ECS tubes, brackets seals etc.) age and require a comprehensive look at the need to potentially replace the entire system or increase inspections before the fleet reaches the end of the life extension. By performing the SLAP in advance of the Service Life Extension Program (SLEP) decision point, decision makers can understand all of the costs and risks associated with a SLEP program in advance of the decision. NOTE: SLAP is only required to be executed on safety-critical parts.

The Program Office may wish to include other components based on cost, obsolescence, reliability drivers, etc.

C.5.6 Fuel characteristics and compatibility.
Fuels used in military and commercial aircraft have been well categorized in numerous government standards. For further information about fuels, consult the standards listed in TABLE C- I.

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TABLE C- I. List of common fuel types with corresponding standard and basic properties of each fuel type.

Common Designation	NATO Designation	Base	Standard(s)	Freeze Point C (F)	Flash Point C (F)	Density @ 15C (kg/L)	Notes
Avgas			ASTM D910	≤-58 (-72)		0.690 - 0.715	
F-24	F-24	Kerosene					Same base fuel as JetA with additives in JP8
F-65							50/50 blend of No. 2 Diesel and JP-5 or JP-8
JAA				≤-40 (-40)			Jet A with additives
Jet B		Kerosene	ASTM D6615	≤-50 (-58)		0.751 - 0.802	Used in cold climates. Similar to JP-4
JetA		Kerosene	ASTM D1655	≤ -40 (-40)	≥38 (100)	0.775 - 0.840	
JetA-1		Kerosene	D1655, DEF STAN 91-91	≤ -47 (-53)	≥38 (100)	0.775 - 0.840	
JP-4	F-40	Kerosene/ Gasoline	MIL-DTL-5624, DEF STAN 91-86	≤-58 (-72)		0.751 - 0.802	In use by USAF from 1951 - 1996
JP-5	F-44	Kerosene	MIL-DTL-5624, DEF STAN 91-86	≤-46 (51)	≥60 (140)	0.788 - 0.845	
JP8							
JP8	F-34	Kerosene	MIL-DTL-83133, DEF STAN 91-87	≤ -47 (-53)	≥38 (100)	0.775 - 0.840	JetA-1 with SDA, CI/LI, FSII, AO, MDA
JP-8+100	F-37	Kerosene					JP-8 additized with +100 thermal stability improvider additive (NATO S-1749)
Mogas			ASTM D4814			0.715 - 0.770,	
No 1 Diesel	F-44	Diesel	ASTM D975		≥38 (100)	0.775 - 0.840	
No 2. Diesel	F-54	Diesel	ASTM D975		≥52 (126)	0.820 - 0.840	
TS-1		Kerosene	GOST 10227	≤-60 (-76)	≥28 (82)	0.787	

C.5.7 Aircraft fuel compatibility with fuel tank materials.

Aircraft fuel compatibility with metals used in fuel tanks requires extensive testing and evaluation since reactions may require extended times to identify compatibility issues. Material degradation in a fuel tank can lead to catastrophic failures such as loss of an aircraft, loss of aircraft availability, or extensive maintenance costs since repair can require removal of fuel tank structure.

The Air Force Research Laboratory (AFRL) has conducted extensive fuel compatibility testing on a variety of fuels and metallic, polymer, and electrical materials. It is highly recommended these reports be consulted when reviewing materials used for fuel tanks, when evaluating new fuel types/additives, or when there is a need to conduct compatibility testing. Many of these reports are restricted access, but can be obtained with a valid need-to-know and proper clearance from:

AFRL/RX
2977 Hobson Way, Bldg. B653

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Wright-Patterson AFB, Oh 45433

A typical fuel compatibility summary table is shown on FIGURE C- 4 followed by specific examples of material reactions with certain fuel types.

Metallic Test Results

Material	Light Optical Evaluation		Microstructural Evaluation	
	JP8	P44 Additive	JP8	P44 Additive
304SS	Non-uniform staining	Clean	No Degradation	No Degradation
17-4 PH	Same as reference	Clean	No Degradation	No Degradation
440 SS	Light deposit	Black discoloration	No Degradation	No Degradation
Ti - 8Al - 1V – 1Mo	Light staining	Clean	No Degradation	No Degradation
Ti – CP – 70	Light staining	Light staining	No Degradation	No Degradation
Ti – 3Al – 2.5V	Light staining	Clean	No Degradation	No Degradation
4130 IVD Coating	Light Staining	Clean	No Degradation	No Degradation
Alloy Steel Fastener TYII, Class 2	Light staining	Clean	No Degradation	No Degradation
A286 Fastener, Ag Plate	Light Staining	Clean	No Degradation	No Degradation
CPM 10V	Black deposit	Black deposits	No Degradation	No Degradation
INCO 625	Clean	Clean	No Degradation	No Degradation
INCO 718	Slight discoloration	Clean	No Degradation	No Degradation
Nitralloy 135	Slight discoloration	Clean	No Degradation	No Degradation
IN 200 Ni	Black deposit	Light staining	No Degradation	No Degradation
Monel 400	Same as reference	Black deposits	No Degradation	No Degradation
Waspalloy	Light staining	Clean	No Degradation	No Degradation
Lead	Light staining	Black deposits	No Degradation	No Degradation
268 Brass Sheet	Black dense deposit	Light staining	No Degradation	No Degradation
TAP MS 285	Black dense deposit	Black deposits	No Degradation	No Degradation
Mag Wire Type 1	Dark Staining	Clean	No Degradation	No Degradation
7075-T6 Chromic Acid Anodize Type I	Clean	Light staining	No Degradation	No Degradation
7075-T6 Sulfuric Acid Anodize TY IIB	Clean	Clean	No Degradation	No Degradation
7075-T6 Chromate Conversion Coated Class IA	Clean	Clean	No Degradation	No Degradation
7050-T74	Clean	Clean	No Degradation	No Degradation
2024-T3 Bare	Clean	Clean	No Degradation	No Degradation
6061-T6 Bare	Clean	Clean	No Degradation	No Degradation
5052-H34 Bare	Clean	Clean	No Degradation	No Degradation
356 T6	Clean	Clean	No Degradation	No Degradation
AZ91 T6	Clean	Black deposits	No Degradation	No Degradation
CU/NI 90/10	Clean	Light staining	No Degradation	No Degradation
Sn 60 Pb 40 Solder	Light staining	Brown staining	No Degradation	No Degradation

FIGURE C- 4. Typical fuel compatibility summary table from AFRL/RXS 07-085 (limited distribution). P44 is a low temperature fuel additive.

C.5.7.1Cadmium.
As an example, AFRL has reported bare Cadmium plated fasteners exhibit degradation inside fuel tanks when exposed to JP-8 fuel. The interaction between cadmium and sulfur has been studied extensively. Cadmium is often used to prevent corrosion of steel hardware by virtue of the galvanic series: the cadmium is anodic to the iron and protects it by sacrificially corroding. Cadmium is also known to react with sulfides (typically found in fuel) to produce a reaction product. This initially appears as a yellow or red discoloration as seen on FIGURE C- 5.

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FIGURE C- 5. Example of cadmium plated fasters in a fuel tank that have been degraded by fuel (see AFRL/RXS 10-097).

In one aircraft, this can potentially impact over 250,000 wing fasteners. The mitigation approach was to add sealant caps to each fastener to isolate fasteners from fuel, which requires many hours to be added to the PDM process. Other Cadmium-plated components such as bolts and structural parts have been damaged by exposure to fuel. Complete materials lists need to be reviewed for the presence of bare cadmium surfaces in new and existing designs. Traditional approaches to prevent Cadmium sulfur reactions in fuel have been to prevent contact by the use of a barrier or sealant applied during manufacturing and maintenance.

C.5.8 Limitations on use of protective metallic coatings - MIL-STD-1568.

MIL-STD-1568 addressed this issue. The following are excerpts from the standard that provide specific guidance on this issue.

C.5.8.1Cadmium.

Limitation on use of protective metallic coatings. Soft surface coatings such as Cadmium, Nickel-Cadmium, and Aluminum cannot be used for sliding or wear applications. Cadmium plated surfaces cannot be used in applications where surface temperature exceeds 232 °C (450 °F). Cadmium cannot be used on functional fuel systems components that can come into contact with fuel during operations of the aircraft. Cadmium plated fasteners, used in areas where contact with fuel can occur, needs to be overcoated with an approved fuel tank sealant and coating over the sealant.

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C.5.8.2Silver.
Silver is known to react with residual sulfur that is commonly found in jet fuel. The presence of moisture greatly accelerates the corrosion reaction between silver and sulfur and over time will expose copper. Once copper is exposed, conductive residues can form around terminals and other electrical terminations causing leakage current and resistance shorts between connections. This issue is extensively discussed in APPENDIX C.13 and APPENDIX G under fuel probes and interconnections. Also, see FAA report DOT/FAA/AR-03/61, Silver-Sulfur Deposits on Fuel Quantity Indication System and Attendant Wiring and AC No: 25.981-1C, Fuel Tank Ignition Source Prevention Guidelines. Silver plated components should not be used inside fuel tanks; only nickel plated terminals are recommended.

C.5.8.3Solders.
Tin, lead, and most solders (lead/tin and silver/tin alloys) will also react with jet fuel in a similar manner as silver and are not recommended for use in fuel tanks. Electrical terminations should be crimped and mechanical connections swaged or welded.

For additional guidance on material compatibility, see MIL-STD-1568. The following are relevant excerpts from this standard:

1. Environmental Sealing – Detail Requirements.
   a. Integral fuel tank coatings will meet the requirements of SAE AMS-C-27725.
   b. Sealants used in integral fuel tanks will conform to SAE AMS-S-8802 or SAE AMS3276.

2. Fastener Installation – Detail Requirements.
   a. Fasteners in integral fuel tanks will be installed with wet sealant conforming to SAE AMS-S-8802 or SAE AMS3276.

C.6 REFERENCES

1. AFRL-PR-WP-TR-2000-2015, Fuel and Fuel System Materials Compatibility Test Program for a JP-8 +100 Fuel Additive.

C.7 FAILURE MECHANISM DESCRIPTIONS

The following section provides short single-page descriptions of AFS components, failure modes and test assessments. The single-page descriptions identify specific examples, common failure modes, potential causes, failure mitigation strategies and other information pertaining to the AFS component, failure mode or test assessment.

C.7.1 Fuel boost pump 53

C.7.2 Separation of lines carrying flammable fluids 54

C.7.3 Fuel tank FQIS 56

C.7.4 Nozzles 58

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Check the source to verify that this is the current version before use.

C.7.5 Fuel bladder/fuel cell 60

C.7.6 Fuel shutoff valve actuator 62

C.7.7 Actuator, electromechanical linear (LVT) 64

C.7.8 A/R master switch 65

C.7.9 Hose assemblies 66

C.7.10 Topcoat degradation 68

C.7.11 Fuel shutoff valve 70

C.7.12 Valve, fuel pressure regulator 72

C.7.13 Valve assembly, fuel swing check (transfer valve) 73

C.7.14 Valve, swing check (CWT override pump) 74

C.7.15 Valve, swing check (A/R pump) 75

C.7.16 Actuator, electromechanical linear (Valves 1&2) 76

C.7.17 Inerting system and flammability 78

C.7.18 Refueling boom signal coil system 80

C.7.19 Failure mode: connector corrosion 82

C.7.20 APU pump 83

C.7.21 Failure mode: electrical arcing at housing interfaces 85

C.7.22 Failure mode: nozzle leaks 87

C.7.23 Component: O-rings 89

C.7.24 Failure mode: FOD 91

C.7.25 Reference JSSG-2009 93

C.7.26 Bypass valve 95

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C.7.1 Fuel boost pump.

EXAMPLE FAILURE	COMMON FAILURE MODES
Screw dislocated from original fit position and migrated into motor stator assembly. Through arcing, the screw welded to the stator. This is an uncommon scenario. 	Fuel pressure loss, or fuel ignition sources. Most maintenance write-ups for boost pumps are related to electrical contacts external to the pump, and are not addressed here. Otherwise, pump issues are usually related to electrical defects associated with the wires inside the pump, or bearing issues. **POTENTIAL CAUSES** Electrical insulation breakdown wiring attached too and part of the motor winding, FOD (common on initial flight tests), contact corrosion, winding failures, bearing wear & fatigue, water ice buildup on screens, pressure changes/ pressure ripple (caused by pump or other system components) and cavitation can also cause impeller fatigue (low likelihood of occurrence (only applies to vane type pumps).

EQUIPMENT SUSTAINMENT

Common pump failure mode is via the pump controller. Separation of controller from pump allows for individual component replacement.

Periodic evaluation of pump current draw can be used to preemptively identify degradation. Health and Usage Monitoring Systems (HUMS) are feasible.

Estimated/Typical number of flight hours 40,000.

REFERENCES FOR FURTHER READING

a. MIL-STD-810, Environmental Engineering Consideration and Laboratory Tests
b. SAE ARP594, Fuel Pump Thermal Safety Design
c. MIL-DTL-62011, Detailed Specification: Pump, Fuel, Electrically-operated: 24-Volt DC, In-Tank Type, General Specification for
d. Advisory Circular AC 25.981-1C, Fuel Tank Ignition Source Prevention Guidelines
e. ARP594, Fuel Pump Thermal Safety Design
f. APPENDIX C.9 – KC-135 Fuel Tank Explosion

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C.7.2 Separation of lines carrying flammable fluids.

EXAMPLE FAILURE	COMMON FAILURE MODES
Chafing is a common problem between fuel pump lines. Marriage clamps can be used but do not solve the problem. Chafing may still occur between lines. The lower photo shows a marriage clamp not preventing chafing between the two hydraulic lines, which power a fuel pump.  	Fuel pressure output when 80 psi or higher can lead to hydraulic leaks, or fuel leaks. Electrical control can lead to perceived problems with fuel lines. **POTENTIAL CAUSES** Complex hydraulic line routing, chafing, B-nuts, clamps, fittings, electrical control wires, jam-nuts, torque, seals, pre-loading, and other mechanical issue.
FAILURE MITIGATION STRATEGIES 1. Maintenance procedures should require a minimum separation distance for hydraulic lines (see No. 5). 2. Reduce preload by specifying exact bending. Preloading occurs when installers adjust the initial line position to the needed operational position. 3. Routinely inspect lines for chafing. From T.O. 1C-135-6WS-13, "chafed areas exceeding 15 percent of wall thickness shall be repaired". 4. From T.O. 1C-135-2-5-1: "Maintain minimum tube, hose, and wire clearance to prevent chafing and the possibility of fire. Do not twist or kink flexible hoses during installation. While tightening hose nuts, use an additional wrench to keep swaged fittings from turning." "Fuel tubing shall be installed to provide 0.25 inch (line-to-line) clearance." 5. From installation drawing: "Design objective clearance is 0.20 inch minimum except that the design objective clearance is 0.10 inch	**OTHER INFORMATION** 1. Electrical control that handles hydraulic activation of pumps may be at fault for failure reports. 2. Pumps that are submerged during operation needs to be qualified to run dry (limits apply). 3. No thermal sensor is needed on hydraulic-powered fuel pumps. 4. Pumps can be overhauled. 5. Some pumps are cooled via hydraulic fluid.

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minimum where tube is located and held by a support so that no rotation or axial motion of tube is allowed to reduce that clearance". 6. Periodically inspect bearing shaft play for compliance with design tolerances, which could cause loss of pressure output and overheating/ignition near end of life.	

EQUIPMENT SUSTAINMENT

Electronics are 40k hours MTBF.

Some components may have cycle limits: e.g. flow limiters may have a 20,000 pressure cycles limit.

Periodic inspection of pumps (10k hours) is recommended.

Failure Indication Signs

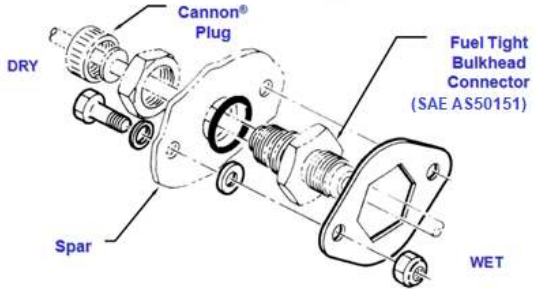
Hydraulic fluid leaks, loss of hydraulic reservoir volume.

Failed ground inspection, pressure tests.

Bearing noise, low fuel output pressure.

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C.7.3 Fuel tank FQIS.

EXAMPLE FAILURE	COMMON FAILURE MODES
In some designs, fuel may wick up the probe harness and penetrate the electrical connection at the fuel tank wall. The figure below shows a configuration with the probe on the right side (inside tank). Typical Spar Penetration for Electrical Connector  	Connectors/potting may not be fuel proof. The Integrated Fuel Management Panel (IFMP) incorrect quantity indication is a common maintenance issue. Capacitance error can occur due to a change in the dielectric caused by fuel incursion in the electrical feed-through connector causing intermittent or varying fuel quantity. **POTENTIAL CAUSES** Fuel leaks, connector faults, splicing. Component supplier design and process changes. Probe contamination through moisture/dirt. Issues are expected to occur throughout life-cycle.
FAILURE MITIGATION STRATEGIES 1. From T.O. 1C-135-6WS-11, "Inspect all fuel quantity probes and compensators (body and wing) for corrosion, cracks, damage, and general condition, loose mounting bolts, and frayed or broken bonding jumpers. Disconnect electrical connectors at fuel probe and compensators at tank locations and inspect for corrosion, fuel seepage, general condition, and clean connector. Perform an electrical capacitance check on each terminal. Apply a thin and even coat of fuel sealant to Fuel Probe Connector." 2. Define qualification requirements in design drawing. Recommended: Weight, dielectric, shock, temperature, vibration, fuel resistance, and capacitance. 3. Capacitance errors are generally positive indication errors (fuel quantities indicated/displayed are higher than actual). Control by design drawing is recommended. a. Avoid use of silver plated wiring and electrical contacts in tanks. Use nickel plated wiring, crimp connections and gold plated electrical contacts.	**OTHER INFORMATION** 1. Original manufacturer may no longer produce the fuel probes under the original Specification Control Drawing (SCD). New vendors can be approved through reverse engineering, analysis, and test, and must require qualification to avoid reliability issues. 2. Newer probes are designed to operate at higher frequencies and lower voltages (as low as 3.3VDC).

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4. Fuel contact can cause solder joint corrosion. Break in wire nickel/tin plating can cause wire conductor corrosion. Seal wires against fuel. 5. Understand thresholds utilized by the Integrated Fuel Management Panel (IFMP) to display dashes in quantity fields. 6. Two areas of the system may not be compliant with SFAR-88: boost pump power wires bundled with fuel probe wires; and lack of lightning protective shielding on wires. 7. Investigate present commercial alternative design.	

EQUIPMENT SUSTAINMENT

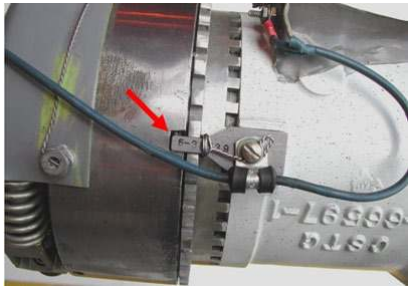
1. On some platforms, the FQIS (not individual probes) is a mission critical component.
2. Each tank can have multiple probes.
3. Fuel probes can be repaired, although it is not recommended due to a reduced life when re-installed on aircraft.
4. Use PDM inspection and testing of fuel system tank unit arrays including capacitance and resistance values.
5. If the FQIS system is being replaced as part of a life extension program or reliability issue, also replace fuel quantity wiring and ensure FQIS wiring is separated adequately from other aircraft wiring.
6. Eliminate water sources from fuel-ground storage systems using filtering, water traps and fuel tank sumping.

REFERENCES FOR FURTHER READING

1. T.O. 1C-135-6WS-13, Technical Manual Overhaul Instructions Depot Maintenance
2. T.O. 1C-135-2-5-1, Technical Manual Maintenance Instructions

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C.7.4 Nozzles.

EXAMPLE FAILURE	**COMMON FAILURE MODES**
	Unexplained inflight aborts and malfunctions of the refueling communications signal coils. Fuel leaks and spray on the receiver aircraft. Indications: Negative contact (missed contacts), delayed disconnects (termination of refueling), aborted missions, and brute force disconnects. **POTENTIAL CAUSES** Degradation of O-rings and seals, corrosion. Centering Mechanism cannot correct for excessive tolerance play.
FAILURE MITIGATION STRATEGIES 1. Establish a maintenance specification for maximum allowable alignment tolerance for nozzle centering. This should require nozzle centering to return nozzle Signal Coil centerline to within ±2.5 degrees of Boom axial centerline at the 6 o'clock position. 2. A basic functional specification is needed to measure Nozzle centering and control growth of rotational play with a maximum allowance for misalignment. 3. Machine modification of Shock Absorber Recoil Assy., Nozzle Centering Mechanism P/N 50-1438 to achieve a positive centered position eliminating free-play dead-spot of 0.100 plus inch arc length, approximately 2 degrees of rotational play. 4. Centering Mechanism is a term to help clarify piece parts requiring attention. The Centering Collar is part of the Shock Absorber Recoil Assembly. A minor machine tool modification to the Recoil Tube Centering STOP could effectively eliminate the undesired Free-play, creating a Positive Center. 5. Design modification and substitution of Nozzle Installation Coupling Nut Locknut Retainers P/N 6-38539-1 to eliminate 0.080-inch excessive clearance play. **6.** Revise maintenance requirements and procedures. Create Adapter Fixture for Nozzle Centering Checkout.	**OTHER INFORMATION** 1. Tolerance stack up in: a. Recoil b. Centering Mechanism c. Nozzle rotational play misaligns the Signal 2. Coils. a. Other factors effecting Signal Coil at contact b. Coil surface corrosion c. Coil separation gap d. Receptacle roller wear

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EQUIPMENT SUSTAINMENT

1. Failures occur throughout life of component.
2. Leakage has limits, but in some cases the leak was within limits, yet still caused a write-up; so, the allowable may be too high.

REFERENCES FOR FURTHER READING

1. Presentation, KC-135 Aerial Refueling Nozzle Centering Mishaps, by Tarrell N. Towle MSgt, 157MXG, Pease ANGB, NH
2. MS24354 - Drogue Cone, Nozzle and Reception Coupling - Type MA-2, Flight Pressure Refueling System, Assembly of. (Inactive for new design)
3. MIL-PRF-81975 - Couplings, Regulated, Aerial Pressure Refueling Type MA-2, Type MA-3 and Type MA-4

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C.7.5 Fuel bladder/fuel cell.

<table>
<tr>
<td>EXAMPLE FAILURE
The failure occurred between the interconnect fitting and the bladder. Failures at this location are common.</td>
<td>COMMON FAILURE MODES
Failures are most common at upper vent interconnects. Bladder manufacturing processes can change over time, resulting in a smaller cell, which causes additional loading on interconnects.

POTENTIAL CAUSES
1. Desire to reduce creasing can result in intentional size reductions, since creasing can also cause leaks.
2. Curing or material changes can result in additional shrinkage and smaller size.
3. O-ring failures and wall punctures after installation are common.</td>
</tr>
<tr>
<td>FAILURE MITIGATION STRATEGIES
1. Dimensional measurement at the manufacturer, as part of the Acceptance Test Plan.
2. Control of manufacturing technique can avoid variations in molds, using digital definition.
3. Control of manufacturing technique can vacuum-form around difficult shapes.
4. Instructions to maintenance personnel to limit preloading to specific values.
5. Backing boards can fill in gaps between structures to increase support.
6. O-ring retention by fittings, as opposed to loose O-rings, reduces installation error.</td>
<td>OTHER INFORMATION
1. Platforms that use the fuel as a heat sink have found shorter service life.
2. Self-sealing bladders are thicker than non-sealing bladders. Crash resistant bladders have thicker walls than self-sealing. Thicker bladders increase risk of installation damage.
3. If bladder does not leak within first 10 hours following installation, rare to find service issues afterwards.

INDICATION SIGNS
1. Fuel leaks can appear in drains or elsewhere on the airplane if cavities are not sealed.
2. Fuel gages can indicate loss of fuel in extreme cases.</td>
</tr>
<tr>
<td colspan="2">EQUIPMENT SUSTAINMENT
1. Some platforms (F-15) have a five-year replacement plan for bladders.
2. Removal of bladders can cause damage and result in more breaches.
3. Helium can be used to detect leaks.
4. Bladders can be repaired.
5. Bladder material degradation may require replacement after 20 years - material dependent.</td>
</tr>
</table>

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6. MTBF for leaks: 86k hrs. MTBF for tears: 150k hrs.

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C.7.6 Fuel shutoff valve actuator.

EXAMPLE FAILURE

Electrical Failure: Failure of the wiring can damage the valve and create an ignition hazard.




Mechanical Failure: Microswitches or stripped gears can cause poor performance.




COMMON FAILURE MODES

1. Electrical actuator fails to move valve sufficiently, or creates ignition hazard. When equipment nears wear out or end of life system can leak fuel inside the cavity.
2. Valve typically does not fail; failures are typically associated with the actuator or actuator control systems. Valve can fail open, closed, or in an intermediate position.

POTENTIAL CAUSES

Wear of wires / electrical components, FOD, micro-switches, wear on mechanical parts (gears), and corrosion of input connection.

FAILURE MITIGATION STRATEGIES

1. On-aircraft diagnostic test depend on location, and consist of energizing the valve and viewing manual actuator arm position, or checking pressure/flow.
2. Preventative maintenance includes visual inspection of valves for security, loose connections, corrosion, cracks, leaks, loose mounting

OTHER INFORMATION

1. Wire problems are commonly written up against the component.
2. Criticality of shutoff valve is closely associated with the number of engines on the vehicle.

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bolts, frayed or broken bonding jumpers, damage and general condition. 3. Qualification testing shall include explosion proof test per MIL-STD-810, Method 511.5, Procedure I, Explosive Atmosphere or Procedure II, Explosion Containment, as appropriate; as well as locked-rotor overheating test. 4. Replacement of micro-switches is recommended during any overhaul. 5. T.O. 1C-135-6WS-14, 1C-135-6WC-2 & 1C-135-6WC-1 list maintenance requirements for each valve type.	

EQUIPMENT SUSTAINMENT

1. Overhaul or replacement on a periodic schedule is recommended.
2. Conduct electrical dielectric test on electrical equipment and run a component case pressure test.
3. Selected qualification testing can be used to evaluate in-service component condition.
4. Failures are likely to occur at end of the life-cycle when insulation degrades and stress fatigue develops.

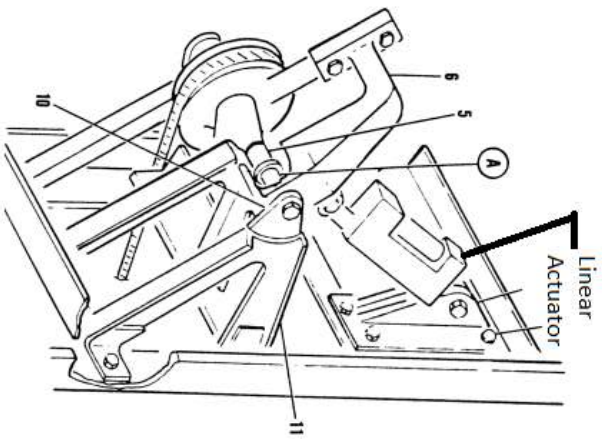
REFERENCES FOR FURTHER READING

1. Submerged & Unsubmerged Overhaul T.O.: 8D1-14-52-3, 6J15-10-66-363, 6J15-10-171-3, 6R9-10-25-3
2. Transfer Valve Overhaul T.O. 6J15-10-76-33
3. T.O. 1C-135-6WS-14, Programmed Depot Maintenance Specification
4. T.O. 1C-135-6WC-2, Periodic Inspection Workcards USAF Model All C-135 Aircraft
5. T.O. 1C-135-6WC-1, Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All 135 Aircraft
6. APPENDIX C.10 – F-16 Main Fuel Shutoff Mishaps

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C.7.7 Actuator, electromechanical linear (LVT).

Actuator Diagram



COMMON FAILURE MODES

1. A quadrant operated by an electrical actuator provides remote control capability of the valves on ARR aircraft. When the master refuel switch is in the ON position, the actuator may be energized through the manifold valves switch on the ARR panel.

2. These valves are cabled together so that when No. 1 (see T.O. 1C-135(K)R-2-5GA-1 Figures. 1-2) is open, No. 2 is closed and when No. 2 is open, No. 1 is closed.

POTENTIAL CAUSES

1. Electrical controls, wiring, electrical solenoid, valve jammed due to FOD or contamination can cause failure.
2. Possible to occur later in life cycle.

FAILURE MITIGATION STRATEGIES

1. T.O. 1C-135-6WS-14 requires the aircrew "accomplish an operational check of aerial refueling receiver (ARR) system on aircraft equipped with a refueling receptacle IAW T.O. 1C-135-2-6, paragraph 11-48".
2. T.O. 1C-135-6WC-1 card 1-007 requires for ARR aircraft, preflight inspection: "Manifold Valves Switch Positioned to Wing Refuel/Refuel Position."

OTHER INFORMATION

Often wire problems are written up against the component.

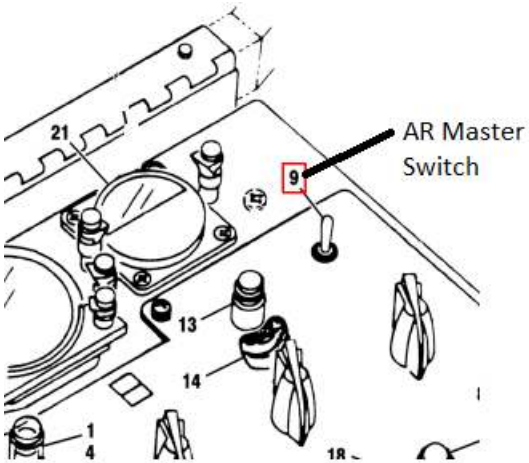
This linear actuator P/N 38092-3-1 is shown in IPB T.O. 1C-135-4-5, Fig. 51, but it does not show what it controls. It actuates Valves 1&2 similarly to the Boeing 38092-4-1 linear actuator, at STA 850-860 per LTV drawing. 8535-17413.

EQUIPMENT SUSTAINMENT

1. Overhaul or replacement on a periodic schedule is recommended.
2. Conduct electrical dielectric test on electrical equipment and run a component case pressure test.
3. Selected qualification testing can be used to evaluate the component condition.
4. Failures are likely to occur at end of life-cycle when insulation degrades and stress fatigue develops.

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C.7.8 A/R master switch.

FAILURE INDICATION SIGNS 1. Ready light does not illuminate. 2. Electrical functions of the Aerial Refueling System (A/R) does not operate. 3.  	**COMMON FAILURE MODES** The switch is not prone to failures, but loss of the conductive path results in loss of aircraft electrical power. **POTENTIAL CAUSES** Corrosion, dirt, electrical controls, wiring damage can result in loss of conductive path and loss of electrical power. Failure more common as component reaches end of life late in the life cycle.
FAILURE MITIGATION STRATEGIES 1. T.O. 1C-135-6WS-14 requires to "perform an operational check of the A/R system on all aircraft" per the indicated T.O.s. 2. T.O. 1C-135-6WC-2 has no requirements for this switch. 3. T.O. 1C-135-6WC-1 has no requirements for this switch.	**OTHER INFORMATION** 1. The A/R master switch (toggle type) when positioned to ON will supply power to the A/R electrical controls and to the A/R amplifier.
REFERENCES FOR FURTHER READING 1. T.O. 1C-135-6WS-14, "Programmed Depot Maintenance Specification". 2. T.O. 1C-135-6WC-2, "Periodic Inspection Workcards USAF Model All C-135 Aircraft". 3. T.O. 1C-135-6WC-1, "Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft".	

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C.7.9 Hose assemblies.

FAILURE INDICATION SIGNS

1. Leaking, chaffing, oil canning.
2. For rubber hoses, finding rubber particles in filters.

Hose Assembly



COMMON FAILURE MODES

Failure of these hoses is usually found as a leak, although a rupture is possible. Particles can detach and lodge in downstream components. Leaks in the vicinity of ignition sources, such as hot air ducts or electrical equipment are a fire risk.

POTENTIAL CAUSES

Surge pressure, fatigue, wear, and preload can cause breakdown. Chafing of hoses and cyclic twisting of hoses. Also, internal delamination of rubber from wire braiding can be a cause. Improper manufacturing, defective materials, or use beyond expected life can cause delamination. Failures are more likely later in life cycle.

FAILURE MITIGATION STRATEGIES

1. Only use T.O. specified replacement hoses. Alternative hoses require engineering approval and need to be replaced with the proper hose type as soon as available. PTFE type hoses are recommended.
2. Do not install hoses without an assembly or installation date (or date is illegible). Rubber hoses with an installation date older than 57 months at the projected sale date of the aircraft should be replaced. All hoses, rubber and Teflon®, shall be inspected for leaks, corrosion, kinking, abrasion, cuts or separation in cover or braiding.
3. Service life for aircraft hose assemblies is typically on condition or hard-time replacement as defined by the Maintenance Engineering document. Acceptance, Shelf, and Service Life terms should also be defined.

OTHER INFORMATION

1. A Work Unit Code (WUC) does not exist for fuel hoses as a class, or APU fuel hose P/N AE2463509H0140, for example.
2. PTFE type hoses are not affected by typical aviation fluids.

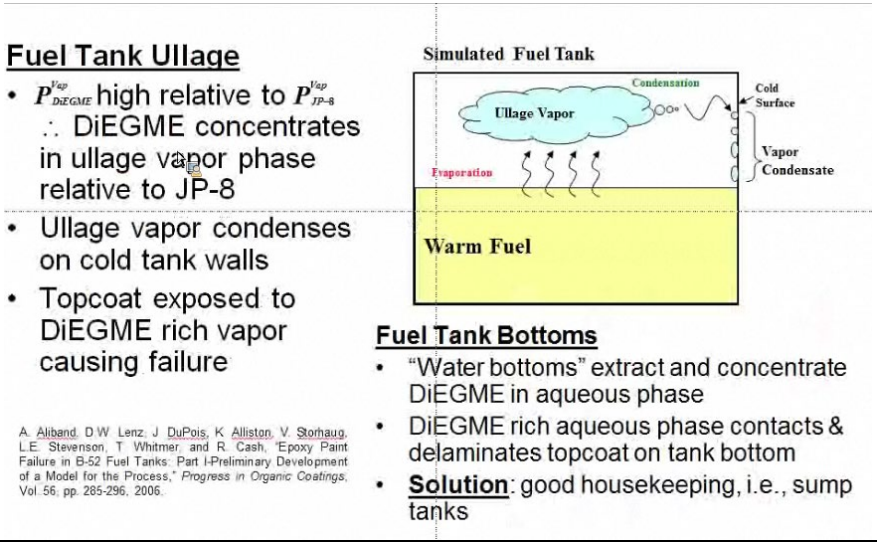
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4. Typically, PTFE hoses do not have a Shelf Life limitation. Rubber hoses typically have a maximum of 10-yr. shelf life. 5. Inspect all air, fuel and hydraulic hoses for the following: (A) Installation Date or date of assembly. Rubber hoses without an installation date (or date is illegible) need to be replaced. Rubber hoses with an installation date of assembly date older than 10 years should be replaced. (B) Obvious defects or damage. 6. Emergency procedures turn off potential ignition sources in the event of leaks. 7. Change design to preclude pressure spikes, and shroud fuel lines in fuselage areas. Currently, surge relief is provided by the Bypass Valve, and Boom Surge Boot. 8. If the life of hoses does not meet expectations, then verify the service temperatures.	

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C.7.10 Topcoat degradation.

Degradation process of topcoats



COMMON FAILURE MODES

1. Fuel tank "topcoat" of paint becomes dislodged, is ingested by pumps, and plugs filters or engine igniters. Causing unprotected fuel tank components show corrosion.
2. The dissolved topcoat residue commonly referred to as 'Apple Jelly' for its color and texture.

POTENTIAL CAUSES

1. Excessive anti-icing inhibitor concentration. Wear of cadmium topcoat exposes silver for erosion. Fuel additives interact with existing fuel system components.
2. 'Apple Jelly' is a mixture of fuel additives (including DiEGME) and water vapor.

FAILURE MITIGATION STRATEGIES

1. To eliminate 'Apply Jelly' it is necessary to reduce DiEGME concentration, water and/or both from the fuel tank.
2. Ensure that the fuel used does not interact with fuel system components.
3. Verify that fuel used has verified additives for aircraft use.
4. Use of fuel system icing inhibitor (FSII) dissolves water in fuel, and is thus anti-microbial.
5. FSII ≥0.04 percent is effective at preventing solidification and blockage even with high total concentrations of water (>288 ppm).
6. Concentrations > ~0.4 percent can promote topcoat failure.
7. DiEGME is concentrated in vapor condensate (factor of ~3 to 8), while TriEGME is diluted in vapor condensate (factor of ~2 to 30).
8. Recommend 0.04 percent concentration as target.

OTHER INFORMATION

1. Use of biocides in fuels is limited.
   a. Very unhealthy.
   b. Fuel cannot be removed, it needs to be burned off.
2. Can cause problems to maintenance personnel.

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EQUIPMENT SUSTAINMENT

1. Topcoat can be removed, and bare aluminum can be utilized. This does leave the aluminum exposed to water, which feeds microbes, which produce acid, which corrodes the aluminum.
2. Preferred solution is to retain topcoat, but reduce DiEGME concentration.

REFERENCES FOR FURTHER READING

1. http://www.wpafb.af.mil/news/story
2. US Air Force Research Laboratory, Propulsion Directorate, Turbine Engine Division, Fuels Branch; cooperative agreement F33615-03-2-2347
3. Determination of the Minimum Use Level of Fuel System Icing Inhibitor (FSII) In Jp-8 That Will Provide Adequate Icing Inhibition and Biostatic Protection for Air Force Aircraft. Report ID: AFRL-RZ-WP-TR-2009-2217
4. Loss of Fuel System Icing Inhibitor Additive During the Transport and Storage of JP-8. Report ID: AFPET-TR-14092009

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C.7.11 Fuel shutoff valve.

FAILURE INDICATION SIGNS 1. Fuel leaks. 2. Inoperable valve. **Example valve** 	**EQUIPMENT DESCRIPTION** Fuel shutoff control for refueling or dump. Includes the Dump Valve, Line Valve (both near STA 1240), ARR Manifold Valves (aft of forward tank) and (forward of STA 344). Most failures are fuel leaks, followed by electrical failures resulting in the valve remaining closed. **POTENTIAL CAUSES** Leaks are caused by aging seals, relative motion of mating tubing, abuse, and assembly procedure. Electrical failures are caused by aging installation, fatigue, and wear. Possible to occur later in life cycle.
FAILURE MITIGATION STRATEGIES 1. T.O. 1C-135-6WS-13 paragraph 4.59, "Wing Tank Fuel Lines, Valves and Pumps" states, "Inspect the electrically operated wing fuel valves and electrical wire and wire tubing for security, loose connections, corrosion, cracks, leaks, loose mounting bolts, frayed or broken bonding jumpers, damage and general condition. Resistance checks each jumper to ensure resistance is 0.1 milliohms (0.0001 ohms) or less." Then, a sub-paragraph includes, "Center Wing Fuselage Compartment Equipment Bay. Electrical Valves, Pressure Switches, and Pressure Regulator." This includes the ARR Manifold Valve on those KC-135 models with this valve (these models do not have the shutoff valve on the side of the flight deck). 2. T.O. 1C-135-6WC-2 includes Card 1-123, which lubricates ARR Manifold Valve manual override linkage. Card 1-035 checks	**OTHER INFORMATION** The difference between the A/R Line Valve and the Dump Valve is function only. The valve is called a Dump Valve when no refueling capability is provided.

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"Freedom of Operation of Manual Override Handle" for the same valve. This card also checks "fuel dump valve handle for proper stowage. Dump valve rod assembly for proper attachment at dump valve manual override handle." 3. Card 1-050 checks "inspect aerial refueling line valve dump valve drip pan assembly for: A. Serviceability and security. B. Petcock in closed position. C. Obstructions. D. Drain hose for deterioration and elasticity. Replace hose if it is deteriorated, cracked, and/or brittle." Card 1-127 lubricates the Dump Valve. 4. T.O. 1C-135-6WC-1 includes inspections for the Dump Valve in Cards 3-018 and 4-006. The "Line Valve" is not included. The Manifold Valve (Main Valve) is included in Card 1-007 and 2-002. The Manifold Valve forward of STA 344 is not separately identified. 5. Based on the above, checks are run for the ARR Manifold Valve and Dump Valve, but they could be extended to the Line Valve.	

REFERENCES FOR FURTHER READING

1. T.O. 1C-135-6WS-13, "Programmed Depot Maintenance Specification".
2. T.O. 1C-135-6WC-2, "Periodic Inspection Workcards USAF Model All C-135 Aircraft".
3. T.O. 1C-135-6WC-1, "Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft".

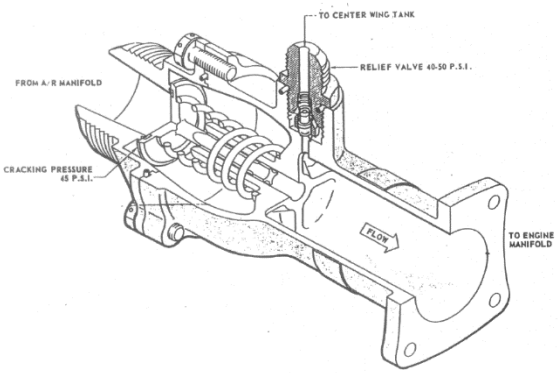
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Check the source to verify that this is the current version before use.

C.7.12 Valve, fuel pressure regulator.

FAILURE INDICATION SIGNS

1. Fuel leaks.
2. Excessive or low pressure in engine feed manifold.
3. Noise.

Fuel pressure regulator diagram



Source: T.O. 1C-135(K)A-BAC-2-5-3-15.

COMMON FAILURE MODES

The valve is not prone to failures, but leaks plus ignition in this area are a fire/explosion hazard, raising the importance of this item. Failures consist of coupling leaks, cracks and spring failure.

POTENTIAL CAUSES

1. Leaks are caused by aging seals, abuse, and assembly procedure. Cracks and splits are caused by surge pressure, and fatigue.
2. Possible to occur later in life cycle.

FAILURE MITIGATION STRATEGIES

1. Inspect the electrically operated wing fuel valves and electrical wire and wire tubing for security, loose connections, corrosion, cracks, leaks, loose mounting bolts, frayed or broken bonding jumpers, damage and general condition. Resistance checks each jumper to ensure resistance is 0.1 milliohms (0.0001 ohms) or less.
2. The manifold is not overhauled.
3. Change design to dampen oscillations, and to prevent closure with each wave peak.

OTHER INFORMATION

1. This valve reduces pressure from the A/R pumps from about 80 psi to about 35 psi; since the valve has a 45 psi, cracking pressure (may vary).
2. The downstream end connects to the engine feed system.
3. A relief valve, which relieves at 45-50 psi is incorporated on the downstream side of the poppet, and is plumbed into the CWT.

REFERENCES FOR FURTHER READING

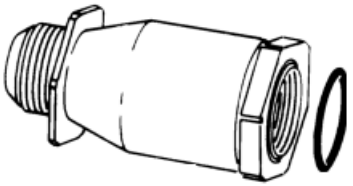
1. T.O. 1C-135-6WS-13, “Programmed Depot Maintenance Specification”.
2. T.O. 1C-135-6WC-2, “Periodic Inspection Workcards USAF Model All C-135 Aircraft”.
3. T.O. 1C-135-6WC-1, “Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft”.

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C.7.13 Valve assembly, fuel swing check (transfer valve).

<table>
<tr>
<td>FAILURE INDICATION SIGNS
1. Uncommanded fuel transfer to tanks.
2. Reduced transfer rate when plugged.

Fuel swing check valve diagram

Source: T.O. 1C-135-4-5 (IPB).</td>
<td>COMMON FAILURE MODES
The valve is not prone to failures, but may leak. Failures consist of coupling leaks or Foreign Object Damage (FOD) blockage.

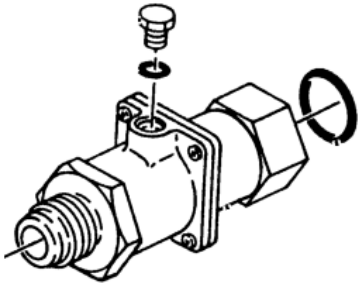
POTENTIAL CAUSES
1. Leaks are caused by aging seals, abuse, and assembly procedure. FOD could interfere with swing door movement.
2. Possible to occur later in life cycle.</td>
</tr>
<tr>
<td>FAILURE MITIGATION STRATEGIES
1. The valve is typically not overhauled.
a. T.O. 1C-135-6WS-13 has no inspections for this valve.
b. T.O. 1C-135-6WC-2 and 1C-135-6WC-1 do not include inspections for this valve.</td>
<td>OTHER INFORMATION
1. This main-to-body tank gravity transfer check valve allows flow into the refuel manifold for gravity transfer.
2. Prevents reverse flow when the manifold is pressurized.
3. Designed as a backup for the transfer valves in the same lines.</td>
</tr>
<tr>
<td colspan="2">REFERENCES FOR FURTHER READING
1. T.O. 1C-135-6WS-13, “Programmed Depot Maintenance Specification”.
2. T.O. 1C-135-6WC-2, “Periodic Inspection Workcards USAF Model All C-135 Aircraft”.
3. T.O. 1C-135-6WC-1, “Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft”.</td>
</tr>
</table>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:46Z
Check the source to verify that this is the current version before use.

C.7.14 Valve, swing check (CWT override pump).

FAILURE INDICATION SIGNS
1. Uncommanded fuel transfer to tanks.
2. Reduced transfer from tanks when plugged.

Swing Check Valve Diagram



Source: T.O. 1C-135-4-5 (IPB).

COMMON FAILURE MODES
The valve is not prone to failures, but may leak.

POTENTIAL CAUSES
1. Failures consist of coupling leaks or Foreign Object Damage (FOD) blockage.
2. Possible to occur later in life cycle.

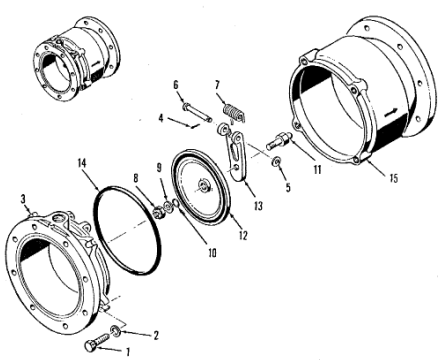
FAILURE MITIGATION STRATEGIES
1. The valve is not overhauled.
   a. T.O. 1C-135-6WS-13 has no inspections for this valve.
   b. T.O. 1C-135-6WC-2 and 1C-135-6WC-1 do not include inspections for this valve.

OTHER INFORMATION
1. This check valve is on the Override pump outlet lines, to prevent loss of pressure when the pump is not operating.
2. It may also be used in place of the Pressure Regulator Valve (the reg. was replaced by the check valve via T.O. 1C-135(K)A-896A (dated 11/10/1960) which stated the purpose was "to eliminate the possibility of restricted fuel flow from body tanks to engines due to a malfunction of the fuel pressure regulator."
3. Those few airplanes use a low-flow hydraulic flow restrictor to drive A/R pumps, resulting in lower pressure output.

REFERENCES FOR FURTHER READING
1. T.O. 1C-135-6WS-13, “Programmed Depot Maintenance Specification”.
2. T.O. 1C-135-6WC-2, “Periodic Inspection Workcards USAF Model All C-135 Aircraft”.
3. T.O. 1C-135-6WC-1, “Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft”.

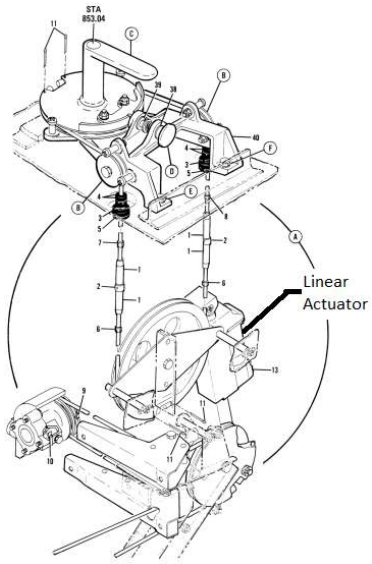
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Check the source to verify that this is the current version before use.

C.7.15 Valve, swing check (A/R pump).

FAILURE INDICATION SIGNS 1. We have seen broken butterflies on the boom fuel pressure regulator, deformed fuel flex couplings in the wheel well (near the SPR), broken slides on the boom by-pass valve, broken fuel level control valves (for the body tanks) and deformed A/R manifold sections. Such damage may all be caused by surge pressure. 2. Deformation of the A/R venturi occurred in the form of bulges in seven of the 13 inspected aircraft, indicating pressure spikes are common. **Component Diagram** 	**DESCRIPTION** These valves allow fuel flow from pumps, but prevent backflow when pumps are off. The cracking pressure is <10 incesh fluid pressure. The figure at left is from T.O. 6A9-6-2-3. Failures consist of flapper seal leaks, cracks, and spring fatigue. **POTENTIAL CAUSES** Leaks are caused by aging seals, abuse, and assembly procedure, FOD, corrosion, dirt and wear. Cracks and splits are caused by surge pressure, and fatigue. **WHEN IN LIFE CYCLE THIS HAPPENS** Later in life.
FAILURE MITIGATION STRATEGIES 1. T.O. 1C-135-6WS-13 has no requirements for this valve. 2. T.O. 1C-135-6WC-2 has no requirements for this valve. 3. T.O. 6A9-6-2-3 is the overhaul manual for this valve. 4. T.O. 1C-135-6WC-1 has no requirements for this valve. Change design to preclude pressure spikes. Currently, this function is provided by the Bypass Valve, and Boom Surge Boot.	**OTHER INFORMATION** The cracking pressure is <10 inches fluid pressure.

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C.7.16 Actuator, electromechanical linear (Valves 1&2).

FAILURE INDICATION SIGNS 1. Uncommanded fuel transfer as indicated by tank gages. 2. Manual override handle positioned midway, or not moving to commanded position. 3. Fill time is slow. **Actuator Diagram**  Source: IPB T.O. 1C-135-4-5 Fig. 48	**COMMON FAILURE MODES** A quadrant operated by an electrical actuator provides remote control capability of the valves on ARR aircraft. When the master refuel switch is in the ON position, the actuator may be energized through the manifold valves switch on the ARR panel. These valves are cabled together so that when No. 1 (see T.O. 1C-135(K)R-2-5GA-1 Fig. 1-2) is open, No. 2 is closed and when No. 2 is open, No. 1 is closed. The diagram at left shows the hand crank override. The actuator is not prone to failures. The hand crank provides backup. **POTENTIAL CAUSES** 1. Electrical controls, wiring, electrical solenoid, valve jammed due to FOD or contamination can cause failure. 2. Possible to occur later in life cycle.
FAILURE MITIGATION STRATEGIES 1. T.O. 1C-135-6WS-14 requires to “perform an operational check of the A/R system on all aircraft” per the indicated T.O.s. 2. T.O. 1C-135-6WC-2, Card 1-123, requires lubrication of the Manual Override ARR Manifold Valve. These are performed every 12 or 24 months (etc.) depending on the aircraft.	

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3. T.O. 1C-135-6WC-1 card 1-007 requires for ARR aircraft, preflight inspection: "Manifold Valves Switch Positioned to Wing Refuel/Refuel Position". Later, the flight crew switches from this position to the opposite.	

REFERENCES FOR FURTHER READING

1. T.O. 1C-135-6WS-14, "Programmed Depot Maintenance Specification".
2. T.O. 1C-135-6WC-2, "Periodic Inspection Workcards USAF Model All C-135 Aircraft".
3. T.O. 1C-135-6WC-1, "Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft".

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C.7.17 Inerting system and flammability.

INERTING SYSTEM TYPES

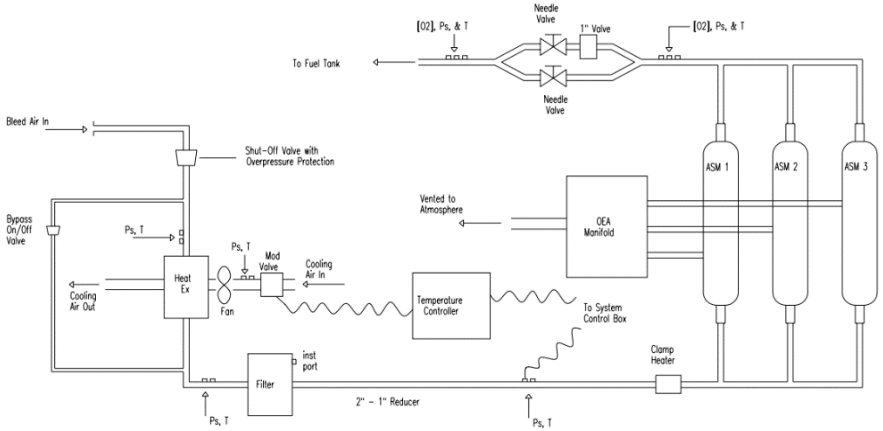
Nitrogen inerting is done through a couple of methods:

1. Hollow fiber membrane.
2. Using the compressed air from the bleed air system, a filter is used to separate the nitrogen. The remaining components of air are pushed outboard and the nitrogen is pushed into the tank.
3. The filters that are used in this process are very sensitive to temperature change and prolonged exposure to high or low temperatures can cause degradation.
4. Nitrogen the simplest.
5. Commonly made from Polysulfone for the hollow fiber membrane. The hollow fiber membrane is essentially a filter that concentrates nitrogen in the bleed air sending it to the fuel tank.
6. Ozone will degrade fibers in the Air Separation Module (ASM), but can be reduced with use of a catalyst to react the ozone prior to reaching the ASM.
7. Pressure Swing Molecular Sieve ASM.
8. Another method for nitrogen inerting is the use of pressure swing molecular sieve ASM technology. These systems rely on surface adsorption of oxygen to concentrate nitrogen for use in inerting fuel tanks. These systems have a more limited flow capacity compared to hollow fiber membrane ASMs. Because of the lower flow capacity, this technology is usually not selected for new applications. ASMs Ozone will degrade the fibers in the ASMs, but can be reduced with the use of catalysts to convert the ozone and remove it from flow path.

Halon inerting:

1. Used on the F-16. Halon inerting is only activated for combat operations.
2. The advantage is low weight and reliability.

Inerting System Diagram



Source: http://www.fire.tc.faa.gov/pdf/03-58.pdf

OTHER INFORMATION

1. FAA requires tank temp and type with fuel not being with probability of tank not being inert.
2. USAF requirements strictly inter independent of temp.
3. FAA developed tool provides a Monte Carlo assessment of fuel tank ignition.

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Check the source to verify that this is the current version before use.

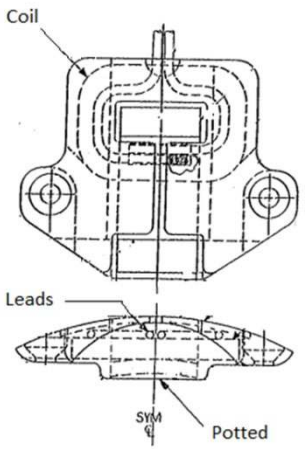
FAILURE MITIGATION STRATEGIES

1. The inerting system has no moving parts and thus is typically reliable.
2. Filters should be regularly assessed for degradation/replacement.
3. Sensors are difficult to implement.

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C.7.18 Refueling boom signal coil system.

Boom Signal Coil System Diagram	**COMMON FAILURE MODES** Unexplained inflight aborts and malfunctions of the refueling communications signal coils. Negative contact (missed contacts), delayed disconnects (termination of refueling), aborted missions, & brute force disconnects. **POTENTIAL CAUSES** Corrosion or dirt in contacts, terminal broken, gouged, signal coil broken, shorted, broken shield, wire broken, nut loose, coil contacts rusty, connector damage, brush and track contacts in Boom.
FAILURE MITIGATION STRATEGIES 1C-135-6WS-14 references the following: Maintenance specifications in TO 1C-135-2-6 & 1C-135(K)R-2-6JG manuals for inspections. The above includes: a. Signal Coil Resistance Checkout. b. Signal Coil Polarity Checkout. 1C-135-6WC-2 references “Resistance Checkout per 2-6 Series Technical Order.” T.O. 1C-135-6WC-1 states, “Boom signal coil face for corrosion and cleanliness, and clear coating for Condition (coil face needs to be completely covered with clear coating) per 2-6 series technical orders.” This is a post-flight requirement. 1. Consider design changes to protect the signal coil, contacts, and wire.	**OTHER INFORMATION**





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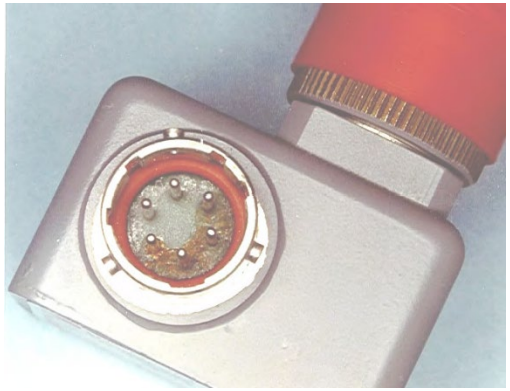
2. CPC could be used to prolong service life.	

REFERENCES FOR FURTHER READING

1. T.O. 1C-135-6WS-14, “Programmed Depot Maintenance Specification”.
2. T.O. 1C-135-6WC-2, “Periodic Inspection Workcards USAF Model All C-135 Aircraft”.
3. T.O. 1C-135-6WC-1, “Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft”.

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C.7.19 Failure mode: connector corrosion.

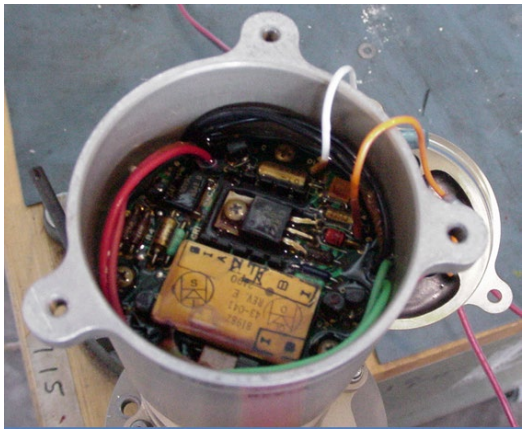
DESCRIPTION	**POTENTIAL CAUSES**
1. Corrosion of tin plated electrical pins mated with gold-plated sockets. Fretting corrosion led to five aircraft crashes (over $100M) when shorting resulted in uncommanded closure of the main fuel shutoff valve. 2. Fuel spray on receiver aircraft or at other times during operation. **CONNECTOR CORROSION** 	Corrosion products of the corroded tin plated steel pins provide a conductivity path between pins A to B and C to B that drives the valve to close. **INDICATIONS, SIGNS, OTHER INFORMATION** Presence of corrosion or moisture inside connector or un-commanded movement of the valve. Check insulation resistance between isolated electrical pins (apply 500 VDC and verify resistance is greater than 5M ohms.

FAILURE MITIGATION STRATEGIES

1. Permanent solution is to eliminate dissimilar metal tin and gold interface and use a gold plated pin and socket.
2. Preventative maintenance is to use a corrosion inhibiting lubricant spray, per MIL-L-87177 Grade B to slow down corrosion reaction, which needs to be re-applied every month until a permanent solution is implemented.

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C.7.20 APU pump.

FAILURE INDICATION SIGNS	COMMON FAILURE MODES
1. Pump inoperable or intermittent, as indicated by low-pressure output. 2. Circuit breaker trips. 3. Noise. **APU Pump Interior** 	Failure is most often due to tripped circuit breakers, or low pressure. **POTENTIAL CAUSES** 1. Chafing of wires, electrical insulation breakdown, contact corrosion, bearing wear & fatigue, water ice buildup on screens. Water hammer and cavitation can also cause impeller fatigue. FOD-caused failures. 2. Possible to occur later in life cycle.
FAILURE MITIGATION STRATEGIES 1. T.O. 1C-135-6WS-13 has no requirement for APU Fuel Boost Pump inspections or tests. 2. T.O. 1C-135-6WC-2, Card 1-099 includes inspections: "remove the APU fuel boost pump per 2-16 Series Technical Orders, paragraph 3-31 instructions and test per paragraph 3-31C (TOLO ASSEMBLY P/N KC49781-501; DUKES PUMP P/N 1500-00-59). Repair as necessary." However, this test only applies to those pumps mounted in the wheel well, which are no longer installed (per TCTO 1C-135-1618). No periodic tests are currently performed. 3. APU pumps are not overhauled. **4.** Consider addition of periodic pressure tests, for both output pressure and containment.	**OTHER INFORMATION** 1. This single pump provides fuel pressure & flow from Left Inboard Main Wing tank to APU 1 & 2. 2. Provides bleed air and electrical power for engine start, HVAC, etc. **3.** Photo is of an alternate part P/N 458-50147-501.

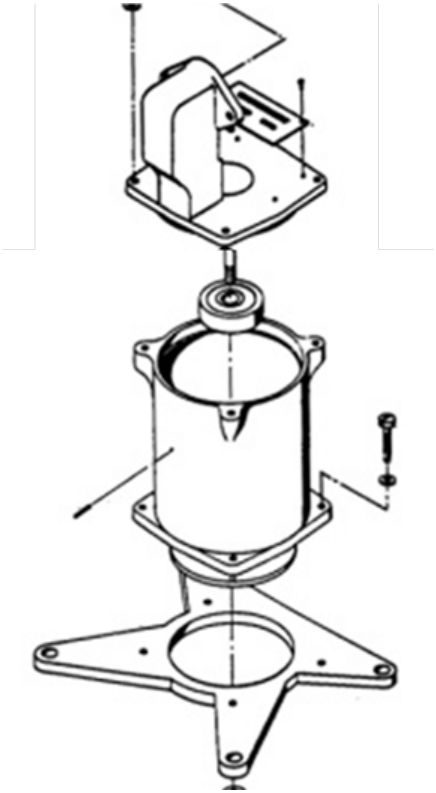
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REFERENCES FOR FURTHER READING

1. T.O. 1C-135-6WS-13, “Programmed Depot Maintenance Specification”.
2. T.O. 1C-135-6WC-2, “Periodic Inspection Workcards USAF Model All C-135 Aircraft”.

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C.7.21 Failure mode: electrical arcing at housing interfaces.

DESCRIPTION In the event of an electrical fault with the component in a fuel environment, design intent is that the current be carried through the housing to the structure so that the circuit breaker trips. If the resistance is above 10 milliohms at any interface overheating and arcing is possible which could ignite flammable vapors. **Housing Interface Diagram** 	**FAILURE INDICATION SIGNS** Localized discoloration on bonding surfaces. **POTENTIAL CAUSES** 1. Corrosion / contamination at faying surfaces: a. Nonconductive Coatings. b. anodized surfaces. c. organic coatings. 2. Failure likely to occur later in service life.
FAILURE MITIGATION STRATEGIES Insulation resistance and electrical bonding test:	

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a. Insulation resistance: Apply 500 volts or higher and measure leakage current (min. of 0.1 mA @500 VDC. b. Bonding resistance: use a four-point probe test by applying max. short circuit current and measuring the voltage drop and calculate the resistance which should be 10 Mohms or the specified value in the T.O. (high voltage i.e. 4-point probe test at overhaul using)	

SPECIAL CONSIDERATIONS

1. The accuracy of the four-point probe test depends on using an outer pair of probes to impress a high current and another inner pair of probes to measure the resulting voltage. In this way, the effect of resistance in the voltage measuring leads is minimized.

2. The measurements may vary with humidity. Best practice would be to track resistance by S/N and record the measurement conditions (humidity, etc.).

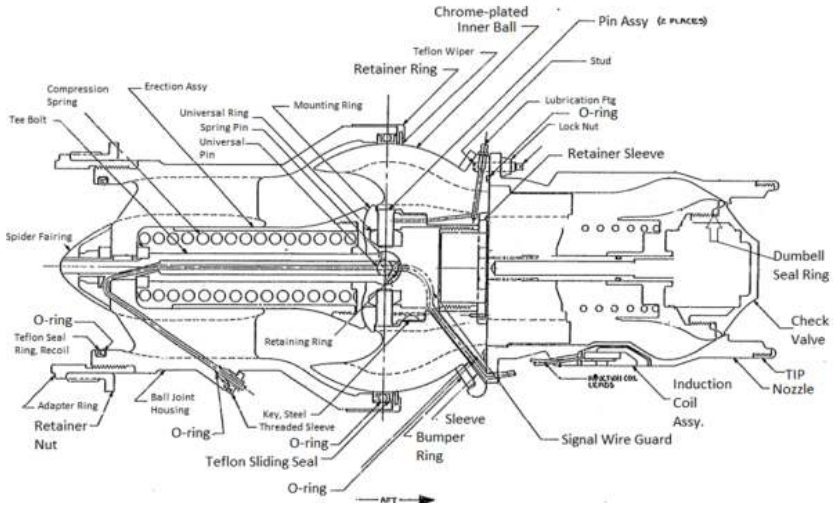
REFERENCES FOR FURTHER READING

Events, AFSAS

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C.7.22 Failure mode: nozzle leaks.

Nozzle Diagram



COMMON FAILURE MODES

Fuel spray on receiver aircraft or at other times during operation.

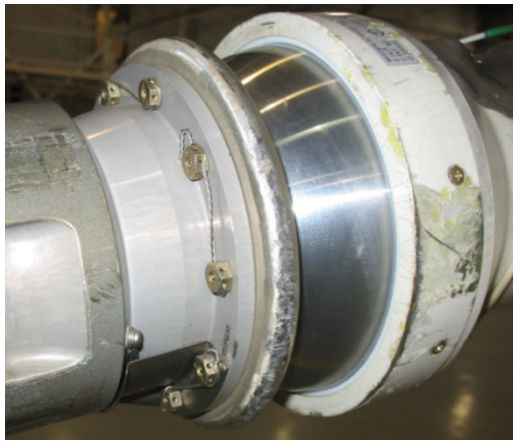
POTENTIAL CAUSES

1. Nicks, scratches, gouges, contamination, and corrosion on the Inner Ball create leakage paths. O-rings and other seals can also contribute.
2. Failure likely to occur later in service life.

FAILURE MITIGATION STRATEGIES

1. Overhaul nozzle.
2. Design changes could be considered. Consider flexible metal bellows, or ball surface hardening.
3. Per T.O. 1C-135-6WS-14, “If boom is to be replaced, retain original boom removed until receipt of replacement boom. Upon receipt of replacement boom, remove the nozzle assembly and ruddervators from original boom and install on replacement boom.” Boom Nozzle Assembly Checkout per 1C-135-2-6 & 1C-135(K)R-2-6JG. Test Tanker 2, ACC, AMC and AFRC aircraft only: Upon aircraft arrival, if there is less than 50 percent of the 2000-hour inspection on the small tip fuel nozzle, inspect for tip wear.
4. Per T.O. 1C-135-6WC-2, inspect BOOM NOZZLE ASSEMBLY (PER T.O. 6A5-3-13):
   A. Nozzle Coupling Nut Centering Swivel for Binding and/or Loose On Tube.

OTHER INFORMATION



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B. Nozzle Universal Ball Joint Cone Action for Binding and Positive Centering. C. Nozzle for Excessive Rotational Play Between Ball and Housing. Per T.O. 1C-135-6WC-1, Prior to Inflight Refueling of F-4 Aircraft: Apply Lubricant VV-P-236 To Boom Nozzle. Includes T.O. 1C-135-6WC-2 checks A, B, & C.	

EQUIPMENT SUSTAINMENT

1. Part supply has been an issue for this assembly.
2. One KC-10 apparently flew outside the acceptable envelop during the refueling resulting in a situation where the boom was overstressed. The bolts holding the nozzle to the boom sheared and the nozzle stayed with the receiver aircraft. At some point, the bolts separated from the nozzle head and impacted the #2 engine. Possible mitigation actions could include a design change that would preclude the bolt heads from separating from the nozzle if it shears off the boom.

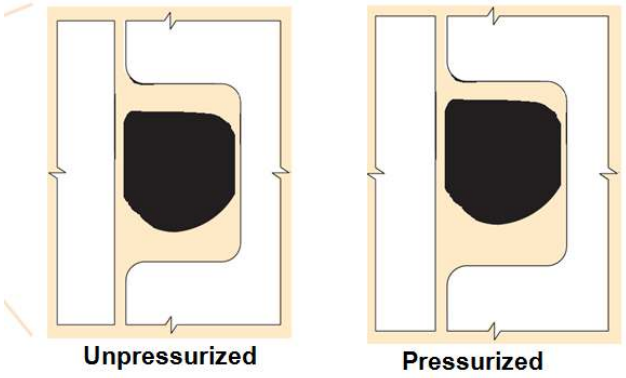
REFERENCES FOR FURTHER READING

1. T.O. 1C-135-6WS-14, “Programmed Depot Maintenance Specification”.
2. T.O. 1C-135-6WC-2, “Periodic Inspection Workcards USAF Model All C-135 Aircraft”.
3. T.O. 1C-135-6WC-1, “Workcards, One Set, Preflight/Postflight/Hourly Post-Flight Inspection Workcards, USAF Model All C-135 Aircraft”.

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C.7.23 Component: O-rings.

EXAMPLE FAILURE



Compression Set, Hardened, Shrunk and/or Worn

COMMON FAILURE MODES

Physical damage due to maintenance action, loss of elastic properties.

POTENTIAL CAUSES

Excessive wear by matting and de-mating can reduce performance. Long-term stress under compression can cause a 'set' and result in leakage. Exposure to heat can make the O-ring stiffer and lose ability to seal a connection. Fluid exposure can degrade performance and may be reactive.

FAILURE MITIGATION STRATEGIES

1. Ensure the O-ring used matches the operational needs of the environment. Different O-ring material types are qualified to certain environmental conditions.
2. Inspect for leakage and repair when leakage is found. If O-rings remain fuel wetted, there is no need to replace them on a periodic basis.

If a coupling or component connection containing an O-ring is removed, the O-rings should be changed at that time. Reuse of O-rings invites opportunity for subsequent leaks.

OTHER INFORMATION

1. There are several types of O-ring materials. These include silicone, fluorosilicone, and a variety of thermoplastics.
2. Wide-cut fuels (such as JP-4) swell seals. If a seal has been in use with wide-cut fuel, and the fuel type is switched to JP-8, for example, the seal will shrink and potentially leak. Potential for seal leakage is a consideration when evaluating alternative fuels using the MIL-HDBK-510 approach.

EQUIPMENT SUSTAINMENT

1. Replace all O-rings when overhauling components.
   a. In some applications, replacement at regular intervals may be warranted (depending on O-ring material selected). For example, fuel lines in fuselage areas, where fuel lines are not fully enclosed in a drained shroud may need "hard time" replacement.

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REFERENCES FOR FURTHER READING

1. **SAE-AMS-P-5315** (Nitrile) Packing, Preformed, Hydrocarbon Fuel Resistant
2. **MIL-DTL-25988** (Fluorosilicone) Rubber, Fluorosilicone Elastomer, Oil-And Fuel-Resistant, Sheets, Strips, Molded Parts, And Extruded Shapes
3. **SAE ARP6179** Evaluation of Gas Turbine Engine Lubricant Compatibility with Elastomer O-Rings
4. **SAE ARP568** Uniform Dash Numbering System for O-Rings

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C.7.24 Failure mode: FOD.

DESCRIPTION	**COMMON CAUSES**
Foreign Object Debris (FOD) is any object alien to a system. Examples of FOD include: a. Degraded fuel foam entering components outside the fuel tank. b. Metal shaving from worn mechanical components and gears. c. Swarf from drillings. d. Screws, buts, bolts, etc. e. Broken component fragments. f. Work materials, metal shavings. g. Personnel Protective Gear. Although FOD can be benign, it can and frequently does cause damage to fuel system components, engines, or auxiliary power units. For example, FOD in the fuel pump may clog up a filter or prevent the pump from closing. 	1. Regular examination for wear of system components is necessary to identify degraded items that may break away from a super structure and create FOD. 2. The loosening of equipment through vibration (e.g. fasteners, nuts, etc.) may also generate FOD. Careless maintenance practice is the primary cause, resulting in fasteners, gloves, and work materials being left in fuel tanks.

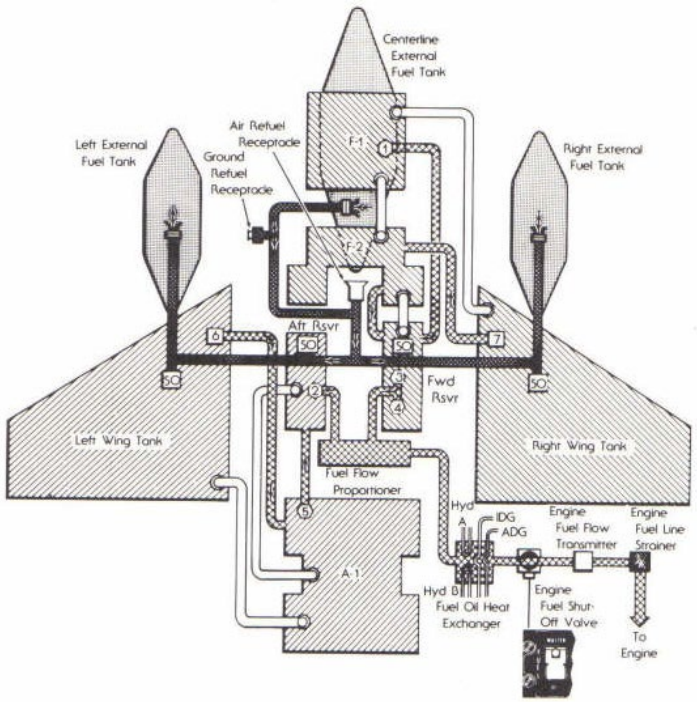
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FAILURE MITIGATION STRATEGIES	OTHER INFORMATION
1. Common mitigation techniques include the use of filters in the system and proper installation of equipment. 2. Use of lock nuts/washers in vibration areas. 3. Replacement of gears and other mechanical components at end of life. 4. Maintenance procedures to prevent FOD, and to cross-check work areas.	1. From T.O. 1-1-3: a. Failure to remove caps, plugs or cover plates from the aircraft prior to refuel may result in severe aircraft structural damage or personnel injury/death. An in-process inspection (IPI) needs to be accomplished prior to fuel tank/cell closures to minimize FOD hazards and to ensure materials, tools, and equipment taken into the fuel tank/cells has been removed. The IPI needs to be documented in the aircraft's AFTO Form 781A. b. At least one TCTO has been proposed to inspect for FOD.

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C.7.25 Reference JSSG-2009.

Fighter Fuel System Diagram.



COMMON FAILURE MODES

1. Bladder damage from low temperature insertion through access doors.
2. Fuel quantity indication system: connector seal leaks, connector corrosion.
3. Foam: disintegration resulting in fuel contamination.
4. Fuel feed system: pump inlet plugging due to FOD, or water ice.
5. Fuel tankage: topcoat pealing, resulting in fuel FOD. External fuel tanks.

SYSTEM

1. Some new aircraft have requirement for constant flow of fuel. This is necessary to sink heat from engines, avionics, hydraulic, and generator systems.
2. Additional heat may impact long-term system reliability (such as seals).
3. Separation of EWIS from fuel system components more difficult.

PUMPS

1. Some new pumps run on 270VDC and are significantly more complex with on-pump controllers.

BLADDERS

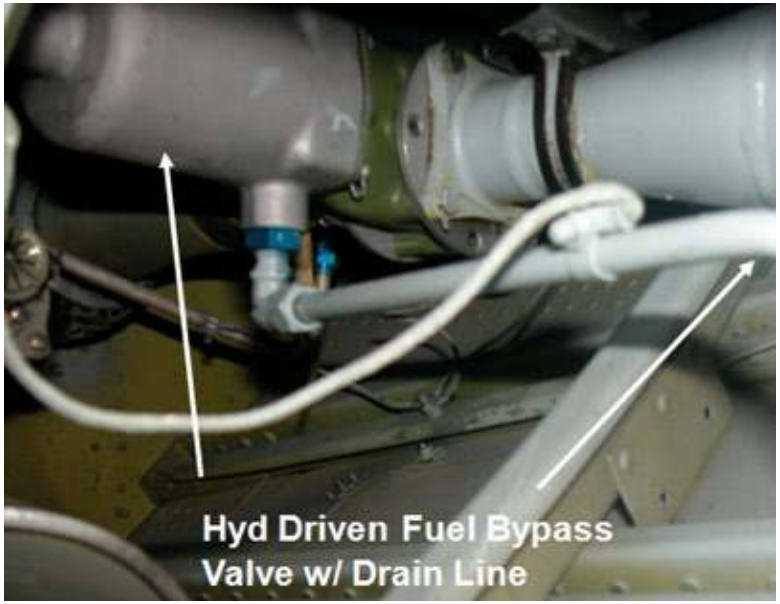
1. Typical integral tank fuel leak.

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2. Self-sealing bladders can minimize the fuel leakage from ballistic damage.
3. Bladders in complex geometry tanks can significantly reduce the likelihood of bladder repair.
4. As fighters have higher stresses, performance requirements are higher on these bladders.
5. F-35 AF variant has 11 fuel tanks spread through the aircraft.
6. Fuel bladders on fighters are not pliable; can incur damage if not prepped (soaked in warm water).
7. Installation of bladders is difficult through access portals.

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C.7.26 Bypass valve.

Installed Bypass Valve  	**COMMON FAILURE MODES** 1. Slow fuel offloading rate. 2. Leakage at hydraulic drain port, on the external airplane skin. 3. Uncommanded fuel transfer to the Aft tank. 4. Unable to retract Boom on a refueling aircraft. **POTENTIAL CAUSES** 1. Valve pin may fail due to fatigue over service life. Internal hydraulic leaks or spring failure. 2. Failure typical later in service life.
FAILURE MITIGATION STRATEGIES 1. T.O. 1C-135-6WC-2 calls for inspection of hydraulic lines for chafing, and in one case to do so per T.O. 42E1-1-1, which in paragraph 22 of the NAVAIR 1-1A-20 section requires a minimum clearance of 1/16 inch for rigid mount points, or ¼ inch where relative motion may exist. 2. Monitor hydraulic drain for leakage. 3. Provide mechanical bypass relief line and valve as backup. 4. Operate A/R Pump in Aft tank to transfer fuel that leaks to that tank from a partially open Fuel Bypass Valve. 5. Pressure test hydraulic ports.	**OTHER INFORMATION** 1. Valve is hydraulically actuated, and spring-loaded closed. 2. Solenoid-operated control valve directs hydraulic fluid to open the valve during boom retraction. 3. When valve fails to open, a retracting boom can over-pressurize the line; broken components have resulted. 4. Closure failure allows un-commanded fuel transfer. 5. The fuel in the boom is removed when the boom is retracted. 6. There is no electrical connection to indicate failure.

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EQUIPMENT SUSTAINMENT

1. The valve allows fuel to flow back to the aft body tank when the boom is telescoped in. It also prevents fuel from flowing back into the tank during a refueling operation. It is spring loaded closed.
2. There is no valve position indication, and opens from 48-52 psi fuel pressure. The valve is hydraulically operated by the Fuel Bypass Control Valve, which is solenoid-operated.

REFERENCES FOR FURTHER READING

1. 1C-135-6WC-2, “Periodic Inspection Workcards, USAF Model All C-135 Aircraft”.
2. T.O. 42E1-1-1, “Operational Supplement Technical Manual, Aviation Hoses and Tube Manual, Organizational, Intermediate and Depot Maintenance”.
3. AFRL/RXS 09-026.
4. ARFL Failure Report # - 788179.

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C.7.27 Fuel tube failure.

EXAMPLE FAILURE	COMMON FAILURE MODES
Tube flange fractured causing separation at connection point, which allowed fuel to pour out of fuel system. The photos below show tube separation and the flange that came off.  	Analysis showed the fuel tube fracture contained multiple origin fatigue that progressed via high stress/low cycle conditions. Although fatigue was found, the progression constituted only about 10 percent of the overall fracture surface, making overstress the predominant fracture mode. **POTENTIAL CAUSES** Poor fuel tube installation design, which inadequately supported fuel tubes resulting in loads being transferred through the couplings. In addition, variations in tube geometry and lack of adjustable bracketry contributed to the failures by forcing the tube to be installed so that there was axial misalignment and preload to cause the lip stress to be reacted by only a small portion of the lip inside the coupling.
FAILURE MITIGATION STRATEGIES 1. Immediate one-time fleet wide inspection of tubes followed by a recurring, periodic flight-hour based inspection of all tube flanges for cracking. 2. Provide adjustable mounting brackets to better accommodate small geometric variations between fuel tubes and prevent pre-loading during installation. 3. Redesign the fuel tube installation to include more tube supports and new couplings that are better suited for transmitting loads which are induced by vibration fluid dynamics (e.g. pressure surges).	**OTHER INFORMATION** 1. All flanges were inspected and cracked flanges replaced as an interim step. a. Initial inspections were accomplished with dye penetrant inspection method. b. An eddy current tool was developed to improve crack detection capability, which allowed for longer inspection intervals. 2. All fleet fuel tubes were replaced with new.

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4. Verify tubes are properly aligned to prevent exceeding permissible coupling angular misalignment die to aircraft flexing throughout the flight envelope.	

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C.8 TEST ASSESSMENTS

The following descriptions of test assessment methods are not all inclusive. These are presented as a few of the more common assessment techniques and some of the considerations when applied.

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C.8.1 Test assessment: dielectric assessment.

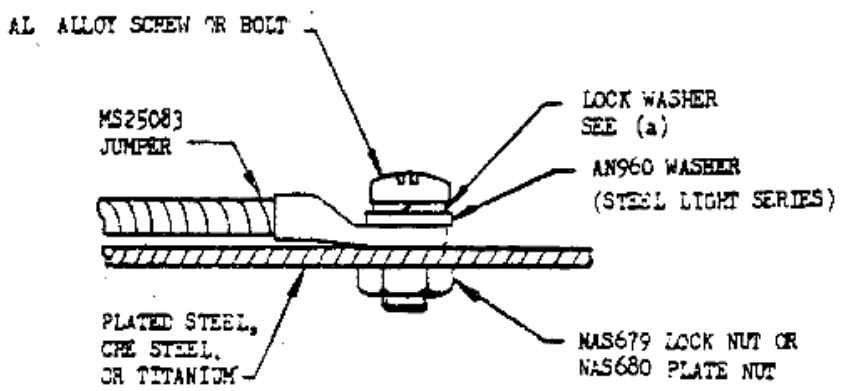
Dielectric Test Equipment Example.	**Use of dielectric testing.**
	1. Dielectric testing can be used to determine if there is loss of insulation integrity between a wire, another wire, ground, or other conductive surface. The applied voltage should be at or below the rating for the component. 2. An aircraft application requires review of potential ignition sources and limiting of energy discharge level to below the ignition threshold level. 3. This testing will not be able to identify insulation cracks where the conductor is not shorted.
WHAT IS IT? Dielectric testing assesses the ability for an insulated system (e.g. wire or winding) to maintain a voltage without insulation breakdown (conduction of electricity through on the surface the insulation).	**OTHER INFORMATION** 1. Alternating Voltage (VAC) is more stressful than direct voltage (VDC). 2. Typical values are 100V, 500V, 1,000V, or 1,500V depending on component specification. 3. The leakage current should be limited to 1 milliamperes or less to minimize component damage. 4. When fuel may be present, the energy applied by the tester in the event of a failure, needs to be well below the ignition level (0.20 millijoules) for the system under test.
REFERENCES FOR FURTHER READING a. **ECA 186-4E** Passive Electronic Component Parts, **Test** Methods for; Method 4: **Dielectric Test** (Withstanding Voltage). b. **ASTM D149** Standard **Test** Method for **Dielectric** Breakdown Voltage and **Dielectric** Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies.	

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c. **ASTM D3755** Standard **Test** Method for **Dielectric** Breakdown Voltage and **Dielectric** Strength of Solid Electrical Insulating Materials under Direct-Voltage Stress.

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C.8.2 Test assessment: bonding.

EXAMPLE  Example of proper bonding: source MIL-STD-464.	**Use of bonding testing.** 1. The use of the proper equipment is necessary to ensure sufficient bonding current is applied during the test. The rated current should be used since a lower level does not evaluate ability to protect against an electrical fault. 2. Testing should be performed on equipment prior to and after aircraft installation using T.O. procedures. 3. Testing should be performed when equipment, such as electrical pumps, are installed. In explosion prone areas and those with fire hazards, the resistance values between the equipment case and structure should not exceed the values defined in MIL-STD-464.
WHAT IS IT? An electrical bonding test evaluates the ability of grounded equipment to sustain a high current short without arcing or overheating to level that could ignite fuel. Electrical bonding also provides a low resistance path so a human is not shocked or adjacent electrical components are not damaged when a metal housing or surface is shorted to a power source. In particular, a bonding test examines the contact resistance between the equipment and the ground return.	**OTHER INFORMATION** 1. This test needs to be performed at the rated maximum fault current value for the power supplied to the component in question. Use of a multimeter or a lower current than the maximum fault current does not provide establish that an adequate a good indication of the quality of the electrical bond exists. 2. If a failure event occurs, poor electrical bonding can result in overheating, arcing, damage to system components, and an electrically charged surface that can pose a danger to personnel. It can also cause fuel tank ignition/explosion and subsequent loss of aircraft.

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C.9 CASE STUDY ON KC-135 FUEL TANK EXPLOSION

C.9.1 Introduction.

Electrical arcing inside fuel tanks continues to be a primary fuel system safety concern. There are a number of electrical components such as pumps, relays, fuel probes, switches, and wiring that have sufficient energy to ignite a fuel tank when a fuel and air mixture exists inside the tank. There have been a number of military and commercial aircraft accidents attributed to electrical arcing inside of a fuel tank (see paper Aircraft Systems Design: Lessons Learned from SFAR-88 for commercial aircraft examples). An electrical arc can generate temperatures from 5,700 °C (10,292 °F) to over 14,000 °C (25,232 °F) depending on the available arc current (ref paper: Fires due to Electric Arcing, by Vytenis Babrauskas, Ph.D. Fire Science and Technology). After several high profile accidents attributed to arcing in aircraft fuel tanks, the aerospace community has developed a number of references such as SFAR-88 to address these concerns (see case study on TWA 800, A.3.6.4). These documents have also been applied in varying degrees to military aircraft.

This discussion concerns a KC-135 accident that occurred in 1993 in which six USAF airmen were killed and the aircraft destroyed by a fuel tank explosion and ensuing fire (see FIGURE C- 6). The airmen were performing routine maintenance to troubleshoot an increase in the amount of fuel in the forward fuel tank which was reported after the previous night's mission. Mishap investigators determined ignition occurred as a result of uncontained arcing in an electrically driven center tank boost fuel pump which was activated during the maintenance operation. The cause of the failure was further attributed to an improper overhaul process that occurred at an Air Logistics Center (ALC).



FIGURE C- 6. Remains of the KC-135 after the aircraft exploded.

The Boeing KC-135 Stratotanker is a military aerial refueling aircraft derived from the Boeing 707 airliner and became operational in 1957, making the mishap aircraft 36 years old at the time of the accident (tail number 57-1470). As of 2014, there were 415 operational KC-135 aircraft in the USAF fleet with the current retirement planned for 2040. Maintaining requires constant review of engineering designs, system performance, and mission needs. The challenge is how to maintain

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safe and airworthy fuel systems over large time spans as the aerospace community develops new understandings on fuel system failure mechanisms and causes. Updating older aircraft fuel systems to meet new requirements based on lessons learned requires considerable understanding of the existing fuel system design, performance, reliability, and mission needs.

C.9.2 Investigation.

Mishap investigators determined the explosion originated from the aircraft center fuel tank. Further analysis, identified the fuel boost pump (see FIGURE C- 7) inside the tank as the most likely ignition source. The pump was suspected to only be partially submerged in fuel, with the connector and explosion emmanating from the flame arrestor hole and into a flamable fuel vapor/air mixture.

Red dot (see FIGURE C- 8) is the 1/32 inch flame arrestor hole that was added to mitigate against pressure building up in the pump in the event of arcing or another pressure creating incident. The electrical connector and wiring is visible on the left. During overhaul, the connector is cut off to access the pump interior. This exemplar boost pump was provided for teardown and analysis.

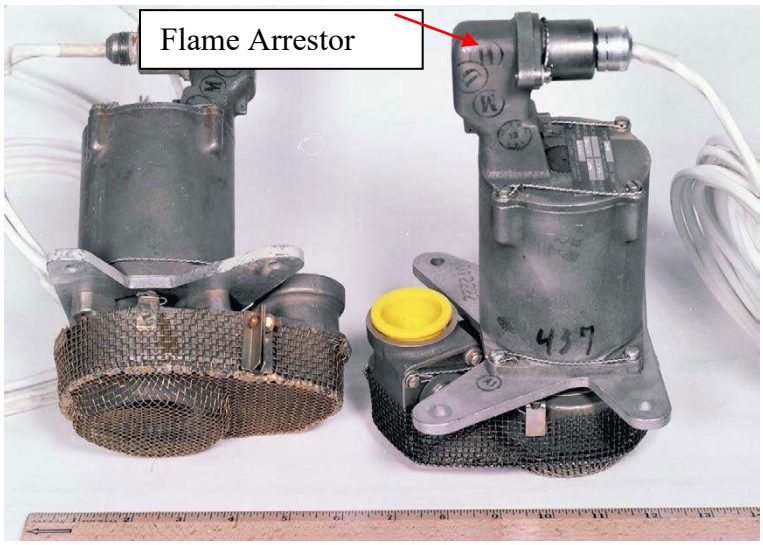


FIGURE C- 7. Example of the boost fuel pump suspected to have initiated the aircraft explosion.

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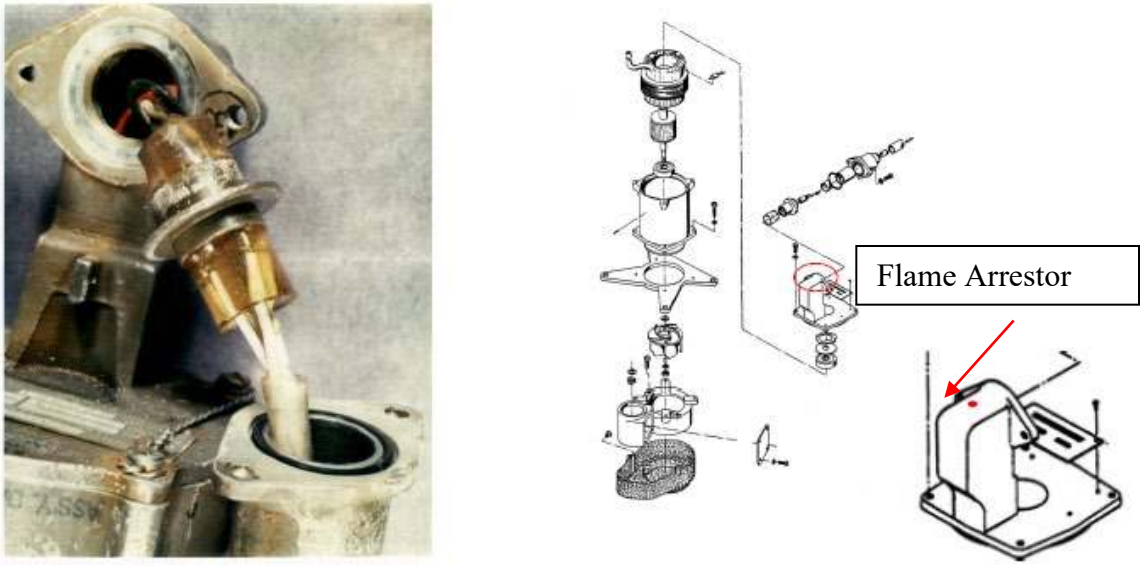


FIGURE C- 8. Exemplar boost pump.

Analysis of the mishap pump showed an excessive length of wiring was used during the last overhaul of the pump at the depot. The wiring was twisted and pushed up against the internal pump housing in order to allow the connector to be removed for a future depot repair (see FIGURE C- 9). Note how the wiring is pushed up against the aluminum housing.

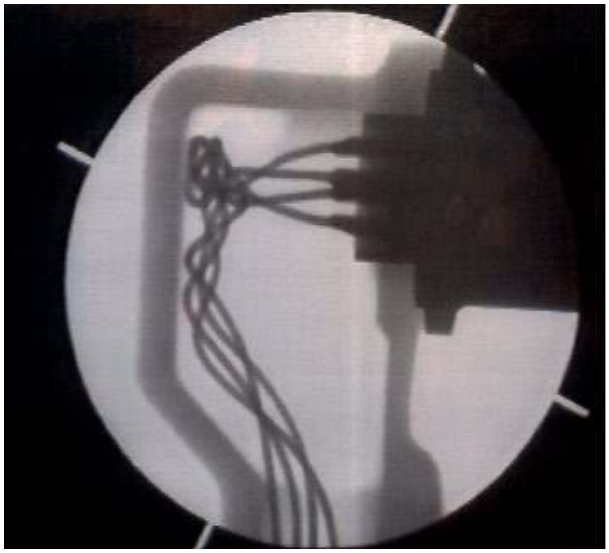


FIGURE C- 9. Radiograph of a fuel pump removed from inventory.

The extra wiring created a chafe condition inside the pump housing. Excessive forces applied to the wiring resulted in cold flow of the Teflon© insulation leading to an exposed conductor that contacted the electrically grounded pump housing and resulted in a high temperature arc. Over time and under pressure, Teflon© insulation will cold flow (creep at or near relatively low temperatures); thinning the insulation and eventually exposing the conductor. New wire insulations such as Tefzel© have been developed which don't exhibit this condition. The wire

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conductor was one leg of the $115V_{rms}$ 400 Hz. 3-phase power used to operate the pump. A short circuit to the aluminum housing (electrical ground) could result in a 270 A_{rms} current fault. One 400 Hz power cycle (2.5 milliseconds) at $115V_{rms}$ volts and $270A_{rms}$ amperes would produce 58 joules of energy; far in excess of the energy required to ignite a fuel/air mixture. Typically, 10 millijoules of energy or less is considered an ignition level for modern aircraft and post SFAR-88 certified aircraft. It was postulated the arc melted and enlarged the flame arrestor hole; allowing molten material to escape and ignite the air/fuel mixture in the tank.

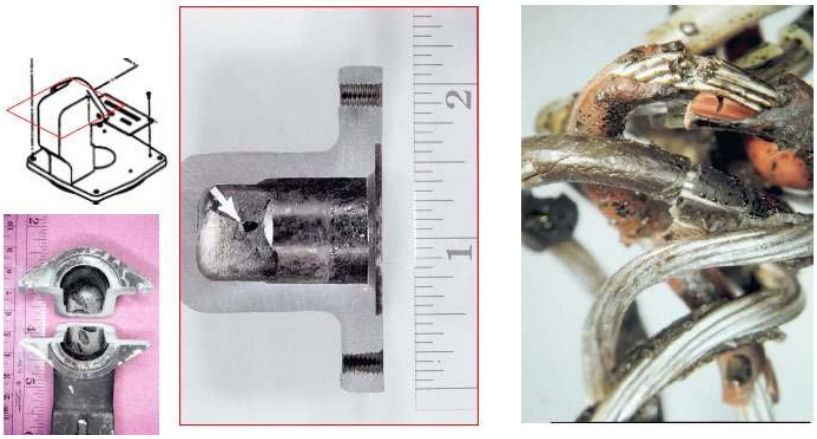


FIGURE C- 10. Mishap pump from exploded aircraft.

White arrow (see FIGURE C- 11) shows melting that enlarged the flame arrestor hole which was filled with a nugget of aluminum and copper. Red circle shows wire conductor arc site on one of the 115VAC phases of the motor. The fuel pump was protected by a standard thermal circuit breaker, which is not capable of protecting against an arc fault event. The breaker was rated at 7.5 ampere, which can exceed its rating, by 1000 percent for 100 milliseconds without tripping.

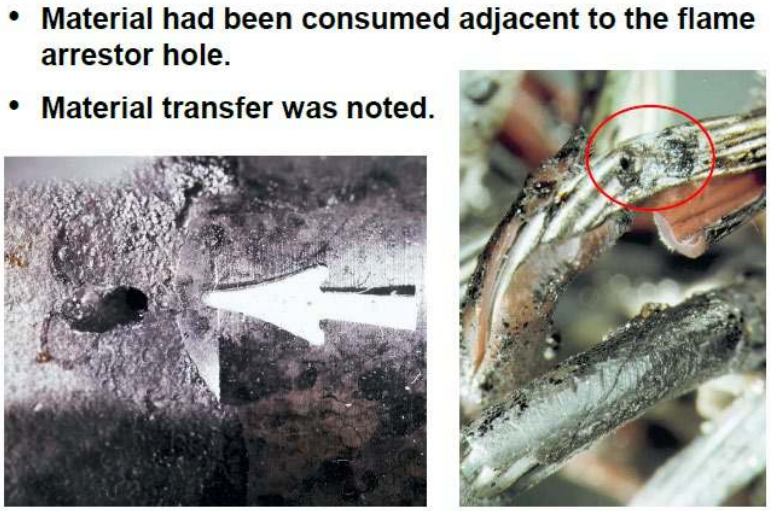


FIGURE C- 11. Arced copper wire and enlarged flame arrestor hole from the mishap pump which contained a nugget of aluminum and copper.

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Since this scenario was difficult to accept, laboratory testing was conducted to simulate a wire arcing to an aluminum housing at the site of a flame arrestor hole (1/32 inch) made of the same material as the mishap pump housing with approximately the same thickness as the mishap pump housing at the flame arrestor site. Test conditions were $115V_{rms}$ with a power source capable of 270 A_{rms} (380 A_{peak}) and revealed molten material could indeed be ejected from the arrestor hole and a flammable mixture ignited.

With a plausible failure scenario developed, the next issue addressed was how the condition developed and identification of corrective actions to prevent future occurrences. An independent team visited the depot pump overall facility to determine why extra wiring was added to the pump connector. The wire length between the pump stator and connector is not specified in the Boost Pump Technical Order (T.O.) repair manual, however, it was clear the depot overhaul staff at some point had decided to use a longer length of wiring without engineering review or documenting in the manufacturing procedures. These changes had apparently been made to reduce repair times and minimize scrapping pump components by the depot repair staff. The wire length included by the OEM would require the stator to be scrapped after removing the connector several times in an overhaul process. Leaving additional length of wiring allowed for more overhauls of the pump without replacing the stator. While this change seemed reasonable, it needed to be assessed for impact on safety and reliability. It was noted the extra wiring had to be twisted and force fit to be placed in the pump housing (see FIGURE C- 12). This local change unintentionally created a condition which resulted in the wiring being pushed tightly up against the interior pump housing. As noted in the mishap investigation above, it was postulated arcing to the internal pump housing resulted in molten material being ejected from the flame arrestor and igniting the fuel tank which in-turn led to the explosion killing six airmen and destroying an aircraft. Left image shows high speed camera frame capturing molten material ejected by the arc. Right image is a high speed camera frame showing ignition of JP-4.

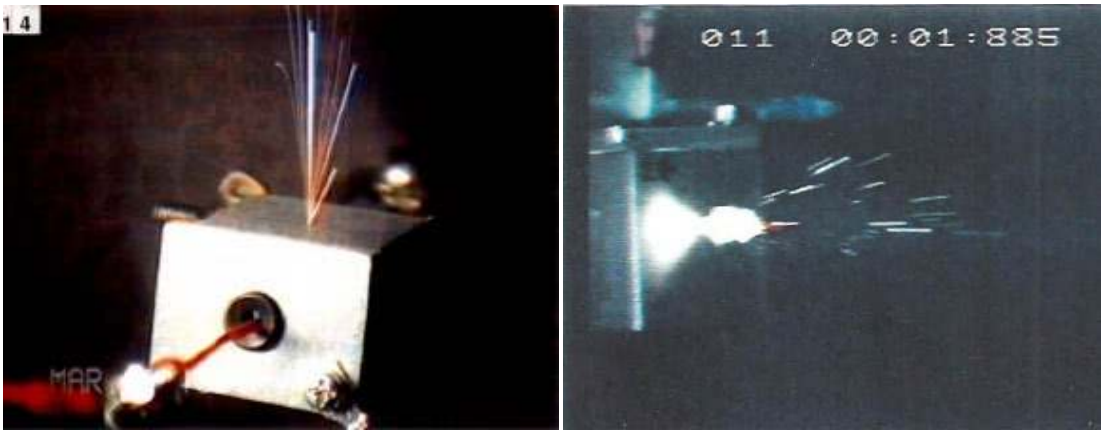


FIGURE C- 12. Lab simulation of an arc inside an aluminum housing (same alloy and thickness as the mishap pump housing) with a 1/32 inch flame arrestor hole.

C.9.3 Recommendations.

Careful review of the depot repair facility showed there were numerous deviations from the boost pump T.O. manual overhaul processes with several possibly contributing to an internal arcing

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condition or impact pump reliability. Some deviations were made by the OEM/pump supplier while others were made by the depot overhaul staff. The pump overhaul process was being signed-off as following the T.O. overhaul process when this was clearly not the case. In addition, the depot overhaul staff observations on pump failures were not being communicated to the engineering staff.

1. The potting compound used to seal the connector was specified as polysulfide in the T.O. yet, the depot changed the potting material to an epoxy; the change was undocumented and inspection revealed a variable depth of material and different curing process between various pumps.
2. Insulation sleeves were not installed over the pump wiring by the depot as specified by the T.O.
3. The depot places the pump in a 150 °C (302 °F) oven to remove the stator, this step is not listed in the T.O. and could damage the wiring since the stator is typically reused.
4. The T.O. specified external leakage to be performed while the pump is mounted in its operating position. The depot overall process does not specify pump orientation.
5. A number of configuration issues were noted during the pump overhaul process review. The pump drawing calls out a 19-strand wire conductor, yet, the OEM supplies the pump stator with a seven-strand conductor. A seven-strand conductor (used in the mishap pump) is much stiffer than the 19-strand conductor is and, therefore more likely to chafe against a structure.
6. The pump T.O. process for measuring bonding resistance was modified almost 10 years before the mishap to allow for simpler and much lower fault current test (five amperes versus the original T.O. requirement of 50 amperes). Bonding is described more fully in SAE AIR5128. Verifying the electrical bonding resistance is essential to prevent arcing if there is a current fault between the pump housing and aircraft ground and it is typically conducted at a high current to ensure there is no degradation at the pump interfaces. Laboratory testing has shown low current measurements are ineffective in verifying the bonding resistance when a potential high current fault is possible. A high current is used to simulate a fault current running through the pump and being properly shunted to ground. A slight increase in resistance at a high current can cause ohmic heating and arcing between interfaces; creating a potential fuel ignition source.

7. The depot staff indicated there were a number of pump failure modes and authorized repairs they had observed, yet there was no formal mechanism in place to inform engineering staff. Depot staff reported the connector potting material was degraded/softened in field returned units. Depot staff have noted stators and impellers have been replaced in the field, which is not an authorized procedure. The depot staff reported the most common pump failure mode is bearing wear. If bearings completely fail, then the rotor and impeller can contact the housing creating high temperatures and damaging the pump and creating potential ignition condition.

C.9.4 Observations.

Additional observations from this incident:

1. This mishap supports the assumption that a fuel tank needs to be considered flammable and mitigation strategies needs to be considered in any safety analysis. The mishap follows a well-documented pattern in most accidents. Several unrelated events

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occur which led to a mishap when the events overlap. For this incident, there were a number of events that occurred at the same time.

a. Unauthorized pump over haul process changes placed a wire in a chafe condition.
b. Wire arcing enlarged a flame arrestor hole and allowed molten material to escape.
c. The fuel tank had a low level of fuel that exposed the pump.
d. A flammable fuel/air ratio existed when the pump was activated.
e. The pump electrical circuit was protected by a thermal circuit breaker that was not designed to protect against an arc fault event.

1. Mitigation strategies exist for each of these events. It would be the system program's responsibility to review and determine which events should be addressed. A comprehensive program such as MECSIP as described in this handbook helps a program develop an assessment process to address the above issues through regular field failure reviews, manufacturing assessments, and verification of configuration control.

2. The investigators postulated the deviations from the T.O manual were the response to a program at the time of this mishap that encouraged technicians to take ownership, look for opportunities to improve repair processes, reduce repair costs, and inspect their own work quality. This is most easily addressed by having periodic independent repair and T.O. reviews. Repair facility operations need to be monitored to ensure they have not deviated from the approved processes and T.O.s need to be updated to allow for process improvements as recommended by MECSIP.
3. It was clear fuel pump repair technicians were not communicating their repair processes to the cognizant engineering authority and the engineering authority was not reviewing the overall process to ensure the correct repair processes were being followed. This can be addressed by ensuring there is regular and documented communication between engineering and depot repair staff and there is independent verification that T.O. procedures are being followed as recommended by MECSIP.
4. There was no effective process in place for technicians to convey their observations of pump failures and possible improvements to the overhaul process to system and component engineers. As above, regular communication between engineering and depot staff can allow technicians to report observations and possible improvements in the repair process and ensuring a change is reviewed for impact on safety and reliability. A system needs to be in place to proactively identify emerging failure modes as recommended by MECSIP.
5. Technicians did not recognize the significance of the changes they were making to the T.O. Regular training is needed to explain the consequences and importance of following T.O. repair processes and reviewing consequences of failures.
6. Fuel system components need to be regularly reviewed to look for opportunities to introduce changes to improve safety and reliability. This could be accomplished by updating Failure Modes and Effects Criticality Analyses (FMECAs) to reflect recent findings from mishaps, review of high failure rate lists, and operational changes as recommended by MECSIP and this handbook.

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C.10 F-16 MAIN FUEL SHUTOFF VALVE MISHAPS

C.10.1 Introduction.

A number of F-16 aircraft have been lost due to suspected uncommanded closure of the fuel system Main Fuel Shut Off Valve (MFSOV) as a result of improper materials selection and safety critical single point failure condition. In most incidents (at least six documented mishaps) the pilot was forced to eject and the aircraft impacted the ground due to partial or complete fuel starvation of this single engine aircraft. Uncommanded closure of a fuel valve has occurred on other aircraft fuel systems due to design flaws, manufacturing defects, use of improper materials, poor maintenance practices or aging of the component.

C.10.2 Background.

Fuel flows through the F-16 aircraft electric main fuel shutoff valve with a travel time of 2-4 seconds during flight. When activated by the FUEL MASTER switch, at military power (afterburner) the engine flames-out in approximately 6 seconds. At IDLE, once the valve is closed the engine flames-out in approximately 45 seconds. The crew has a very short time to react to a closed valve and typically ejects once the engine loses power. This illustrates the criticality of a failure in the fuel system and the challenge of recovering since a backup system may not exist. This is also a good example of a major difference in design philosophy of a military versus commercial aircraft. Military aircraft design accepts considerably more risk than commercial aircraft and assumes that certain failure mechanisms/modes can be managed through use of military and commercial certified parts, maintenance, or through inspections. The FAA typically assumes a failure and the commercial design authority needs to demonstrate through modeling or testing that a component failure will not result in catastrophic failure or loss of a safety critical aircraft system such as the fuel system.

C.10.3 Investigation.

In F-16 mishap investigations where the MFSOV was recovered, the MFSOV has been found in the fully or partially closed position at the time of recovery. A review of the control valve system in flight shows that only an electrical signal will close the MFSOV. This valve is used to facilitate strainer cleaning and disconnection from the engine for certain maintenance operations. Additionally, there is an actuating switch in the cockpit (Fuel Master Switch) so that the pilot could close the valve in event of an engine fire or emergency.

A FMECA review shows the MFSOV could be closed when the pilot activates the fuel master switch or uncommanded via an electrical short in the switch, wiring, or valve connector. A fish bone would be a good approach to show all the possible conditions that could close the valve.

This would typically include probable and less likely failure conditions. For the valve in question, some less probable failure conditions would be mechanical failure of the valve.

In a recent mishap, it was established the pilot did not place the Fuel Master Switch in the cockpit to the OFF position, based on based on collaborating evidence leaving one of the possible causes of engine shutdown an electrical short in the valve control circuit. It was noted the valve initially was not fully closed since the engine lost power, but did not flame-out until after several minutes. The pilot could not have commanded a partially closed MFSOV, as the only option for the pilot to command is fully open or fully closed. Additionally, the fuel master switch in the cockpit requires

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three movements to activate, break the safety-wire on the red guarded cover, lift and move the guarded cover, and then activate the switch.

C.10.3.1 Connector assessment.

A review of maintenance records and past mishaps has established the MFSOV electrical connector (see FIGURE C- 13) is susceptible to corrosion which can compromise the electrical isolation between the connector pins as shown below.



FIGURE C- 13. Fuel off valve connector showing corrosion on the tin-plated steel electrical pins. Note the internal O-ring is intact.

The electrical leakage path between the normally isolated contact pins can impress sufficient valve resulting in engine fuel starvation. Under this scenario the valve failure is due to moisture entering the connector and fretting corrosion between the tin plated contacts and gold plated sockets. The extensive corrosion is accelerated by the presence of moisture and having tin plated pin contacts inserted into gold plated copper alloy sockets. Gold and tin form a strong galvanic couple and the tin plated contact pin will rapidly corrode when a small amount the moisture is present. Fretting corrosion results in loss of the tin coating and exposure of the steel pins to moisture leading to galvanic corrosion of the steel (rust) and production of ionic materials. The ionic materials electrically shorted the isolated connector contacts, which can result in closure of the valve when several pins are shorted by the presence of the conductive corrosion products when moisture is present.

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The extensive corrosion damage is shown on FIGURE C- 14, below; also note the O-ring is missing which is intact on FIGURE C- 13.

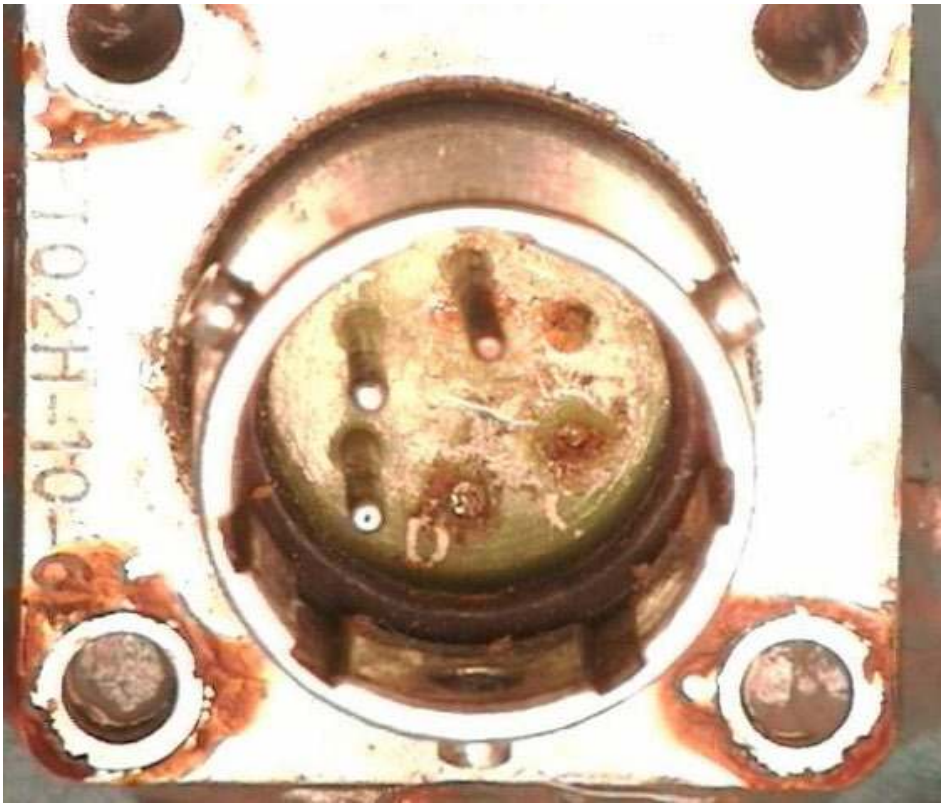


FIGURE C- 14. Three of the six tin plated steel pins have been corroded and broken away from the connector. Note the internal O-ring is missing (see photo above).

C.10.4 Mitigation.

The mitigation strategy was to inspect the connector every six months and apply a Corrosion Preventative Compound (CPC). The CPC acts as a water displacement film and protects surfaces from corrosion. This is a temporary solution and the CPC needs to be re-applied every 100 hours as prescribed by the Technical Order Manual and a Time Compliance Technical Order (TCTO). There have been recent reports showing that CPCs sprayed internally in connectors can degrade electrical connections and damage electrical insulations and protective seals (see Corrosion Preventative Compounds (CPCs) Effect on Aircraft Electrical Wiring Components, dated 2010).

The T.O. procedures have not always been followed correctly, as noted in a recent mishap investigation on an F-16 aircraft (see AIB Report F-16C, T/N 92-003886 22 July 2012). This illustrates a common maintenance issue, that over time, repetitive maintenance actions may not be followed correctly as the rationale is lost and short cuts are developed. While this is primarily a training issue on proper application of T.O. procedures it also is related to verification that the T.O. was accomplished correctly. Another lessons learned is that aircraft maintenance may be performed by airman, USAF depot personal, or contractors and each has a different process for verification that a given procedure was performed correctly and by a properly certified technician.

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C.10.5 Summary.

To summarize, this case-study shows the criticality of the aircraft fuel system and how reviewing maintenance/mishap data and identifying and understanding failure conditions and critical paths can improve aircraft safety, availability, and reliability and reduce operating costs.

Additional information about interactions of dissimilar metals can be found in MIL-STD-889.

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C.11 B-1B MISHAP CAUSED BY FAILURE OF A FUEL SYSTEM CONTROL PANEL

C.11.1 Introduction.

On the morning of 18 February 1998, a B-1B bomber took off from its base in Texas to conduct a routine training mission over the central U.S. About three and one-half hours later, as the bomber's four-man crew was commencing the final portion of the mission, the pilot was alerted to a possible Auxiliary Drive System failure on engine no. 3. Following standard procedure, he initiated a shutdown of that engine using the Fire Warning Extinguisher Panel (FWEP) (see FIGURE C- 15).



FIGURE C- 15. Fire warning extinguisher panel (FWEP).

As soon as the switch was enabled to cut-off fuel to the engine, the fire warning lights for the remaining three engines immediately illuminated. Thick smoke filled the cockpit. All four engines shut down without action by the crew, and in-turn, disabled the B-1B's electrical and hydraulic systems. With the aircraft uncontrollable, the crew had no choice but to eject and the abandoned bomber impacted on open farmland about five miles northeast of Marion, Kentucky. None of the airmen suffered any serious injuries, and the aircraft crashed in a relatively unpopulated area with no loss of life on the ground. Loss of aircraft was estimated at $310 million. As a result of the accident the entire B-1B fleet was grounded until the cause could be determined.[1]

C.11.2 Accident investigation.

Air Force flight safety investigators were on scene within a matter of hours. The B-1B does not have an on-board voice recorder, but it has a Crash Data Recorder (CDR), and the CDR was recovered. Unfortunately, the CDR had been heavily damaged by the impact, so only about a third of its memory remained intact. Nonetheless, by combining aircrew interviews, CDR data and other information, the investigators were able to establish a reasonably detailed sequence of events. Within just over a minute after completing the final step in shutting down engine no. 3—depressing an engine fire pushbutton on the FWEP —all the other engines flamed out for lack of fuel. Since the FWEP controlled the emergency fuel shut-off valves for all four engines, suspicion immediately fell upon this fuel system component. The FWEP had been retrieved from the wreckage, and it quickly became the subject of an intense examination and the USAF Safety Investigation Board asked for AFRL Materials Directorate, Materials Integrity Branch technical assistance in analyzing the FWEP. Fortunately, the FWEP had been thrown from the impacting

[1]Rpt, AFSC/JA, "Aircraft Accident Investigation Report," 30 Mar 98, [Extract], **(SD ACC-1)**.

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aircraft and was mostly intact. Analysis of the FWEP revealed extensive damage in critical areas of the device's printed wiring board (PWB).

C.11.3 Circuit card evaluation.
Lab analysis and testing determined that a short circuit most likely occurred outside of the circuit board and the high current over heated a 28VDC copper trace on the FWEP PWB. The overheated trace carbonized the glass epoxy PWB insulation and initiated sustained arcing that extensively damaged the circuit card. All six-fuel cut-off valves were controlled by relay circuits on the circuit card (one for each of the four engines and one for the right side and left side APU), furthermore, the fuel cut-off valve relay controls lines were all adjacent to the shorted trace.

The sustained arcing and carbonization resulted in loss of insulation integrity and propagation of the failure into multiple circuits. The loss of the epoxy's electrical isolation effectively applied 28VDC to the fuel off valve control relay traces. This resulted in the un-commanded closure of all the fuel cut-off valves.

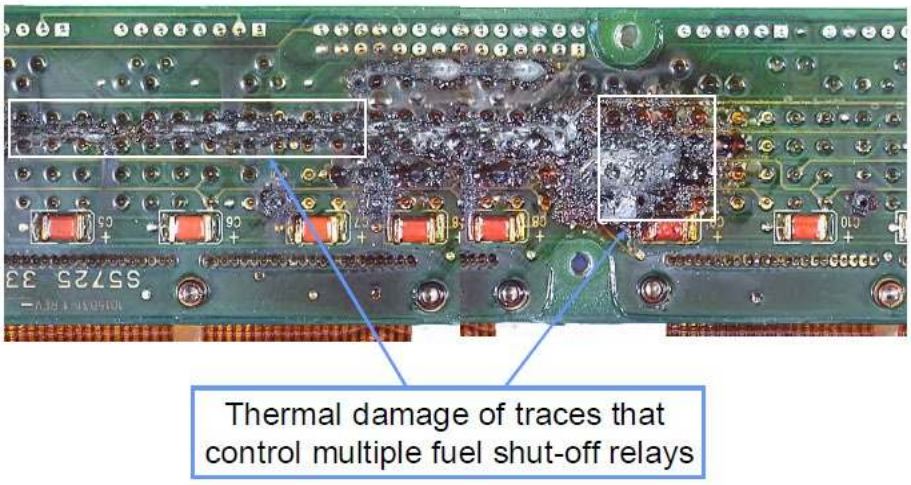


FIGURE C- 16. Mishap PWB with melting and arcing of copper traces and carbonized epoxy which resulted in 28VDC being applied to the fuel cut-off valve control relays.

As the circuit board arced and burned, many of its copper circuit pathways melted together, generating an uncontrolled cascade of electrical shorts so that multiple paths were applying 28VDC to close the fuel cut valve relays. This corresponds to the crew's reported thick smoke in the cockpit signals. Thus a single point failure condition was able to create an uncommanded and irreversible closure of the fuel cut-off values for all four engines.

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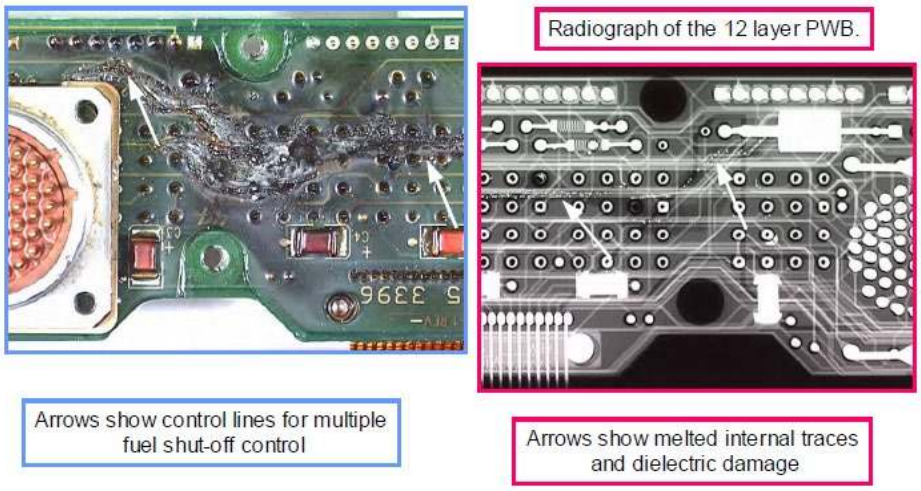


FIGURE C- 17. Close-up of the carbonized PWB and location of traces that control the fuel cut off relays.

Resistive paths across the degraded epoxy were created and the circuit protection failed to operate since the circuit protection opens only when there is a complete/hard short as opposed to a soft/resistive short. The single point failure condition described above were duplicated in the lab[2] resulting in immediate modifications to remaining B-1B aircraft.

A laboratory simulation duplicated the mishap failure using one-trace currents that did not trip the 3A circuit protection. The arcing continued for 30 seconds before the circuit breaker finally opened. During this time the PWB arced and carbonized and produced vast quantities of smoke and fire. Most of the critical fuel control valve copper traces had also been compromised and were shorted to 28VDC which would have activated the relays and cut fuel off to the engines.

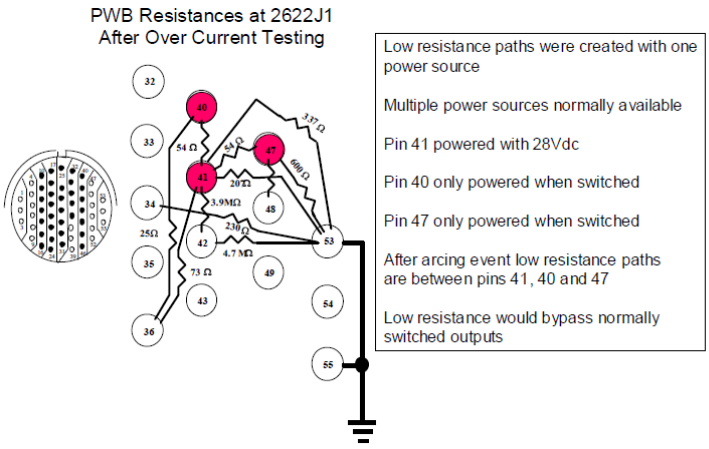


FIGURE C- 18. Diagram of assessed connector.

[2] *Ibid*; Cockpit Electrical Fires Attributed to Printed Wiring Boards, SAE Safety Conference 2001

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The FWEP had been designed to provide significant improvements over its previous design. The original FWEP panel was point-to-point wired (contained no circuit card) using mechanical switches that frequently failed, yet provided adequate separation between the critical circuits. The new FWEP design replaced the mechanical switches with more reliable solid-state switches that were on a single multilayer PWB. This eliminated any separation between the safety critical fuel valve control circuits and sources of 28VDC that could activate the relays and close the fuel cut off valves. There are thirteen 28VDC power sources entering the circuit card from two connectors protected by multiple 3-ampere thermal circuit breakers. Since the designer and cognizant government authority considered the new design a direct form, fit, and function replacement, no safety analysis was done on the new design. As noted above, the fuel cut off valve control lines were all adjacent to each other and were also next to 28VDC power which activates the cut off valves, thus a single point failure condition existed.

The lab was given an opportunity to review the design process for the mishap FWEP. It was learned the FWEP system designer was aware of the single point failure condition; however, he had reviewed failure rates for PWB's and determined that a shorted PWB trace causing voltages to be applied to critical circuit was an improbable failure condition. A MIL-HKB-217 reliability analysis showed the most common failure on a PWB was an open trace with occurrence calculated at 1.2 failures every 10 million hours. Arcing and carbonization of the PWB was not a listed typical failure condition. While commonly used for this purpose, MIL-HKB-217 was last updated in 1995 and should not be used to determine failure rates of specific components.

According to MIL-STD-882, System Safety standard, one failure in one million hours or less is considered an improbable event. The FWEP designer therefore concluded, the existence of a safety critical single point failure condition was acceptable. This is a flawed assumption and shows the designer did not fully understand how to conduct a proper risk analysis. The problem is that reliability models assume proper design practices are followed, there are no manufacturing defects and that secondary failures such as external shorts and arcing are not part of the reliability analysis. The designer should have prepared a Failure Mode, effects and Criticality Analysis (FMECA), which would have shown the serious vulnerability of the single point failure condition that had been created. Placing 13 power sources and all safety critical circuits on a single PWB without adequate separation is also a serious violation of proper engineering design practices. Furthermore, the government office responsible for procuring the new FWEP failed to recognize the new design was not a simple form, fit, and function replacement of the existing FWEP.

The new design used completely new technology with unique vulnerabilities, which should have required an independent safety review of the design. The principal vulnerability was a single point failure condition in the new design that could cause uncommanded cut off of the fuel supply to all aircraft engines.

C.11.4 Actions taken.
Since this incident, there have been several other electrical designs that have had single point failures that have cascaded through a circuit card and caused complete loss of a safety critical system. The take away is that high levels of complexity, integration, and electrical power can lead to complete system failure. While each incident is unique, two recurring themes are inadequate separation of critical circuits/paths and arcing. A designer has to assume insulation between critical circuits will be compromised when they are on the same circuit card and model or conduct

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actual testing to determine if arcing would compromise the critical paths. The designer has an option to employ circuit protection that can trip when arcing is detected by using arc fault detection technologies. The designer has to weigh the advantages and disadvantages of using this technology since false tripping the arc detection devices could disable a critical system. See the references cited at the end of this case study for more discussion on arc fault detection.

C.11.5 Summary.

Failure to follow proper design and review processes compromised the B-1B aircraft fuel system safety and resulted in the loss of a $310 million aircraft and impacted the entire fleet of aircraft during a time when the B-1B was operating in an active conflict zone. All remaining aircraft had to be retrofitted with the original FWEP boxes prior to return to service. This case-study shows the importance of conducting a through failure analysis, taking the time to fully understand a design, the need to fully review system vulnerabilities and risk, and understanding how a design impacts a complex aircraft system.

The B-1B mishap and its findings were briefed to USAF Command resulting in mandatory implementation of Operational Safety, Suitability, and Effectiveness (OSS&E) process as a requirement for maintaining aircraft airworthiness (see AFI63-1201). The OSS&E plan states organizations responsible for Air Force systems or end-items need to use a disciplined engineering process throughout the system's life-cycle to ensure that activities such as operational use, configuration changes, maintenance repairs, and part substitutions do not degrade system or end-item baselined characteristics over their operational life.

This has far ranging implications for how USAF aircraft fuel systems are designed, procured, upgraded, modified, and maintained. Each fuel system component needs to be assessed for impact on aircraft airworthiness and when modified or upgraded any changes in the design or technology need to be also assessed for impact on aircraft airworthiness. This can only be accomplished if the system or component design and manufacturing processes have been fully documented and an independent FMECA and risk analysis has been accomplished. The B-1B mishap emphasized the need for locating all potential single point failure conditions and conducting modeling or actual testing to determine its impact on aircraft safety.

C.11.6 References.

1. SAE AS5692 - Arc Fault Circuit Breaker (AFCB), Aircraft, Trip-Free Single Phase 115VAC, 400Hz – Constant Frequency.
2. SAE AS6019 – Arc Fault Circuit Breaker (AFCB), Aircraft, Trip-Free 28VDC.

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C.12 NIMROD CASE STUDY

C.12.1 Introduction.

On 2 September 2006, an RAF Nimrod XV230 was on a routine mission over the Helmand Province in Southern Afghanistan when, only minutes after completing air-to-air refueling, the aircraft suffered a catastrophic mid-air fire which led to the total loss of the aircraft and the death of all 14 on board. Following an initial mishap investigation, a broader independent review was initiated in late 2007.

C.12.2 Mishap investigation.

The Nimrod accident board concluded the most likely cause of the accident was an inadvertent fuel overflow from number one tank into the starboard No. 7 Tank Dry Bay during air-to-air refueling, which ignited on contact with a hot pipe. The XV230 was lost because the fire broke out in an inaccessible part of the aircraft, which had no fire protection. Going beyond the tragedy of loss of life, this accident caused a major review of aircraft safety in the UK since the crew had no chance of controlling the fire and the conditions that caused the fire had existed for decades. As noted in other case studies, catastrophic aircraft failures that occur outside the crew's control call into question basic airworthiness and how it is maintained. The XV230 accident resulted in a wide ranging inquiry that reviewed the air worthiness process in the RAF.

This in depth review concluded design flaws in the Nimrod introduced in 1969, 1979 and 1989 all played a crucial part in the loss of the XV230. There had been a number of previous incidents and warning signs potentially relevant to the suspected failure causes that were never addressed by the cognizant program authority. It was further concluded, the safety of the Nimrod program was undermined by a general complacency since it had successfully flown for 30 years and the safety program had become a paperwork exercise.

As noted, the loss of the aircraft was attributed to a fire, which was initiated when aviation fuel came into contact with a high temperature ignition source. The most likely source of ignition was the high temperature Cross-Feed/Supplementary Conditioning Pack duct (Cross-Feed/SCP duct) and the most likely initial seat of the fire was the dry bay located forward of the starboard No. 7 tank which contained numerous fuel pipes adjacent to the Cross-Feed/SCP duct. The source of fuel was more difficult to determine due to in air breakup. A number of fuel sources were identified and are ranked below in probability:

a. The escape of fuel during Air-to-Air Refueling (AAR) occasioned by an overflow from No. 1 tank (from the No. 1 tank blow-off valve and/or vent pipe connection) source.
b. A leak from the fuel line coupling leading to the pooling of fuel in the starboard No. 7 Tank Dry Bay.
c. A hot air leak from the Cross-Feed/ Supplementary Cooling Pack (SCP) duct causing damage to adjacent fuel seals.

According to the investigating board, each of these three sources of fuel is potentially relevant to the cause of the loss of the XV230. Each raises significant issues regarding the airworthiness of the aircraft over many years and raises serious questions about the management of airworthiness by the Nimrod program.

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C.12.3 Design flaws.
The Nimrod suffered from fundamental design flaws, which played a crucial part in the loss of the XV230. Defective design occurred at three stages:

First, the original fitting of the Cross-Feed duct to aircraft from 1969-1975 was to enable the distribution of Auxiliary Power Unit (APU) air, and then engine bleed-air, to the engines for ground starting. It also enabled engines to be shut down in flight and re-started, using hot bleed-air routed from another engine via the Cross-Feed duct. The Cross-Feed duct gave rise to a serious fire hazard, particularly in No. 7 Tank Dry Bay, which is located in the fuselage, in close proximity to fuel pipes and outside any fire zone (see FIGURE C- 19). The following four factors gave rise to two risks:

Factors:

1. Its location at the bottom of a bay resulted in closely packed fuel pipes and couplings;
2. design of the No. 7 Tank Dry Bays, which allowed fuel pooling;
3. inadequate insulation, which was vulnerable to fuel ingress; and
4. absence of fire protection in those bays.

Risks:

1. Fuel leaking from adjacent fuel couplings, or other parts of the fuel system onto exposed or vulnerable parts of the Cross-Feed duct could ignite; or
2. a leak of hot air from the Cross-Feed duct could cause heat damage to the fuel couplings in close proximity and ignite escaping fuel.

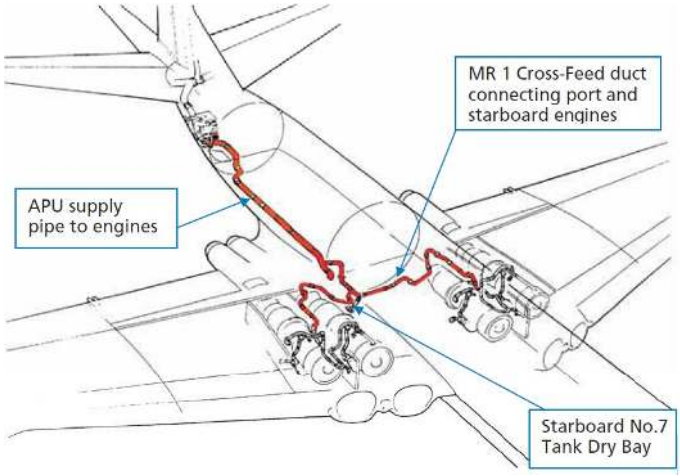


FIGURE C- 19. Cross-Feed duct installed in aircraft between 1969-1975. Also note location of No. 7 dry bay.

Second, the addition of the Supplementary Cooling Pack (SCP) to aircraft from 1978-1984 provided additional cooling for the extra electronic equipment. The SCP required bleed-air taken off the Cross-Feed duct. The fitting of the SCP take-off duct significantly exacerbated the fire hazard posed by the bleed-air system because (see FIGURE C- 20):

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a. The SCP duct was routed through the bottom of the starboard No. 7 Tank Dry Bay, thereby increasing the chances of fuel from fuel couplings above dripping onto it and causing a pool fire at the bottom of No. 7 Tank Dry Bay;
b. immediately aft of the elbow of the SCP duct, an expansion bellows was fitted with a laced muff which was vulnerable to the ingress of fuel
c. the elbow was encased in a fairing proud of the fuselage hull which was vulnerable to fuel tracking along the outside of the fuselage; and
d. the flow of air through the Cross-Feed duct increased the duct temperature whenever the SCP was on.



FIGURE C- 20. [Left] No. 7 tank dry bay (lower panel). [Right] No. 7 tank dry bay (internal drain holes).

Third, the fitting of the permanent Air-to-Air Refueling (AAR) modification to aircraft from 1982-1989 to enable refueling in flight to take place. The addition of AAR capability further increased the fire hazard posed by the Cross-Feed/SCP duct in three ways:

a. The refuel gallery was now pressurized when airborne and undertaking AAR, thereby increasing the chance of leaks in flight contacting the Cross-Feed/SCP duct;
b. it gave rise to the risk that blow-off valves on the side of the fuselage of the aircraft (which can automatically release excess fuel from tanks during refueling to prevent fuel tank over-pressure) might now operate in-flight, and lead to fuel tracking back along the aircraft and re-entering the fuselage through ports, vents and gaps in the aircraft structure; and
c. it gave rise to the risk that fuel would enter the air vent system during AAR in flight and leak (both b and c are because of an existing design problem with asymmetric filling of No. 1 tank).

Previously, such blow-off valves and fuel entering the vent system did not pose a hazard as refueling was only conducted on the ground.

C.12.4 Conclusions from investigating board.

The investigating board focused on why the Nimrod program failed to recognize the above design flaws over many decades. The presence of hot ducting and fuel pipes in both No. 7 Tank Dry Bays, and the threat from fuel from the starboard No. 1 tank blow-off valve entering the fuselage, represented a serious, failure condition to the Nimrod fleet for several decades. Many competent and dedicated engineers had worked on the Nimrod program and had observed fuel pipes and couplings within inches of hot ducting yet took no action. Apparently, the engineers and

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technicians assumed the aircraft was properly designed and delivered in an airworthy condition. The following are examples where the program made incorrect assumptions:

1. The Nimrod program had assessed the potential for fuel system leak as ‘improbable’ yet a review of Nimrod maintenance data showed that there was an average of 40 fuel leaks each year.
2. The program had determined the Cross-Feed Duct was only pressurized during engine start. Testing determined the Cross-Feed Duct is actually pressurized for long periods (at up to 420 °C).
3. The program noted fire detection and suppression systems as mitigation for the No. 7 Tank Dry Bay, yet neither a detection nor suppression existed within the bay.

In summary, this case illustrates how dangerous such fundamental “embedded” design defects can be, and why a robust MECSIP program is critical to maintaining air worthiness and aircraft safety.

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C.13 TWA 800 INFLIGHT FUEL TANK EXPLOSION MISHAP

C.13.1 Introduction.

On 17 July 1996, Trans World Airlines, Inc. (TWA) flight 800, a Boeing 747, crashed in the Atlantic Ocean near East Moriches, New York. The National Transportation Safety Board (NTSB) determined the probable cause of the TWA flight 800 accident was an explosion of the center wing fuel tank (CWT), resulting from ignition of the flammable fuel/air mixture in the tank. The source of ignition energy for the explosion could not be determined with certainty, but, of the sources evaluated by the investigation, the most likely was a short circuit outside of the CWT that allowed excessive voltage to enter it through electrical wiring associated with the fuel quantity indication system.

C.13.2 Investigation.

The Boeing 747 aircraft center fuel tank was nearly empty prior to take-off, allowing excessive heat build-up in tank since part of the aircraft cooling system was under the center tank and generated considerable heat. An NTSB funded study produced the results shown on FIGURE C-21 with respect to center tank heating when mishap aircraft fuel tank conditions were duplicated on a Boeing 747 aircraft. For an almost empty tank the temperature was well above 37.7 °C (100 °F). Jet A is kerosene based and has a flash point over 37.7 °C (100 °F) and an auto-ignition temperature of 210 °C (410 °F).

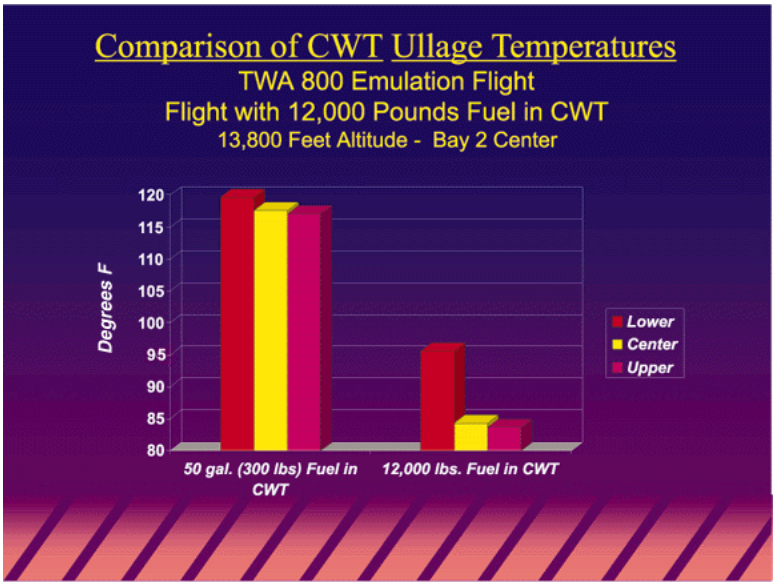


FIGURE C- 21. Comparison of center wing tank ullage temperatures.

NTSB testing also revealed Jet A auto-ignition at various energy (joules) levels and temperatures. At a fuel tank temperature of 48.9 °C (120 °F) auto-ignition occurred at 2.5 millijoules, at fuel tank temperature of 46.1 °C (115 °F) auto-ignition occurred at 25 millijoules, and fuel tank temperature of 40 °C (104 °F) auto-ignition occurred at 1 joule.

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C.13.3 Ignition.

It is theorized that the ignition energy was an electrical arc that occurred between a center fuel probe electrical connections as noted on FIGURE C- 22.

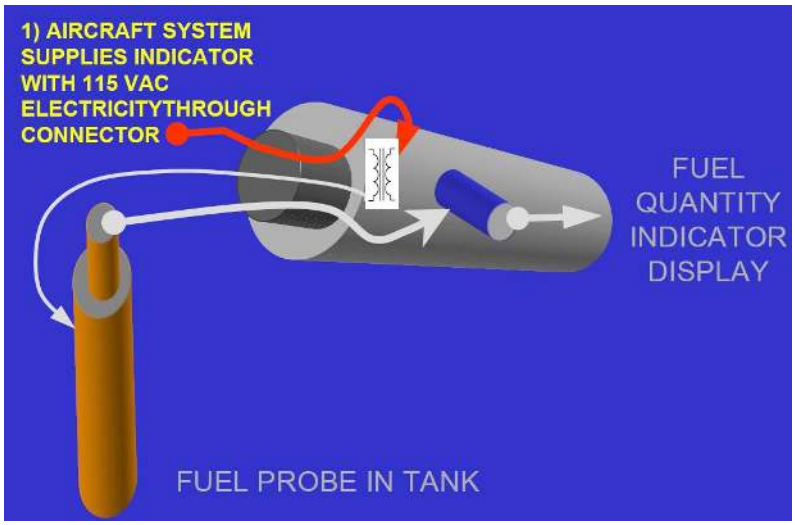


FIGURE C- 22. Diagram of fuel probe system.

The arc may have originated from damaged wiring or a build-up of a conductive residue between fuel probe elements. Damaged fuel probe PTFE insulated conductors were found on retired 747 aircraft. The insulation was damaged by cold flow of the Teflon insulation when it was compressed against a metal clamp (see FIGURE C- 23).

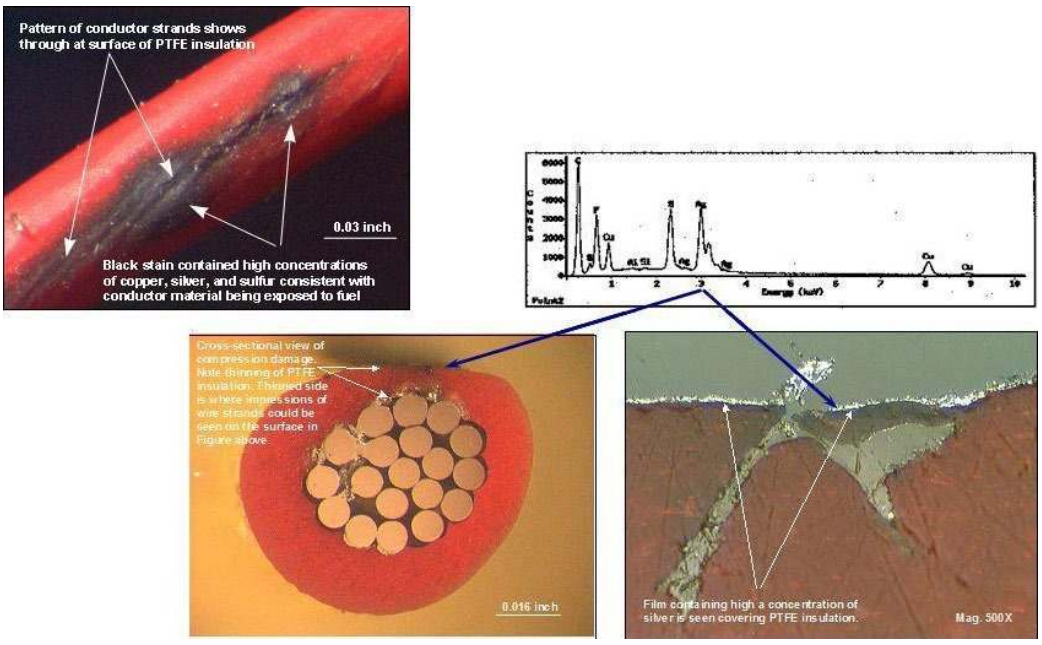


FIGURE C- 23. Photos of damaged wire and analysis.

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The exposed silver plated copper conductors reacted with the fuel and created a semi-conductive residue of copper and silver oxides and sulfides. Several studies funded by the FAA and NTSB and conducted by AFRL/RXSA have shown that fuel will react with silver and form semi-conductive residues, which will degrade a number of electrical properties such as insulation resistance and contact resistance (see Reports AFRL/MLS 99-2, AFRL/MLS 99-33, AFRL/MLS 99-68, and FAA sponsored research paper "Do Conductive Residues on Electrical Wires and Contacts in Aircraft Fuel Tanks Pose a Combustion Hazard? Part I. Formation Mechanism" dated 2001).

Degradation of these electrical properties can change capacitance, which will effect fuel probe measurements and lower the voltage breakdown level between electrical connections. Silver is used as a plating on copper conductors, as plating's on terminals, and on relay/switch contacts. The lesson learned is that silver should not be used on surfaces that come in contact with fuel. Wiring should be nickel plated copper and, terminals should be nickel plated, and relay and switch contacts should be gold plated.

Fuel probe terminals blocks were removed from retired Boeing 747 aircraft and exhibited conductive residues between the silver plated terminals as noted on FIGURE C- 24. Breakdown and arcing occurred when 115 VAC peak voltage (170VDC) was applied to the residues.

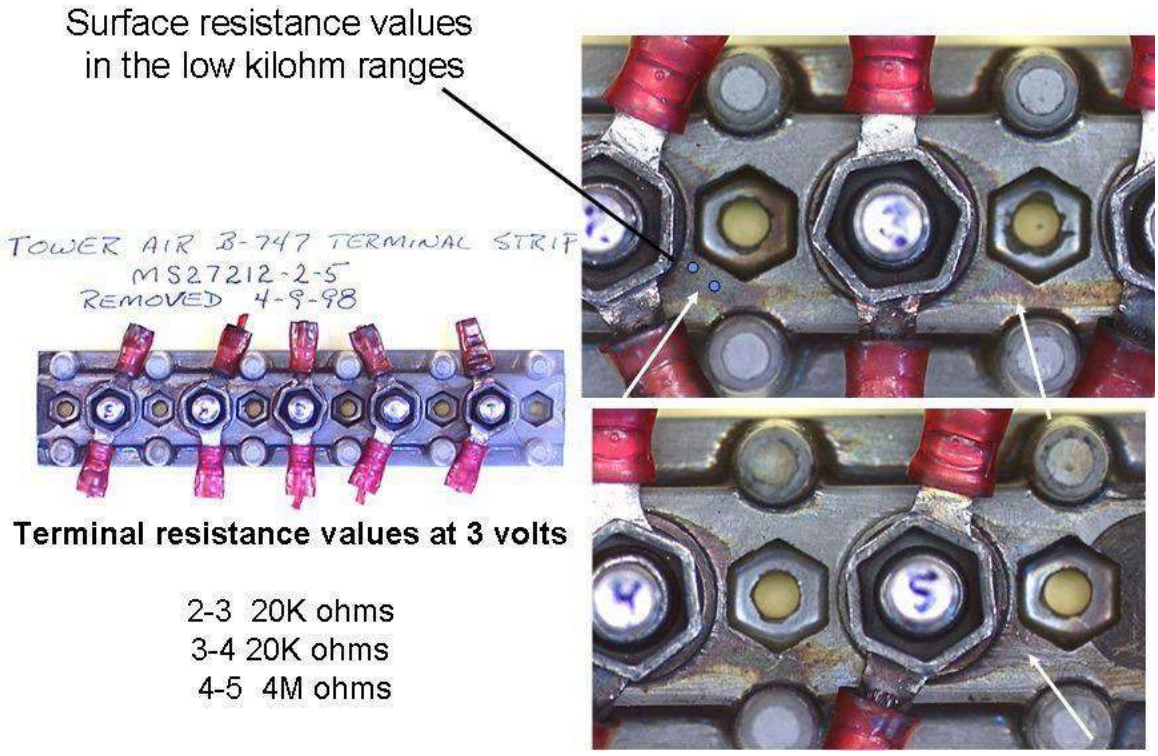


FIGURE C- 24. Fuel probe terminal block removed from retired Boeing747 aircraft.

Contributing factors to the accident were the design and certification concept that fuel tank explosions could be prevented solely by precluding all ignition sources and the design and

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certification of the Boeing 747 with heat sources located beneath the CWT, with no means to reduce the heat transferred into the CWT, or to render the fuel vapor in the tank nonflammable.

C.13.4 Recommendations.
The recommendations from the NTSB included the following:

1. "Examine manufacturers design practices with regard to bonding of components inside fuel tanks and require changes in those practices, as necessary, to eliminate potential ignition hazards."
2. "Review the design specifications for aircraft wiring systems of all U.S.-certified aircraft and (1) identify which systems are critical to safety and (2) require revisions, as necessary, to ensure that adequate separation is provided for the wiring related to those critical systems."
3. "Require the development and implementation of corrective actions to eliminate the ignition risk posed by silver-sulfide deposits on fuel quantity indication system components inside fuel tanks."
4. "[T]he need for improved training of maintenance personnel to ensure adequate recognition and repair of potentially unsafe wiring conditions."
5. "[T]he need for improved documentation and reporting of potentially unsafe electrical wiring conditions."
6. "[T]he need to incorporate the use of new technology, such as arc-fault circuit breakers and automated wire test equipment."

Information taken from NTSB report PB2000-910403.

The findings from this mishap have resulted in the following actions by the aerospace government agencies and aerospace community.

Most active Boeing 747 aircraft have had fuel probes and wiring replaced and rerouted away from power wiring. Separation of fuel probe wiring from power wiring has been accomplished on other aircraft with similar fuel systems and military and commercial aviation standards have included this practice in their specifications; (as an example, refer to SAE AS50881, Wiring Aerospace Vehicle, which is often cited as a contractual requirement for military aircraft and is recognized by the FAA as a means to meet air worthiness compliance requirements. From SAE AS50881, paragraph 3.10.12.3, sensitive circuits fuel probe wiring shall be physically separate from power wiring throughout its entire length including connectors, terminations, and junction boxes).

The FAA issued an operational rule that required all high flammability tanks on commercial transports manufactured after 1991 to incorporate means to limit fuel tank flammability to two percent of the operational time and two percent of the time when the operating temperature is above 26.7 °C (80 °F). Inerting systems are installed in all heated center wing tanks in Boeing and Airbus airplanes manufactured after 1991 that are in US operations. As noted in other sections of this handbook, fuel tank inerting is typically employed on military aircraft; however, this may not be the case for commercial derivative aircraft used in military operations.

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C.14 C-21 TRANSPORT AIRCRAFT FUEL IMBALANCE MISHAP

C.14.1 Case introduction.

This case-study involves crew error, inadequate training, inadequate monitoring of maintenance actions, lack of technical insight when leasing commercial aircraft for military missions, using multiple vendors to support the fleet, a serious fuel system design error, a fuel transfer system component failure, and aging/degradation of fuel system components. On April 17, 1995, the flight crew of a U.S. Air Force C-21A (Learjet 35A) declared an emergency after developing a fuel imbalance. After more than 30 minutes of discussion between the crew and air controllers, an emergency was declared. During an attempted landing, the aircraft flipped upside down due to a massive fuel imbalance (1,800 lbs.) in the left wing compared to the right wing. The weight imbalance made the plane uncontrollable and impact with the ground killed the two pilots and six passengers.

In its final report, the Air Force stated, the accident was caused by a combination of an aircraft mechanical malfunction, a flight-manual deficiency and human error. To quote the official accident report "The investigating officer found that the mechanical malfunction consisted of the right standby [fuel] pump continuing to operate uncommanded after engine start". "This malfunction resulted in fuel being pumped into the left wing and prevented fuel from being transferred to the right wing during normal transfer procedures. This condition caused a fuel imbalance." An excellent summary of the overall accident is available from the Flight Safety Foundation at http://flightsafety.org/ap/ap_feb00.pdf.

C.14.2 Aircraft information.

The C-21A is the U.S. Air Force designation for the Gates Learjet 35A, a light twin-turbofan business aircraft that first flew in 1973. In 1983 the Air Force contracted with Gates Learjet for the lease and logistical support of 80 C-21As, aircraft for personnel transport, cargo transport and medical-evacuation missions. The aircraft has accommodations for two pilots and up to eight passengers. Each of the two Garrett (now AlliedSignal) TFE731-2-2B engines is rated at 3,500 pounds static thrust. Maximum usable fuel capacity is 931 gallons (6,051 lbs.). Fuel is stored in two wing-tip tanks, two internal wing tanks and one fuselage tank.

C.14.3 USAF findings.

A quick summary of the overall USAF report findings follow: The crew did not fully understand how the fuel system functioned and made incorrect assumptions about what was causing the fuel to transfer to the left wing tank. The Air Force C-21A flight manual did not contain an available checklist for correcting a fuel imbalance which was available for commercial aircraft (published in 1990) since the USAF did not contract for manual updates. Because the crew did not have a checklist or flight-manual on the fuel imbalance condition, they misanalysed the malfunction and allowed their airspeed to become too slow for the aircraft's configuration.

C.14.3.1 Crew.

The crew had correctly turned off the transfer pumps during the mishap flight, which moves fuel between the wings. They were unaware the right standby fuel continued to run as a result of a postulated stuck pump control relay. Subsequent interviews with other USAF pilots and maintenance personnel revealed some pilots and maintenance personnel were aware of the stuck

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pump relay issue and had pulled the pump circuit breaker to stop the pumps as was recommended in the civilian aircraft manual.

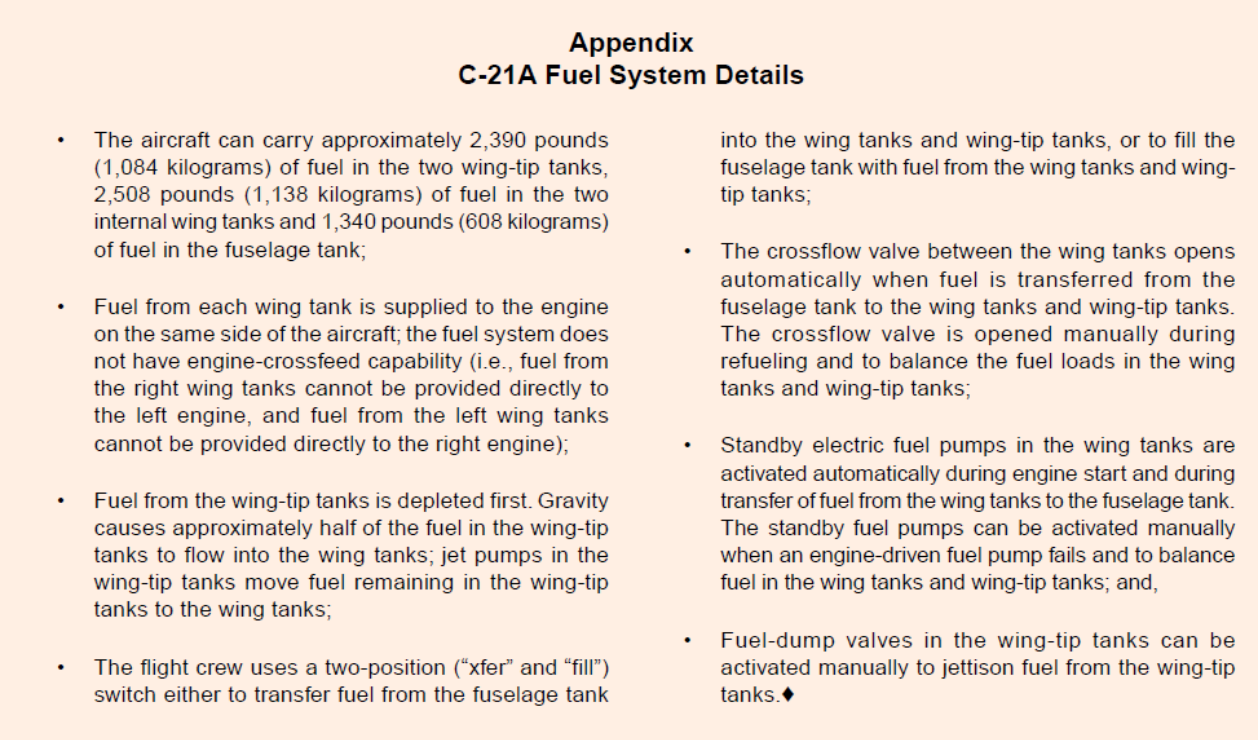
C.14.3.2 Maintenance personnel.

Maintenance personnel, which were contractors from a logistics and maintenance company, had unstuck relays in the past by hitting the relay to break the contacts loose. This had become routine and did not result in replacement depending on the time available to perform maintenance. It is worthwhile to note that the previous flight of the mishap aircraft, had a reported problem with the standby pumps, which was reported by the mishap crew. From the USAF report "in the flight prior to the mishap…the (mishap) crew requested a full load of fuel and told maintenance technicians that they had been unable to transfer fuel from the wing tanks to the fuselage tank. The maintenance technician told the crew that replacement of the fuselage-tank switch had not corrected the problem and that he was going to try to correct the problem by replacing the fuel-control relay panel." When told this was a long process, the mishap crew elected to not replace the relay panel and to address it at another time since the problem appeared intermittent.

C.14.4 Recommendations.

Maintenance should have placed the aircraft into an "out of commission" status and not given the pilot the option to fly the aircraft, considering the unresolved issue with the standby pump system. Every discrepancy needs to be recorded as completely as possible, so while maintenance personnel troubleshoot the underlying problem causing the reported discrepancy, it can be resolved. Troubleshooting requires looking at the entire system and not just one component of an aircraft. Once a pilot turns the aircraft over to maintenance, maintenance personnel are "responsible and accountable" for providing full and complete documentation for the maintenance of the aircraft to the aircrew prior to the next flight. Proper protocol would have been for maintenance to call engineering to determine if the aircraft was flyable. In researching this mishap, is was also learned that three months earlier the mishap aircraft had the fuel panel replaced using a panel from another aircraft. While not uncommon, this procedure was not cleared by engineering nor documented. This only became known after litigation against the maintenance contractor and was reported by the mechanic who was unaware that this required documentation and engineering approval. Before continuing, it would be helpful to review the C-21 fuel system operation. FIGURE C- 25 is an excerpt from the safety foundation report.

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Appendix
C-21A Fuel System Details

- The aircraft can carry approximately 2,390 pounds (1,084 kilograms) of fuel in the two wing-tip tanks, 2,508 pounds (1,138 kilograms) of fuel in the two internal wing tanks and 1,340 pounds (608 kilograms) of fuel in the fuselage tank;
- Fuel from each wing tank is supplied to the engine on the same side of the aircraft; the fuel system does not have engine-crossfeed capability (i.e., fuel from the right wing tanks cannot be provided directly to the left engine, and fuel from the left wing tanks cannot be provided directly to the right engine);
- Fuel from the wing-tip tanks is depleted first. Gravity causes approximately half of the fuel in the wing-tip tanks to flow into the wing tanks; jet pumps in the wing-tip tanks move fuel remaining in the wing-tip tanks to the wing tanks;
- The flight crew uses a two-position ("xfer" and "fill") switch either to transfer fuel from the fuselage tank into the wing tanks and wing-tip tanks, or to fill the fuselage tank with fuel from the wing tanks and wing-tip tanks;
- The crossflow valve between the wing tanks opens automatically when fuel is transferred from the fuselage tank to the wing tanks and wing-tip tanks. The crossflow valve is opened manually during refueling and to balance the fuel loads in the wing tanks and wing-tip tanks;
- Standby electric fuel pumps in the wing tanks are activated automatically during engine start and during transfer of fuel from the wing tanks to the fuselage tank. The standby fuel pumps can be activated manually when an engine-driven fuel pump fails and to balance fuel in the wing tanks and wing-tip tanks; and,
- Fuel-dump valves in the wing-tip tanks can be activated manually to jettison fuel from the wing-tip tanks.♦

FIGURE C- 25. Reprinted from Flight Safety Foundation, Accident Prevention, Fuel Imbalance Cited in Learjet 35A Control Loss, Vol. 57 No. 2, prepared by FSF Editorial staff.

For the C-21, fuel needs to be moved between the right and left tank to feed each engine. The standby pumps are automatically turned on during engine start and then can be manually controlled by a switch, which activates a control relay. In the case of the mishap aircraft, the standby pumps were engaged automatically during engine start and the right pump apparently remained "on" when the pump control relay contacts stuck together and did not "open" when the relay solenoid was de-energized after engine start. The pilots and post mishap analysis confirmed the standby pump switches were in the "off" position just prior to the mishap and after mishap recovery.

C.14.5 Component assessment.
Many of the components associated with the C-21 fuel system were received by the AFRL failure analysis lab. This included the right and left standby pumps and associated switches and control relays.

C.14.5.1 On-site evaluation.
When parts are recovered, there is a tendency to actuate switches, rotate pumps and motors, disassemble components and even try to run a mishap part. Many times an OEM or vendor will want a mishap part to be operated to show it was not a contributor to an accident. This should be avoided since an anomaly or failure site could be destroyed when a part is actuated, rotated, or operated. Always attempt to conduct non-destructive tests. Only then open the part to reveal the internal components.

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C.14.5.2 Pump.

The C-21 mishap was no different and there was pressure to try to operate the recovered components that were still intact. Standard failure analysis protocol was followed and analysis revealed the mishap pump bearings had considerable pre-mishap damage from wear/corrosion which results in higher electrical current since current is directly proportional to the force required to rotate the pump rotor. The pump vendor had disassembled the mishap standby pump while the mishap right hand standby pump was still intact (see FIGURE C- 26).

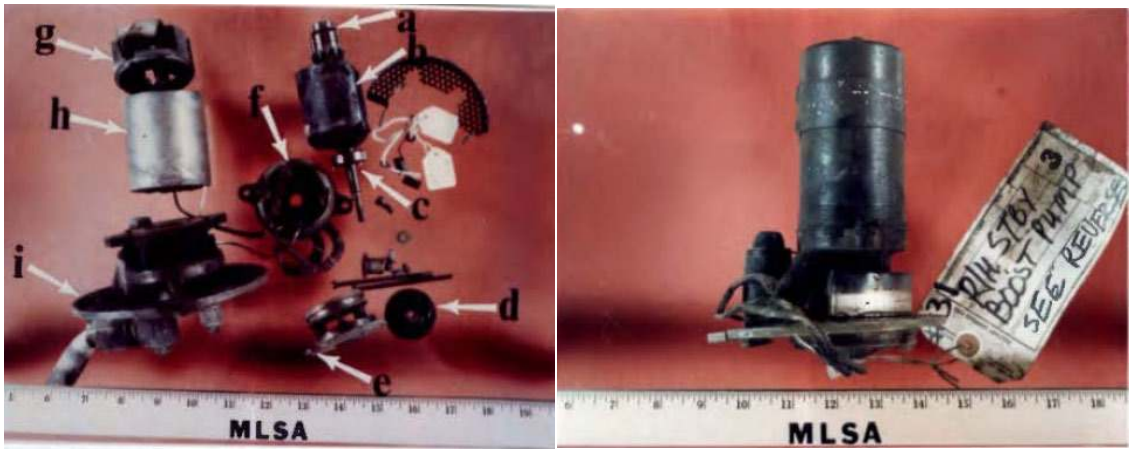


FIGURE C- 26. [Left] Left hand standby pump. [Right] Intact mishap right hand standby pump.

Pitting and evidence of degradation to both the inner and outer pump bearing races were found. The bearings from the right hand pump had suffered from severe degradation by pitting, surface damage, and galling. The inner race was found to have suffered more severe damage than the outer race. Heat tinting was found on the outer race indicating the bearing achieved a temperature of approximately 426.7-537.8 °C (800-1000 °F). The bearings appeared to be in good condition on both standby pumps.

C.14.5.3 Fuel pump control relay.

The right side standby fuel pump control relay is shown on FIGURE C- 27.

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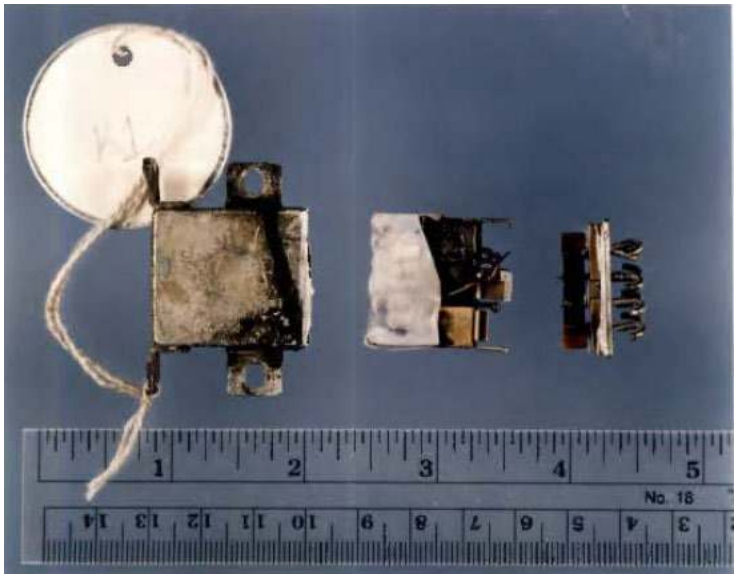


FIGURE C- 27. Right side standby fuel pump control relay.

Examination revealed major arcing damage to the internal contacts that power the right and left standby fuel pumps (see FIGURE C- 28). Two sets of normally open contacts were connected in parallel to turn-on the pump. Energizing the relay closes the normally open contacts and allowing the current to be shared between the two contact pairs. The problem is that one contact pair will always mate and or de-mate first, which will take the entire start-up current or the arcing that occurs when the relay is switched off. Contacts from the right pump control relay exhibited a material build-up or hillock on the movable contact and a corresponding pit in the stationary contact. Scanning electron micrographs of the contacts show melting, metal transfer, and smearing of surfaces, which suggest the contact, could have been stuck together which would have powered the pump when the relay was de-energized (see FIGURE C-29).

NOTE: Black arrow (lower left terminal) (see FIGURE C- 28), which is the stationary normally open contact and the movable contact with materials build-up. [Right] Close-up of the mishap right pump control relay contacts. Note pitting and hillock o the contact pair surface which is suggestive of a stuck contact.

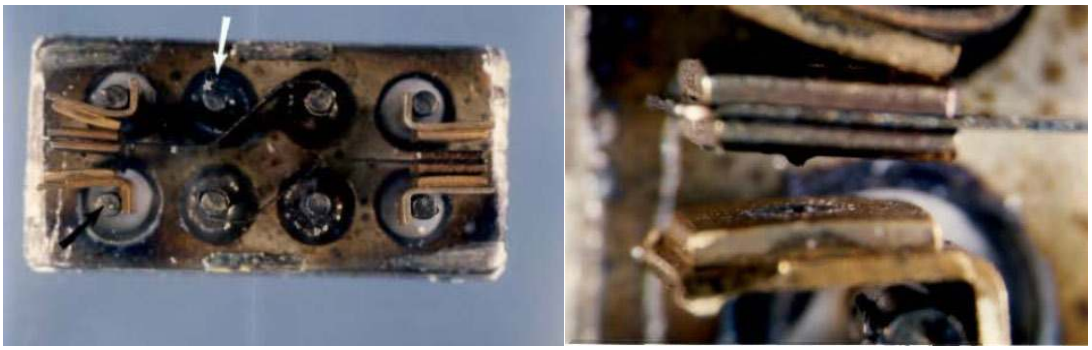


FIGURE C- 28. [Left] The mishap right pump control relay.

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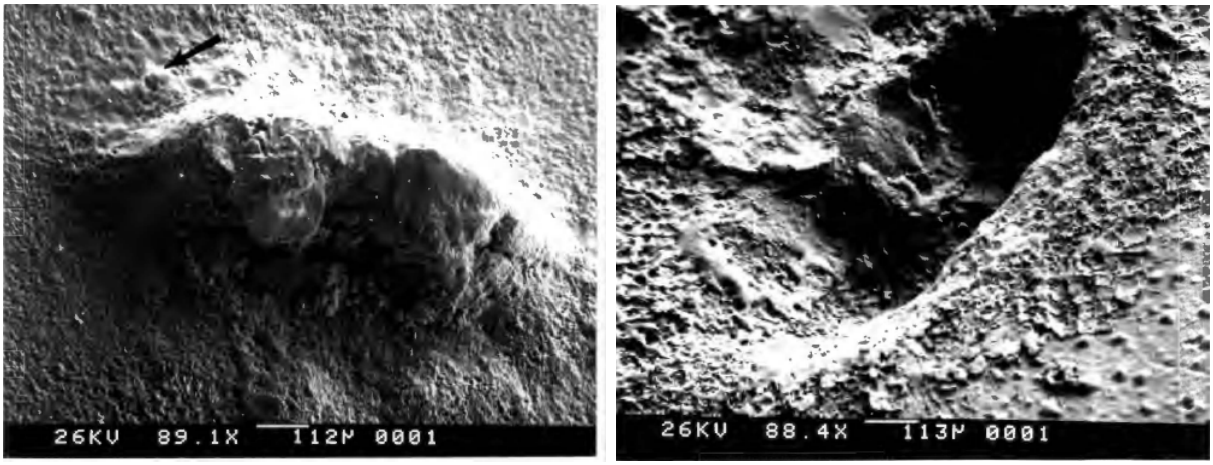


FIGURE C- 29. Scanning electron micrographs of the mishap right pump control relay contacts showing melting of the contacts but no evidence of welding.

It was postulated the contacts were stuck together as a result of interlocking of the irregular surfaces and the attraction of the opposite contact surfaces. A relay exhibiting this type of bond would most likely break apart if tapped with a tool or from a hard impact. This was reported that maintenance had used this technique to un-stick a relay on the C-21 fuel control panel. Recalling that a stuck contact would keep the pump operating even when the relay was de-energized.

The standby pump relays are rated at a continuous load current of 10 amperes and as noted earlier are configured to divide the current between the two sets of "NO" contacts. Testing of field obtained pumps off an aircraft yielded typical currents in the range of 5 amperes. When the pumps were loaded by either fuel or a degraded bearing condition, considerably higher currents can be drawn than the typical running current. Current measurements on C-21 aircraft indicate 35 amperes can be drawn on pump start-up. This current would last for less than 100 milliseconds. A degraded pump could draw even higher currents for extended periods.

In a degraded pump condition, relay contacts would be expected to be severely degraded and reach a point where contacts could bond or stick together as a result of localized melting. Contact degradation would accelerate as material builds up on one contact pair. This contact would mate earlier than the second set and, therefore, be subjected to the total start-up current accelerating degradation.

C.14.5.4 Component assessment conclusion.

The conclusion was that the relay was improperly sized for the standby pumps and violates standard design practice, which prohibits using paralleled contacts in a switch or relay to meet current rating requirements. One challenge was to find where this is mentioned in a reference, since organizations want to cite a document to justify a change that requires funding. The best choice would be a military specification or FAA document that prohibits the practice, which was not found. Today, there is a reference to not using contact pairs in parallel to meet a current rating requirement.

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C.14.6 Design evaluation.
In accordance with MIL-HDBK-454, paralleling contacts, discusses the parallel circuit design. Paralleling contacting is when two relays are placed in parallel to handle the current that one of them cannot handle alone. The problem with this type of design is that mechanical switching occurs at relatively slow switching speeds. Therefore, for a brief instant, only one relay carries the full current load. Further, switching speeds tend to slow with age, amplifying the affect over time. The preferred method is to use a single relay of sufficient current handling capability. If dual relays are used in parallel in increase reliability, each relay should be capable of handling the full current load.

C.14.7 Maintenance evaluation.
The C-21 had a maintenance culture of accepting that pumps degraded and relays had to be replaced or hit to release a stuck relay. The issue was tolerated as a nuisance and not a safety issue since it was a known problem in the fuel system. There was also the issue of lack of communication between the maintenance contractor, the aircraft manufacturer, the pump vendor, the crew training company, the flight crew, and the small government office that managed all the support and procurement contracts. This is best illustrated by the earlier discussion of maintenance allowing the mishap aircraft to fly without resolving the fuel transfer issue that was reported in a prior flight. There should have been procedures that required maintenance to call engineering for disposition of the aircraft. Also, as noted, maintenance had replaced the panel in the mishap aircraft without getting approval or notifying the cognizant engineer.

Once all the lab analysis was completed and the mishap and accident report were issued, the aircraft manufacturer and responsible government office determined a future pump failure that would cause a fuel imbalance was unlikely since crews were now trained to pull the pump circuit breaker if a pump continued to run when in the OFF position. This was unacceptable to some stakeholders in the Air Force and as an advisor to several interested parties, the AFRL failure analysis group proposed a program to analyze relays removed from C-21 aircraft with the understanding that if significant contact damage were found, the relays would be replaced with properly sized relays on the fleet of remaining 78 C-21 aircraft. When suggesting an engineering study, it is critical to obtain a clearly written test plan with agreed upon decisions based on the study findings from all stakeholders. In the case of this engineering study, the program office would fund replacement of the improperly sized relays if relays removed from active aircraft exhibited significant contact damage. Lab analysis revealed that the 12 standby pump relay contacts examined from six fuel panels all exhibited moderate to severe contact damage and were considered at their end of life.

C.14.8 Conclusions.
At this point, over the objections of the aircraft manufacturer, the Program Office agreed to fund the replacement of the pump relays with properly sized devices. The required additional analysis may seem surprising. However, Program Offices, OEMs, and crewmembers typically address a design or maintenance issue by relying on their personal history to make major decisions when it impacts safety or reliability. If they have no experience with a failure condition personally or within their respective company/organization, they are likely to dismiss it as unlikely to occur again or as improbable. This is why serious failures need to be reported to the aerospace

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community and documented in handbooks, specifications, and in programs such as MECSIP. This is critical to maintaining safe and reliable aircraft.

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C.15 FOAM IN FUEL TANKS SUMMARY

C.15.1 Introduction.

The function of fuel tank foam baffle systems (see FIGURE C- 30) is to suppress explosion reactions and prevent overpressure damage to the fuel tank. This is accomplished by controlling the rate of burning after ignition of fuel vapors through energy removal from the combustion process by the foam through heat absorption and mechanical interference. Fuel tanks containing foam are protected from damaging explosion when pierced or ruptured by ground fire, arcing as the result of wiring/component malfunction, or as a result of lightning strike. The foam also prevents fuel sloshing, which can have an adverse impact on safety[3].

Foam is specified in MIL-PRF-87260 (conducting foam). This document provides extensive guidance on specifying and testing electrically-conductive open-celled foam material.

In addition, foam can significantly impact maintenance in several ways. One of the main impacts is on the removal/replacement of in-tank components. If the foam is out of the tank for an applicable length of time, special handling and wrapping is required. Foam, which is no longer usable is difficult to dispose of without environmental damage. Aging fire retardant reticulated foam has been reported as one of the number one issues impacting USAF aircraft fuel systems for those aircraft equipped with foam[4].

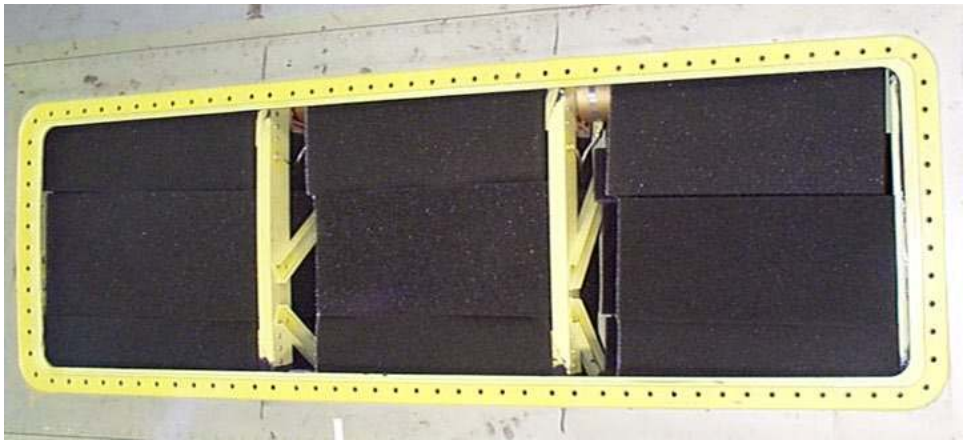


FIGURE C- 30. Typical foam installation in a fuel tank.

The color of the foam helps determine the age and type of foam. Typical colors include blue, orange, yellow, red, or charcoal grey.

The original Type I foam (orange) is a polyester-based, coarse-pore polymer, first installed in 1965. In 1970, lower-density Type II (yellow, Coarsepore) and Type III (red, fine-pore) polyester polyurethane foams were developed. In 1974, Type IV (dark blue, coarse-pore) and Type V (light blue, fine-pore) polyether polyurethane foams were introduced. Each of these foams have unique

[3] *History of Fire Suppression in Aircraft, Chapter 2,* Donald P. Bein Naval Air Systems Command,

[4] Report on Fuel System Problems at USAF Air Logistics Centers, 9 May 2005

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properties and many of the older foams have exhibited degradation and require replacement. References on the foams are provided at the end of this summary. A comprehensive report on the older foam types can be found in ADA061450 Static Electricity Hazards in Aircraft Fuel Systems report. After the development and incorporation of fuel tank foams in the 1970s, the Air Force discovered the materials used for the foam were susceptible to hydrolytic degradation. Subsequently, better materials were developed; producing what is commonly called blue foam. Today the grey foam as specified in MIL-PRF-87260 is used in many applications.

The following are examples of problems which have been reported on military aircraft with fuel tank foam:

a. On attack aircraft, fuel tank foam has reached its life limit on several platforms. The aging foam shrinks and pieces of the foam break off and can clog filters and pumps. It is also compressing so the foam "puzzle" does not fit together anymore resulting in space inside the tank, thus defeating the purpose and effectiveness of the foam. The present solution is to replace as needed.
b. Transport aircraft have experienced flame outs on taxi due to foam. The foam was designed for Type 3A fire protection - absorb ballistic shockwave. In addition to the hostile fire issue, the foam also provides protection from "static discharge". The foam is deteriorating and scheduled replacement effort is needed.

There are considerable costs associated with the maintenance procedures, storing, and reinstalling removed foam. Maintenance personnel report foam is usually so damaged during removal that much of it needs to be replaced. Recommend maintaining foam as a life limited item and providing clarification of procedures and cost/benefit analysis of current replacement schedule of every third PDM and not to exceed 18 years.

Inspect explosion suppressant foam in fuel tanks for deterioration when foam-filled fuel tanks are drained 30 days or more and exposed to high heat or high humidity environment (relative humidity 100 percent, 32.2 °C (90 °F) or higher).

After fuel tank entry, with more than 25 percent of tank foam disturbed or replaced, and prior to first flight, perform engine run for a minimum of 15 minutes, remove, clean, and reinstall water strainer.

Every third PDM and not to exceed 18 years, replace charcoal explosive suppressant foam (see MIL-PRF-87260, Class 2, Grade I).

Some fighters have fire retardant foam glued to the skin/structure inside the tanks and it is changed every six years even though it has been rated as good for 20-year life. Consequently, these aircraft do not have the foam problems cited by attack and transport aircraft.

Recommended actions:
a. Establish the life limit for the current fire retardant foam used in aircraft fuel tanks.
b. Recommend a change interval so the problems with shrinkage, break-up, and "slumping" can be avoided.

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c. Determine the survivability impacts of "slumping" foam.
d. Determine whether development of a new generation of foam is feasible or indicated or whether the current foam should simply be replaced at the end of its service life. Replacement of foam with an inerting system is an option, however integrating an inerting system would require complete recertification.

C.15.2 References.

a. History of Fire Suppression in Aircraft, Chapter 2, Donald P. Bein Naval Air Systems Command.
b. Report on Fuel System Problems at USAF Air Logistics Centers, 9 May 2005.
c. Aviation Rulemaking Advisory Committee, Foam Task Group 4, 17 July 1998.
d. Advisory Group for Aerospace Research & Development, Report No. 771, Advisory Group For Aerospace Research And Development.
e. MIL-F-87260B, Foam Material, Explosion Suppression, Inherently Electrically Conductive, For Aircraft Fuel Tank and Dry Bay Areas.
f. MIL-B-83054B, 17 May 1978, Baffle and Inerting Material, Aircraft Fuel Tank.
g. MIL-B-87162 Baffle Materials, Explosion Suppression, Expanded Aluminum Mesh, for Aircraft Fuel Tank.
h. ADA061450 Static Electricity Hazards in Aircraft Fuel Systems.
i. ESL-TR-84-63 Effective Disposal of Fuel Cell Polyurethane Foam.
j. SAE Aerospace Information Report, AIR4170, 12 February, 1991, Reticulated Polyurethane Foam Explosion Suppression Material for Fuel Systems and Dry Bays.
k. AFWAL-TL-80-2043 Evaluation of Exploxsafe Explosion Suppression System for Aircraft Fuel Tank Protection.

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C.16 B-1B FUEL LEAK

C.16.1 Introduction.

On 20 August 2014 at approximately 0245, after fueling a B-1B TN 85-0081 (see FIGURE C- 31) with 160,000 lbs. (18,000 gallons of fuel), the fuel started to spill out of the entire right nacelle. The fuel hoses were still connected and the fuel truck was depressurizing when the spill began.

The fuel continued to spill for approximately 45 minutes. It was difficult to contain the fuel, but all means were used. Multiple spill kits were used and all the materials that Petroleum Oil Lubricants (POL) and the fire department had were used as well.



FIGURE C- 31. Fuel spill B-1B TN 85-0081.

C.16.2 Background.

The B-1B right hand engine nacelle contains a complex fuel tube installation to facilitate ground refueling and engine and APU fuel supply. In addition to the tubes, the installation also includes flow meters, check valves, shut-off valves, and dual single-point refueling nozzles (see FIGURE C- 32). Several of the connections utilize W700-series flexible fuel couplings (see FIGURE C- 33) which require the connecting parts to have a flange. These flanges may be part of a ferrule swedged onto a tube or can be machined into the component (see FIGURE C- 34 and FIGURE C- 35).

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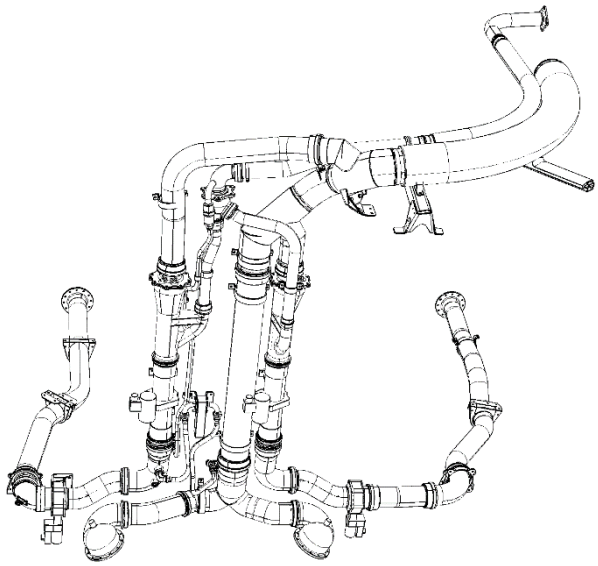


FIGURE C- 32. B-1B fuel tube configuration representation.

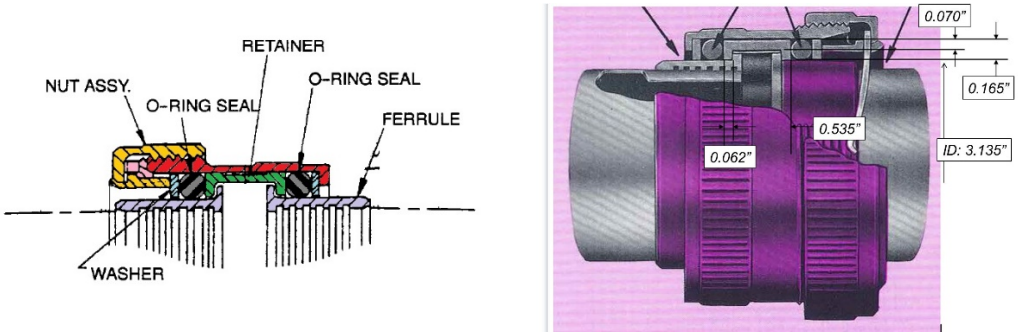


FIGURE C- 33. W700-series flexible fuel couplings.

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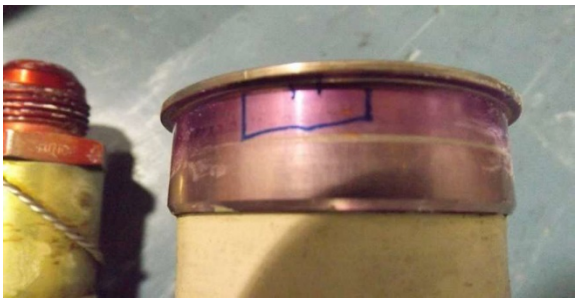


FIGURE C- 34. Tube with sweged fitting ferrule.



FIGURE C- 35. Tube with machined integral ferrule.

C.16.3 Investigation.
The source of the fuel spill was determined to be at one of the 3-inch engine feed tubes. The tube ferrule lip had fractured allowing the tube to separate from the coupling. This allowed fuel to flow out of the aircraft unrestrained resulting in a major fuel spill (see FIGURE C- 36).



FIGURE C- 36. Fuel tube.

The fuel tube was sent for metallurgical failure analysis to help identify the root cause(s) of the failure on 21 August 2014.

Analysis showed the fuel tube fracture contained multiple origin fatigue that progressed via high stress/low cycle conditions. Although fatigue was found, the progression constituted only about 10 percent of the overall fracture surface, making overstress the predominant fracture mode.

1. **FINDINGS OF SIGNIFICANCE:**
   a. Ferrules met the material specifications for alloy and temper.
   b. Slanted wear marks were noted on both ferrules from the fuel tube. The angle of wear was approximately 3.9 degrees.
   c. The fatigue initiation site was located diametrically opposite to the slanted wear pattern.

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d. Although fatigue initiated and progressed for about a 1-inch span and through the ferrule wall thickness, the predominant mode of cracking on the remaining 90 percent of the fracture was ductile overload.
e. Fatigue quickly exhibited unstable crack growth as observed by the variable (non-uniform) striation sizes.

2. ROOT CAUSE OF THE FAILURE IS BELIEVED TO BE:

a. Poor fuel tube installation design which inadequately supported fuel tubes resulting in loads being transferred through the couplings.
b. Variations in tube geometry and lack of adjustable bracketry contributed to the failures by forcing tube to be installed so that there was axial misalignment (see FIGURE C-37) and preload to cause the lip stress to be reacted by only a small portion of the lip inside the coupling.

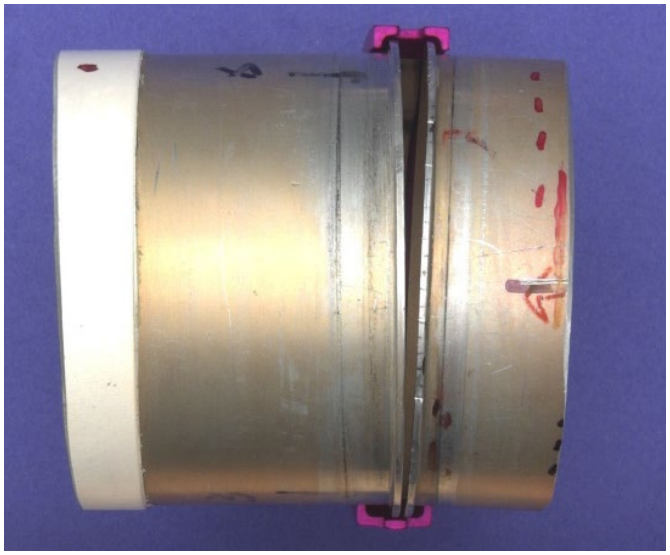


FIGURE C- 37. Axial misalignment.

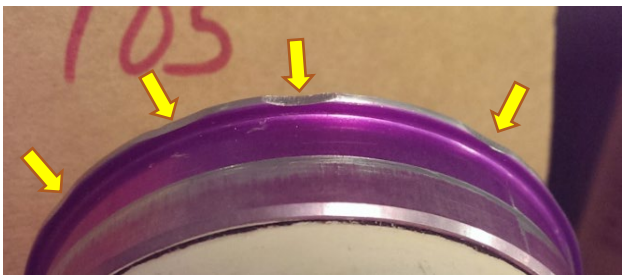
C.16.4 Mitigation.

All nacelle fuel tubes and fittings which had a flange lip were deemed to be at-risk. Immediate off-aircraft inspections were accomplished using dye-penetrant inspections. Any tube with a crack indication was discarded. Tubes not found to have cracks were re-installed. All tubes were additionally inspected for lip wear, or "scalloping" (see FIGURE C- 38). No tech data limits existed for this, so damage allowable thresholds were developed.

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Integral Machined Ferrule



Swaged Ferrule

FIGURE C- 38. Scalloping.

Probability of crack detection was very low with FPI inspection, so a more capable inspection technique was developed using custom eddy current probes. The EC inspection method improved the re-inspection interval by 300 PERCENT. The inspection interval was determined using a crack growth model developed specifically for the lip geometry, material, and loads.

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APPENDIX D

TASK 4 – TEARDOWN ANALYSIS GUIDANCE

D.1 SCOPE

D.1.1 Scope.

The information presented in this section has been compiled from AFRL protocols for failure analysis or teardown of components.

D.2 COMPONENT HANDLING OVERVIEW

The aging analysis objective is to understand and document the hardware condition and capture relevant evidence to build the case to determine the failure mode. To achieve that goal, it is not enough to know what happened and why. The right evidence collection and documentation is necessary to demonstrate the value of the operations. This brief description of the rules of thumb which steer typical analysis practices.

D.2.1 Rules of thumb.

The following two items are critical when attempting to understand the component condition:

1. Preserve evidence, and
2. identify and document anomalies.

The following are recommended practices for performance of component assessment.

D.2.2 Typical practices.

1. Understand the overall function prior to any disassembly.
    a. Examine an intact unit, or disassembled undamaged unit.
    b. Review diagrams and schematics, offsite ahead of time of possible.
    c. Discuss interfaces with other systems.

2. Perform nondestructive inspections first.
    a. Visual examination of exterior, radiography etc.
    b. Compare to new and similarly used 'exemplar' components. Do not perform a functional test if it may alter relevant internal damage. Collect relevant samples, identifying their location and protecting from cross contamination. Mark orientations and photograph before disassembly.

3. Label components with proper part designations.
    a. Scribe forward or upwards to indicate precise alignments if interfaces are potential suspects, do not disassemble there.
    b. Leave fittings intact, cut wires and tubes far from any damage.
    c. Do not alter damaged locations.
    d. It can confuse laboratory interpretation.
    e. If you inadvertently create new damage, mark it as new.

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D.3 AFS COMPONENT AGING ANALYSIS

1. Selection of components for further analysis.
    a. This should be based on the results of the maintenance data analysis.
    b. Critical path analysis and overall risk analysis.
    c. Other factors such as known service life limits of components.
    d. Service life should be included as a consideration for component selection.

2. Determine the objectives of analysis.
    a. Should the assessment be:
        1. Remaining life projection?
        2. Functional Check?
        3. Materials and manufacturing process analysis?

3. Select the type of testing to be performed based on the component function and analysis objective. Teardown analysis may include one or more of the following:
    a. Perform NDE/NDI.
    b. Perform functional tests if no damage is expected from the evaluation.
    c. Material analysis.
    d. Dielectric testing.
    e. Insulation resistance.
    f. Load analysis.
    g. Accelerated aging (stress testing).

D.4 PROJECT FLOW FOR FAILURE ANALYSIS

1. Background Research.
    a. Drawing / Process / Materials Specifications: Follow set quality assurance provisions, military and industry standards, and vendor specifications for all equipment.
    b. Previous Failure History: Study previous reports and investigations to assure past failures are not repeated and the repairs made are still in good condition.
    c. Materials Research: Study and use the Aerospace Structural Metals Handbook, ASM Manuals, and Metallic Materials Properties, Development and Standardization (MMPDS) MMPDS-09 to ensure the proper knowledge of the materials is applied.
    d. Process Research: Study and use relevant ASM Manuals to assure the proper processes are being followed.

2. As received overall photography: when receiving the materials, photograph the equipment including the shipping container.
3. Visual examination.
4. Stereo optical examination.
5. Area of interest photography.
6. Contamination / residue documentation and sampling/corrosion.
7. Cleaning: Have a parts washer inspect and clean the parts before installation back on the aircraft.

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8. Dimensional inspections: Inspect the thicknesses and radii of the part. Also note any degree of deformation.
9. Non-destructive inspections: These may be used as means for non-destructive assessment of component(s) under investigation.
    a. Radiography: Measures the differences in radiation absorption among different areas and finds inclusions and cracks.
    b. Ultrasonic: Uses high frequency sonar to detect surface and near surface defects and inclusions. It can also measure the thickness of the material and pinpoint the position of the defects.
    c. Dye penetrant: Locates and identifies cracks or open defects in all non-porous materials.
    d. Magnetic particle: Uses a magnetic field and iron powder to detect surface and near surface defects and cracks.
    e. Eddy current: Eddy current detects flaws (cracks, defects, inclusions) in conductive materials and measures conductivity, magnetic permeability, and physical dimensions.
    f. Macro etching: Such as using Blue Etch Anodize (SAE AMS2642 or SAE AS7108) to reveal alpha case in titanium and Temper Etch per MIL-STD-867 to reveal grinding burns in high strength steels.
    g. Surface SEM exam: An electron microscope is used to check for patterns of anodize cracking.
10. Fractography.
    a. Cleaning: Replicate cracks to investigate full-sized cracks in the metal and clean the surface using Acetone/Ultrasonic and/or Alconox.
    b. Stereo Optical.
    c. Scanning Electron Microscope (SEM).
    d. Energy Dispersive Spectroscopy (EDS): Uses X-ray excitation to check verification of finish systems material and coverage and checks for segregation in the material and containments.
11. Hardness measurements/ Microhardness traverse.
12. Bulk conductivity measurements.
13. Compositional or chemical analysis techniques.
    a. Scanning Electron Microscopy Energy Dispersive X-Ray (EDX).
    b. Fourier Transform Infrared (FTIR).
    c. Gas Chromatography (GC/GC-MS).
    d. Liquid chromatography.
    e. Mass Spectrometry (MS).
    f. X-ray Photoelectron Spectroscopy (XPS/ESCA).
    g. Auger/SIMS.
14. Metallurgical examination.
15. Dimensional verification of parts and wear of surfaces.
16. Other testing.
    a. Mechanical properties: Measure the yield, fatigue, Stress Corrosion Cracking (SCC), plane strain fracture toughness, and maximum elongation limit of the equipment.

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D.5 FAILURE ANALYSIS PROCESS FOR A FUEL PUMP

The following is an example of a failure analysis process that was applied for a fuel pump. The failure analysis process and benefits from the analysis could be greatly enhanced if a baseline unit is available for comparison of material properties.

D.5.1 Conduct nondestructive evaluation.

1. Inspect visually and document any overheating/burning/discoloration.
2. Radiograph assembly.
3. Measure connector/CB/wiring input resistances between phases and ground using a 4-point probe milliohm meter use MIL-STD-202-303 or equivalent procedure.
4. Measure the connector/CB/wiring Insulation Resistance (IR) between phases and ground. Start at 10VDC and proceed to 50VDC, 100VDC and 500VDC. Use MIL-STD-202-302 or equivalent procedure, stop test if measurement is less than 200 MΩ.
5. If no failures are detected in the IR test conduct dielectric breakdown test between all phases and case (1000 VAC or maximum level allowable by pump specification) use MIL-STD-202-301 or equivalent test method. Repeat input resistance test over operating temperature range and monitor for changes. If a change is detected initiate root cause analysis.
6. Measure bonding resistance between metal interfaces using a 4-point probe and the input current required by the design specification.
7. Review data for anomalies or measurements outside of expected values.
8. Visually inspect and document components associated with the fuel pump: switches, circuit breaker, wiring, and connectors.
9. If no anomalies detected, consider a functional test while monitoring the current draw while the pump is under load.
10. Disassemble pump and inspect seals and other parts for arcing, discoloration, degradation, corrosion, or mechanical damage or cracking. Check wiring, windings, bearings, stator, and rotor for damage.

D.5.2 Pump failure analysis teardown:

1. Disassemble intact pump and inspect for any overheating/burning/discoloration, document and radiograph. Examine radiograph for melting or opens in the motor wiring/windings.
2. Possible failure modes - short then open in phase A.
3. Connector/wiring short.
4. Shorted phase wire to housing.
5. Short in winding to laminate.
6. FOD/Contaminate.
7. Insulation failure.

D.5.3 Conduct electrical tests:

1. Re-measure wiring input resistances between phases and ground.
2. Re-measure insulation resistance between phases and ground, use lowest voltage that causes a failure or no more than 500VDC.
3. If IR test passed at 500VDC measure dielectric breakdown between phases and case (1000VAC).

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D.5.4 Review results and proceed with tear down.

1. Remove connector, inspect exposed wiring for mechanical/thermal/electrical damage, if no damage noted proceed with process for opening case and exposing stator windings, bearings, and the rotor. Inspect components and document for discoloration, overheating, wear, or other types of damage.
2. Recheck for electrical open/short, if no electrical fault location is found, radiograph windings with rotor removed. Stator may need to be cut in half for detailed visual and radiographic inspection.
3. Locate open in wiring/windings and document.
4. Specifically look for cold working of wiring, reduced cross-section from wear or evidence of a manufacturing anomaly. Also examine for pitting/melting in laminate and carbonization of insulation in and around the failure site. Examine stator and rotor surfaces for wear, carbonized, or thermally degraded materials.
5. Inspect materials for crazing and micro-cracking at high magnification and conduct a materials analysis on the various pump insulation and dielectric materials (epoxy, insulation varnish on wiring, insulation between conductive surfaces and other organic materials). Specifically examine for evidence of aging/degradation using analysis techniques such as Infrared Spectroscopy, Thermal Mechanical Analysis (TMA), Differential Scanning Calorimetry (DSC), or hardness testing.

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APPENDIX E

TASK 5 – RISK ASSESSMENT, A RELIABILITY CENTERED MAINTENANCE (RCM) APPROACH

E.1 SCOPE

E.1.1 Scope.

Reliability Centered Maintenance (RCM) is a corporate level maintenance strategy that is implemented to optimize the maintenance program. Ideally, a RCM program provides maintenance strategies that can improve the performance, availability, safety, and reliability of an aircraft. Using RCM, maintenance strategies can be optimized for an aircraft or a sub-system of an aircraft such as the fuel system. There are many books, articles, and standards that explain how to implement a RCM program. Some references are given at the end of this section.

There are four principles that guide a RCM program.

1. Preserve system function.
2. Identify failure modes that can affect the system function.
3. Prioritize the failure modes.
4. Select applicable and effective tasks to control the failure modes.

Equipment reliability and availability are achieved by reducing the probability of system failure. This is achieved by considering the function of the equipment, possible failure modes and failure consequences. Once this knowledge base is developed, cost-effective maintenance techniques can be applied.

E.2 RCM INPUTS

When properly implemented, RCM identifies which inputs and outputs of a system that most contribute to the overall aircraft performance, availability and reliability. SAE-STD-JA1011 provides the minimum criteria for certifying that an RCM has been implemented. As an example, for an aircraft fuel system, seven questions need to be asked:

1. What are the functions and desired performance standards for each component of the fuel system?
2. How can each fuel system component fail to fulfill its functions?
3. What are the failure modes for each functional failure?
4. What causes each of the failure modes?
5. What are the consequences of each failure?
6. What can and/or should be done to predict or prevent each failure?
7. What should be done if a suitable proactive task cannot be implemented?

E.3 EVALUATION EXAMPLE

In developing this handbook, a RCM analysis on the fuel systems of an USAF aircraft was reviewed. The reader should review this report to gain insight into the RCM process and how it

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can be used to optimize maintenance practices and improve aircraft safety, lower costs, and improve aircraft availability and reliability. RCM analysis is data intensive and requires detailed engineering knowledge of the aircraft fuel system, impact of a failure, and detailed understanding of maintenance processes.

Excerpts from the cited USAF Report are given below with system specifics removed and general information added to aid in understanding.

RCM analysis is based on data extracted from the Reliability Maintainability Information System (REMIS, G099), Programmed Depot Maintenance Scheduling (PDMSS, G097), AFMC Form 202 records, Technical Orders, Technical Assistance Request (TAR) records, Joint Deficiency Reporting System (JDRS), and the Air Force Safety Automated System (AFSAS). In addition to the data, interviews were conducted with Subject Matter Experts (SMEs), including fuel system engineers, equipment specialists, and maintainers.

E.3.1 Scope.

For the purpose of this RCM analysis shown here, the Fuel System includes controls, storage, venting, refueling and defueling, distribution, water removal, dumping, quantity and pressure indication, and fire and explosion protection. The aerial refueling system was not part of the analysis.

E.3.2 Failure count and impact.

There were 102 failure modes analyzed in the Failure Modes, Effects, and Criticality Analysis (FMECA). From these failure modes, 46 Preventive Maintenance (PM) change recommendations generated.

The analysis recommended total improvements that could result in an estimated annual fleet cost avoidance of $1,443,000, including saving 7,970 Maintenance Labor Hours (MLH) if fully implemented. The recommendations primarily consist of changes to Programmed Depot Maintenance (PDM) specifications, TO's, Inspection Work Cards, and other support technical data that would be submitted via the AFMC Form 252. Other recommended actions include re-working the Work Unit Code table for the Fuel System, identifying components to be serially controlled, identifying safety critical components and noting time change components.

The RCM report determined the Fuel System was a high driver in aircraft down time and the third highest driver of non-mission capable time as noted on FIGURE E- 1.

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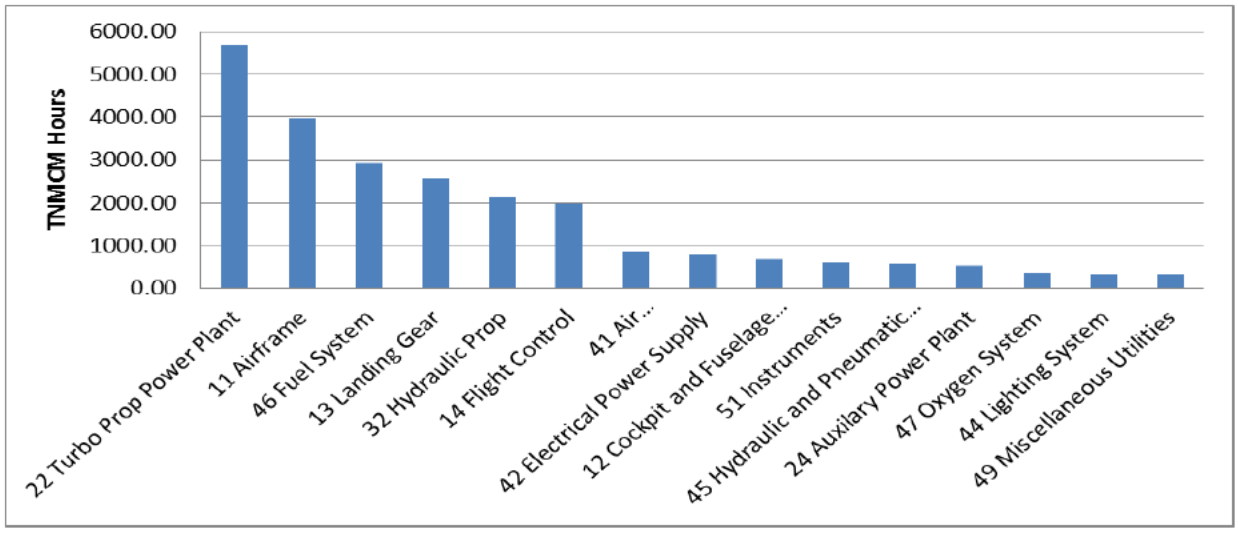


FIGURE E- 1. Average monthly total non-mission capable maintenance (TNMCM) time.

With respect to the fuel system, the subsystem components responsible for non-mission hours were fuel tanks, fuel lines, and instruments such as fuel probes are given on FIGURE E- 2.

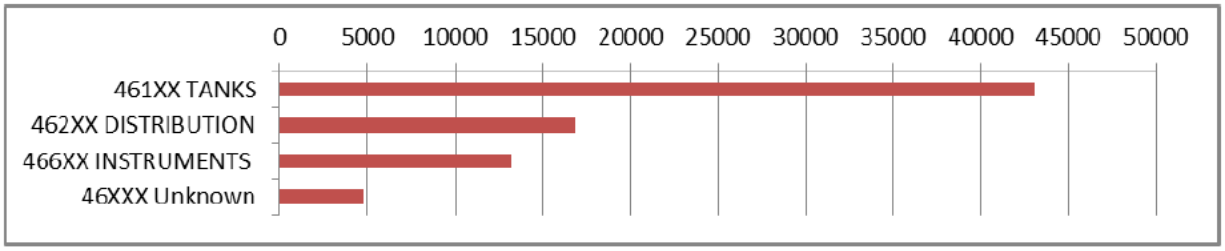


FIGURE E- 2. Fuel system total non-mission capable maintenance (TNMCM).

The FMECA analysis was augmented by interviews with maintenance staff, engineering, and vendor data. Within the fuel system, 102 failure modes were identified and documented. Failure modes were derived from a functional analysis of the fuel system based on review of the General Systems Technical Order, Illustrated Parts Breakdown (IPB), current Preventive Maintenance (PM) tasks, interviews with Tactical Airlift Division Engineers and Equipment Specialists, Organizational and Depot level maintainers, and failure data (REMIS, PDMSS, 107s, 202s, deficiency reports, and safety reports).

E.3.3 Mission impact.

Another important data set used to assess priority of maintenance actions was mission abort data. This indicates something has failed on the aircraft that makes it unsafe or incapable of completing its assigned mission. A mission non-abort failure is one which has occurred either in-flight or on the ground, that is not essential to completing the mission, but will cause unscheduled maintenance after the aircraft lands. It is important to note that an aircraft accident is rarely caused by a single

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failure, but rather by a chain of unforeseeable events, so even failures that do not immediately result in an abort need to be minimized.

The number of in-flight and ground aborts due to failures in the fuels system has a direct impact on the mission capability of the aircraft as a whole. Indication system failures are the leading cause of mission aborts and Fuel Storage is the next highest as seen on FIGURE E- 3. FIGURE E- 4 further breaks down mission aborts by failure mode.

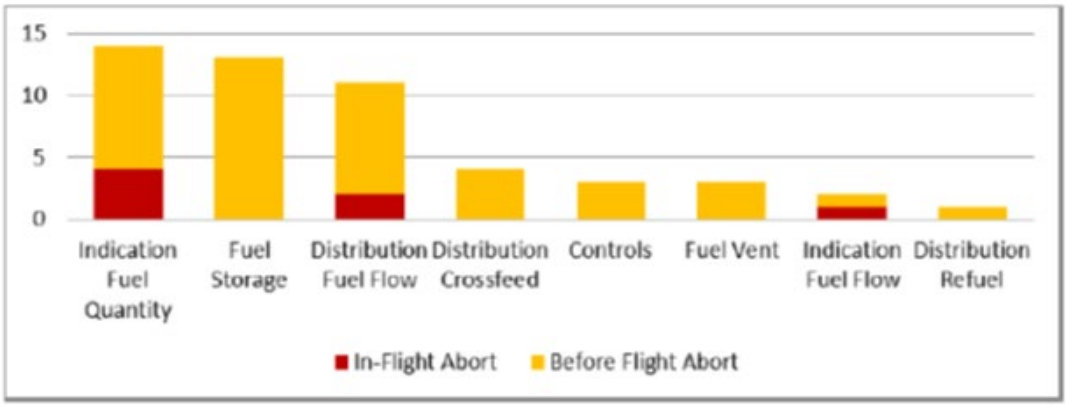


FIGURE E- 3. In-flight and ground aborts caused by failure of system (2009 – 2010).

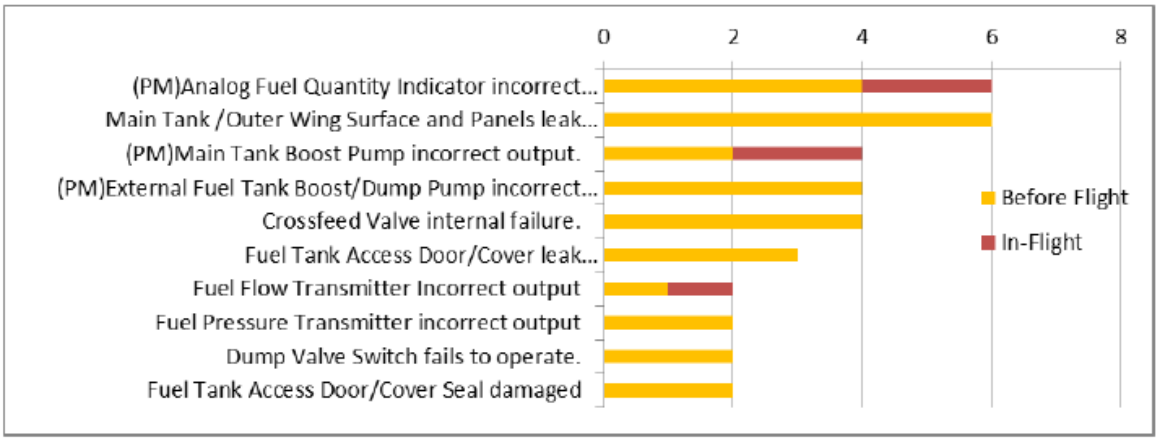


FIGURE E- 4. Top ten drivers of aborts by failure mode (2009 – 2010).

E.4 MAINTENANCE ACTION CODE

The maintenance action code is an indicator of the effectiveness and efficiency to manage the consequences of failure. Examining when a failure or maintenance requirement was discovered and the type maintenance associated with the action, such as scheduled or unscheduled maintenance, can provide additional indication as to the reliability and maintainability of a system. FIGURE C- 36 shows that, of the two years of maintenance data analyzed, 49 percent of all maintenance actions were Unscheduled Maintenance. Another 45 percent were accomplished during Depot. FIGURE E- 5 and FIGURE E- 6 display prominent components and maintenance action codes.

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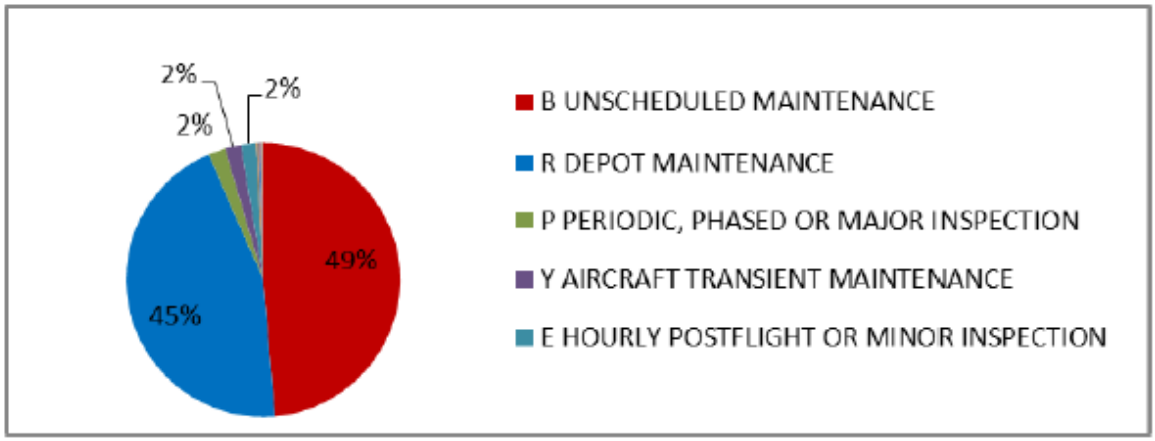


FIGURE E- 5. Type of maintenance.

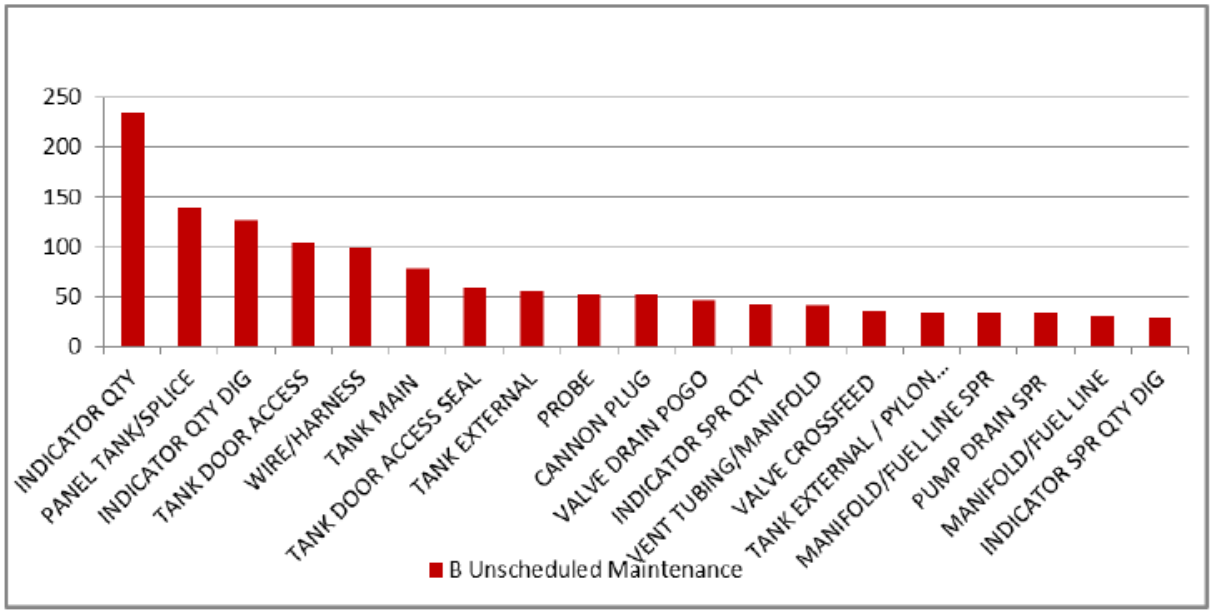


FIGURE E- 6. Prominent component unscheduled maintenance.

E.5 TOTAL FAILURE EVALUATION

Another metric of system reliability is the total failure count in the system. Though maintenance downtime and safety ramifications cannot be directly derived from this value; it proves to be a good indicator of trouble areas. The system's failures can be seen on FIGURE E- 7, showing that Fuel Storage and Indication systems as the highest drivers of failures. FIGURE E- 8 shows distribution of prominent failure modes.

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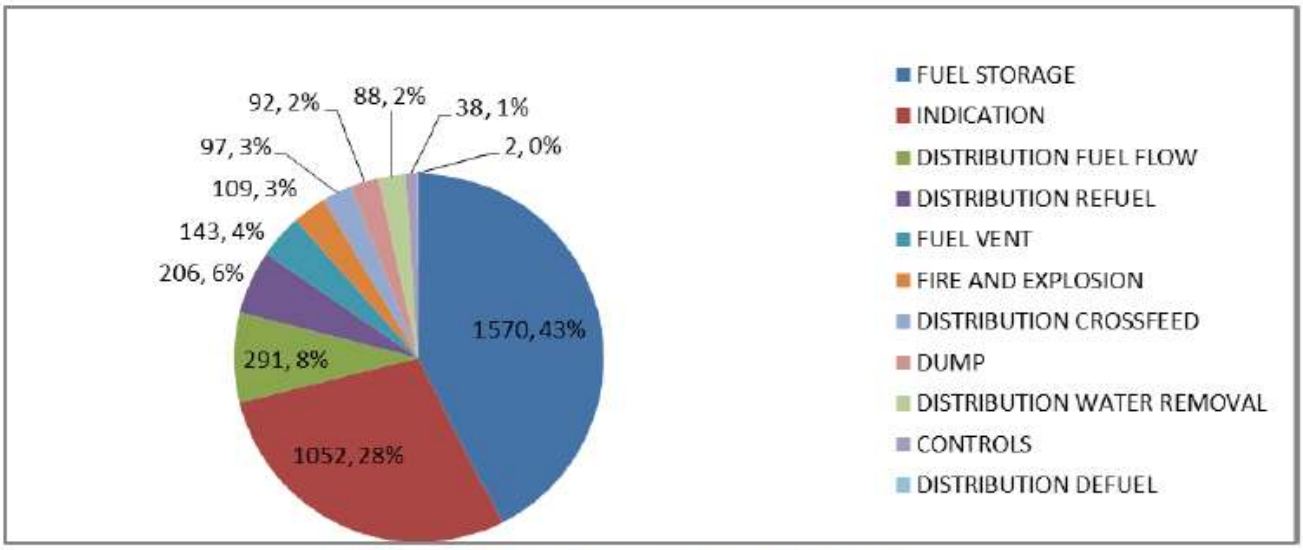


FIGURE E- 7. Distribution and count of failures.

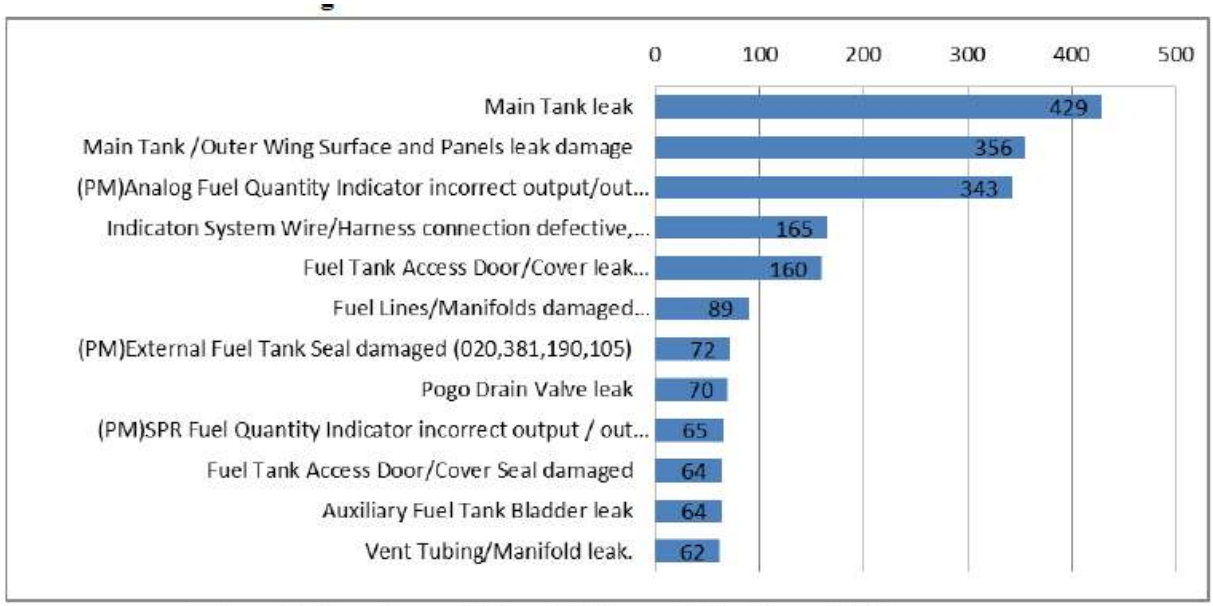


FIGURE E- 8. Prominent failure modes and number of failures.

By summing the FMECA cost of each analyzed failure mode, the total cost of the analyzed Fuel System components during the analysis period is approximately $28 per flight hour. There were several failure modes that were high drivers in this cost. FIGURE E- 9 highlights the prominent cost per flight hour among failure modes of the fuel system. The External Tank Pylon is the highest driver of cost component at $2.95 per flight hour. The second highest cost failure mode is the Indication System Wiring at $2.24 per flight hour.

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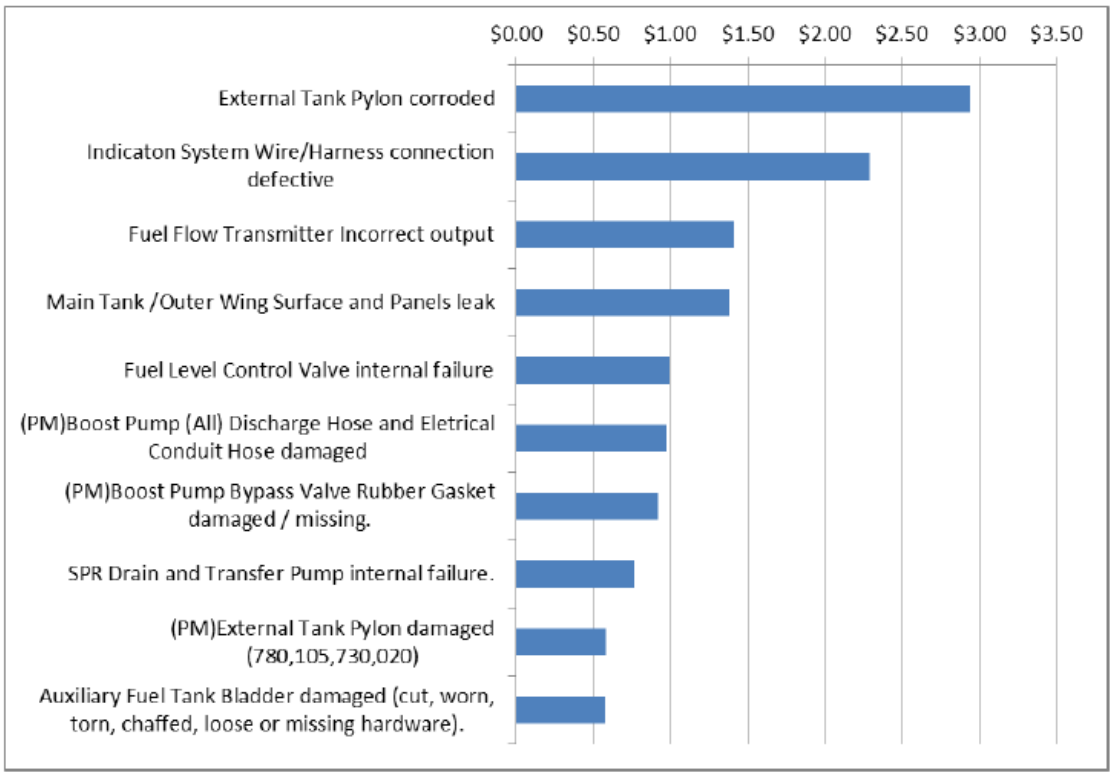


FIGURE E- 9. Top ten failure modes by cost per flight hour.

Accurate Maintenance Data Documentation (MDD) in accordance with T.O. 00-20-2 is used to support USAF equipment maintenance, reliability and maintainability. The key to documenting correct and reliable maintenance data is the accuracy of Work Unit Codes in each maintenance record. In accordance with MIL-PRF-38769(USAF), the Work Unit Code (WUC) is used to identify the specific assembly or part within an end item on which an action was performed. FIGURE E- 10 points out the low percentage of WUC accuracy extracted in the analysis data. Only 23 percent of the components were correctly identified, and 47 percent were misidentified. FIGURE E- 11 highlights the WUC accuracy variance among components.

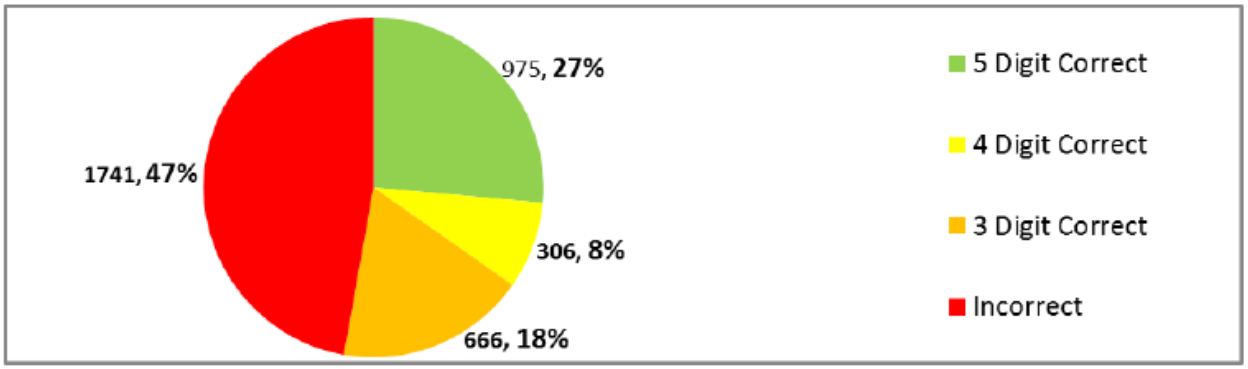


FIGURE E- 10. System work unit code accuracy.

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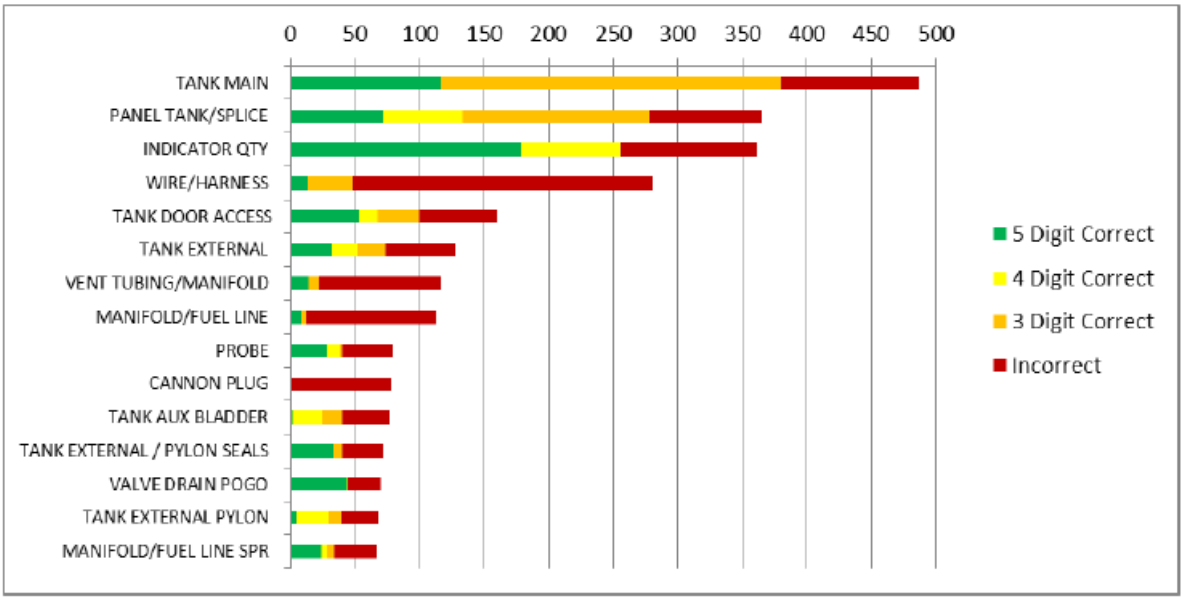


FIGURE E- 11. Component work unit code accuracy.

FIGURE E- 12 is an excerpt from the aircraft FMEC analysis. The detailed analysis can be viewed in the full report.

FMI	Failure Mode Description	Current PM Interval	Recommended PM Interval	Current Cost Per FH	Recommend PM Cost Per FH	FRI Info	FRI	In-Flight Abort	Before Flight Abort	MTBF (Flight Hours)
01 - A - 01	Boost Pump Switch fails to operate.	No PM	No PM	0.01	0.01	Occasional / Marginal	13	0	0	46461.6
01 - A - 02	Dump Valve Switch fails to operate.	No PM	No PM	0.02	0.02	Remote / Marginal	16	0	2	104539
01 - A - 03	Fuel Pressure Signal Switch fails to operate.	No PM	No PM	0.03	0.03	Occasional / Marginal	13	0	0	32165.7
01 - A - 04	Crossfeed Manifold Primer Switch fails to operate.	No PM	No PM	0	0	Occasional / Marginal	13	0	0	69692.3
01 - A - 05	Crossfeed Valve Selector Switch fails to operate.	No PM	No PM	0	0	Remote / Marginal	16	0	0	418154
01 - A - 06	Crossfeed Separation Switch fails to operate.	No PM	No PM	0	0	Remote / Marginal	16	0	0	418154
01 - A - 07	SPR Panel Off-load Switch fails to operate.	No PM	No PM	0	0	Remote / Marginal	16	0	0	418154
01 - A - 08	SPR Panel Ground Transfer Switch fails to operate.	No PM	No PM	0	0	Remote / Marginal	16	0	0	418154
01 - A - 09	SPR Panel Refuel Control Valve Fill Switch fails to operate.	No PM	No PM	0.02	0.02	Remote / Marginal	16	0	1	139385
01 - A - 10	SPR Panel Master Selector Switch fails to operate.	No PM	No PM	0	0	Remote / Marginal	16	0	0	418154
02 - A - 01	(PM)External Tank cracked.	540 Days	540 Days	0.21	0.21	Remote / Critical	12	0	0	418154
02 - A - 02	(PM)External Tank corroded.	540 Days	540 Days	0.04	0.08	Remote / Critical	12	0	0	418154
02 - A - 03	(PM)External Tank bent/dent/distorted	540 Days	540 Days	0.15	0.09	Remote / Critical	12	0	0	418154
02 - A - 04	(PM)External Tank damaged (105, 425,020,730).	540 Days	540 Days	0.15	0.22	Remote / Critical	12	0	0	418154
02 - A - 05	(PM)External Tank leak	540 Days	540 Days	0.21	0.1	Occasional / Critical	8	0	1	83630.8

FIGURE E- 12. FMECA compilation spreadsheet.

A Failure Risk Index (FRI) can be used to categorize and prioritize failure modes. Failure severity categories are defined to provide a qualitative measure of the most reasonable failure mode resulting from subsystem, component failure or malfunction. FIGURE E- 13 shows the different

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combinations of failure frequency and criticality. The values are on FIGURE E- 14 and were used on FIGURE E- 13 to determine the level of analysis required.

PROBABILITY	CATEGORY			
	Catastrophic	Critical	Marginal	Negligible
NAVAIR 00-25-403 Figure 4	Class A Death, System Loss, Property Damage > $1M	Class B Severe, Major System Damage Leading to Property Damage $200K - $1M	Class C Minor Injury, Minor System Damage leading to Mission Degradation, Property Damage $10K - $200K	Class E or below Less than Minor Injury, Less Than Minor System Damage Leading to Unscheduled Maintenance 2K- $10K
FREQUENT $>10^{-3}$ MTBF: 1 to 1,000	1	2	4	11
PROBABLE 10^{-3}-10^{-4} MTBF: 1,000 to 10,000	3	6	9	15
OCCASIONAL 10^{-4}-10^{-5} MTBF: 10,000 to 100,000	5	8	13	18
REMOTE 10^{-5}-10^{-6} MTBF: 100,000 to 1,000,000	7	12	16	19
IMPROBABLE $<10^{-6}$ MTBF: > 1,000,000	10	14	17	20

FIGURE E- 13. Failure risk index (FRI).

Failure Risk Index Value	Risk Level	Level of Analysis
1, 2, 3, 4, 5, 6, 7,	High	Analysis required
8, 9, 10**, 11, 12, 13	Medium	Analysis required
14, 15, 16, 17, 18, 19, 20	Low	FMECA only - Unless PM is associated with the Failure Mode then detailed RCM analysis required

** Hazards that have frequencies below 1 in 1,000,000 are acceptable regardless of hazard category; however, management review of potentially catastrophic failure modes is required before accepting failure mode as "improbable"

FIGURE E- 14. Level of analysis matrix.

E.6 REFERENCES

1. C-130 Mechanical Equipment & Systems Integrity Program Reliability Centered Maintenance Analysis Fuel System, Contract Number: FA8504-07-D-0002-0017, Document No. 151700-12020-F1, 24 July 2012.

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2. SAE-STD-JA1011, Evaluation Criteria for Reliability-Centered Maintenance (RCM) Processes.

3. Reliability-Centered Maintenance, AD/A066 579, F.S. Nowlan, United Airlines, San Francisco, California, 1978.

4. Guidelines for The Naval Aviation Reliability-Centered Maintenance Process, NAVAIR TO 00-25-403, 01 July 2005.

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APPENDIX F

TASK 6 – ACTION PLAN

F.1 SCOPE

F.1.1 Scope.

Once the AFS risk assessment is completed (Task 5), actions need to be taken to address those components posing a risk to aircraft airworthiness. This task centers on the actions that can be taken to address issues that have been identified and the continuing improvement of the risk assessment action plan through updating the current maintenance plan periodically.

F.2 RISK MITIGATION STRATEGIES

The actions selected for implementation are dependent upon multiple factors including project, platform, engineering, and risk tolerance constraints. The selection of a mitigation strategy needs to consider each of these. The following sections highlight some available methods to address the risks identified during the assessment.

A reduction in physical failure risk can be achieved by implementing one or more strategies. Methods for reduction of physical failure risk include but are not limited to:

F.2.1 Materials change.

If components are identified to have negative impacts on AFS reliability, one option is for material changes of the components. The changes will depend on the component type and function, but may include a change in plating/sealing materials or entire components. Before any changes are made, it is recommended that fuel compatibility is verified to avoid future issues.

F.2.2 Design changes.

F.2.2.1 Development and validation of physical failure mitigation strategies.

Components creating unacceptably high failure severity values may need to be considered for design changes (e.g. B-1 fuel panel). Identify and develop a mitigation strategy for those components deemed to have too high a risk index if the failure severity was associated with physical failure.

If a redesign or component changes are considered, the risk assessment for the affected components should be re-evaluated with different solutions to determine how the risk reduction will have different impacts on the system. Validation and verification of the mitigation solution should ensure that:

1. Hazardous failure conditions are extremely remote.
2. Catastrophic failure conditions are extremely improbable and do not result from a single event or failure.
3. This mitigation solution does not introduce any new potential failure conditions.

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Following the selection of a mitigation strategy, the damage assessment and associated risk should be reexamined. This process can be performed multiple times or until an acceptable level of risk has been achieved.

F.2.3 Replacement.

If design changes are not possible or an acceptable level of risk has not been achieved, then replacement should be considered for AFS components that have reached end of life criteria, may soon experience performance degradation, or to achieve risk targets. The need for replacement is based on the individual platform needs and constraints.

F.2.4 Maintenance changes.

From the risk assessment, the safety critical fuel system components are identified and correlated with aging and maintenance hot-spots. This information, combined with the aircraft zone breakdown and inspection checklist developed, provides a framework to generate or augment an existing AFS inspection program. Maintenance efforts can be directed to:

1. Monitor components that are exhibiting deterioration found during periodic inspection. Corrective actions can be taken more quickly if failures occur sooner and in higher numbers than predicted by the degradation models.
2. Monitor maintenance hot-spot areas.

F.3 COMPONENT REPAIR PROCEDURE CHANGES

In order to better identify the causes of component failures, additional inspection practices could be added to repair shops handling AFS equipment. These inspections would look for the reason for component failure and not simply look to get the component operational. The inspection techniques identified in Task #4 could be used as a basis for recommended practices.

The benefits of including this would be to help identify component wear before too many platforms are impacted.

F.4 FORECASTING REPLACEMENT

The results of the component degradation assessment performed in Task 4, provides a means to forecast AFS reliability. The benefit of this is degrading components can be scheduled for replacement in advance. AFS components that do not have an immediate need for replacement and are not projected to impact aircraft safety or reliability can be scheduled for future maintenance cycles.

F.5 PERIODIC UPDATE PROCESS

Following the completion of Tasks 1-6, it is important to track the progress and the fleet level impacts. To do this, the following actions should be considered.

F.5.1 AFS component reassessment.

Determine if/when additional fleet inspection and fuel system component comprehensive materials/electrical properties assessment should be performed.

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F.5.2 Develop assessment metrics.
The overall goal of the assessment is to improve the aircraft's fuel system. This should be measured and tracked to determine if there has been a noticeable improvement based on the changes implemented from the action plan. The data gathered in Task 2 should be used as a baseline for determining the improvements in system safety and reliability. Comparisons should be normalized to fleet size and usage to provide relevant analyses.

F.5.3 Action plan implementation assessment.
Review the action plan that was established and determine the implementation of mitigation strategies, maintenance plans, and other actions defined in Task 6.

F.5.4 Periodic reassessment.
Continued aircraft use will lead to additional aging and possible degradation of AFS components. The following are possible circumstances under which additional iterations through the AFS risk assessment process may be performed.

F.5.5 Change in risk tolerances.
The action plan formed in Task 6 is based on project specific objectives and constraints; over time, these may change. Periodic reassessment will determine if the risk tolerances made in the original assessment are still valid or if changes are needed to satisfy current objectives.

F.5.6 Design life review.
An aircraft AFS service life extension assessment can be evaluated using this handbook and Task 6; reassessments should be scheduled prior to projected impact on aircraft airworthiness.

F.5.7 System upgrades.
Upgrade or modification to aircraft fuel systems (such as avionics, hydraulics, etc.) can impact the AFS risk assessment assumptions. During these upgrades or modifications, changes in the fuel system loads, routing and/or separation distance may affect the physical or functional failures on the aircraft. The proposed changes should be integrated with the fuel system risk assessment model.

F.5.8 Mission change.
This may include changes in equipment. Furthermore, a change in aircraft mission profile may also impact the risk tolerance. The decisions made on risk mitigation strategies proposed during the original risk assessment effort many need to be reexamined to determine if the actions will be necessary to achieve risk level for the new mission profile.

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APPENDIX G

FUEL TANK IGNITION SOURCE SUPPRESSION

BY SFAR-88

G.1 SCOPE

G.1.1 Scope.

Special Federal Aviation Regulation (SFAR)-88 is a brief two-page regulation which was effective 12/9/2002 and can be found in its entirety at: SFAR-88. This regulation resulted from the TWA 800 accident investigation and subsequent review of in- service airplane fuel tank explosions (both military and civilian), service difficulties, and Airworthiness Directives. This review uncovered unforeseen failures in aircraft fuel tank systems resulting in over 100 ADs on in service civilian airplane models worldwide. The SFAR required a retrospective design review of in service airplanes utilizing revised regulatory standards contained in FAR section 25.981 that included consideration of manufacturing variability, wear, corrosion and foreseeable maintenance errors when showing the fuel tank system was fail safe. Detailed listing of considerations for preventing ignition sources in the fuel tank system is provided in FAA Advisory Circular 25.981-1C. It is recommended this be considered as part of military AFS assessment. Although this guidance was originally developed for commercial aircraft (there is no military equivalent), this is the best available process for fuel tank ignition risk reduction. The major points included are:

1. Applicability (transport category type certificate holders…).
2. Compliance.
    a. Conduct a safety review of the airplane fuel tank system to determine that the design meets the requirements of Secs. 25.901 and 25.981(a) and (b) including needed design changes.
    b. Develop all maintenance and inspection instructions.
    c. Submit a report which provides substantiation, and contains all maintenance and inspection instructions.

NOTE: This appendix is intended as a resource and not a replacement for the SFAR-88 regulation and the guidance documents. It is highly recommended the source documents be fully reviewed to gain a full understanding of fuel tank ignition suppression.

G.2 INSTALLATION – FEDERAL AVIATION REGULATION (FAR) 25.901

Federal Aviation Regulation (FAR) 25.901 “Installation” is under the Powerplant, General, heading and currently states in part: “the major components of the installation needs to be electrically bonded to the other parts of the airplane (grounding and bonding system – consideration of composite construction),” and “it needs to be established that no single failure or malfunction or probable combination of failures will jeopardize the safe operation of the airplane except that the failure of structural elements need not be considered if the probability of such failure is extremely remote. ”

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G.3 FUEL TANK IGNITION PREVENTION – FEDERAL AVIATION REGULATION (FAR) PART 25.981

FAR Part 25.981 "Fuel tank ignition prevention" is under 'Powerplant, Fuel System, heading' and may be found in its entirety at FAR Part 25.981. It currently states in part: (FAR Part 25.981(a)), ignition sources are not allowed, depending on auto-ignition temperature of the fuel, and probabilities are included: *an ignition source could not result from each single failure, from each single failure in combination with each latent failure condition not shown to be extremely remote (failure rate – 10^{-7}), and from all combinations of failures not shown to be extremely improbable (failure rate – 10^{-9}). The effects of manufacturing variability, aging, wear, corrosion, and likely damage needs to be considered.*; (FAR Part 25.981(b)) no fuel tank Fleet Average Flammability Exposure on an airplane may exceed 3 percent of the Flammability Exposure Evaluation Time (FEET), but this does not apply (FAR Part 25.981(c)) to a fuel tank if means are provided to mitigate the effects of an ignition of fuel vapors within that fuel tank such that no damage caused by an ignition will prevent continued safe flight and landing; (FAR Part 25.981(d)) critical design configuration control limitations (CDCCL), inspections, or other procedures needs to be established.

While SFAR-88 does not require 25.981(d), it is still part of the FARs, and the intent should be considered in assessing the air worthiness of military aircraft. The equivalent mitigation in military applications is mandatory inspections.

G.3.1 Advisory circulars (AC).

Advisory Circulars (ACs) have been published to present acceptable means of compliance with the above regulations. Specifically, AC 25.981-1C "Fuel Tank Ignition Source Prevention Guidelines" and AC 25.981-2A "Fuel Tank Flammability" are the critical publications for SFAR-88 compliance. These ACs list other ACs and FARs which further describe requirements and compliance. Per AC 25.981-1C, "the inspection interval should be established based upon the standard practices defined in AC 25.1309." This 10E-9 criterion of AC 25.1309 is fundamentally at odds with the MIL-STD-882 probabilities typically used for military analysis. AC 25.1309-1A states Minor failure conditions may be probable, major failure conditions needs to be improbable (between 10E-5 and 10E-9), and catastrophic failure conditions needs to be extremely improbable. Extremely improbable failure conditions are those so unlikely they are not anticipated to occur during the entire operational life of all airplanes of one type. For quantitative assessments, Extremely improbable failure conditions are those having a probability on the order of 10^{-9} or less. Compare that to the MIL-STD-882 definition where a probability of E "Improbable" (10^{-6} or better) is generally acceptable. Thus, it may not be practical to fully implement SFAR-88 principles in military aircraft; rather, lessons learned from the commercial world will likely be implemented incrementally, as funding becomes available.

Compliance with the above requirements entails applying fuel ignition safety concepts to the categories listed below. The concepts have been applied to a varying extent to military aircraft. Some military aircraft have fully implemented FAA design requirements and even obtained FAA FAR Part 25 certification, such as the VC-25 and C-32 aircraft. Most, if not all, commercial derivative military aircraft have been evaluated for these requirements to some extent. For existing military aircraft, it may be more practical to retroactively implement SFAR-88 requirements in the way that it has been for legacy commercial aircraft, since existing design and new design have

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different requirements, which also depend on aircraft size (see 25.981-2A "Fuel Tank Flammability" Table 1, and FAR 26.5 Table 1). However, all requirements are included below.

NOTE: In the following section, italics are used for quoted words. Underlines are added to indicate key concepts.

G.4 ELECTRICAL CONNECTORS

AC 25.981-1C lists the following concerns for fuel pump connectors:

a. *Electrical arcing at connections within electrical connectors due to bent pins or corrosion.*

b. *Fuel leakage and subsequent fuel fire outside of the fuel tank caused by corrosion of electrical connectors inside the pump motor which led to electrical arcing through the connector housing (connector was located outside the fuel tank).*

c. *Selection of improper insulating materials in connector design resulting in degrading of the material because of contact with fuel that is used to cool and lubricate the pump motor.*

AC 25.981-1C lists the following concerns for Fuel Quantity Indicating System (FQIS) connectors:

a. *Conductive or semi-conductive (silver, copper or cadmium) deposits at electrical connectors inside fuel tanks.*

AC 25.981-1C adds other issues for connectors:

a. *Sealing of electrical connectors - additional attention should be given of sealed harnesses (model D harnesses).*

b. *Higher power circuits within the LRU should not be located on the same… electrical connector with intrinsically safe circuits or wiring.*

c. *Use of sealant in connectors that is not compatible with the fuel may allow leakage into the connector and the possibility of a fire near the connector… qualification test results should be reviewed to establish that the sealant is compatible with the fuel and to determine if a design life or periodic inspections for the pump connector are needed.*

d. *Electrical connections… are typically located either inside or outside the fuel tank in areas of the airplane where the presence of flammable fuel vapors should be assumed because no secondary sealing of fuel is provided.*

e. *Arcing at the pump electrical connections should be assumed with fuel present in the safety assessment. The energy release during an arcing condition may be limited by installation of arc/ground fault protection features. The pump connector should be shown to contain any resultant arcing or fire and maintain all surface temperatures below the auto-ignition temperature of the fuel.*

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A common issue with older aircraft is electrical connectors are often located with the component, in a fuel tank.

G.5 COMPONENTS

G.5.1 Conduits, wiring, and systems.

AC 25.981-1C adds other issues for wiring:

a. *Higher power circuits within the LRU should not be located…in a wire harness…with intrinsically safe circuits or wiring. Fuel probe wiring shall be physically separate from power wiring throughout its entire length including connectors, terminations, and junction boxes.*

b. A failure mode is *wear of Teflon sleeving and wiring insulation on wires in metallic conduits located inside fuel tanks, allowing arcing from wire through the conduits into fuel tank ullages.*

c. *Inspections have shown that protective sleeving has been missing, improperly installed, or the wrong sleeving material used, resulting in damage to the insulation. For these reasons, use of protective sleeving on wiring would not by itself be adequate for showing compliance.*

d. A failure mode is *damage to insulation on wiring routed adjacent to the fuel tank exterior surfaces that resulted in arcing to the metallic fuel tank surface.*

e. *Conduits containing float switch wiring failing due to freezing of water that entered the conduit, allowing fuel leakage into the conduit and along the airplane front spar resulting in an engine tailpipe fire.*

f. High energy *wiring should not be routed through the metallic conduit inside the fuel tank or adjacent to fuel tank surfaces such that damage, inappropriate maintenance, or other failure/wear conditions could result in arcing to the conduit or metallic tank surface and development of an ignition source in the fuel tank. If metallic or other conductive conduit materials are used, the single failure of electrical arcing of the wiring to the conduit, adjacent tank surfaces, or structure should be assumed to occur. In addition, circuit protective features or other features should be incorporated to preclude development of an ignition source in the fuel tank. Methods that may be used to address this foreseeable failure condition include the use of circuit protective features such as dual conduits, thick wall conduit, and/or fast-acting ground fault interrupter (GFI) circuit breakers.*

g. *All wiring conducting levels of power that exceed intrinsically safe levels (e.g., fuel pump power supply) into or through a fuel tank should be evaluated assuming arcing to adjacent surfaces such as metallic conduits or wing surfaces, unless fail-safe protective features are provided.*

h. *Advisory Circular 25-8 addresses the use of metallic conduit as an acceptable means for routing of electrical power within airplane auxiliary fuel tanks.* [However]…,*these past*

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practices would not meet the fail-safe requirements of § 25.981 for any fuel tank, unless additional fail-safe features were incorporated.

i. *Wiring designs used on transport airplanes vary significantly between manufacturers and models; therefore, it is not possible to define a specific universal separation distance or the characteristics of physical barriers between wire bundles to protect critical wiring from damage. Separation requirements for wiring and other components of EWIS are contained in § 25.1707 , (the regulatory definition of EWIS is provided in § 25.1701).BV Advisory Circular AC 25.1701-1, Certification of Electrical Wiring Interconnection Systems on Transport Category Airplanes, paragraph 5D, contains guidance on determining adequate separation distance between EWIS and airplane systems and structures.*

j. *If nonmetallic conduit is used, compatibility with fuel should be demonstrated. The nonmetallic conduit should be evaluated for the effects of aging due to heat, corrosion at the connecting fittings, and resistance to heat damage from internal shorts of wires routed within the conduit.*

k. *Splices in fuel system wiring have been allowed as a standard repair procedure. The acceptability of splices will be based upon the system design and fail-safe features.*

l. *Silver can combine with sulfur or water and form silver-sulfide or oxide deposits between exposed conductors (terminal block connections, etc.). The silver-sulfide deposits reduce the resistance between conductors and can ignite fuel vapor when exposed to very low levels of electrical energy. If use of silver in electrical components and wiring in the tank is determined to be critical, it should be defined as a critical design configuration control limitation.*

m. *Steel wool has been used as a cleaning tool to remove corrosion and clean parts inside the fuel tanks. Steel wool creates small conductive filaments that can cause ignition sources.*

n. *Laboratory testing has shown that the minimum ignition energy in an electrical spark required to ignite hydrocarbon fuel vapor is 200 microjoules. Therefore, for electrical or electronic systems that introduce electrical energy into fuel tanks, such as fuel quantity indicating systems, any electrical arcs or sparks that are created into any fuel tank should be less than 200 microjoules during either normal operation or operation with failures…Systems with a maximum energy of 20 microjoules are considered technologically feasible. Normal systems operations at minimum ignition energies of up to 50 microjoules would be acceptable. Under failure conditions, the system should not have an ignition energy which would exceed 200 microjoules.*

o. *A maximum of steady-state current of 25 milliamperes RMS is considered an intrinsically safe design limit for electronic and electrical systems that introduce electrical energy into fuel tanks* [for steel wool filament heating]. *For failure conditions, limit the current to 50 milliamperes RMS and for lightning induced transients to 125ma peak current.*

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p. *Surfaces that come within 50 degrees of the auto-ignition temperature of the fuel air mixture for the fluid as ignition sources. The FAA has historically accepted 204.4 ℃ (400 °F) for maximum surface temperatures for kerosene type fuels… The auto-ignition temperature of these fuels varies inversely with the ambient pressure.*

q. *Wiring that does not adhere to aircraft manufacturer's standard wiring practices (i.e., wires bent back along themselves with bend radius less than defined in the aircraft manufacturer's standard wiring practices, multiple splices lying next to one another, etc.)* may result in arcing.

r. *Electrical Generator Power Feeder Cables* can arc to *a pressurized fuel line resulting in fire adjacent to the fuel tank.*

s. *Wiring should also be protected from lightning and High Intensity Radiated Fields (HIRF) induced transients.*

t. Consider the *number, type, fire resistance, and location of support devices along the wire path of the intrinsically safe wire*, along with *amount of slack wire resulting from wire bundle build tolerances.*

u. *If wiring is routed inside the metallic conduit inside the fuel tank, redundant protection means would be required to prevent an arc of the wiring to the conduit from penetrating into the fuel tank. A practical means to achieve this is to provide sleeving of the wiring and an independent means such as a GFI or arc fault circuit breaker. Providing multiple layers of sleeving alone would not be considered acceptable since wear could defeat the multiple layer protection.*

G.5.2 Circuit boards.

AC 25.981-1C adds other issues for circuit boards:

a. *Higher power circuits within the LRU should not be located on the same circuit board…with intrinsically safe circuits or wiring.*

b. *There should be a physical barrier between circuit boards to isolate the intrinsically safe circuits from the effects of broken components or fire within the LRU. If limiting devices are installed on the same circuit board in series with system circuitry to limit the amount of power or current transmitted to the fuel tank, there should be three inches between the traces, unless the manufacturer can justify a smaller separation on the basis that the effects of fire on the circuit board will not compromise the intrinsically safe circuit(s).*

G.5.3 Circuit breakers.

AC 25.981-1C suggests the following on circuit breakers:

a. *Service experience has indicated that thermal mechanical circuit breakers installed in the fuel pump circuits have not been shown to preclude arcing of electrical wiring through*

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metallic barriers such as conduit, fuel pump housings, electrical connectors, or the tank wall into the fuel tank on some airplane designs. Evidence suggests that arcing from the wiring to metallic surfaces may not result in a hard short, which would trip the circuit breaker and may result in intermittent low level arcing that gradually arcs through the metallic barrier into the fuel tank. For these failure conditions circuit protective devices such as Arc Fault Circuit Breakers and Ground Fault Interrupters may be needed to provide fail safe features necessary to demonstrate compliance.

b. *Transient suppression/energy limiting devices. If failure of the device is latent and this feature is needed to maintain fail-safe features, periodic checks will be needed.*

Thermal circuit breakers should not automatically reset, since resetting would perpetuate the hazard.

G.5.4 Pumps.

AC 25.981-1C suggests the following issues related to fuel pumps:

a. *Friction Sparks. Service experience has shown that pump inlet check valves, inducers, nuts, bolts, rivets, fasteners, lockwire, roll pins, cotter pins, drill chips, and manufacturing debris, etc., have been inducted into fuel pumps and contacted the impeller resulting in the possibility of metallic deposits on rotating and stationary components within the pump. This condition has resulted in creation of friction sparks and should be an assumed failure condition when conducting the system safety assessment. Fail-safe features as described below in paragraph 10B(2) [*flame arrestor in the inlet line] *have been used to mitigate this hazard.*

b. *Pump inlet case degradation, allowing the pump inlet check valve to contact the impeller.*

c. *Failure of one phase of the stator winding during operation of the fuel pump motor together with subsequent failure of a second phase of the motor windings, resulting in arcing through the fuel pump housing.*

d. *Arcing due to the exposure of electrical connections within the pump housing that have been designed with inadequate clearance to the pump cover.*

e. *Omission of cooling port tubes between the pump assembly and the pump motor assembly during pump overhaul.*

f. *Extended dry running of fuel pumps in empty fuel tanks, which was contrary to the manufacturer's recommended procedures…dry operation of pumps may result in surface temperatures above the auto-ignition temperature of the fuel or expose the pump to dry run operation where debris from the fuel tank could enter the impeller and cause sparks. Manufacturers recommended procedures have not been shown to be adequate in preventing dry run operation. Therefore, additional fail-safe features are necessary to preclude dry run operation of airplane fuel pumps*, such as keeping the inlet submerged, auto-shutoff, or inlet flame arrestor.

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g. *Use of steel impellers that may produce friction sparks if debris enters the pump.*

h. *Debris lodged inside pumps.*

i. *Electrical connections within the pump housing have been designed with inadequate clearance or insulation from the metallic pump housing, resulting in arcing.*

j. *Thermal switches aging over time resulting in a higher trip temperature.*

k. *Flame arrestors falling out of their respective mounting.*

l. *Internal wires coming in contact with the pump rotating group, energizing the rotor and arcing at the impeller/adapter interface.*

m. *Poor bonding across component interfaces.* (Specify maximum resistance.)

n. *Insufficient ground fault current protection capability.*

o. *Poor bonding of components to structure.*

p. *Loads from the airplane fuel feed plumbing transferred into the pump housing resulting in failure of the housing mounts and subsequent failure of the pump case, which defeated the explosion proof capabilities of the pump.*

q. *Premature failure of fuel pump thrust bearings allowing steel rotating parts to contact the steel pump side plate.*

r. *Single phase operation of three-phase electrical fuel pumps* may be hazardous. *In many cases fuel pump motors are protected by a (single) three-phase circuit breaker. In several instances, resetting of circuit breakers has resulted in arcing inside the fuel tank and the development of an ignition source. Therefore, the fuel pump circuit should also preclude development of an ignition source if the breaker is reset or forced in by a mechanic. Features such as non-resettable breakers or fuses are design features that would meet the fail-safe requirements.*

s. *Two-phase operation of three-phase electrical fuel pumps* may be hazardous. *Failure of a single phase of a multiple-phase fuel pump may significantly increase the load on the remaining phases of the pump and generation of heat in the pump…Automatic protective means, such as arc/ground fault interrupters or other means, should be provided to shut down the pump when a single electrical phase failure occurs. Periodic inspections or maintenance of these features may be required.*

t. *The design should not include features that result in spraying of fuel, such as fuel pump motor cooling flow return to the tank.*

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u. Wear of pump bushings, bearings, and seals may significantly affect the performance of fuel pumps and degradation of features necessary to maintain the explosive proof qualification.

G.5.5 Motor-operated valves.
AC 25.981-1C suggests some of the following which could be applied to valves:

a. *Electrical connections within the motor housing have been designed with inadequate clearance* or insulation from the metallic…housing, resulting in arcing.

b. Internal wires coming in contact with the rotating group, energizing the rotor and arcing.

c. *Poor bonding across component interfaces.* (Specify maximum resistance.)

d. Insufficient ground fault current protection capability.

e. Poor bonding of components to structure.

f. Loads from the airplane fuel feed plumbing transferred into the motor housing resulting in failure of the housing mounts and subsequent failure of the… case, which defeated the explosion proof capabilities.

g. Micro-switch failures.

G.5.6 Fuel quantity probes.
AC 25.981-1C suggests the following on probes:

a. Conductive or semi-conductive corrosion (copper or silver sulfur deposits) causing reduced breakdown voltage in FQIS wiring.

b. Damage to FQIS wire insulation resulting in reduced breakdown voltage as a result of wire clamping features at electrical connections on fuel quantity probes.

c. Contamination in the fuel tanks creating an arc path for low levels of electrical energy between FQIS probe walls (steel wool, lock wire, nuts, rivets, bolts, and mechanical impact damage to probes).

G.5.7 Float switches.
AC 25.981-1C suggests the following on float switches:

a. Float switch wire chafing observed which might have provided potential for subsequent electrical short to the conduit.

b. Float switch sealing failure has allowed fuel/water to egress into the switch, compromising switch operation in an explosive environment.

c. Temperatures should be determined.

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G.5.8 Fuel tubes and couplings.
AC 25.981-1C suggests the following on couplings:

a. *Poorly conducting…couplings that may become electrical arc sources when exposed to electric currents, including lightning currents.*

b. *Insufficient clearances between tubes and surrounding structure.*

c. *Intermittent electrical bonding in flexible couplers.*

d. *Bonded couplers unable to conduct expected lightning or power fault currents without arcing.*

e. *Early generation self-bonding flexible fuel couplers did not have multiple bonding paths. Thus these bonding couplers exhibited single-point failures which caused loss of function…clam shell type self-bonding metallic couplings in composite fuel tanks have shown arcing could occur during lightning conditions if a coupling was improperly latched.*

G.5.9 Bonding & straps.
AC 25.981-1C suggests the following on bonding:

a. *Corrosion of bonding strap wires resulting in failure to provide required current paths.*

b. *Inappropriately attached connections (loose or improperly grounded attachment points).*

c. *Static bonds on fuel system plumbing connections inside the fuel tank worn due to mechanical wear of the plumbing from wing movement, and corrosion.*

d. *Corrosion of bonding surfaces near fuel tank access panels that could diminish the effectiveness of bonding features.*

e. *Aging of self-bonding fuel system plumbing connections resulting in higher resistance bonding.*

f. *Missing bonds.*

g. *Loose or intermittent contacts between bond straps and other conductive components.*

h. *Failure of bonding jumpers is generally considered a latent failure, since there is no annunciation or indication of the bonding failure. The airplane fleet fuel tank inspections that occurred as a result of the TWA 800 investigation showed that failure of bonding jumpers, due to damage, wear, or manufacturing errors, was not unusual. Based on this, it would be difficult to show that failure of a single bonding jumper is extremely remote or extremely improbable. Therefore, the electrical bonding jumper or other bonding provisions would need to consider the consequences of these latent failures. This may result in designs that incorporate electrical bonding redundancy, if the failure of a single*

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electrical bonding feature could create a fuel ignition source. Additionally, manufacturers would need to consider the use of appropriate maintenance to detect failed bonding jumpers.

i. *There is no specific FAA guidance on maximum resistance or impedance of airplane electrical bonding provisions. Electrical bonding within a fuel system should be tailored to the performance requirements of a particular airplane design. The electrical bonding should consider the electrical sources, such as electrical faults, fuel static electrification, and lightning. The electrical bonding should also consider the specific airplane fuel system design, which would include the structure material used (aluminum, carbon fiber composites, fiberglass composites, etc.), the configuration of the fuel system (routing of fuel tubes, wires, and hydraulic tubes), and the electrical bonding concept (intentional isolation, self-bonding fittings, separate bonding jumpers, etc.).* See SAE AIR5128 for more guidance on this.

G.5.10 Pneumatic system.
AC 25.981-1C suggests the following on the pneumatic system:

a. *Leakage of air from ducting located near fuel tanks due to duct failure resulting in undetected heating of tank surfaces above the auto-ignition temperature.* This bleed air can also impinge on fuel tubes.

b. *A maximum allowable surface temperature of 204.4 °C (400 °F), with a transient excursion up to 260 °C (500 °F) for a maximum duration of two minutes, has been accepted for certain pneumatic system installations.*

c. *The maximum internal tank temperatures that can occur during normal and failure conditions should be determined.* This should apply to fuel tubes as well.

G.6 FUEL AND ADDITIVES.
AC 25.981-1C suggests the following on Fuels:

a. *Without a static dissipater additive, typical Jet-A fuel has a low electrical conductivity. A static dissipater additive will increase the charging rate of fuel but at the same time greatly improve the conductivity of the fuel to rapidly dissipate the developed charge.*

b. *Water contamination…increases the charging tendency of the fuel. Fuel tank inerting is one method of precluding this ignition source. However, for transport airplanes, this phenomenon is usually addressed by incorporating an active water removal scavenge system for the fuel tanks and periodically sumping the tanks to remove the collected water.*

AC 25.981-2A suggests the following on Fuels:

a. *The definitions of LFL* (Lower Flam. Limit) *and UFL* (Upper Flam. Limit) *define the fuel temperature at which a fuel tank can be expected to be flammable. From the definitions, it can easily be seen that fuel flash point is key. Currently, Jet-A and Jet-A1 are the predominant fuels used in commercial aviation. Because of this, wing tanks are commonly*

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not flammable as the fuel temperature is typically below the LFL. However, the heat input to any tank can push tanks fueled with Jet-A/A1 into the flammable range.

G.6.1 Flammability.

AC 25.981-1C suggests the following on flammability:

a. See *FAA Document DOT/FAA/AR-98/26, Review of the Flammability Hazard of Jet-A Fuel Vapor in Civil Transport Aircraft Fuel Tanks, June 1998.*

b. *Due to the difficulty in predicting fuel tank flammability, the FAA has always assumed that a flammable fuel air mixture exists in airplane fuel tanks and has required that no ignition sources be present. The system safety analysis should be prepared considering all airplane in-flight, ground, service, and maintenance conditions, assuming that an explosive fuel air mixture is present in the vapor space of fuel tanks and vent systems at all times, unless the fuel tank has features that mitigate the effects of tank ignition (e.g., polyurethane foam).*

AC 25.981-2A suggests the following on flammability:

a. *This advisory circular (AC) provides information and guidance on compliance with the airworthiness standards for transport category airplanes about limiting the time a fuel tank may be flammable or mitigation of hazards from flammable fuel air mixtures within fuel tanks.*

b. Existing design and new design have different requirements, which also depend on aircraft size.

c. *See for reference these documents and others which can be obtained as described in the AC where not linked below:*

1. SAE AIR 5128, “Electrical Bonding of Aircraft Fuel System Plumbing Systems” (January 1997);
2. SAE AIR 4170A, “Reticulated Polyurethane Safety Foam Explosion Suppressant Material for Fuel Systems and Dry Bays” (November 1998);
3. SAE AIR 1662, “Minimization of Electrostatic Hazards in Aircraft Fuel Systems” (October 1984);
4. MIL-B-83054, Baffle and Inerting Material, Aircraft Fuel Tank (March 1984), (Note: this reference provides an extensive list of military specifications about the use of polyurethane foam.)
5. FAA Document DOT/FAA/AR-98/26, "A Review of the Flammability Hazard of Jet A Fuel Vapor in Civil Transport Aircraft Fuel Tanks" (June 1998);
6. “Effects of Fuel Slosh and Vibration on the Flammability Hazards of Hydrocarbon Turbine Fuels Within Aircraft Fuel Tanks,” Technical Report AFAPL-TR-70-65 (November 1970), Edwin E. Ott;
7. FAA Document DOT/FAA/AR-00/19, “The Cost of Implementing Ground-Based Fuel Tank Inerting in the Commercial Fleet,” (May 2000);

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8. FAA Document DOT/FAA/AR-01/6, "Inerting of a Vented Aircraft Fuel Tank Test Article with Nitrogen Enriched Air" (December 2000);
9. FAA Document DOT/FAA/AR-TN02/79, "Limiting Oxygen Concentrations Required to Inert Jet Fuel Vapors Existing at Reduced Fuel Tank Pressures" (April 2003);
10. FAA Document DOT/FAA/AR-07/30, "Jet A Volatility Survey," July 2007;
11. FAA Document DOT/FAA/AR-05/8, "Fuel Tank Flammability Assessment Method User's Manual" (May 2008).

d. Table 1 incorporates by reference these (and others): *14CFR Part 25 Appendix M, Fuel Tank System Flammability Reduction Means; 14CFR Part 25 Appendix N, Fuel Tank Flammability Exposure and Reliability Analysis; Part 26, subpart D, Fuel Tank Flammability; FAR Part 26.33, Holders of Type Certificates, Fuel tank safety - If above 7 percent, normally emptied…needs to develop service instructions – See FAR Part 25.981 above for 3 percent limit – see AC Table 2 for clarification.*

e. The accepted level for tank inerting used by the military is to reduce the oxygen concentration in the tank ullage to less than 9 percent…the applicant may establish the acceptable oxygen concentration.

f. The use of appropriate foams to fill the fuel tank and thereby control the pressure rise following an ignition of the fuel vapor/air mixture has been demonstrated to be effective by the United States Air Force (USAF) and other military forces. Foams are in use on several civilian transport category airplane types.

g. Address the potential for, and the effects of, degradation of the foam, from any environmental effects and long term aging, on both the airplane and engine fuel systems.

Some aircraft use ventilation to reduce flammability in enclosed spaces adjacent to fuel tanks, such as "dry bays." While this method can be effective in flight, there may be no motive force while on the ground. Thus, an analysis of probabilities may not be significantly improved by occasional venting. Also, an analysis would be needed to show the amount of venting is adequate for the amount of expected fuel vapor generation.

G.6.2 Fuel heating.

AC 25.981-1C suggests the following on fuel heating:

a. Submerged heat exchangers should be evaluated for maximum temperature.

AC 25.981-2A suggests the following on flammability:

a. In general, heat sources should not be located in or near fuel tanks, and heating from other sources, such as hydraulic heat exchangers or rejection of heat from engine systems, should be avoided unless features are incorporated to maintain fuel tank flammability equivalent to an unheated wing fuel tank.

G.7 LIGHTNING & EMI

AC 25.981-1C suggests the following on Lightning:

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a. *See FAR 25.954 Fuel system lightning protection. See AC 20-53B Protection of Aircraft Fuel Systems Against Fuel Vapor Ignition Caused by Lightning. See AC 20-136A Protection of Aircraft Electrical/Electronic Systems against the Indirect Effects of Lightning. See ARP 5412A, Aircraft Lightning Environment and related test waveforms. See ARP 5414A, Aircraft Lightning Zoning. See K. E. Crouch, "Aircraft Fuel System Lightning Protection Design and Qualification Test Procedures Development," FAA report DOT/FAA/CT-94/74, September 1994. See F. A. Fisher, J. A. Plumer, and R. A. Perala, "Aircraft Lightning Protection Handbook," FAA report DOT/FAA/CT-89/22, September 1989.*

b. *Optical detection methods consist of subjecting a fuel tank to a simulated lightning strike while a specific camera/lens/film configuration is positioned near the fuel system component, system or fuel tank with the shutter open. The test is passed if no spark or arc is visible on the developed film.*

c. Intrinsically safe wiring should be protected from lightning by separation, shielding, and transient suppression devices. This is especially important for FQIS wiring and probes.

d. *The failure of other system components may also affect EMI protection. Consequently, the effect of any anticipated failure on the continued environmental protection should be considered. For example, a normally high impedance electrical load that fails due to an electrical short, or a normally low impedance load that fails to an open state, can cause a significant redistribution of the lightning currents within a wire bundle; hence, the voltage induced at components within the fuel tank.*

e. *Standardized tests, such as those in RTCA DO-160, Sections 19, 20, and 22, are not sufficient alone, without evaluation of the characteristics of the specific electromagnetic environment and induced transient levels assigned to systems installed within a particular airplane model, to show that appropriate standardized RTCA document DO-160 test categories, procedures and test levels are selected.*

The "Aircraft Lightning Protection Handbook" DOT/FAA/CT-89/22 provides a method for determining zones. See also SAE ARP5414 "Aircraft Lightning Zone" for more recent methods. The most vulnerable zone would be the wing tips, where aircraft skin is thin and can allow penetration, and there may be no flame arrestor on the vent scoop to protect from swept lightning and ground fire.

G.7.1 Flow-induced static electricity

AC 25.981-1C suggests the following on static electricity:

a. *Electrostatic charges are generated in liquid hydrocarbons when they are in motion with respect to another surface such as fueling hoses, filters, nozzles, fuel tank structure and aircraft plumbing...Flowing fuel can generate an electrical charge especially through fuel filtration...If the fuel conductivity is low, the relaxation time for dissipation of the electrical charge is long. As a consequence, the fuel may accumulate an electrical charge inside an aircraft fuel tank. This electrical charge may produce a high potential on the fuel surface*

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and an electrical discharge to structure. This is particularly a concern when large unbonded objects are located inside an airplane fuel tank.

b. *Flow velocities of 6 to 7 meters per second are considered acceptable after the discharge port is covered with fuel. The guidelines also indicate that the flow velocity should be held to less than 1 meter per second until the discharge port is covered with fuel. This criterion may be met by incorporating multiple refueling discharge ports, lowering the flow velocity through the use of piccolo tubes that distribute the fuel at low velocities in the tank, and locating them at or near the bottom of the tank.*

c. *Use of a non-conductive type of reticulated polyurethane foam in only a portion of a fuel tank system that allowed electrostatic charge build-up and arcing in the unprotected portion of the system.*

d. *Spraying of fuel through refueling nozzles located at the upper portion of the tank.*

e. *Bonding straps are used on fuel components and plumbing lines to allow the charge to dissipate to the tank structure. During refueling, the airplane is bonded to the refueling vehicle with a separate bonding wire to provide an electrical path back to fuel filtration, which is the principal electrostatic charge generator.*

f. *Polyurethane reticulated foams used for ignition suppression within fuel tanks and other nonconducting objects may accumulate and retain charge. These items may have to be treated with antistatic additives to prevent charge accumulation.*

G.8 ADDITIONAL MITIGATIONS AND TESTING

AC 25.981-1C suggests the following mitigations:

a. *Any components located in or adjacent to a fuel tank needs to be designed and installed in such a manner that, during both normal and anticipated failure conditions, ignition of flammable fluid vapors will not occur. Compliance with this requirement is typically shown by a combination of component testing and analysis. Testing of components to meet the appropriate level of explosion-proof requirements may be carried out for various single failures, and combinations of failures, to show that arcing, sparking, auto-ignition, hot surface ignition, or flame propagation from the component will not occur. Testing of components has been accomplished using several military standards and component qualification tests. For example MIL-STD-810, Method 511, Procedure I and II, defines one method that has been accepted for demonstrating that a component is explosion proof. Section 9 of RTCA Document DO-160 has also been accepted for demonstrating that airborne equipment is explosion proof.*

b. *Fuel pumps should be located such that electrical power for the pumps is routed outside the fuel tanks.*

c. *Propagation of fire through the airplane vent system into the fuel tanks is addressed by regulations such as 25.975 "Fuel tank vents and carburetor vapor vents,"* which states in

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part, *No vent or drainage provision may end at any point…where the discharge of fuel from the vent outlet would constitute a fire hazard. This regulation was amended to include a requirement to prevent propagation of fire from outside the vent into the fuel tank system for a minimum of 2 minutes and 30 seconds. Lightning induced ignition of fuel vapors at the fuel tank vents is addressed in FAR Part 25.954.* Therefore, Pods and vent scoops (which should have flame arrestors) should not constitute a fuel tank ignition source. However, other referenced items such as uncontained engine debris, engine fires, explosives, and crash fires, are covered elsewhere for military airplanes. *The possibility of localized heating due to external fires needs to be considered.*

d. See AC 25.981-1C for System Safety Analysis methods and assumptions.

e. *A safety review of flammable fluid leakage zones adjacent to fuel tanks will be needed to determine that the design complies with the requirements of §§ 25.863(a) and FAR Part 25.981.* This applies to "dry bay" areas. FAR Part 25.863 is titled "Flammable fluid fire protection," and is effectively incorporated by reference.

f. Aging fuel line couplings seals/ O-rings. In certain instances, materials used in fuel line couplings may lose flexibility and harden with age.

g. If failure of an arc/ground fault protective feature and/or a thermal fuse (closed) or transient suppression *is latent and this feature is needed to maintain fail-safe features, periodic checks would likely be needed.*

h. Acceptable methods of providing visible means would include color coding of the wiring and labeling…Industry and regulatory authorities are currently developing standards for color coding of wiring for various systems…Pink wiring has been used by some manufacturers for fuel tank system wiring. Use of this color as a standard for fuel tank system wiring is recommended by the FAA.

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APPENDIX H

ELECTRICAL BONDING INSIDE FUEL TANKS

H.1 SCOPE

H.1.1 Scope.

For the purposes of this discussion, electrical bonding refers to the joining of metal parts together to form an electrically conductive path which has the capacity to safely conduct any fault current likely to be imposed on it. With respect to fuel tanks, proper bonding minimizes the possibility of electrical arcing or resistive heating when an electrical current passes through a bonded interface which can occur when there is an electrical fault, static discharge, or lightning strike. An electrical bond can also provide a ground path or electrical shielding for an electrical component. Typically, fuel pumps, valves, switches, fuel probes, mechanical components, fuel tubes, couplings, and fittings are electrically bonded to a metal structure or strap and shunted to a ground reference point.

H.2 BONDING STANDARDS

For many years, MIL-B-5087, Bonding, Electrical, and Lightning Protection, for Aerospace Systems, had been the standard for electrical bonding. This specification has been cancelled and superseded by MIL-STD-464, Department of Defense Interface Standard, Electro-magnetic Environmental Effects. MIL-B-5087 classified bonds according to the purpose of the bond and sets specific resistance or impedance requirements and is still a useful reference document. MIL-STD-464 has taken a different approach and only has general requirements based on the purpose of the electrical bond. MIL-STD-464 allows the OEM more flexibility to set electrical bonding requirements.

H.2.1 MIL-STD-464.

The relevant statements in MIL-STD-464 which address electrical bonding are given below. Comments in brackets were added for clarification and are not part of the standard.

H.2.1.1 Mechanical interfaces.

The system electrical bonding needs to provide electrical continuity across external mechanical interfaces on electrical and electronic equipment, both within the equipment and between the equipment and other system elements, for control of E3 [static discharge, lightning, and other imposed currents] such that the system operational performance requirements are met. For instances where specific controls have not been established for a system and approved by the procuring activity [typically defined by the OEM], the following direct current (DC) bonding levels needs to apply throughout the life of the system [this a lifetime requirement].

a. 10 milliohms or less from the equipment enclosure to system structure, including the cumulative effect of all faying surface interfaces [mechanical bond interfaces].

b. 15 milliohms or less from cable shields to the equipment enclosure, including the cumulative effect of all connector and accessory interfaces [mechanical bond interfaces].

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c. [Lower values may be needed, depending on aircraft shield frequency and attenuation requirements].
d. 2.5 milliohms or less across individual faying interfaces [mechanical bond interfaces] within the equipment, such as between subassemblies or sections.

Compliance will be verified by test, analysis, inspections, or a combination thereof.

H.2.1.2 Shock, fault, and ignitable vapor protection.
Bonding [bond interfaces] of all electrically conductive items subject to electrical fault currents needs to be provided to control shock hazard voltages and allow proper operation of circuit protection devices. For interfaces [bond interfaces] located in fuel or other flammable vapor areas, bonding needs to be adequate to prevent ignition from flow of fault currents. Compliance will be verified by test, analysis, or a combination thereof.

H.3 MIL-STD-1798.

Aircraft fuel tank bonding requirements are also derived from MIL-STD-1798, this military standard describes the general process to achieve and maintain the physical and functional integrity of the mechanical elements of aircraft systems.

As noted in APPENDIX C.9, a KC-135 was destroyed by a fault current in a fuel pump and the loss of a Boeing 747 aircraft in flight (TWA 800) was attributed to a fuel tank explosion (see APPENDIX C.13). After the KC-135 incident there was a review of how electrical bonding was verified at the depot responsible for overhauling fuel pumps and other fuel tank components. The review determined there was a lack of understanding on why electrical bonding is important and, in some cases; electrical bonding was not being properly verified after overhaul or as part of inspections.

H.4 COMMERCIAL APPROACH TO BONDING

The same issue was also present in the commercial airline industry and after the Boeing 747 incident. The FAA subsequently developed new requirements and guidance for ensuring proper electrical bonding in aircraft fuel tanks as a mitigation strategy against fuel tank explosions. The FAA requirements and guidance are described in SFAR-88 (see pertinent SFAR-88 regulations here).

After the TWA 800 accident, the FAA required large aircraft operators (14 CFR Part 25) and other aircraft types to comply with SFAR-88. The SFAR-88 regulations require airworthiness certificate holders to reduce fuel tank ignition sources and the flammability exposure in fuel tanks. This is typically accomplished through a risk analysis followed by mitigation approaches against fuel tank explosions. Ensuring proper electrical bonding is a critical part of this assessment.

There is no specific FAA guidance on maximum resistance or impedance of aircraft electrical bonding provisions. The approach is to identify and tailor electrical bonding to the performance requirements of a particular aircraft design. The FAA guidance states electrical bonding should consider the electrical sources, such as electrical faults, fuel static electrification, and lightning. The certification authority should also consider the specific airplane fuel system design, which

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would include the structure material used (aluminum, carbon fiber composites, fiberglass composites, etc.), the fuel system configuration (routing of fuel tubes, wires, and hydraulic tubes), and the electrical bonding concept (intentional isolation, self-bonding fittings, separate bonding jumpers, etc.).

After the FAA published the fuel tank safety requirements, many military system program offices were tasked to conduct similar assessments on aircraft under their authority. Since 2006, the Air Force has also moved more to an airworthiness approach similar to the FAA for certification of military developed aircraft. The military community has attempted to apply the FAA guidance and the cognizant fuel system engineer should have access to a fuel tank safety analysis which was most likely conducted by the aircraft or fuel system OEM. Commercial derivative aircraft are typically certified through the FAA and an SFAR-88 assessment should be available for review by the system program office responsible for the aircraft. These can be highly proprietary documents and maybe be difficult to obtain.

H.4.1 Bonding measurements.

The integrity of an electrical bond is determined by measuring the resistance through the bonding interface. As noted earlier, the required resistance value is determined by the application and potential failure condition that needs to be mitigated (see SAE AIR5128 on typical values). Electrical bonding for a fault current path can be the most challenging requirement to meet and maintain. A typical example would be the mechanical attachment points of a fuel pump. Many pumps use 115 VAC three phase power and, if the power source is shorted to the housing, hundreds of amps can potentially flow through the pump housing bonding interface. The bonding resistance is determined by the potential fault current. Typical resistance values range from less than 0.1 milliohms to 2.5 milliohms.

H.4.2 Understanding bonding.

Most interfaces are an aluminum alloy that visually appears to be flat smooth surfaces. The faying surfaces are usually held together by metal bolts or screws.

The current flow through the interface is through the attachment hardware (bolts/screws) and faying surfaces. The disparity of the faying surface can result in less than 10 percent of the area actually conducting the fault current so most of the current can flow through the attachment hardware.

This why it is important to use a high current to evaluate the bonding interface. Historical documents report that FOD, such as safety wire, have been found between faying surfaces. Laboratory testing has shown the safety wire can overheat and arc when high currents are placed between the faying surfaces. Using a low current or precision voltmeter does not provide sufficient current to locate a safety wire caught between the faying surfaces. Corrosion and other debris have also been found in the thread attachment interfaces which can degrade these current paths (see FIGURE H- 1).

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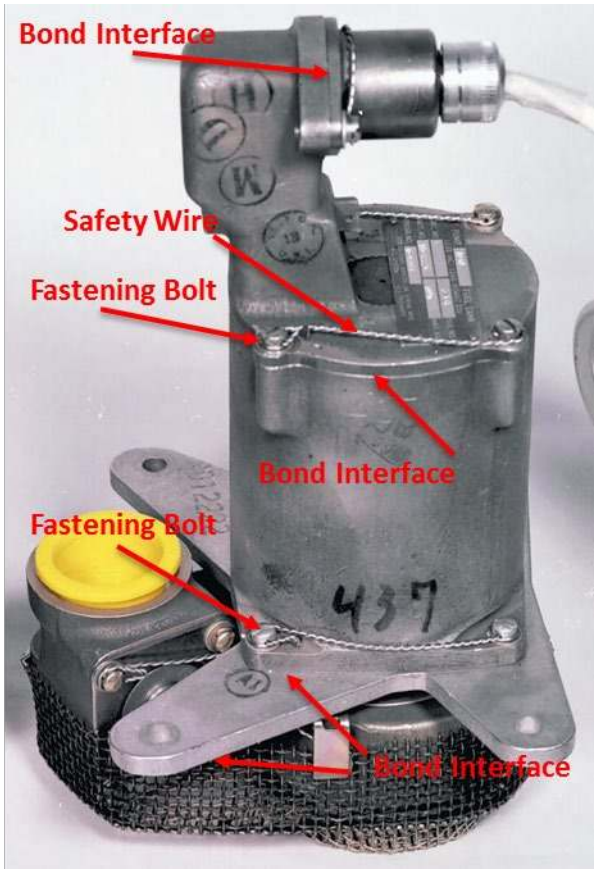


FIGURE H- 1. Typical fuel pump with multiple bonding interfaces. Note fastener bolts and safety wire (potential FOD).

H.4.3 Need for high current.

Verifying the electrical bonding resistance is essential to prevent arcing if there is a current fault between the pump housing and aircraft ground. A high current fault, typically 10 amperes or more, can degrade the pump bond interfaces and lead to arcing. Laboratory testing has shown low current (less than 100 mA) measurements are ineffective in verifying the bonding resistance when a potential high current fault is possible. Imposing a high current simulates a fault current running through the pump and being properly shunted to ground. A slight increase in resistance at a high current can cause ohmic heating and arcing between interfaces, creating a potential fuel ignition source. Typical measurement currents can range from 1 to over 50 amperes. See FIGURE H- 2 for calculating resistance based on fault current.

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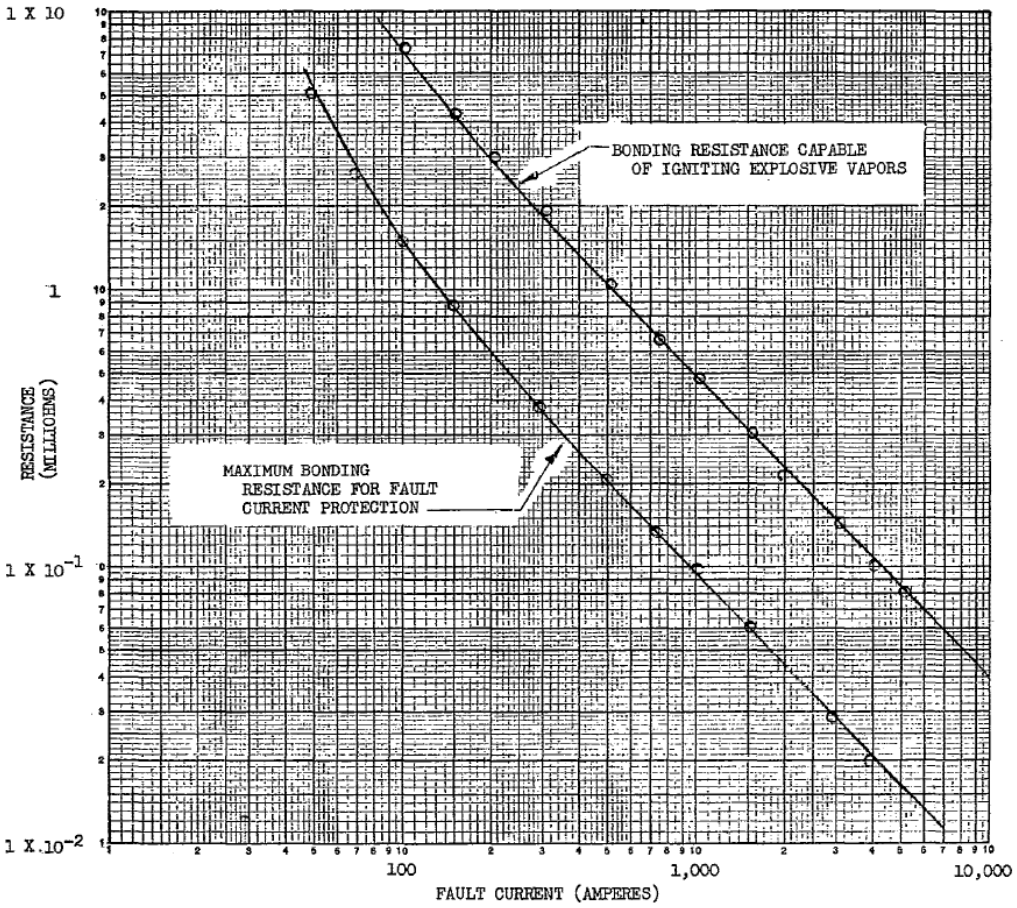


FIGURE H- 2. Fault current vs maximum allowed resistance for bonding between equipment and structure.

In addition to evaluating the effect of high current, bonding resistance cannot be reliably measured using an ohmmeter or voltmeter due to the very low resistance value being measured. These instruments use a low voltage (1.5 to 9 Volts) to measure resistance and typically produce unstable and unrepeatable measurements. The proper method to measure the bonding interface is to use a four-point probe system (see FIGURE H- 3).

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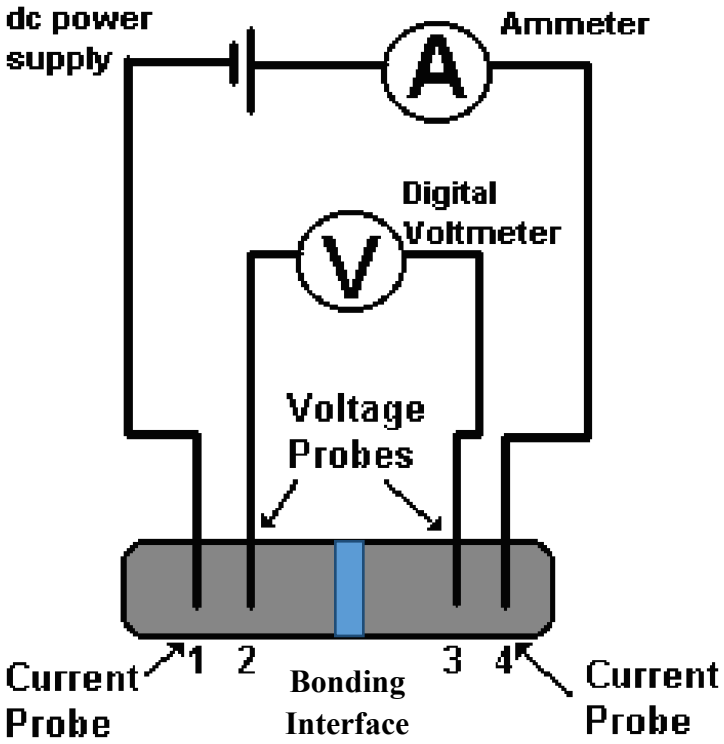


FIGURE H- 3. Four-point probe setup for accurately measuring the bonding resistance.

This technique uses two probes that inject a known current (typically 1 to 50 amperes) between the bonding interfaces. Two probes are then placed between the bonding interface and the two current probes to measure the voltage drop across the bonding interface. This effectively eliminates the probe resistance and provides enough of a voltage for a repeatable measurement. Resistance is calculated by dividing the voltage by the current.

Test methods can be found in Technical Orders, Standards, and bonding meter manuals. A detailed procedure for making a four-point probe measurement can be found in ANSI/EIA-364. This document describes a procedure for measuring the resistance between mated electrical connectors and can be used as guidance in making a four-point probe bonding measurement. Guidance for making a four-point probe measurement can also be found in ASTM B539.

H.4.4 High current applications.
For measuring less than 1 milliohm, injecting 1 ampere would require measuring 1 millivolt which can be a difficult measurement considering noise and accuracy of a typical voltmeter (needs to be better than 0.1 millivolt). Injecting 10 amperes would increase the voltage drop to 10 millivolts which would be a reasonable value to measure. There are commercially available bonding testers/meters that perform a four-point probe measurement and directly display resistance. These systems can inject various currents (typically up to 10 amperes) to improve accuracy of the measurement. When validating a design, the calculated fault current should be used for measuring

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the bonding resistance. After any maintenance or repair, the bonding resistance should be measured to verify it is within specification and yields stable results.

When on an aircraft, highly sensitive four-point probe meters are available that can measure the bonding resistance using low currents (less than 100 mA). These meters are also used in component overhaul facilities since they are much more convenient and safer to use than injecting a high current into a component to measure the bonding resistance.

As noted earlier, these systems are not adequate to verify a design or manufacturing/overhaul process and should not be used when conducting a teardown assessment or life extension evaluation. A low current meter (less than 1 ampere) may not detect a degraded bond interface. While a fault current through a bonding interface is rare, it has occurred and needs to be addressed. Some OEMs and subject matter experts believe a low current measurement is adequate and using a high current offers no value. The cognizant engineer will need to carefully review the system in question, measurement conditions, field performance, and risk assessment when determining the acceptable measurement technique that will be required to validate an electrical bond interface.

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APPENDIX I

TANKER AIRCRAFT FMECA ITEM EXAMPLES

I.1 SCOPE

I.1.1 Scope.

In these examples, all capital letters in the Failure Mode and Local Effects columns indicate where words were copied directly from a REMIS report. End effects here are generally a duplicate of Next Higher Effects; the End Effects are needed in IRCMS, but otherwise may not be required.

I.1.2 Introduction.

Aircraft maintenance data recorded in REMIS provides the basis for reliability analysis. This data is downloaded periodically**.** Records are then sorted into WUC order and validated, or "scrubbed", to assure each has the properly assigned WUC (see TABLES I- I through I- XI).

a. Item ID: Unique identifier for the record.
b. Item name: the failed component being evaluated.
c. Function: a brief description of the component functions.
d. Function Failure: the method by which the component failed to operate.
e. Failure Mode: the stressor or degradation factor leading to functional failure.
f. Local Effects: the immediate impact of component failure.
g. Next Higher Effects: this may be thought of at the system level.
h. End Effects: the highest level effect of the component failure. This may be thought of at the aircraft level.
i. Failure Detection: the method by which the failure was detected.
j. Severity: on a scale from 1-4 with '1' being the most severe, this is a qualitative failure severity assessment impact on the aircraft.

Effects; the End Effects are needed in IRCMS, but otherwise may not be required.

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TABLE I- I. Gravity transfer valve.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
1	Gravity Transfer Valve (Fuselage)	Gravity Transfer Valve (fuselage). These 3 electric motor-operated fuel shutoff valves allow fuel to flow from the CWT to the Fwd tank, and from the Upper Deck tank to the Aft tank.	Fuel Leak External	Caused by age, sealing, wear	Ignition hazard. Fuel gages show a loss overboard, for large leaks.	With ignition source and lack of dry bay air flow, may cause fire. Mission fuel shortage.	With ignition source and lack of dry bay air flow, may cause fire. Mission fuel shortage.	Ground Inspection, walk-around	1
2	Gravity Transfer Valve (Fuselage)	Gravity Transfer Valve (fuselage). These 3 electric motor-operated fuel shutoff valves allow fuel to flow from the CWT to the Fwd tank, and from the Upper Deck tank to the Aft tank.	Failed to Open Fully	Caused by circuit breaker trip, short, gears, corrosion, STUCK CLOSED	Unable to transfer CWT fuel to Fwd tank or Upper Deck fuel to Aft tank at full rate.	Mission fuel shortage. Increased landing roll distance if needed for fuel dump.	Mission fuel shortage. Increased landing roll distance if needed for fuel dump.	Fuel gages, inspection. Note: there is no valve position indication (IMFP lights show switch position only).	3
3	Gravity Transfer Valve (Fuselage)	Gravity Transfer Valve (fuselage). These 3 electric motor-operated fuel shutoff valves allow fuel to flow from the CWT to the Fwd tank, and from the Upper Deck tank to the Aft tank.	Failed to Close Fully	Caused by electrical shorts, insulation failure, bearings, FOD, seals, leaks, gears	Fuel flows between associated tanks, depending on fuel surface level, shifting CG.	May need to run A/R Pump to empty Fwd tank. Aft tank vent may fill. CG shifts.	May need to run A/R Pump to empty Fwd tank. Aft tank vent may fill. CG shifts.	Fuel gages show transfer, and CG display may be altered.	3

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TABLE I- II. Pump assembly.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
4	Pump Assy, Fuel Booster, Elec Driven	Provide fuel pressure and flow from main tanks.	Fuel Leak	Caused by age, wear, abuse, seals.	Reduced fuel flow and pressure	In conjunction with an ignition source, such as the electric motor or leads, may be an ignition hazard.	In conjunction with an ignition source, such as the electric motor or leads, may be an ignition hazard.	Pressure and flow tests	1
5	Pump Assy, Fuel Booster, Elec Driven	Provide fuel pressure and flow from main tanks.	Damage or missing parts	Caused by abuse, low component strength, pre-load.	Unknown	Unknown	Unknown	Inspection	4
6	Pump Assy, Fuel Booster, Elec Driven	Provide fuel pressure and flow from main tanks.	Pump Inop or Low Pressure	Caused by connections, R1-12 RELAY, RELAY R15-1, R1-13 RELAY, RELAY R1-16, circuit breaker, FOD, SWITCH dirty, TERMINAL ON B2 CONTACT OF RELAY, #1 GCU (Generator Control Unit).	low pressure, intermittent, EXCESSIVE DEPLETION OF #4 MAIN TANK with Manifold Valve open, PRESSURE LIGHTS WOULD FLICKER.	With loss of other pumps, may lead to engine failure.	With loss of other pumps, may lead to engine failure.	Pressure light indication, circuit breaker trip, inspection.	3
7	Pump Assy, Fuel Booster, Elec Driven	Provide fuel pressure and flow from main tanks.	Bearing Failure	Caused by age, stress accumulation, wear, FOD damage, air in the system.	Noise, surging, vibration, HOWLING TYPE SOUND.	With loss of other pumps, may lead to engine failure.	With loss of other pumps, may lead to engine failure.	Noise, vibration	4

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TABLE I- III. Hose assembly.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
8	Hose Assys	Conduct fuel from AR Manifold to Feed Manifold, or from APU pump to APU.	Fitting leaks	Cause: aging seals, abuse, assembly procedure.	Leaks in the Boomer compartment are personnel and fire hazard.	In the presence of an ignition source, may cause fire.	In the presence of an ignition source, may cause fire.	None	1
9	Hose Assys	Conduct fuel from AR Manifold to Feed Manifold, or from APU pump to APU.	Hose leaks	Cause: Surge pressure, fatigue, wear, preload.	Leaks in the Boomer compartment are personnel and fire hazard.	In the presence of an ignition source, may cause fire.	In the presence of an ignition source, may cause fire.	None	1
10	Hose Assys	Conduct fuel from AR Manifold to Feed Manifold, or from APU pump to APU.	APU Fuel Line burst	Cause: Surge pressure, fatigue, wear, preload.	This short line is just aft of STA820 (the fwd end of the main gear wheel well). Leaks here may be a hazard with hot brakes. This short line is between STA 600 and 620 (between the Fwd Body and Center Wing tanks).	In the presence of an ignition source, may cause fire.	In the presence of an ignition source, may cause fire.	loss of APU fuel pressure	1
11	Hose Assys	Conduct fuel from AR Manifold to Feed Manifold, or from APU pump to APU.	AR to Feed Manifold Line burst	Cause: Surge pressure, fatigue, wear, preload.	Leaks here may be a hazard with hot brakes.	In the presence of an ignition source, may cause fire.	In the presence of an ignition source, may cause fire.	loss of AR Manifold pressure	1

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TABLE I- IV. Line valves.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
12	Line Valves (AR Manifold)	Fuel shutoff control for refueling or dump. Includes the Dump Valve, Line Valve (both near STA 1240), Manifold Valve (aft of fwd tank) or (fwd of STA 344).	External leak Failed closed	Cause: Seals, O-rings.	Ignition hazard	Fire hazard in the presence of ignition sources.	Fire hazard in the presence of ignition sources.	visual, odor	1
13	Line Valves (AR Manifold)	Fuel shutoff control for refueling or dump. Includes the Dump Valve, Line Valve (both near STA 1240), Manifold Valve (aft of fwd tank) or (fwd of STA 344).	Other	Cause: Electrical, insulation, switches.	Cannot load or offload fuel	If manual open does not work, may not be able to dump fuel.	If manual open does not work, may not be able to dump fuel.	Fuel quantity and flow gages	2
14	Line Valves (AR Manifold)	Fuel shutoff control for refueling or dump. Includes the Dump Valve, Line Valve (both near STA 1240), Manifold Valve (aft of fwd tank) or (fwd of STA 344).	Switch Position	Cause: electrical,	Pops circuit breaker	If manual open does not work, may not be able to dump fuel.	If manual open does not work, may not be able to dump fuel.	Electrical issues	2
15	Line Valves (AR Manifold)	Fuel shutoff control for refueling or dump. Includes the Dump Valve, Line Valve (both near STA 1240), Manifold Valve (aft of fwd tank) or (fwd of STA 344).	Light Burnout	Cause: Indicator light.	No switch position indication	None	None	Visual	4

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TABLE I- V. Fuel pressure regulator.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
16	Fuel Pressure Regulator	Reduces pressure from the AR pumps from about 80 psi to 35 psi, when used to supply engines. A relief valve which relieves at45-50 psi is incorporated on the downstream side of the poppet, emptying to the CWT.	Fails closed	Cause: corrosion, dirt, contamination.	Cannot use body fuel in engines.	Fuel shortage for mission.	Fuel shortage for mission.	Fuel quantity gages, inspection.	4
17	Fuel Pressure Regulator	Reduces pressure from the AR pumps from about 80 psi to 35 psi, when used to supply engines. A relief valve which relieves at45-50 psi is incorporated on the downstream side of the poppet, emptying to the CWT.	Cracking pressure low	Cause: incorrect or weak spring.	Slightly reduced pressure to engine.	May cause engine effects and fatigue downstream pressure components.	May cause engine effects and fatigue downstream pressure components.	Inspection.	4

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TABLE I- VI. Valve assembly, fuel swing check.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
18	Valve Assy, Fuel Swing Check (Xfr Vlv)	This main-to-body tank gravity transfer check valve allows flow into the refuel manifold for gravity transfer, but prevents reverse flow when the manifold is pressurized.	Leaks external.	Cause: aged or damaged seals, such as the inlet and outlet fittings. The housing develops a crack.	Since the valve is located inside a tank, leaked fuel is safely captured.	A pressurized manifold could leak into the wrong tank during fueling, if the transfer valve is open.	A pressurized manifold could leak into the wrong tank during fueling, if the transfer valve is open.	Fuel gages	4
19	Valve Assy, Fuel Swing Check (Xfr Vlv)	This main-to-body tank gravity transfer check valve allows flow into the refuel manifold for gravity transfer, but prevents reverse flow when the manifold is pressurized.	Leaks internal.	Cause: aged or damaged seals, FOD.	None, when transfer valve is closed.	A pressurized manifold could leak into the wrong tank during fueling, if the transfer valve is open.	A pressurized manifold could leak into the wrong tank during fueling, if the transfer valve is open.	Inspection	4
20	Valve Assy, Fuel Swing Check (Xfr Vlv)	This main-to-body tank gravity transfer check valve allows flow into the refuel manifold for gravity transfer, but prevents reverse flow when the manifold is pressurized.	Failed closed.	Cause: FOD, corrosion.	Failure to transfer fuel	May have less fuel to provide to receiver aircraft, or aid in balancing.	May have less fuel to provide to receiver aircraft, or aid in balancing.	Inspection	4

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TABLE I- VII. Actuator.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
21	Actuator, Electro Mechanic Al Linear (Ltv)	This linear actuator is shown in IPB T.O. 1C-135-4-5, Fig. 51, but it does not show what it controls It is unclear if or how this actuator is used.	Fails to extend fully	Cause: electrical controls, wiring, electrical motor, valve jammed due to FOD or contamination.	Unknown	Unknown	Unknown	Unknown	
22	Actuator, Electro Mechanic Al Linear (Ltv)	This linear actuator is shown in IPB T.O. 1C-135-4-5, Fig. 51, but it does not show what it controls It is unclear if or how this actuator is used.	Fails to retract fully	Cause: electrical controls, wiring, electrical motor, valve jammed due to FOD or contamination,	Unknown	Unknown	Unknown	Unknown	
23	Actuator, Electro Mechanic Al Linear (Valves 1&2)	A quadrant operated by an electrical actuator provides remote control capability of the AR-to-Refuel Manifold, and Aft Tank Gravity valves on ARR aircraft. The master refuel switch needs to be ON at the ARR panel. These valves are cabled together.	Fails to extend fully	Cause: electrical controls, wiring, electrical motor, valve jammed due to FOD or contamination.	When extended, valves are in flight configuration (#1 closed, #2 open). No effect from #1 with no other failures, but a partially or fully closed #2 will slow gravity transfer to the aft tank.	If other valves leak, fuel could transfer to other tanks when AR pumps are on.	If other valves leak, fuel could transfer to other tanks when AR pumps are on.	Fuel gages, handle position if checked	4
24	Actuator, Electro Mechanic Al Linear (Valves 1&2)	A quadrant operated by an electrical actuator provides remote control capability of the AR-to-Refuel Manifold, and Aft Tank Gravity valves on ARR aircraft. The master refuel switch needs to be ON at the ARR panel. These valves are cabled together.	Fails to retract fully	Cause: electrical controls, wiring, electrical motor, valve jammed due to FOD or contamination.	Goal: #1 open, #2 closed. If #1 is only partly open, fueling may be slow. If #2 is only partly closed, there is unwanted transfer to the aft tank during fueling.	If other valves leak, fuel could transfer to other tanks when AR pumps are on.	If other valves leak, fuel could transfer to other tanks when AR pumps are on.	Fuel gages, handle position if checked	4

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TABLE I- VIII. APU fuel pump.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
25	APU Fuel Pump	Provide Fuel Pressure & Flow from Left Inbd Main Wing tank to APU 1 & 2, which provide bleed air and electrical power for engine start, HVAC, etc.	Pump Inop or Low Pressure	Cause: INOP, Connections, circuit breaker POPPED, internal electrical and mechanical issues, temperature.	Low pressure, intermittent ops. or inop.	May lead to APU failure, resulting in lack of bleed air and electricity for ground ops.	May lead to APU failure, resulting in lack of bleed air and electricity for ground ops.	Low pressure fuel light on the APU FUEL control panel illuminates when pressure downstream of the filter & bypass falls below 6.5 psi.	4
26	APU Fuel Pump	Provide Fuel Pressure & Flow from Left Inbd Main Wing tank to APU 1 & 2, which provide bleed air and electrical power for engine start, HVAC, etc.	Loss of Ignition Containment	Cause: seal failure with arcing, or screen failure with FOD.	Fire hazard.	With flammable mixture and ignition source, tank fire, explosion.	With flammable mixture and ignition source, tank fire, explosion.	Fuel pressure light.	1
27	APU Fuel Pump	Provide Fuel Pressure & Flow from Left Inbd Main Wing tank to APU 1 & 2, which provide bleed air and electrical power for engine start, HVAC, etc.	Fuel Leak	Cause: age, wear, abuse, seals.	Reduced pressure output.	With low fuel level and fuel spray, and ignition may cause fire.	With low fuel level and fuel spray, and ignition may cause fire.	Inspection.	4

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TABLE I- VIII. APU fuel pump – Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
28	APU Fuel Pump	Provide Fuel Pressure & Flow from Left Inbd Main Wing tank to APU 1 & 2, which provide bleed air and electrical power for engine start, HVAC, etc.	Missing Parts	Cause: poor maintenance.	Unknown.				4
29	APU Fuel Pump	Provide Fuel Pressure & Flow from Left Inbd Main Wing tank to APU 1 & 2, which provide bleed air and electrical power for engine start, HVAC, etc.	Damage	Cause:	Abuse, strength, pre-load.	May lead to loss of pressure, or with low fuel qty. and loss of containment, could cause fire.	May lead to loss of pressure, or with low fuel qty. and loss of containment, could cause fire.	None	4
30	APU Fuel Pump	Provide Fuel Pressure & Flow from Left Inbd Main Wing tank to APU 1 & 2, which provide bleed air and electrical power for engine start, HVAC, etc.	Bearing Failure	Cause: bearing wear, FOD damage, air in the system.	Noise, surging, vibration.			Noise	3

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TABLE I- IX. Bladders.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
31	Bladders/Cells	Fuel Storage.	Fuel Leak (general).	Caused by age, sealing, tank FOD, cell size, wear, vent ftg threaded Inserts.	Less fuel available, spray on Boomer Window & Receiver, fire hazard.	With ignition source, may cause fire. Mission fuel shortage.	With ignition source, may cause fire. Mission fuel shortage.	Fuel gages, boomer window, inspection.	1
32	Bladders/Cells	Fuel Storage.	Upper Vent Fitting, Fabric Tears.	Caused by cell too small.	Less fuel available, spray on Boomer Window & Receiver, fire hazard.	With ignition source, may cause fire. Mission fuel shortage.	With ignition source, may cause fire. Mission fuel shortage.	Fuel gages, boomer window, inspection.	1
33	Bladders/Cells	Fuel Storage.	Interconnect (general).	Caused by seal, O-ring, tears.	Leaks, maintenance difficulties.	With ignition source, may cause fire. Mission fuel shortage.	With ignition source, may cause fire. Mission fuel shortage.	Fuel leak at drain, inspection.	1
34	Bladders/Cells	Fuel Storage.	Corrosion.	Caused by moisture, dissimilar metals, susceptible materials, D RINGS.	Less fuel available, spray on Boomer Window & Receiver, fire hazard.	With ignition source, may cause fire. Mission fuel shortage.	With ignition source, may cause fire. Mission fuel shortage.	Potential fuel leak, inspection.	1
35	Bladders/Cells	Fuel Storage.	Fuel Leak from Access Door Seal.	Caused by poor sealing.		With ignition source, may cause fire. Mission fuel shortage.	With ignition source, may cause fire. Mission fuel shortage.	Fuel gages, boomer window, inspection.	1

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TABLE I- X. Boom equipment.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
36	Boom Equipment	Connect tanker to receiver for transferring fuel.	Ruddervator	Cause: Paint, damage from snapped cable, corrosion, screws, rivets, pealing tape & decal, delamination, drain hole clogged, gouges, cracks, PCU, worn shaft, bent, hail, loose Joe-Bolts, rigging, LOCK PAWL SPRINGS BROKEN, tail cone, END CAP, Jackshaft, rig.	Shaking, uncontrollable, dump actuator not engaging, BOOM LAGGED AND BECAME erratic, stiffness, binding, buffet, LOCKED.	Could impact receiver aircraft.	Could impact receiver aircraft.	Inspection, operation.	3
37	Boom Equipment	Connect tanker to receiver for transferring fuel.	Nozzle Assy	Cause: fuel leak, won't self-center, fasteners.	Loss of fuel, spray on receiver, can't connect.	Unable to refuel, or fuel spray on receiver.	Unable to refuel, or fuel spray on receiver.	Inspection, operation.	3
38	Boom Equipment	Connect tanker to receiver for transferring fuel.	Signal Coil	Cause: corrosion or dirt in contacts, TERMINAL BROKEN, GOUGED, SIGNAL COIL BROKEN, SHORTED, BROKEN SHIELD, WIRE BROKEN, REPLACED NUT, COIL CONTACTS RUSTY, CONNECTOR.	No signal, can't refuel, DELAYED DISCONNECT, INTERPHONE RECEIVES OK BUT DOES NOT TRANSMIT, INTERPHONE INTERMITTENT.	Arcing in the vicinity of fuel has ignition potential	Arcing in the vicinity of fuel has ignition potential	Inspection, operation.	3

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
39	Boom Equipment	Connect tanker to receiver for transferring fuel.	AZIMUTH PANTOGRAPHING DRUM	Cause: DRUM CRACKED, DRUM PULLEY MIC'D BAD, INDEX PLATES ROTATED, DRUM COLLAR WORN, BOOM PIVOT SUPPORT FITTING INSIDE DIAMETER worn, bearings, rigging, LOCKS ROTATED.	BOOM FLYS AT 25' ELEVATION DURING FLIGHT, DRUM BEARINGS GRINDING, BOOM TRAILS AT 32 DEG AND 6 DEG RIGHT ALSO SHOWS LEAD AND LAG GOING FROM RIGHT TO LEFT CONTROLS VERY STIFF/UNSAFE.	Boom could impact receiver aircraft.	Boom could impact receiver aircraft.	Inspection, operation.	3
40	Boom Equipment	Connect tanker to receiver for transferring fuel.	FUEL DUMP ACTUATOR	Cause: CANNON PLUG D56-3 ON S BUNDLE, BAD THREAD PROTRUSION, CAM GUIDE, corrosion, circuit breaker, broken wire on connector.	Function: depress poppet valve on Nozzle. NOT ENGAGING, unable to dump fuel.	Could increase landing roll, or flight duration.	Could increase landing roll, or flight duration.	Observe no fuel dumping, fuel gages.	3
41	Boom Equipment	Connect tanker to receiver for transferring fuel.	PRESSURE RELIEF VALVE, BOOM HYDRAULIC	Cause: B-nut, internal leaks	Telescope valve function: allow force on the Nozzle to extend and retract boom. Hoist valve relieves due to the boom hoist control valve leaking internally through the emergency hoist port. Uncommanded boom extension. CHATTER NOISE WHILE CONNECTED.	Excess relief could drain right hydraulic system.	Excess relief could drain right hydraulic system.	Inspection, operation.	3

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
42	Boom Equipment	Connect tanker to receiver for transferring fuel.	CABLE, HOIST, BOOM	Cause: CABLE SNAPPED, CABLE GUIDE TUBE MOUNTING BRACKET IS BROKEN, BOOM LATCH CABLE, WORN SPACER ON PULLEY INSIDE TAIL AREA, CHAFING INTO SUPPORT SRUCT.	Function: lift hoist. Sudden movements of the hoist lever may damage hoist cable. Never position BOOM HOIST lever to FREE WHEEL with boom unlatched unless boom is supported.	If the cable breaks in flight, usually can't latch boom.	If the cable breaks in flight, usually can't latch boom.	Inspection, operation.	4
43	Boom Equipment	Connect tanker to receiver for transferring fuel.	HYDRAULIC FLOW REGULATOR, BOOM AND AR PUMP	Cause: leaks, not regulating.	A regulator in the boom hoist circuit limits the rate of lowering the boom. Another regulator in the pressure line to the cable tension motor will prevent cable tension motor overspeed if the boom hoist cable breaks. Plus others. Noise, pressure drops.	Reducing pressure in left or right hydraulic system.	Reducing pressure in left or right hydraulic system.	Inspection, pressure gages.	4
44	Boom Equipment	Connect tanker to receiver for transferring fuel.	BOOM FORK ASSEMBLY		Function: retain Boom and allow pivoting.	Boom could depart aircraft.	Boom could depart aircraft.	Inspection, operation.	2
45	Boom Equipment	Connect tanker to receiver for transferring fuel.	SLIDING GLAND SEAL, BOOM	Cause: CRACKED, BEARING Cause: seals	Function: seal fuel between outer and inner Boom tubes.	Loss of fuel.	Loss of fuel.	Inspection, operation.	4

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
46	Boom Equipment	Connect tanker to receiver for transferring fuel.	ELEVATION CONTROL UNIT, BOOM	Cause: LINKAGE ROD END TERMINAL BEARING BROKEN, adjustment, rigging	Function: transmit boom elevation. Inaccurate, reading constant, lags, POPS PDI LIGHT C/B	Disconnects.	Disconnects.	Inspection, operation.	4
47	Boom Equipment	Connect tanker to receiver for transferring fuel.	SIGNAL AMPLIFIER, BOOM	Cause: LOOSE CONNECTOR, etc.	Function: Energizes the made relay; Applies power to the operating envelope limit switches, provided the elevation and extension limit switches are in ACTIVE; Illuminates the MADE light and the pilot BOOM ENGAGED light; Operates the aerial refueling pumps.	Supplies power to the director lights. INTERPHONE NOT TRANSMITTING, INOP, DISCONNECT LIGHT STAYS ON WHEN IN "READY" MODE.	Disconnects.	Inspection, operation.	3
48	Boom Equipment	Connect tanker to receiver for transferring fuel.	FLEXIBLE COUPLING, BOOM	Cause: leaks, COUPLING OUT OF DATE, R2... IAW 6JG-3	Function: fuel tube coupling just upstream of Boom.	Ignition hazard on the ground.	Ignition hazard on the ground.	Inspection, operation.	3
49	Boom Equipment	Connect tanker to receiver for transferring fuel.	BOOM HOIST FLOW REGULATOR	Cause: Hoist Reg. LEAKING	Function: limits the rate of lowering the boom.	Ignition hazard on the ground	Ignition hazard on the ground	Inspection, operation.	3

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
50	Boom Equipment	Connect tanker to receiver for transferring fuel.	Azimuth Control Unit	Cause: ROD DISCONNECTED AT BOOM FORK, adjustment.	Function: BOOM AZIMUTH CONTROL UNIT connects drive rod to transmitter. In failure of Control Unit, indicator follows transmitter control arm when moved in one direction only. GAUGE SPINS.	Azimuth unknown.	Azimuth unknown.	Inspection, operation.	4
51	Boom Equipment	Connect tanker to receiver for transferring fuel.	SURGE BOOT, BOOM	Cause: SERVICING PORT BROKEN, LEAKING	Function: fuel surge pressure suppression (50 psi charge).	Lack of suppression can lead to major ruptures.	Lack of suppression can lead to major ruptures.	Inspection.	3
52	Boom Equipment	Connect tanker to receiver for transferring fuel.	LATCH EYE, BOOM	Cause: BOLTS TORQUED, ADJUSTED BOOM LATCH EYE FOR APPROPRIATE CLEARANCE	Function: metal eye hook attaches to hold Boom to the fuselage. WOULD NOT UNLATCH WHILE FLYING APPROX 320 KTS SLOWED TO 3 0 KTS AND BOOM UNLATCHED.	Fueling may not be possible.	Fueling may not be possible.	Inspection, operation.	4
53	Boom Equipment	Connect tanker to receiver for transferring fuel.	TAILCONE, BOOM	Cause: POPPED RIVETS, REQUIRES PAINT, STRIKE PLATE WORN, BIRDS NEST, CHAFFING, DENTED, CORROSION, ICE SHIELD, CRACKED	Function: protect equipment, reduce drag.	Parts could detach and impact receiver.	Parts could detach and impact receiver.	Inspection, operation.	4

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
54	Boom Equipment	Connect tanker to receiver for transferring fuel.	SHOCK ABSORBER RECOIL ASSY, BOOM	Cause: RECOIL BOLTS, LEAK AND PRESSURE CHECK, SEALANT, BOOM AFT RETAINING BOLTS, SHEAR BOLTS, RECOIL SPRING EXCESSIVELY WORN, CRACKED, SPRING GUARD PLATE	Function: absorbs receiver connect impact. Recoil spring wear ALLOWs ADAPTER TO ROTATE.	Damage to receiver or Boom on subsequent connects.	Damage to receiver or Boom on subsequent connects.	Inspection.	4
55	Boom Equipment	Connect tanker to receiver for transferring fuel.	TELESCOPE CONTROL VALVE, BOOM HYD	Cause: body cracked, seal, BOOM CREEPS, extends RETRACT UNCOMMANDED, GRINDING NOISE , BROKEN CLAMP	Function: controls hydraulic flow to telescope system.	Unable to refuel.	Unable to refuel.	Inspection, operation.	3
56	Boom Equipment	Connect tanker to receiver for transferring fuel.	CABLE - TELESCOPING, BOOM	Cause: wear, lubrication.	Function: pulls telescope in or out.	Unable to refuel, may impact receiver.	Unable to refuel, may impact receiver.	Inspection, operation.	3
57	Boom Equipment	Connect tanker to receiver for transferring fuel.	PUMP CONTROL VALVE, BOOM HYD AND AR PUMP	Cause: LEAK, B-NUT, SOLENOID LOOSE, LEAKING HYD FLUID THRU SOLENOID TO CANNON, INOP.	Function: 4 A/R or body fuel pump control valves control the flow of hydraulic fluid to the 4 A/R fuel pumps. They are electrically operated solenoid valves which open when energized and are normally closed when deenergized. Telescoping Motor	Unable to run pump; if 2 pumps are inop in aft tank, unable to empty tank.	Unable to run pump; if 2 pumps are inop in aft tank, unable to empty tank.	Inspection, operation.	3

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
58	Boom Equipment	Connect tanker to receiver for transferring fuel.	MOTOR, BOOM TELESCOPING (also hoist motor and hoist cable tension motors)	Cause: HOIST LEAKING OUT DRAIN HOLE, B-NUT ON CABLE TENSION MOTOR, TELESCOPE DRIVE MOTOR LEAKING AT SHAFT, PRESSURE TEST, EXTEND PORT FITTING LEAKING ON TELESCOPE MOTOR, B-NUT ON CABLE TENSION MOTOR	Function: reversible hydraulic motor for retracting and extending the boom. Hoist motor raises and lowers Boom. BOOM CHATTERS WHILE IN CONTACT, BOOM CREEPS	Unable to refuel, or lift/lower Boom.	Unable to refuel, or lift/lower Boom.	Inspection, operation.	3
59	Boom Equipment	Connect tanker to receiver for transferring fuel.	RUDDERVATOR LOCK SYSTEM, BOOM	Cause: LOCK PAWL SPRINGS, LATCH PLATE WORN, LOCK ARM, PAWL SPRING BRACKET, PAWL SPRING EYELET WORN, ADJUSTED, LATCH CAM GUIDE ROLLER BOLT STRIPPED, TORQUE TUBE GOUGED Cause: LEAKING, TRACKS BROKEN, GEARBOX MOUNT SCREW BROKEN.	Function: The ruddervator lock system is used to prevent high speed overbalance of the ruddervators while the boom is stowed. Hoisting of the boom will depress the lock quadrant. The lock quadrant is spring-loaded to the unlock position.	FAILED TO GET GOOD RUDDERVATOR LOCK INDICATION.	High speed overbalance could damage Ruddervator.	Inspection, operation.	4
60	Boom Equipment	Connect tanker to receiver for transferring fuel.	TELESCOPING DRIVE ASSY, BOOM	Cause: LEAKING, TRACKS BROKEN, GEARBOX MOUNT SCREW BROKEN.	Function: The motor is connected to the telescoping drive assembly and drives the boom chain. Enables loop to either extend or retract the boom.	Unable to refuel, may impact receiver.	Unable to refuel, may impact receiver.	Inspection, operation.	3

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
61	Boom Equipment	Connect tanker to receiver for transferring fuel.	INNER TUBE, TELESCOPING, BOOM	Cause: SLIDING GLAND HAS CUT SEALS, NOZZLE RETAINER RING REPLACED.	Function: Inner Assy. allows extension.	Ignition hazard on the ground	Ignition hazard on the ground.	Inspection, operation.	3
62	Boom Equipment	Connect tanker to receiver for transferring fuel.	PRESSURE RELIEF VALVE, BOOM HOIST	Cause: LEAKING, RELIEVING EARLY.	Function: provides hydraulic pressure relief for: emergency hoist reservoir; hoist; filter.	Loss of hydraulic fluid	Loss of hydraulic fluid.	Inspection, operation.	3
63	Boom Equipment	Connect tanker to receiver for transferring fuel.	EXTEND & RETRACT FLOW REGULATOR, BOOM HYD	Cause: LEAKING, fails flow test, JAM-NUT, B-NUT.	Function: A 7.5 to 8.5 gpm flow regulator limits the rate of extension of the boom. The second flow regulator, allows a flow rate of 18.5 to 20.5 gpm, limits the rate of retraction.	Ignition & personnel hazard	Ignition & personnel hazard.	Inspection, operation.	3
64	Boom Equipment	Connect tanker to receiver for transferring fuel.	RUDDERVATOR POWER UNIT, BOOM	Cause: FILTER, RIGGED, PCU LEAKING, FLARE CRACKED.	Function: Each ruddervator is individually positioned by a hydraulic power unit, which is a combination control valve and actuator. BOOM IS BINDING. AUDIBLE CLICK HEARD BELOW36DEG & 24DE.	Ignition hazard on the ground	Ignition hazard on the ground.	Inspection, operation.	3

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
65	Boom Equipment	Connect tanker to receiver for transferring fuel.	RUDDERVATOR CONTROL STICK & QUICK DISCONNECT SWITCH	Cause: ADJUSTED, RIGGED, CLEANED AND LUBED DISCONNECT SWITCH, STICK FEEL SPRING.	Function: Control Stick is the handle that moves the Ruddervators. The control stick contains two thumb switches and a trigger switch. The trigger switch is the AIR disconnect switch. (The thumb switches are the interphone and microphone switches.)	Unable to refuel.	Unable to refuel.	Inspection, operation.	4
66	Boom Equipment	Connect tanker to receiver for transferring fuel.	SIGHTING DOOR ACTUATOR, BOOM HYD	Cause: LEAKING, B-NUT, UNION, MANUAL RELEASE BENT or BROKEN, ROD END LOOSE, MISSING RELEASE (sic.) KNOB, UPPER AND LOWER JAMBNUTS, ADJUSTED ACTUATOR ROD END, CORROSION.	Function: The door is closed when the piston is fully extended. The actuator receives hydraulic pressure from the boom operator sighting door control valve.	Ignition & personnel hazard.	Ignition & personnel hazard.	Inspection, operation.	3
67	Boom Equipment	Connect tanker to receiver for transferring fuel.	CABLES ASSY, RVRA, RVRB, RVLA, AND RVLB RUDDERVATOR, BOOM	Cause: RIGGED, CLEANED INSPECTED AND LUBED.	Function: Control cables are the primary means of transmitting motion from the control area to control surfaces of the aircraft. The cables are generally attached to the control drums and quadrants by means of a terminal swaged to the end of the cable.	LAG, BINDING.	Unable to refuel.	Inspection, operation.	4

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TABLE I- X. Boom equipment - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
68	Boom Equipment	Connect tanker to receiver for transferring fuel.	CONTROL VALVE, BOOM HOIST	Cause: ROD END MISSING GREASE FITTING, LEAK.	Function: manually regulate the speed and elevation of the boom. When the boom hoist control lever is held to the RAISE position, system pressure is applied to the raise side of the boom hoist motor.	Operation of the hand pump (Figure 9-2), will supply pressure to the emergency pressure port of the boom hoist control valve.	Ignition & personnel hazard.	Inspection, operation.	3
69	Boom Equipment	Connect tanker to receiver for transferring fuel.	STOWAGE SHOCK ABSORBER, BOOM	Cause: RECOIL ASSY SHEAR BOLTS, DENTED, WHEEL IS BROKEN, LEAKING INTERNALLY, LEAKING AIR, DOUGHNUT WORN AND REQ SEALANT, RE-SEALED RUBBER CUSHION, RING SEAL, RUBBER STOP... TORN LOOSE, WILL NOT HOLD NITROGEN CHARGE, CRACKED.	Function: prevent any damage to the boom and/or the boom chock during stowage by hydraulically decelerating the upward force of the boom during an inadvertent loss of boom control from a lower limit auto disconnect.	Boom or aircraft could be damaged.	Boom or aircraft could be damaged.	Inspection, operation.	4
70	Boom Equipment	Connect tanker to receiver for transferring fuel.	BOOM HOIST EMERGENCY RESERVOIR	Cause: SIGHT GLASS CHIPPED, FILTER, SEAL, AIR BUBBLE IN SIGHTING GAGE, RELIEF VALVE, BOTTOMB-NUT, T-FITTING.	Function: supply hydraulic fluid to raise the boom in an event of a failure to the aircraft pressure supply system.	Ignition and personnel hazard.	Ignition and personnel hazard.	Inspection, operation.	3
71	Boom Equipment	Connect tanker to receiver for transferring fuel.	RESTRICTOR-BOOM HYD, ctl valve filter, hoist filter	Cause: FILTER BOWL, FILTER.	Function: restricts hyd. flow in the sighting door actuator line, or a vent line. Filters prevent particulate damage. No REMIS items for Restrictor.	ignition and personnel hazard, slower ops.	ignition and personnel hazard, slower ops.	Inspection, operation.	3

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TABLE I- XI. FQIS.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
72	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL TANK UNIT, NO. 1/4 RESERVE.	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTL Y SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
73	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL TANK UNIT, NO. I /4 MAIN.	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTL Y SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
74	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL TANK UNIT, NO. 2/3 MAIN.	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTL Y SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4

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TABLE I- XI. FQIS - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
75	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL COMPENSATORNO. 1/4 MAIN.	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTL Y SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
76	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL TANK UNIT, CENTER WING.	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTL Y SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
77	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL COMPENSATOR, NO. 1 RESERVE	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTLY SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
78	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL COMPENSATOR AND TANK UNIT, FWD BODY	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTLY SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4

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TABLE I- XI. FQIS - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
79	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	TANK INTERFACE UNIT (TIU)	Cause: ADJUSTMENT COVER BOTTOM SCREW BACKING IS BROKEN, HARNESSES, TIU CLEANED, ADJUSTMENT, J1 CONNECTOR BACK SHELL, SOLDER SLEEVE, TRIM CAPACITOR, GROUND WIRES BROKEN, J1 CANNON PLUG, loose shell	WILL NOT CALIBRATE, WILL NOT SHOW FULL, INTERMITTENTLY DASHES, BIT CODE 3, FSAS FMAC CH.B INOP BIT CODE 4, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
80	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL COMPENSATOR, NO. 2/3 MAIN	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTLY SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
81	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL COMPENSATOR, CENTER WING	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTLY SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4

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TABLE I- XI. FQIS - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
82	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL COMPENSATOR AND TANK UNIT, AFT BODY	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED.	INTERMITTENTLY SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications	Inspection, operation	4
83	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	FUEL TANK/COMP UNIT UPPER DECK	Cause: LEAKING FUEL THROUGH CANNON LUG, CANNON PLUG RETAINER FITTING, CALIBRATED, SPLICE ON SHORTING PLUG, ADJUSTED	INTERMITTENTLY SHOWING DASHES, CIRCUIT BREAKER POPPED, bad reading.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications.	Inspection, operation.	4
84	Fuel Quantity Indication System (FQIS)	Indicates fuel quantity in each tank. Tank Units indicate either fuel level or compensate for fuel temperature.	IFMP	Cause: PSD-60 test set DISCONNECTED, RESEATED J1 COUPLING MECHANISM, COLD SPLICE, STRAIN RELIEF ON FSB BUS B, SOLDER SLEEVE, BUS COUPLERS RM7532 & RM7533, BIT CODE 7	Function: convert fuel qty to capacitance, & adjust capacitance for temperature, plus other fuel mgmt. Tank won't fill, TOTALIZER INTERMITTENT, display DASHES, BIT CODE 3, FSAS... NO-GO ASTERISK WAS ALWAYS ON SCREEN	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications.	Fuel balance errors, mission duration, results in Loss of CG and Totalizer Indications.	Inspection, operation.	4
85	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level.	Tank Unit/Compensator or Circuit Wiring Open	Cause: Aircraft, tank unit harness, or tank unit interconnect failure.	Individual tank inaccurate or fuel quantity or loss of indication, Totalizer inaccuracy. In operating range fuel indication is less than actual for Tank Unit.	May affect mission duration.	May affect mission duration.	Fuel gages.	4

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TABLE I- XI. FQIS - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
86	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level.	Tank Unit/Compensator or Circuit Wiring Shorted	Cause: Aircraft wiring, tank unit harness, tank unit interconnect failure, physical damage, or gross fuel contamination	Loss of individual tank and totalizer fuel quantity.	May affect mission duration.	May affect mission duration.	Fuel gages.	4
87	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level.	Tank Unit/Compensator or Circuit Wiring Low Resistance	Cause: Aircraft, tank unit harness, tank unit interconnect failure, physical damage, gross fuel contamination, fuel harness connector leakage.	Loss of individual tank and totalizer fuel quantity.	May affect mission duration.	May affect mission duration.	Fuel gages.	4
88	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level.	Tank Unit/Compensator Connector Leakage.	Cause: Deterioration of sealant/barriers in harness and connector.	Fuel leak.	Ignition hazard.	Ignition hazard.	Inspection.	4
89	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	Tank Capacitance Values out of range High	Excessive water in tank or open Hi Z shield in interconnect wiring	Inaccurate or loss of individual and totalizer fuel quantity.	May affect mission duration.	May affect mission duration.	Fuel gages.	4
90	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	IFMP single channel loss	Cause: Internal component failure.	None - Automatic selection of backup channel	If both channels fail, fuel quantity can be estimated.	If both channels fail, fuel quantity can be estimated.	Light indication.	4

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TABLE I- XI. FQIS - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
91	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	IFMP dual channel loss	Cause: Internal component failure or source power loss.	Loss of all fuel quantity indication.	Likely to terminate mission.	Likely to terminate mission.	Fuel gages.	3
92	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	IFMP display driver loss	Cause: Internal component failure.	Loss of individual tank fuel quantity.	May affect mission duration.	May affect mission duration.	Fuel gages.	4
93	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	IFMP Individual Display Failure	Cause: Internal component failure.	Loss of individual tank fuel quantity.	May affect mission duration.	May affect mission duration.	Fuel gages.	4
94	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	FMAC Single Channel Loss	Cause: Internal component failure.	None - Automatic selection of backup channel.	If backup fails, loss of all tank indications.	If backup fails, loss of all tank indications.	Fuel gages.	4
95	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	FMAC Dual channel or complete failure.	Cause: Internal component failure or source power loss.	Loss of all fuel quantity indication.	Likely to affect mission duration.	Likely to affect mission duration.	Fuel gages.	3

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TABLE I- XI. FQIS - Continued.

Item ID	Item Name	Function	Functional Failure	Failure Mode	Local Effects	Next Higher Effects	End Effects	Failure Detection	Severity
96	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	FMAC individual tank circuit failure.	Cause: Internal component failure.	Loss of individual and totalizer fuel quantities.	Fuel flow indication and other tank quantities.	Fuel flow indication and other tank quantities.	Fuel gages.	4
97	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	TIA Tank Unit Trimmer Circuit Open	Cause: Capacitor or interconnect failure.	Individual tank inaccurate fuel quantity, Totalizer inaccuracy.	May affect mission duration.	May affect mission duration.	Fuel gages.	4
98	Fuel Quantity Indication System (FQIS)	Tank Unit/Compensator - senses capacitance for a given fuel level. Integrated Fuel Management Panel - The IFMP provides 10 individual tank displays and a total fuel quantity display.	TIA Tank Unit Trimmer Circuit Shorted	Cause: Capacitor or interconnect failure.	Loss of individual tank and totalizer fuel quantity.	May affect mission duration	May affect mission duration.	Fuel gages.	4

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CONCLUDING MATERIAL

Custodian:

Army - AV

Navy - AS

Air Force - 11

Preparing activity:

Air Force - 11

(Project SESS-2021-004)

Review activity:

Army - CR

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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